

Asymmetric Weighted Earliness-Tardiness: Scheduling with a Nonrestrictive Common Due Date

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Results at a Glance

Part I: Intractability. Positive-integer asymmetric AWET at the boundary nonrestrictive due date is strongly NP-complete through a unary-polynomial exact-cover compiler.

Part II: General approximation. An anchored SDP and deterministic marginal threshold give a polynomial-time $(3 + 2\sqrt{2} + \varepsilon)$ -approximation for unrestricted AWET.

Part III: Antithetical ordering. Exact reversal is weakly NP-complete; a two-resource recurrence gives a pseudopolynomial exact algorithm.

Part IV: Separating-tree analysis. For fixed total ratio-order refinements whose ratio permutation is separable, exact separating-tree convolution and geometric trimming give an FPTAS.

Part V: Global synthesis. The concluding Part integrates the exact-hardness, approximation, weak-hardness, and separating-tree regimes.

Organizing thesis. Once common-due-date schedule geometry is reduced to the canonical early-set objective, the computational regime is governed by the structure available in the two Smith-ratio orders. Unrestricted order permits strong NP-completeness and still supports constant-factor approximation; exact reversal produces a weakly hard load-balancing core; and recursive direct/skew compatibility exposes the bounded interface needed for an FPTAS. The four technical Parts are regimes of one quadratic model.

Four unifying mechanisms.

Canonicalization: canonical early-set choice $\rightarrow$ quadratic objective.	Certificates: prefix/load squares and SDP pseudo-marginals.
Interfaces: direct/skew blocks $\rightarrow$ four additive coordinates.	Controlled loss: scale gaps, thresholds, and geometric boxes.

Abstract

Single-machine asymmetric weighted earliness–tardiness scheduling asks how to sequence jobs around a common synchronization date when early and late completion incur unrelated job-dependent penalties. At the boundary nonrestrictive date $d = \sum_j p_j$, a compact V-shaped schedule reduces the continuous-time problem to a quadratic choice of a nonempty early set. We establish four complementary results for this model. First, the positive-integer problem is strongly NP-complete by a unary-polynomial reduction from Restricted Exact Cover by 3-Sets. Second, unrestricted AWET admits a polynomial-time $(3 + 2\sqrt{2} + \varepsilon)$ -approximation based on an anchored semidefinite relaxation and deterministic marginal thresholding. Third, when the earliness and tardiness ratio orders are strict reversals, the problem is weakly NP-complete but has an exact two-resource pseudopolynomial dynamic program. Fourth, for fixed total refinements whose ratio permutation is separable, an exact separating-tree recurrence and coordinated geometric trimming yield an FPTAS. The proofs use different manifestations of the same canonical objective: scale-separated prefix penalties, positive-semidefinite minimum-kernel covariance, a dominant completed load square, and a bounded four-coordinate decomposition interface. Together, the results show that the decisive issue is not merely whether the two ratio orders agree, but whether their interaction can be controlled by a global certificate or compressed into a bounded constructive interface.

Keywords: scheduling; common due date; asymmetric earliness and tardiness; strong NP-completeness; approximation algorithm; semidefinite programming; antithetical ordering; separable permutation; separating tree; FPTAS.

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1 Purpose, audience, and reading paths

This document contains four full theorem chains: unrestricted strong NP-completeness, a general constant-factor approximation algorithm, weak NP-completeness under strict antithetical ordering, and an FPTAS for separable ratio permutations.

The intended audience includes researchers and graduate students in scheduling, combinatorial optimization, approximation algorithms, and computational complexity. Familiarity with NP-completeness, dynamic programming, and elementary semidefinite programming is helpful but not required for the main proof architecture. Definitions precede the mechanisms that use them, and the longer reductions and algorithms include intuition, proof checkpoints, and worked arithmetic.

There are five natural reading paths.

Results path Read this front matter, the principal theorem statements in Parts I–IV, and Part V.

Hardness path Read Part I for strong NP-completeness and Part III for the distinct completed-square reduction at exact ratio reversal.

Algorithmic path Read Part II for the unrestricted constant-factor algorithm and Part IV for the separating-tree FPTAS. These Parts use different certificates: positive-semidefinite covariance in Part II and bounded additive interfaces in Part IV.

Course path Read the canonical objective in Section 2, one hardness chain (Part I or III), one algorithmic chain (Part II or IV), the corresponding worked example, and Part V. This route supports a graduate module without requiring every arithmetic appendix.

Implementation path

Read Sections 12–20 for the anchored-SDP algorithm, Theorem III.11 for the two-resource recurrence, [Sections 36](#) and [37](#) for separating-tree tables and trimming, and the worked examples for exact data flow and reconstruction.

Reader-facing box contract. The colored boxes have nonoverlapping roles. **How to read** gives navigation or a safe skip; **Intuition** explains mechanism; **Proof checkpoint** records what has been established and what remains; **How to use** gives an algorithmic instruction; and **Arithmetic guide** gives a reproducibility rule. A box is used only when it performs one of these functions rather than paraphrasing the adjacent prose.

Terminological convention. A job is *early/on time* when it completes no later than the common due date and *tardy* otherwise. The superscript L labels the tardiness channel. The term *strict antithetical* means that, after jobs are indexed by increasing early ratio, their tardiness ratios strictly decrease. After fixed total tie refinements, the *inversion graph* joins exactly the job pairs whose two ratio orders disagree. The resulting ratio permutation is *separable* precisely when this inversion graph is a cograph.

Style convention. Numbered structural units are capitalized as Parts and Appendices. Compound modifiers are written consistently as *positive-integer*, *common-due-date*, *ratio-order*, *finite-precision*, and *minimum-kernel*. A slash is reserved for the intentional combined category *early/on time*; elsewhere the prose uses ordinary conjunctions.

2 Problem, canonical formulation, and literature context

Why the problem matters

Many production and service systems are organized around a shared synchronization date rather than independent job-specific deadlines. Components may need to be released together to assembly, orders may be coordinated for a common delivery, or preparatory activities may have to be completed near the start of a downstream stage. Finishing too early can create holding, storage, financing, obsolescence, or premature-delivery costs; finishing too late can create service failures, contractual penalties, lost production, or disruption to dependent work. The asymmetric model allows these consequences to differ both from job to job and between the two sides of the target date. It is deliberately stylized, but it isolates the sequencing tradeoff created by heterogeneous early and tardy consequences; broad discussions of these interpretations appear in the common-due-date surveys [3, 9, 18].

The boundary choice $d = \sum_j p_j$ is especially informative. It gives enough aggregate time to process every job by the due date in a compact schedule starting at zero, but it does not make the optimization trivial. A scheduler may still choose to place some processing after d when the resulting reduction in earliness cost elsewhere outweighs the tardiness incurred. Because total processing is fixed, every unit placed after the due date reappears as an equal initial offset before the early block. This conservation relationship leaves a genuine combinatorial question—which jobs should lie on each side—while removing irrelevant calendar slack.

That balance makes the model a useful test case for complexity and approximation. It is simple enough that exchange arguments determine the optimal order on each side of the due date, reducing the continuous-time problem to one canonical early-set objective. Yet the remaining set choice is rich enough to display several sharply different regimes: strong NP-completeness without order restrictions, a universal constant-factor approximation, weak NP-completeness under exact reversal of the two ratio orders, and an FPTAS when the ratio interaction has a separable recursive decomposition.

Accordingly, the classifications below answer three complementary questions. What prevents exact polynomial-time optimization in the general model? What performance guarantee remains possible without structural assumptions? Which recognizable forms of order structure restore substantially stronger algorithms? The answers identify not only complexity boundaries but also the mathematical objects—global certificates and bounded interaction interfaces—that make those boundaries visible.

An AWET instance consists of jobs $j \in \mathcal{J}$ with positive processing times p_j , positive earliness weights w_j^E , and positive tardiness weights w_j^L . All jobs are available at time zero and are processed nonpreemptively on one machine. The common due date is

$$d = P := \sum_{j \in \mathcal{J}} p_j,$$

and a schedule with completion times C_j has objective

$$\sum_{j \in \mathcal{J}} \left(w_j^E (d - C_j)^+ + w_j^L (C_j - d)^+ \right).$$

The equality $d = \sum_j p_j$ is the *boundary nonrestrictive due date*. It permits a compact schedule to finish exactly at d , but it does not eliminate the early–tardy assignment decision: placing processing after d creates the same amount of initial offset before the early block.

This conservation identity turns schedule geometry into the quadratic set objective used throughout the tutorial.

Write

$$r_j = \frac{w_j^E}{p_j}, \quad s_j = \frac{w_j^L}{p_j}.$$

The canonical-schedule theorem says that an optimum may be chosen compact and V-shaped. Consequently, for every nonempty early set E and tardy complement $T = \mathcal{J} \setminus E$, the associated due-date-aligned canonical schedule has cost

$$\Phi(E) = \sum_{\{i,j\} \subseteq E} p_i p_j \min\{r_i, r_j\} + \sum_{\{i,j\} \subseteq T} p_i p_j \min\{s_i, s_j\} + \sum_{i \in T} p_i^2 s_i. \quad (2.1)$$

Moreover, the global optimum is $\text{OPT} = \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E)$. This formula is the common backbone of all four technical Parts. Part I compiles an exact-cover signal into it; Part II relaxes and rounds it; Part III shapes it into an equal-load square; and Part IV decomposes it across separating-tree blocks.

Why the normalization $d = \sum_j p_j$ is used. Put $P = \sum_j p_j$. The familiar condition $d \geq P$ is a simple sufficient condition for the fixed common due date to be nonrestrictive [15]: every compact V-shaped candidate has total processing span P , and subtracting $d - P$ from both the due date and all completion times preserves every earliness and tardiness value while keeping all start times nonnegative. Thus every instance satisfying $d \geq P$ is equivalent, for the present objective, to the normalized case $d = P$.

The condition $d \geq P$ is sufficient rather than necessary. Let $\Phi(E)$ be the canonical early-set objective in (2.1). A sharper instance-dependent sufficient threshold is

$$d_{\text{nr}}(I) := \min \left\{ \sum_{j \in E} p_j : E \in \arg \min_{\emptyset \neq A \subseteq \mathcal{J}} \Phi(A) \right\}.$$

If $d \geq d_{\text{nr}}(I)$, at least one optimal canonical early arm fits before time d , so the due date does not change the unconstrained optimum. This criterion is less concise because it depends on the instance and on an optimal side assignment; the uniform normalization $d = P$ avoids carrying that threshold through every theorem. When the due date is early enough to constrain placement of the V-shaped schedule, the problem is in the restrictive regime studied by Hall, Kubiak, and Sethi [11].

2.1 Common-due-date structure and exact complexity

Early common-due-date models established the V-shaped structural viewpoint and the importance of distinguishing nonrestrictive and restrictive due-date regimes. Kanet studied total deviation about a common due date, while Panwalkar, Smith, and Seidmann treated common due-date assignment [14, 17]. Hall and Posner analyzed weighted symmetric earliness–tardiness at a nonrestrictive common due date; Hall, Kubiak, and Sethi treated the companion restrictive setting [11, 12]. Broad reviews include Baker and Scudder, Gordon, Proth, and Chu, and Rolim and Nagano [3, 9, 18].

For the weighted symmetric nonrestrictive model, Hall and Posner proved weak NP-hardness and gave a pseudopolynomial dynamic program [12]. Hoogeveen and van de Velde recorded the strong-complexity boundary for general asymmetric weights as unresolved [13]. Part I resolves that published question in the exact formulation used here: positive-integer asymmetric weighted earliness–tardiness scheduling at the boundary due date $d = \sum_j p_j$ is strongly NP-complete.

2.2 Pseudopolynomial algorithms, FPTAS results, and approximation

Weak hardness in common-due-date scheduling has long coexisted with pseudopolynomial algorithms and approximation schemes in symmetric or otherwise structured settings. The Hall–Posner recurrence is an early example; later FPTAS results include the schemes of Kovalyov and Kubiak and Kellerer, Rustogi, and Strusevich [12, 15, 16]. These positive results depend on structure absent from unrestricted asymmetric data.

Part II therefore addresses a different boundary. It permits unrelated positive earliness and tardiness weights and obtains a universal constant-factor guarantee rather than an FPTAS. Part III then identifies a strict reversed-order class whose hardness is weak and whose state space is pseudopolynomial, while Part IV obtains an FPTAS under a recursive order-decomposition condition.

2.3 Separable-permutation decomposition and semidefinite methodology

For fixed total refinements of the two ratio orders, Part IV represents their interaction by a ratio permutation. Separable permutations admit recursive direct and skew sums and can be recognized, with a separating tree recovered, in linear time [4]. Equivalently, their inversion graphs are cographs, whose union/join decomposition is represented by the corresponding cotree [5, 10].

Part II uses semidefinite optimization in a different way. Polynomial-time weak optimization and rational bit-model control are standard parts of semidefinite programming [1, 19, 20]. Semidefinite relaxations are also a classical source of approximation algorithms in combinatorial optimization [8]. The rounding here is not hyperplane rounding: it uses an anchored local-consistency relaxation, deterministic marginal thresholding, and a minimum-kernel covariance inequality tailored to the scheduling objective.

Provenance note. The tutorial proves every stated result. Historical priority is claimed only where the comparison with published literature is documented explicitly; otherwise the theorem and its scope are stated without a priority claim over unpublished or independent work.

3 Contribution map and scope

Part	Closest established boundary	New result here	Key mechanism	Not implied
I	Symmetric nonrestrictive weights were weakly NP-hard and pseudopolynomially solvable; published tables left the general asymmetric strong boundary open.	Positive-integer AWET at $d = \sum_j p_j$ is strongly NP-complete and unary NP-hard.	Effective-atom profile compression, separated prefix-square signal, and an anchor-coupled minimum-kernel load lock.	No APX-hardness or no-PTAS conclusion follows from strong NP-completeness alone.
II	FPTAS results were available in symmetric or structured common-due-date settings.	Unrestricted asymmetric AWET has a polynomial-time $(3 + 2\sqrt{2} + \varepsilon)$ -approximation.	Anchored SDP, deterministic marginal thresholding, local early-side consistency, and PSD tardy-side covariance payment.	The theorem does not establish a PTAS or optimality of the constant.
III	The unrestricted problem is strongly NP-complete, but exact reversal had not been classified by the Part I construction.	Strict antithetical AWET is weakly NP-complete and has an $O(nW_E P)$ pseudopolynomial recurrence.	Binary pair tagging, a dominant completed load square, and a uniform residual budget.	Weak hardness does not rule out an FPTAS; exact reversal is separable and belongs to Part IV's structured class after fixed refinements.
IV	Separable permutations have recursive direct/skew decompositions and linear-time recognition.	For fixed total refinements of the two ratio orders, if the resulting ratio permutation is separable, AWET admits an FPTAS.	A four-coordinate nonnegative bilinear interface, exact separating-tree convolution, and coordinated geometric trimming.	Separability is sufficient for this algorithm, not claimed necessary for every approximation scheme.

For the Part III running-time bound, $n = |\mathcal{J}|$ is the number of jobs, $W_E = \sum_{j \in \mathcal{J}} w_j^E$ is the total earliness weight, and $P = \sum_{j \in \mathcal{J}} p_j$ is the total processing time.

Why these results belong together

Each technical Part begins from the same canonical quadratic objective but asks a different question of it. Part I asks how much information the objective can encode; Part II asks how much of its value can be certified by a relaxation; Part III asks what exact reversal does to both hardness and state-space dimension; and Part IV asks when order decomposition compresses all future interactions into a bounded interface. This shared starting point is the

main reason the four results are best read together.

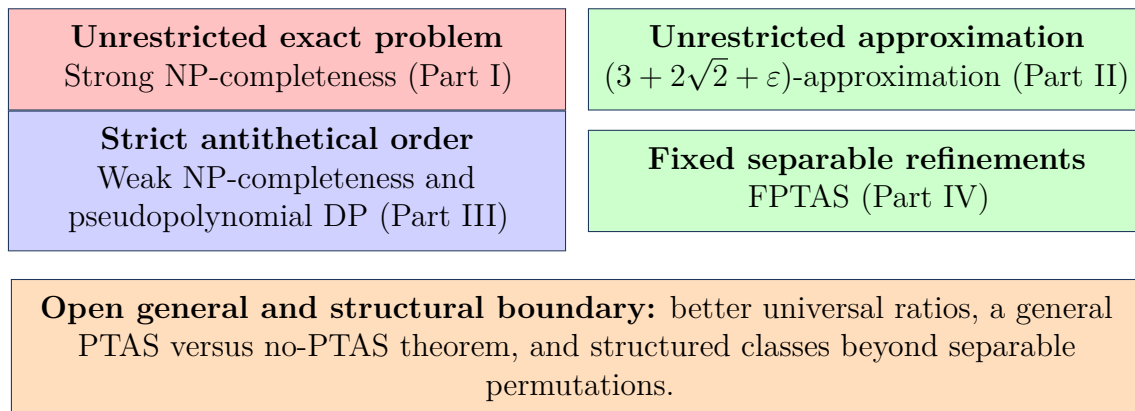


Figure 1: Front-matter landscape: four proved regimes of the canonical early-set objective and the principal open boundary. As in the combined landscape, red marks exact intractability, green marks approximation algorithms, blue marks the mixed weak-hardness/pseudopolynomial regime, and amber marks unresolved boundaries.

Dependency map

Shared formal prerequisite: the canonical compact schedule and early-set objective (2.1).

Part	Logical dependency chain from the shared prerequisite
I	$\implies \text{RXC3} \rightarrow \text{RPE}$ (Lemmas I.8–I.9) $\implies$ coded blocks and scale separation $\implies$ load lock (Lemma I.15) $\implies$ master decomposition (Lemma I.16) $\implies$ Theorem I.2.
II	$\implies$ anchored SDP (Definition II.9) $\implies$ early and tardy payment (Lemmas II.12–II.16) $\implies$ rational realization (Lemma II.20) $\implies$ threshold algorithm $\implies$ Theorem II.3.
III	$\implies$ binary tagging (Lemma III.3) $\implies$ cost identity (Lemma III.6) and residual bound $\implies$ anchor and gap separation $\implies$ weak NP-completeness; independently, the two-resource recurrence gives Theorem III.11.
IV	$\implies$ fixed refinements and ordered separating tree $\implies$ bilinear convolution (Lemma IV.5) $\implies$ exact DP (Theorem IV.7) $\implies$ representative invariant $\implies$ FPTAS (Theorem IV.11).

Figure 2: Logical dependency map. The arrows denote formal dependence, not chronology. [Lemma I.14](#) proves Part I’s scale separation; [Appendix A](#) supplies the coefficient algebra, and [Appendix B](#) the full yes/no arithmetic.

Notation at a glance

Symbol	Meaning	Scope	First definition
p_j, w_j^E, w_j^L	processing time, earliness weight, and tardiness weight	global	Section 2
d	common due date; in the main results, the boundary nonrestrictive date $d = \sum_j p_j$	global	Section 2
r_j, s_j	early and tardiness Smith ratios	global	Section 2
E, T (or $\mathcal{T}$)	early/on-time and tardy job sets	global; Part I uses $\mathcal{T}$ to avoid the scale T	(2.1)
N, G, M, T, H, L	scale hierarchy, load-lock parameters, or kernel levels	reset in Part I	notation reset before Section 4
W_E	total earliness weight $\sum_{j \in \mathcal{J}} w_j^E$	Parts III–IV	Section 3
A, B, Γ, R	master constant, decision threshold, residual budget, and residual	Part I	Section 8.1.9
$\mathcal{A}, \mathcal{B}, \mathcal{U}$	SDP early-pair, tardy-pair, and self-cost accounts	Part II	(15.3)
e_i, t_i, e_{ij}, t_{ij}	SDP pseudo-marginals and pseudo-joint masses	Part II	Section 14
Q, S, U, M, D	tagged total, selected load, residual budget, dominant scale, and due-date scale	Part III	notation reset before Section 25
$\prec_E, \prec_L, \pi$	fixed ratio-order refinements and ratio permutation	Part IV	Section 33
$z_M = (e_M, t_M, u_M, v_M)$	additive separating-tree signature	Part IV	Definition IV.4
λ, H	geometric trimming ratio and separating-tree height	Part IV	Section 37

How to read this step. The contribution table separates theorem novelty from proof-technique novelty. The dependency map shows which formal statements are needed, while the notation table shows where symbols are reset. Readers may now enter any Part without treating the document as one linear 200-page proof.

Part I

Intractability

Strong NP-Completeness of Single-Machine Asymmetric Weighted Earliness–Tardiness Scheduling with a Common Due Date

Part synopsis. This Part proves strong, and hence unary, NP-completeness of positive-integer AWET. Restricted Exact Cover by 3-Sets is first converted to regular prefix exactness. Each source set is then represented by an eight-job block with exactly two balanced states. Separated early-ratio bands turn those states into squared prefix-cover errors; an anchor-coupled minimum-kernel lock enforces the block loads; and integer tardiness offsets correct the remaining logical fields. The final objective is an exact constant plus a dominant prefix penalty and a bounded residual.

Part positioning. Established boundary. The symmetric-weight nonrestrictive model was weakly NP-hard and pseudopolynomially solvable, while the published strong-complexity boundary for general asymmetric weights remained open. **Result.** Positive-integer AWET at the boundary due date $d = \sum_j p_j$ is strongly NP-complete by a unary-polynomial reduction. **Technique.** Effective-atom profile compression and separated ratio gaps carry the logical signal, while an anchor-coupled minimum-kernel quadratic enforces all block loads simultaneously.

Notation reset. Here n is the size of the regular exact-cover and prefix-exactness source, $N = n + 1$ controls the scale hierarchy, and m denotes the number of constructed jobs when needed. The global job data remain p_j, w_j^E, w_j^L , with ratios $r_j = w_j^E/p_j$ and $s_j = w_j^L/p_j$. Scale symbols such as G, M, T, H , and L are local to this Part.

Proof contract.

- **Input:** an n -set Restricted Exact Cover by 3-Sets instance.
- **Output:** an AWET instance with eight state jobs per source set and one anchor, together with a threshold B .
- **Forward invariant:** an exact cover gives balanced block states, zero prefix penalty, and a canonical schedule of cost at most B .
- **Reverse invariant:** every schedule of cost at most B can be replaced without higher cost by an optimal canonical schedule; that canonical schedule has an early anchor, satisfies the load lock, decodes to block states, and annihilates all prefix-cover errors.
- **Size obligation:** every constructed integer is bounded by a fixed polynomial (the written audit uses at most N^{650}), so unary length is polynomial.

Proof navigation for Part I

Part I has two coordinated lanes.

Lane	Main chain	Audit support
Logical lane	RXC3 $\rightarrow$ prefix equations $\rightarrow$ two coded states per source set $\rightarrow$ zero prefix error $\rightarrow$ exact cover	Appendix A tracks the canonical-objective coefficients; Appendix B instantiates both directions.
Numerical lane	scale hierarchy $\rightarrow$ ratio-order separation $\rightarrow$ anchor-coupled load lock $\rightarrow$ residual domination $\rightarrow$ integer decision gap	Lemma I.14 proves the separation inequalities; Appendix B reproduces every scale, field, and threshold value.

The two lanes meet in the master identity. A theorem-level reader may follow the logical lane and consult only the stated numerical lemmas; an arithmetic-audit reader may reproduce the supporting appendices without changing the logical proof. The two fully instantiated

numerical companions are [Sections B.1](#) and [B.2](#): the first follows the forward construction through the 49-job schedule and threshold, while the second repeats the compiler for a closely related no-instance and audits the reverse contradiction.

4 The AWET problem and historical context

4.1 Formal definition

A schedule specifies a permutation of the jobs and a nonnegative start time for each job, subject to nonoverlap and nonpreemption. Because there is only one machine and no release dates, the only combinatorial decisions are the order, the placement of idle time, and which completion lies on each side of the common due date.

Definition I.1 (Asymmetric weighted earliness–tardiness scheduling). An instance of AWET consists of a set $\mathcal{J} = \{1, \dots, m\}$ of jobs. Job j has an integer processing time $p_j > 0$, an integer earliness weight $w_j^E > 0$, and an integer tardiness weight $w_j^L > 0$. The jobs must be processed nonpreemptively on one continuously available machine, which can process at most one job at a time. The machine is available from time zero, and all start times must be nonnegative. The common due date is fixed at

$$d = \sum_{j \in \mathcal{J}} p_j.$$

If C_j is the completion time of job j , define

$$E_j = (d - C_j)^+, \quad T_j = (C_j - d)^+,$$

where $x^+ = \max\{x, 0\}$. The cost of a schedule is

$$F = \sum_{j \in \mathcal{J}} (w_j^E E_j + w_j^L T_j).$$

The decision version of AWET additionally contains an integer bound B and asks whether a feasible schedule of cost at most B exists.

The terminology “asymmetric” refers to the absence of any required relation between w_j^E and w_j^L . A job with $C_j < d$ is *early*; a job with $C_j = d$ has zero deviation and will be

grouped with the early jobs; and a job with $C_j > d$ is *tardy*. We use AWET throughout for both the optimization problem and, when a bound is present, its decision version. Baker and Scudder [3] provide a broad review of single-machine sequencing with earliness and tardiness penalties.

The choice $d = \sum_j p_j$ is important. It equals the length of a compact schedule containing all jobs. Thus, if a compact sequence starts at time zero, its final job completes exactly at d ; if some jobs are placed after d , the same total amount of processing must appear as an initial offset before the early block. This conservation identity is used repeatedly in the set formulation and in the numerical schedule.

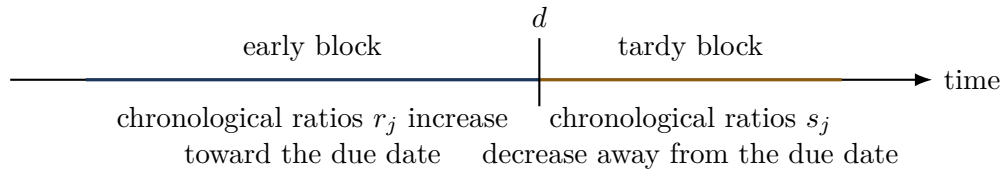


Figure 3: Geometry of a compact canonical schedule. The early block ends at the common due date and the tardy block starts immediately afterward.

For complexity terminology, *strong NP-hardness* means that NP-hardness persists even when every numerical value is bounded by a polynomial in the combinatorial size of the source instance. A reduction with that property also remains polynomial when the target numbers are written in unary. The proof below therefore establishes both strong NP-completeness and unary NP-hardness.

Hall and Posner [12] considered the symmetric-weight case $w_j^E = w_j^L$ with a nonrestrictive common due date. They proved weak NP-hardness and gave a pseudopolynomial dynamic program. Hoogeveen and van de Velde [13] explicitly identified strong NP-hardness of the common-due-date problem with general weights as unresolved; their complexity table lists the general case as open. This tutorial resolves the published strong-complexity question for positive-integer asymmetric weighted earliness–tardiness scheduling at the boundary due date $d = \sum_j p_j$.

4.2 Main result

Theorem I.2. *The decision version of AWET is strongly NP-complete. In particular, AWET is unary NP-hard.*

Membership in NP follows from [lemma I.3](#) and [proposition I.4](#). If any schedule has cost at most the decision bound, then a globally optimal canonical schedule also has cost at most that bound. A certificate may specify its nonempty early set and the canonical orders on the two sides of d ; ratio comparisons use cross multiplication, and all start times, completion times, and objective terms are obtained by polynomially many integer additions and multiplications. The remainder of Part I proves strong NP-hardness.

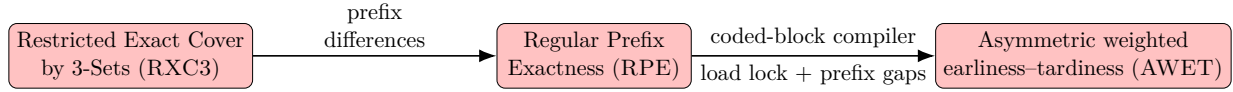


Figure 4: Reduction chain. Both transformations are polynomial. Every numerical value in the second transformation is bounded by a fixed polynomial in the RPE instance size.

5 Overview of the proof

The proof follows the two reductions in Figure 4. Reduction I rewrites Restricted Exact Cover by 3-Sets as an equivalent system of prefix equations. Reduction II compiles that system into a scheduling instance. Each source set becomes an eight-job block consisting of three occurrence pairs and one controller pair. The two balanced orientations of the block encode one binary variable x_v .

The scheduling construction then performs three tasks. First, an effective atom summarizes each complete two-job pair for cross-block calculations; its position places occurrence information in an element-specific early-ratio band, and the gaps between bands produce squared prefix errors. Second, a field correction centers those squares at the required prefix values. Third, an anchor and a positive-definite minimum-kernel penalty force every block of an optimal canonical assignment to use one of its two balanced orientations. The decision threshold then separates zero prefix error from every nonzero prefix error. Keeping these tasks separate is the main organizational principle of the proof.

Main-line map: logic and arithmetic.

Logic	source exact cover $\rightarrow$ prefix equalities $\rightarrow$ block spins $\rightarrow$ signed prefix totals $\rightarrow$ zero squared error.
Arithmetic	positional digits $\rightarrow$ separated ratio bands $\rightarrow$ positive-definite load lock $\rightarrow$ bounded within-band residual $\rightarrow$ strict threshold gap.

Lemma I.14 certifies the band and gap inequalities; Appendix A certifies the field and coupling algebra; Appendix B certifies one full yes-instance and one full no-instance.

6 A canonical set formulation of AWET

For each job, define its earliness and tardiness Smith ratios by

$$r_j = \frac{w_j^E}{p_j}, \quad s_j = \frac{w_j^L}{p_j}.$$

These ratios determine the canonical order once the early-job set is fixed. Integrality is not needed in this section; it is imposed by the reduction later.

Lemma I.3 (Compact canonical schedule). *Every AWET instance has an optimal schedule with the following properties:*

- (i) *there is no idle time between consecutive jobs;*
- (ii) *at least one job completes exactly at d ;*
- (iii) *read outward from d , the jobs completing no later than d are in nonincreasing order of r_j ;*
- (iv) *read outward from d , the tardy jobs are in nonincreasing order of s_j .*

Proof. We establish the four properties in order.

Step 1: remove internal idle time. Choose an optimal schedule with minimum idle time between its first and last jobs. An idle interval wholly before d can be closed by shifting the block immediately to its left rightward; every affected job remains early and becomes no more costly. The symmetric left shift closes an interval wholly after d . If an interval contains d , shift the adjacent blocks toward d . Each operation weakly decreases cost, so an optimal schedule with minimum internal idle time has none. This proves (i).

Step 2: put a completion at the due date. Fix a compact job sequence and let its first job start at $z \geq 0$. Because the sequence has total length d , each completion time has the form $C_j(z) = z + \hat{C}_j$. The total cost is a convex piecewise-linear function of z , being a sum of terms

$$w_j^E(d - C_j(z))^+ + w_j^L(C_j(z) - d)^+.$$

A minimum on $[0, \infty)$ may be chosen at $z = 0$ or at a breakpoint. At $z = 0$ the last job completes at d ; at a breakpoint some job completes at d . An endpoint of any flat minimum interval has the same property. This proves (ii).

Step 3: order the early jobs. Let adjacent early job i be farther from d than adjacent early job j . In the order i, j , the order-dependent contribution is

$$w_i^E p_j = p_i p_j r_i;$$

in the reverse order it is $p_i p_j r_j$. The smaller ratio must therefore be farther from d . Repeated adjacent exchanges prove (iii).

Step 4: order the tardy jobs. Let tardy job i be closer to d than tardy job j . In the order i, j , the order-dependent contribution is

$$w_j^L p_i = p_i p_j s_j;$$

in the reverse order it is $p_i p_j s_i$. Again the smaller ratio belongs farther from d , and adjacent exchanges prove (iv). $\square$

Intuition. The lemma removes the continuous part of the scheduling decision. Once the early-job set is chosen, both sides are packed around d and ordered by their ratios. The reduction may therefore optimize over job subsets rather than over arbitrary start times.

Let E be the set of jobs completing no later than d . To distinguish the tardy-job set from the numerical scale T introduced in [Section 8](#), write $\mathcal{T} = \mathcal{J} \setminus E$. For a prescribed E , the ratio rules determine a due-date-aligned canonical schedule up to ties. The structural lemma guarantees that at least one global optimum is of this form.

Proposition I.4 (Canonical set objective). *For every nonempty $E \subseteq \mathcal{J}$, put $\mathcal{T} = \mathcal{J} \setminus E$. Arrange the jobs of E before d in the early ratio order of [lemma I.3](#), with the last early job completing at d , and arrange the jobs of $\mathcal{T}$ immediately after d in the corresponding tardy ratio order. The cost of this feasible due-date-aligned canonical schedule is*

$$\begin{aligned} \Phi(E) = & \sum_{\{i,j\} \subseteq E} p_i p_j \min\{r_i, r_j\} \\ & + \sum_{\{i,j\} \subseteq \mathcal{T}} p_i p_j \min\{s_i, s_j\} + \sum_{i \in \mathcal{T}} p_i^2 s_i. \end{aligned} \tag{SO}$$

Moreover,

$$\text{OPT} = \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E). \tag{6.1}$$

Proof. Fix a nonempty set E and put $\mathcal{T} = \mathcal{J} \setminus E$.

Feasibility. Place the canonical early block so that its last job completes at d , followed immediately by the canonical tardy block. The first start time is

$$d - \sum_{i \in E} p_i = \sum_{i \in \mathcal{T}} p_i \geq 0,$$

because $d = \sum_{j \in \mathcal{J}} p_j$. Thus this due-date-aligned canonical schedule is feasible.

Its cost. For an unordered pair $\{i, j\} \subseteq E$, the processing time of the job closer to d contributes to the earliness of the job farther from d . The exchange order places the smaller early ratio farther away, so the pair contributes

$$p_i p_j \min\{r_i, r_j\}.$$

There is no early self term because a job's own processing time ends at its completion. An unordered tardy pair similarly contributes $p_i p_j \min\{s_i, s_j\}$. Every tardy job also includes its own processing time in its tardiness, giving the self term

$$w_i^L p_i = p_i^2 s_i.$$

Summing these contributions proves (SO).

Global optimality identity. Let $V = \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E)$. Every schedule just constructed is feasible, so $\text{OPT} \leq V$. Conversely, lemma I.3 supplies a globally optimal schedule with no internal idle time, at least one completion exactly at d , and the two canonical ratio orders. Let E^* be its jobs completing no later than d . Because a completion occurs at d and there is no internal idle time, the early block ends at d and the tardy block starts immediately afterward; no job straddles the due date. The preceding pair calculation therefore gives the schedule's cost as $\Phi(E^*)$. Hence

$$\text{OPT} = \Phi(E^*) \geq V.$$

Together with $\text{OPT} \leq V$, this proves (6.1). □

Proof checkpoint. Equation (SO) is the exact cost of each due-date-aligned canonical schedule, and (6.1) says that minimizing these values gives the true scheduling optimum.

The first term depends only on which jobs are early, and the other two only on which jobs are tardy. This is the point at which the scheduling instance becomes a discrete quadratic optimization problem.

Encode the early–tardy choice by a spin

$$c_i = \begin{cases} +1, & i \in E, \\ -1, & i \in \mathcal{T}. \end{cases}$$

Then $\mathbf{1}\{i \in E\} = (1 + c_i)/2$ and $\mathbf{1}\{i \in \mathcal{T}\} = (1 - c_i)/2$. A product of two indicators expands into a constant, two linear terms, and one quadratic term. Below, these are called the constant term, logical fields, and logical couplings.

7 Reduction I: RXC3 to regular prefix exactness

Gonzalez [7] proved that Restricted Exact Cover by 3-Sets is NP-complete.

Definition I.5 (RXC3). An instance consists of a universe $X = \{e_1, \dots, e_n\}$ and a family $\mathcal{S} = \{S_1, \dots, S_n\}$. Every S_v has three elements, and every element belongs to three sets. Its incidence matrix is

$$A^{\text{inc}} = (a_{kv}) \in \{0, 1\}^{n \times n}, \quad a_{kv} = \begin{cases} 1, & e_k \in S_v, \\ 0, & e_k \notin S_v. \end{cases}$$

Rows represent elements and columns represent sets. A vector $x \in \{0, 1\}^n$ selects column v when $x_v = 1$. The selected sets form an *exact cover* exactly when

$$A^{\text{inc}}x = \mathbf{1}, \quad \text{equivalently} \quad \sum_{v=1}^n a_{kv}x_v = 1 \quad (k = 1, \dots, n).$$

The decision question is whether such a vector exists.

Each row and each column contains three ones, so counting incidences gives $|\mathcal{S}| = |X|$; we use n for both. The intermediate problem keeps the same feasible vectors but replaces the individual row equations by cumulative prefix equations. This form matches the intervals that will arise on the scheduling ratio axis.

Definition I.6 (Regular Prefix Exactness). An instance of RPE is an ordered universe $X = \{e_1, \dots, e_n\}$ together with the same three-uniform, three-regular family $\mathcal{S} = \{S_1, \dots, S_n\}$. Define

$$d_{v,k} = \sum_{\ell=1}^k a_{\ell v} = |S_v \cap \{e_1, \dots, e_k\}|, \quad C_k(x) = \sum_{v=1}^n d_{v,k} x_v,$$

RPE asks whether there is $x \in \{0, 1\}^n$ such that

$$C_k(x) = k \quad (k = 1, \dots, n).$$

Here $d_{v,k}$ counts the entries equal to one among the first k rows of column v , and $C_k(x)$ counts selected incidences in those rows. Thus $C_k(x) = k$ says that the first k elements contain exactly k selected incidences.

Lemma I.7 (Prefix-difference identity). *For every binary vector x and every $k = 1, \dots, n$, with $C_0(x) = 0$,*

$$C_k(x) - C_{k-1}(x) = \sum_{v: e_k \in S_v} x_v.$$

Proof. Adding row e_k changes the cumulative value of column v by

$$d_{v,k} - d_{v,k-1} = a_{kv}.$$

Hence

$$\begin{aligned} C_k(x) - C_{k-1}(x) &= \sum_v (d_{v,k} - d_{v,k-1}) x_v \\ &= \sum_v a_{kv} x_v \\ &= \sum_{v: e_k \in S_v} x_v, \end{aligned}$$

which is the number of selected sets containing e_k . □

7.1 Constructed instance

Given an RXC3 instance, choose any order $e_1, \dots, e_n$ of its elements and retain the same family $S_1, \dots, S_n$.

7.2 The “if” proof

Lemma I.8. *If the RXC3 instance has an exact cover, then the constructed RPE instance is a yes-instance.*

Proof. Let $x_v = 1$ exactly for the subset-columns in an exact cover. Fix a prefix $\{e_1, \dots, e_k\}$. Each of its k element rows is covered by exactly one selected subset-column, so there are exactly k selected incidences in the prefix. Counting the same incidences column by column gives

$$\sum_v |S_v \cap \{e_1, \dots, e_k\}| x_v = \sum_v d_{v,k} x_v = C_k(x).$$

Hence $C_k(x) = k$ for every k , as required. $\square$

7.3 The “only if” proof

Lemma I.9. *If the constructed RPE instance is a yes-instance, then the original RXC3 instance has an exact cover.*

Proof. Let x satisfy $C_k(x) = k$ for all k , and set $C_0(x) = 0$. By [lemma I.7](#), the number of selected subset-columns covering element row e_k is

$$\sum_{v: e_k \in S_v} x_v = C_k(x) - C_{k-1}(x) = k - (k-1) = 1.$$

Thus every element row is covered by exactly one selected subset-column. Equivalently, $A^{\text{inc}}x = \mathbf{1}$, so the selected columns form an exact cover of the universe. $\square$

Intuition. The prefix equations contain exactly the same information as $A^{\text{inc}}x = \mathbf{1}$: successive differences recover the row equations. Their advantage is geometric—the scheduling construction can read cumulative counts from gaps on a one-dimensional ratio axis.

Corollary I.10. *RPE is NP-complete.*

Proof. The transformation retains the same binary variables and computes all values $d_{v,k}$ using $O(n^2)$ additions, so it is polynomial. [Lemmas I.8](#) and [I.9](#) prove feasibility equivalence with RXC3. Membership in NP is immediate: a vector $x \in \{0, 1\}^n$ is a certificate, and all n prefix equations are checked in polynomial time. Since RXC3 is NP-complete, so is RPE. $\square$

8 Reduction II: regular prefix exactness to AWET

Fix an RPE instance $(X, \mathcal{S})$ with $X = \{e_1, \dots, e_n\}$ and $\mathcal{S} = \{S_1, \dots, S_n\}$. We may assume $n \geq 3$, since smaller instances can be decided directly. Set

$$N = n + 1, \quad \kappa(k) = n - k + 1.$$

Thus $N \geq 4$, and earlier elements receive larger reverse levels. The construction creates $8n + 1$ jobs: eight *state jobs* for each source set and one global anchor. All job data are integers.

How to read this step. Read the construction in four layers. First, arithmetic digits give each block exactly two balanced states. Second, early ratios place the three occurrences of each set into the appropriate element bands. Third, integer tardiness offsets supply the linear field needed to center the prefix squares. Fourth, the anchor and minimum kernel force every optimal canonical assignment onto the two-state block manifold. Each layer uses only properties already established by the preceding layers.

Two-lane checkpoint. The logical lane asks only what a balanced block means and how prefix errors decode. The numerical lane proves that every low-cost schedule is balanced and that one nonzero prefix error dominates every residual term. Readers following only the theorem architecture may treat the scale lemmas as certified interfaces and return to the proof of [lemma I.14](#) for the full separation arithmetic.

Table 1: Construction blueprint for one source set S_v .

Component	Encoded information	Later use
Three occurrence pairs	The three elements of S_v	Their effective atoms appear in the three corresponding element bands.
One controller pair	Total signed mass -3	It cancels the three $+1$ occurrence masses below the final element band.
Logical spin σ_v	Select or reject S_v	$x_v = (1 + \sigma_v)/2$ becomes the exact-cover variable.
Target load β_v	Balanced half of the block processing	The global load lock forces every optimum to one of the two logical states.

8.1 Constructed instance

8.1.1 Numerical scales

Use the fixed pair codes

$$Q_0 = 10, \quad Q_1 = 10^2, \quad Q_2 = 10^3, \quad Q_3 = 10^4, \quad q = Q_0 + Q_1 + Q_2 + Q_3.$$

Set

$$G = N^{20}, \quad M = N^{40}, \quad T = N^{50}.$$

The ratio-space scale G is distinct from the tardiness ratios s_i and is used consistently throughout the construction and worked examples. The exponents are deliberately loose; their only purpose is to make the two separations

$$T \gg nM \gg q \gg 6, \quad GM \gg N^2T$$

immediate. Inside a block, the T digit controls the number of unsplit pairs, the M digit controls their aggregate element level, the codes Q_h identify the pairs, and the final constants 1 and 3 determine the orientation. In early-ratio space, the product GM separates consecutive element bands from all block-index variation.

For each v , write

$$S_v = \{e_{k_{v1}}, e_{k_{v2}}, e_{k_{v3}}\}, \quad k_{v1} < k_{v2} < k_{v3}.$$

Define four base processing values

$$P_{v0} = T + Q_0, \tag{8.1}$$

$$P_{vh} = T + M\kappa(k_{vh}) + Q_h, \quad h = 1, 2, 3. \tag{8.2}$$

The controller has no M digit. Each occurrence base records, in decreasing order of scale, a common synchronizing value T , its element level $M\kappa(k_{vh})$, and its pair code Q_h . The one-unit heavy–light difference is added when the two jobs are created. A discrepancy at any scale is too large to be canceled by all smaller scales.

Table 2: Jobs in block v . The occurrence pair h represents element $e_{k_{vh}}$.

Pair	Job	Role	Processing time	ϵ
0	A_{v0}	controller, heavy	$P_{v0} + 3$	-1
0	B_{v0}	controller, light	P_{v0}	$+1$
$h \in \{1, 2, 3\}$	A_{vh}	occurrence, heavy	$P_{vh} + 1$	$+1$
$h \in \{1, 2, 3\}$	B_{vh}	occurrence, light	P_{vh}	-1

8.1.2 The eight-job coded block

For each set S_v , create the eight state jobs in Table 2. The fixed sign ϵ_i specifies one reference orientation; reversing all eight signs gives the other.

Let

$$\mathcal{B}_v = \{A_{v0}, B_{v0}, A_{v1}, B_{v1}, A_{v2}, B_{v2}, A_{v3}, B_{v3}\}$$

be the job set of block v . Put

$$\beta_v = P_{v0} + P_{v1} + P_{v2} + P_{v3} + 3.$$

The block has total processing time $2\beta_v$. Its assignment is *canonical* when some $\sigma_v \in \{-1, +1\}$ satisfies

$$c_i = \sigma_v \epsilon_i \quad (i \in \mathcal{B}_v).$$

Thus σ_v selects one of two complementary orientations. The next lemma proves that these are exactly the zero-signed-load assignments. In either orientation the tardy processing load is β_v .

The two orientations can be read directly from the signs:

σ_v	early controller job	early occurrence jobs	tardy load
$+1$	B_{v0}	A_{v1}, A_{v2}, A_{v3}	$(P_{v0} + 3) + P_{v1} + P_{v2} + P_{v3} = \beta_v$
-1	A_{v0}	B_{v1}, B_{v2}, B_{v3}	$P_{v0} + \sum_{h=1}^3 (P_{vh} + 1) = \beta_v$

The controller difference 3 balances the three occurrence differences $1 + 1 + 1$. Consequently both orientations place exactly half of the block's processing on the tardy side.

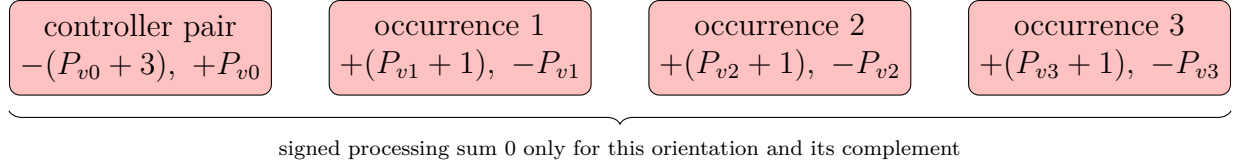


Figure 5: Signed-processing representation of one coded block. The signs shown are ϵ_i ; multiplying every sign by σ_v selects one of the two logical states.

8.1.3 Arithmetic synchronization

Lemma I.11 (Two zero-load states). *For every sign vector $c \in \{-1, +1\}^8$ on block v ,*

$$\sum_{i \in \mathcal{B}_v} p_i c_i = 0 \iff c_i = \sigma_v \epsilon_i \text{ for all } i \in \mathcal{B}_v \text{ and some } \sigma_v \in \{-1, +1\}.$$

Proof. Forward direction. If $c_i = \sigma_v \epsilon_i$, each occurrence pair contributes

$$\sigma_v \left((P_{vh} + 1) - P_{vh} \right) = \sigma_v,$$

while the controller contributes

$$\sigma_v \left(-(P_{v0} + 3) + P_{v0} \right) = -3\sigma_v.$$

The three $+\sigma_v$ contributions cancel the one $-3\sigma_v$ contribution.

Converse: expose the scale digits. Suppose $\sum_{i \in \mathcal{B}_v} p_i c_i = 0$, with arbitrary signs. For pair $h \in \{0, 1, 2, 3\}$ define

$$u_h = \frac{c_{A_{vh}} + c_{B_{vh}}}{2} \in \{-1, 0, 1\}.$$

Thus $u_h = 1, -1, 0$ means, respectively, that the pair is all early, all tardy, or split. Put

$$u_\Sigma = \sum_{h=0}^3 u_h, \quad z_\kappa = \sum_{h=1}^3 \kappa(k_{vh}) u_h.$$

Expanding every processing time by its T -, M -, and Q -parts gives

$$\begin{aligned} \sum_{i \in \mathcal{B}_v} p_i c_i &= 2T u_\Sigma + 2M z_\kappa + 2 \sum_{h=0}^3 Q_h u_h \\ &\quad + c_{A_{v1}} + c_{A_{v2}} + c_{A_{v3}} + 3c_{A_{v0}}. \end{aligned} \tag{8.3}$$

The last line is a residual of absolute value at most 6.

The T digit forces $u_\Sigma = 0$. If $u_\Sigma \neq 0$, then $|2Tu_\Sigma| \geq 2T$. The other terms have combined absolute value at most

$$2M|z_\kappa| + 2q + 6 \leq 6nM + 2q + 6 < T.$$

They cannot cancel the T term, contradicting zero signed load. Hence $u_\Sigma = 0$.

The M digit forces $z_\kappa = 0$. With $u_\Sigma = 0$, a nonzero integer z_κ would give $|2Mz_\kappa| \geq 2M$. The remaining coded and residual terms have absolute value at most $2q + 6 < 2M$. Therefore $z_\kappa = 0$.

The pair codes force every $u_h = 0$. The sequence 10, 100, 1000, 10000 is superincreasing for coefficients in $\{-1, 0, 1\}$. More explicitly, let h^* be the largest index with $u_{h^*} \neq 0$. If $h^* = 0$, the coded sum has magnitude 10; if $h^* > 0$, then

$$Q_{h^*} - \sum_{h < h^*} Q_h \geq 90 > 10.$$

Thus every nonzero vector u satisfies $|\sum_h Q_h u_h| \geq 10$. Its doubled contribution in (8.3) has magnitude at least 20, too large to be canceled by the residual of magnitude at most 6. Consequently $u_h = 0$ for all h , so the two jobs in each pair have opposite signs.

The final residual chooses one orientation. With all $u_h = 0$, equation (8.3) reduces to

$$c_{A_{v1}} + c_{A_{v2}} + c_{A_{v3}} + 3c_{A_{v0}} = 0.$$

If $c_{A_{v0}} = -1$, all three occurrence signs must be $+1$; if $c_{A_{v0}} = +1$, all three must be -1 . Their partners have the opposite signs. These are exactly $c = \epsilon$ and $c = -\epsilon$. $\square$

Intuition. The proof decodes the signed load from largest scale to smallest. The T and M digits remove aggregate imbalances, the superincreasing Q_h digits force every pair to split, and the final six-unit residual selects one of the two complementary orientations.

Let b_v be the total processing time of the tardy jobs in block v . Because the block total is $2\beta_v$, its signed load is $2\beta_v - 2b_v$. Hence lemma 1.11 gives the equivalence

$$b_v = \beta_v \iff c_i = \sigma_v \epsilon_i \text{ for every } i \in \mathcal{B}_v. \quad (8.4)$$

The global load lock will enforce the left side; the arithmetic lemma then supplies the two-state code on the right.

8.1.4 Early ratios and the minimum-kernel field

Define

$$\lambda_v = G + n - v + 1, \quad L_v = 2\lambda_v,$$

so $L_1 > L_2 > \cdots > L_n > 0$. The factor 2 is chosen for the exact cancellation used in [lemma I.15](#). Let

$$K_{uv} = \min\{L_u, L_v\},$$

and write

$$K = (K_{uv})_{u,v=1}^n, \quad \beta = (\beta_1, \dots, \beta_n)^\top, \quad f = K\beta.$$

Thus K is the symmetric *minimum kernel* and $f_v = \sum_{u=1}^n K_{vu}\beta_u$. This choice centers the load-lock quadratic at the target vector β . Every state job $i \in \mathcal{B}_v$ receives early ratio

$$r_i = f_v + \lambda_v p_i. \tag{8.5}$$

Set $w_i^E = p_i r_i$. The common term f_v moves the whole block, whereas $\lambda_v p_i$ turns processing-time differences into ratio separation. The same affine form permits the exact two-job compression used below.

Lemma I.12 (Kernel-field identities). *For $v = 1, \dots, n-1$,*

$$f_v - f_{v+1} = 2 \sum_{u=1}^v \beta_u.$$

Moreover,

$$4T < \beta_v < 5T \quad \text{and} \quad 0 < f_1 - f_n < 10N^2T.$$

Proof. Because the levels L_u decrease with u , compare rows v and $v+1$ of K . For $u \leq v$,

$$K_{vu} = L_v, \quad K_{v+1,u} = L_{v+1},$$

so their difference is $L_v - L_{v+1} = 2$. For $u \geq v+1$, both minima equal L_u , so their difference

is zero. Multiplying by β_u and summing gives

$$f_v - f_{v+1} = \sum_u (K_{vu} - K_{v+1,u})\beta_u = 2 \sum_{u=1}^v \beta_u.$$

In particular, the fields f_v decrease strictly.

Each β_v contains four copies of T plus positive lower-order terms, so $\beta_v > 4T$. For the upper bound,

$$\beta_v \leq 4T + 3nM + q + 3 < 5T,$$

because $nM, q + 3 < T/4$ under the chosen powers and $N \geq 4$. Finally, telescope the first identity:

$$\begin{aligned} f_1 - f_n &= 2 \sum_{v=1}^{n-1} \sum_{u=1}^v \beta_u \\ &< 2n \sum_{u=1}^n \beta_u < 2n \cdot n \cdot 5T < 10N^2T. \end{aligned}$$

□

Intuition. The block shifts f_v vary by only $O(N^2T)$, whereas consecutive element levels will be separated by more than $GM = N^{60}$. Thus the kernel field is large enough for the load lock but too small to change the element-band order.

8.1.5 Effective atoms: compressing complete pairs

In this subsection, a *two-job cluster* means one designated occurrence pair or the controller pair of a coded block. Compression is applied only to complete clusters from different blocks. Within-block terms are kept in the uncompressed objective: they contribute an assignment-independent constant and part of the baseline field a_v^0 . Here a_v^0 denotes the coefficient of σ_v in the full early state-job cost on coded assignments; its explicit formula is given in (8.14).

For this calculation, restrict to the coded assignments already characterized by [lemma I.11](#): $c_i = \sigma_v \epsilon_i$ for $i \in \mathcal{B}_v$. Thus the early indicator of job i is

$$\mathbf{1}\{i \in E\} = \frac{1 + c_i}{2} = \frac{1 + \sigma_v \epsilon_i}{2}.$$

This is an algebraic restriction at this stage, not yet a claim about optimal assignments. The global load lock and the subsequent choice of the anchor's tardiness ratio will justify the restriction for an optimal canonical schedule.

For a cluster g , define its signed processing mass and signed first ratio moment by

$$m_g = \sum_{i \in g} \epsilon_i p_i, \quad \mu_g = \sum_{i \in g} \epsilon_i p_i r_i.$$

Every designated pair has $m_g \neq 0$, so define its *effective ratio*

$$\rho_g = \frac{\mu_g}{m_g}.$$

The bookkeeping pair (m_g, ρ_g) is the *effective atom* of cluster g : a signed point mass m_g placed at ratio ρ_g .

The effective atom is not a new scheduling job and does not replace the two physical jobs inside their own block. Its role is narrower: when two complete clusters are separated in ratio order, their four physical cross-job terms depend only on the signed mass m_g and first moment $\mu_g = m_g \rho_g$. The effective atom therefore preserves those cross-block interactions exactly while reducing the amount of algebra.

For an occurrence pair (v, h) , $h \in \{1, 2, 3\}$,

$$\begin{aligned} m_{vh} &= (P_{vh} + 1) - P_{vh} = 1, \\ \mu_{vh} &= (P_{vh} + 1)(f_v + \lambda_v(P_{vh} + 1)) - P_{vh}(f_v + \lambda_v P_{vh}) \\ &= f_v + \lambda_v(2P_{vh} + 1). \end{aligned}$$

Hence

$$\rho_{vh} = f_v + \lambda_v(2P_{vh} + 1).$$

For the controller pair,

$$\begin{aligned} m_{v0} &= -(P_{v0} + 3) + P_{v0} = -3, \\ \mu_{v0} &= -(P_{v0} + 3)(f_v + \lambda_v(P_{v0} + 3)) + P_{v0}(f_v + \lambda_v P_{v0}) \\ &= -3(f_v + \lambda_v(2P_{v0} + 3)), \end{aligned}$$

so

$$\rho_{v0} = f_v + \lambda_v(2P_{v0} + 3).$$

Thus a block has three occurrence atoms of mass +1 and one controller atom of mass -3; its total effective mass is zero.

For complete clusters g and h from different blocks, define the physical signed cross-interaction

$$I^{\text{phys}}(g, h) = \sum_{i \in g} \sum_{j \in h} \epsilon_i \epsilon_j p_i p_j \min\{r_i, r_j\},$$

and its effective-atom counterpart

$$I^{\text{comp}}(g, h) = m_g m_h \min\{\rho_g, \rho_h\}.$$

These are signed coefficients arising after the early indicators are expanded; they are not separate scheduling costs. In particular, a controller mass is negative even though every physical cost is nonnegative.

Symbolic micro-example: why a two-job cluster becomes one atom. For a cluster $g = \{g^+, g^-\}$ with signed processing masses $+p^+$ and $-p^-$, the definitions give

$$m_g = p^+ - p^-, \quad \mu_g = p^+ r_{g^+} - p^- r_{g^-}.$$

If every ratio in g exceeds every ratio in a second cluster h , each cross minimum selects the ratio in h . Hence the four signed physical terms factor as

$$I^{\text{phys}}(g, h) = \left(\sum_{i \in g} \epsilon_i p_i \right) \left(\sum_{j \in h} \epsilon_j p_j r_j \right) = m_g \mu_h = m_g m_h \rho_h.$$

When the effective-ratio order agrees, this is $m_g m_h \min\{\rho_g, \rho_h\}$. The mass records the net signed processing, while the first moment determines the effective location needed to preserve the interaction.

Lemma I.13 (Exact pair compression). *Let g and h be complete clusters from different blocks. Suppose every physical early ratio in g exceeds every physical early ratio in h . Then*

$$I^{\text{phys}}(g, h) = m_g m_h \rho_h.$$

Consequently, whenever the physical order of the two clusters agrees with their effective-ratio order,

$$I^{\text{phys}}(g, h) = I^{\text{comp}}(g, h).$$

Proof. The lower ratio always belongs to h , so $\min\{r_i, r_j\} = r_j$ for all four cross-job pairs.

The sum therefore factors:

$$\begin{aligned}
 I^{\text{phys}}(g, h) &= \sum_{i \in g} \sum_{j \in h} \epsilon_i \epsilon_j p_i p_j r_j \\
 &= \left(\sum_{i \in g} \epsilon_i p_i \right) \left(\sum_{j \in h} \epsilon_j p_j r_j \right) \\
 &= m_g \mu_h = m_g m_h \rho_h.
 \end{aligned}$$

If the effective order agrees, then $\rho_h = \min\{\rho_g, \rho_h\}$ and the first case gives $I^{\text{comp}}(g, h)$. The symmetric case is identical. $\square$

Let $\mathcal{G}_v$ be the four designated clusters of block v . For jobs $i \in \mathcal{B}_u$ and $j \in \mathcal{B}_v$, with $u \neq v$, the coded early indicators satisfy

$$\mathbf{1}\{i, j \in E\} = \frac{1}{4} \left(1 + \sigma_u \epsilon_i + \sigma_v \epsilon_j + \sigma_u \sigma_v \epsilon_i \epsilon_j \right).$$

Thus the coefficient of $\sigma_u \sigma_v$ records the interaction between the two block choices. The 64 physical cross-job terms can be grouped into 16 cluster-pair interactions, so this coefficient is

$$\frac{1}{4} \sum_{g \in \mathcal{G}_u} \sum_{h \in \mathcal{G}_v} I^{\text{phys}}(g, h).$$

When the physical and effective orders agree, [lemma I.13](#) replaces every $I^{\text{phys}}(g, h)$ by $I^{\text{comp}}(g, h)$. The scale-separation lemma below verifies these order hypotheses for the construction. The prefix-profile calculation then denotes the coefficient by J_{uv} and rewrites it as a profile integral. This is the purpose of effective atoms: they preserve the block-spin couplings exactly while converting the physical cross-block algebra into a one-dimensional signed-mass calculation.

Intuition. The physical pair is retained in the schedule, but its cross-block effect is summarized by one signed point mass. Exactness comes from separation: every minimum in the four-term interaction selects the same lower cluster.

Ratio bands. View the physical ratios r_i and effective ratios ρ_g on a common real line, called early-ratio space. For element e_k , let its *effective element band* be the smallest closed

interval containing all occurrence-atom ratios associated with e_k :

$$\mathcal{R}_k = [\underline{\rho}_k, \bar{\rho}_k].$$

Define the controller band analogously:

$$\mathcal{R}_0 = [\underline{\rho}_0, \bar{\rho}_0].$$

The main-gap lengths are

$$\Delta_k = \underline{\rho}_k - \bar{\rho}_{k+1}, \quad k = 1, \dots, n-1, \quad (8.6)$$

$$\Delta_n = \underline{\rho}_n - \bar{\rho}_0. \quad (8.7)$$

Let

$$W = \sum_{k=0}^n (\bar{\rho}_k - \underline{\rho}_k)$$

be the total band width. We write $\mathcal{R}_k > \mathcal{R}_\ell$ when every point of $\mathcal{R}_k$ exceeds every point of $\mathcal{R}_\ell$. The open intervals between consecutive bands are the *main gaps*. The main gaps carry the exact-cover signal; all within-band intervals, whose total length is W , enter the residual.

Lemma I.14 (Scale separation). *The physical ratio intervals of the two-job clusters are disjoint, and their order agrees with the effective-ratio order. The bands satisfy*

$$\mathcal{R}_1 > \mathcal{R}_2 > \dots > \mathcal{R}_n > \mathcal{R}_0,$$

where $\mathcal{R}_k = [\underline{\rho}_k, \bar{\rho}_k]$. Moreover,

$$\Delta_k > GM \quad (k = 1, \dots, n), \quad W < 15N^3T. \quad (8.8)$$

Proof. We give all separation and width bounds explicitly, so that the proof is self-contained and no asymptotic comparison is needed.

For an occurrence cluster (v, h) representing element e_k , put $a = \kappa(k)$. Its physical early-ratio interval and effective ratio are

$$\mathcal{I}_{vh} = [f_v + \lambda_v(T + Ma + Q_h), f_v + \lambda_v(T + Ma + Q_h + 1)],$$

$$\rho_{vh} = f_v + \lambda_v(2T + 2Ma + 2Q_h + 1).$$

For a controller cluster they are

$$\mathcal{I}_{v0} = [f_v + \lambda_v(T + Q_0), f_v + \lambda_v(T + Q_0 + 3)], \quad \rho_{v0} = f_v + \lambda_v(2T + 2Q_0 + 3).$$

For $N \geq 4$, the definitions $G = N^{20}$, $M = N^{40}$, $T = N^{50}$, and $q = 11110$ imply

$$\begin{aligned} n &< N, & G + N &< 2G, & |\lambda_v - \lambda_w| &< N, \\ N^2M &< N^2T, & 8G(q + 3) &< N^2T, & GM &> 20N^2T, \\ 4G(q + 3) &< T, & 3nM + q + 3 &< T. \end{aligned}$$

For example, $GM > 20N^2T$ is equivalent to $N^8 > 20$, and $8G(q + 3) < N^2T$ is equivalent to $8(q + 3) < N^{32}$; both hold for $N \geq 4$.

Different element levels. Consider occurrence cluster (v, h) for e_k and occurrence cluster (w, h') for e_ℓ , where $k < \ell$. Put $a = \kappa(k)$ and $b = \kappa(\ell)$, so $a - b \geq 1$ and $b \leq n$. The lowest physical ratio of the first cluster minus the highest physical ratio of the second is

$$\begin{aligned} D_{\text{phys}} &= (f_v - f_w) + (\lambda_v - \lambda_w)T \\ &\quad + M\{\lambda_v(a - b) + (\lambda_v - \lambda_w)b\} + \lambda_v Q_h - \lambda_w(Q_{h'} + 1). \end{aligned}$$

Using $f_v - f_w > -(f_1 - f_n) > -10N^2T$, $\lambda_v \geq G + 1$, $|\lambda_v - \lambda_w| < N$, $b < N$, and $\lambda_w < 2G$, we obtain

$$D_{\text{phys}} > (G + 1)M - N^2M - NT - 10N^2T - 4G(q + 3).$$

The adverse terms are less than $13N^2T$, whereas $GM > 20N^2T$; hence $D_{\text{phys}} > 0$. Thus the full physical intervals are strictly ordered by element level.

For the effective points, the analogous subtraction gives

$$D_{\text{eff}} > 2(G + 1)M - 2N^2M - 2NT - 10N^2T - 8G(q + 3).$$

The adverse terms are less than $15N^2T$, and therefore

$$D_{\text{eff}} > 2GM - 15N^2T > GM.$$

The same estimates, with the lower level's element digit set to zero and controller width 3 absorbed in the code bound, show that every occurrence cluster lies above every controller cluster, physically and effectively, with effective separation greater than GM .

The same element level. Suppose two occurrence clusters represent the same element and $v < w$. Then

$$\delta := \lambda_v - \lambda_w = w - v \geq 1, \quad f_v - f_w > 0.$$

The lowest physical ratio in the v -cluster minus the highest in the w -cluster is at least

$$\delta T - 4G(q + 3) > 0,$$

and the effective-point difference is at least

$$2\delta T - 8G(q + 3) > 0.$$

Thus the physical and effective orders agree inside every element band. The controller comparison is easier and follows from the same inequalities with the fixed code Q_0 .

Gap and width bounds. The effective comparison between consecutive element levels, and between the last element level and the controller level, is greater than GM . Hence $\Delta_k > GM$ for every k .

For a fixed element e_k , put $A_k = 2T + 2M\kappa(k)$. For any two effective occurrence points in its band,

$$\begin{aligned} |\rho_{vh} - \rho_{wh'}| &\leq |f_v - f_w| + |\lambda_v - \lambda_w|(A_k + 2q + 1) + 2\lambda_w|Q_h - Q_{h'}| \\ &< 10N^2T + N(2T + 2nM + 2q + 1) + 2(G + N)q \\ &< 10N^2T + 3NT + N^2T < 15N^2T. \end{aligned}$$

For the penultimate inequality, the scale definitions give $2nM + 2q + 1 < T$, so $N(2T + 2nM + 2q + 1) < 3NT$. Moreover,

$$2(G + N)q < 4Gq < 8G(q + 3) < N^2T,$$

using the preliminary inequalities above. Thus the final code-variation term is bounded by N^2T itself. The controller band satisfies the same upper bound. Since there are $n + 1 = N$

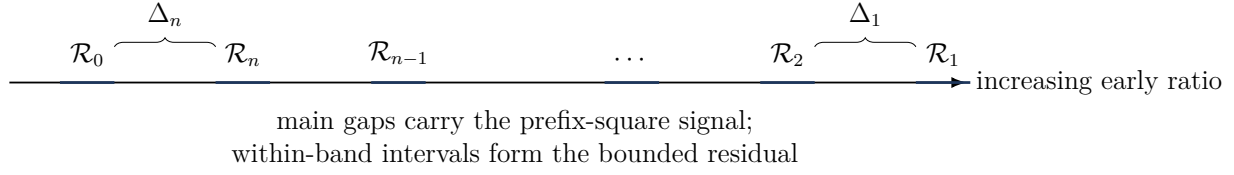


Figure 6: Effective-ratio bands. Element e_1 is highest, element e_n is lowest, and the controller band lies below all element bands.

bands,

$$W < 15N^3T.$$

All complete physical cluster intervals are therefore disjoint, their order agrees with the effective-ratio order, and the hypothesis of [lemma I.13](#) holds. $\square$

8.1.6 Prefix profiles and the main quadratic signal

Write $\mathbf{1}\{\cdot\}$ for an indicator. For block v , define the signed effective-atom profile

$$F_v(t) = \sum_{h=0}^3 m_{vh} \mathbf{1}\{t \leq \rho_{vh}\}. \quad (8.9)$$

Thus $F_v(t)$ is the signed mass at effective ratios at least t . As t moves downward, each occurrence atom adds 1; after the controller atom is crossed, the mass -3 returns the profile to zero. The controller therefore prevents a spurious tail below the last element band.

For distinct blocks u, v , define the cross-block quadratic coefficient generated by early interactions by the physical-job sum below. By [lemmas I.13](#) and [I.14](#), it also has the effective-atom and profile representations

$$\begin{aligned} J_{uv} &:= \frac{1}{4} \sum_{i \in \mathcal{B}_u} \sum_{j \in \mathcal{B}_v} \epsilon_i \epsilon_j p_i p_j \min\{r_i, r_j\} \\ &= \frac{1}{4} \sum_{g=0}^3 \sum_{h=0}^3 m_{ug} m_{vh} \min\{\rho_{ug}, \rho_{vh}\} \\ &= \frac{1}{4} \int_0^\infty F_u(t) F_v(t) dt. \end{aligned} \quad (8.10)$$

The first line ranges over the eight physical jobs in each block. The second ranges over the four effective atoms, whose signed masses are $m_{v0} = -3$ and $m_{vh} = 1$ for $h = 1, 2, 3$.

The final equality uses the exact minimum-kernel identity

$$\min\{a, b\} = \int_0^\infty \mathbf{1}\{t \leq a\} \mathbf{1}\{t \leq b\} dt.$$

Equivalently, sort the effective ratios and sum the constant product $F_u F_v$ over consecutive intervals. The integral notation makes the split into main gaps and within-band intervals explicit.

The factor $1/4$ comes from the product of the two early indicators: the coefficient of $\sigma_u \sigma_v$ for jobs $i \in \mathcal{B}_u$ and $j \in \mathcal{B}_v$ is $\epsilon_i \epsilon_j / 4$. Exact pair compression then gives (8.10).

On the main gap following element e_k , the active occurrence atoms of block v are exactly those corresponding to $S_v \cap \{e_1, \dots, e_k\}$, while the controller is inactive. For $k < n$ this gap is $(\bar{\rho}_{k+1}, \underline{\rho}_k)$; for $k = n$ it is $(\bar{\rho}_0, \underline{\rho}_n)$. Hence

$$F_v(t) = d_{v,k} \quad \text{throughout the } k\text{th main gap.} \quad (8.11)$$

Because a block contains three $+1$ atoms and one -3 atom, $|F_v(t)| \leq 3$ everywhere.

Define the *signed prefix total*

$$Y_k(\sigma) = \sum_{v=1}^n \sigma_v d_{v,k}.$$

This notation is distinct from the source sets S_v .

Symbolic micro-example: why one ratio gap gives a square. Fix the k th main gap, whose length is Δ_k . For two blocks u, v alone, the signed profile total is $Y = \sigma_u d_{u,k} + \sigma_v d_{v,k}$, and their interaction on this gap is

$$\frac{\Delta_k}{4} \sigma_u \sigma_v d_{u,k} d_{v,k} = \frac{\Delta_k}{8} (Y^2 - d_{u,k}^2 - d_{v,k}^2).$$

The subtracted terms do not depend on the spins. Thus the gap length weights the square of the signed profile total; summing over all blocks gives the same interpretation for $Y_k(\sigma)$.

The cross-block quadratic contribution of the k th main gap is

$$\frac{\Delta_k}{4} \sum_{u < v} \sigma_u \sigma_v d_{u,k} d_{v,k}.$$

Using

$$Y_k(\sigma)^2 = \sum_v d_{v,k}^2 + 2 \sum_{u < v} \sigma_u \sigma_v d_{u,k} d_{v,k},$$

this equals, up to the diagonal term $-(\Delta_k/8) \sum_v d_{v,k}^2$ that is independent of σ ,

$$\frac{\Delta_k}{8} Y_k(\sigma)^2. \quad (8.12)$$

Thus the gap length Δ_k weights the square of a signed prefix total. If $x_v = (1 + \sigma_v)/2$, regularity gives $\sum_v d_{v,k} = 3k$, because each of the first k elements belongs to three sets. Therefore

$$\begin{aligned} Y_k(\sigma) &= \sum_v (2x_v - 1) d_{v,k} \\ &= 2 \sum_v d_{v,k} x_v - \sum_v d_{v,k} \\ &= 2C_k(x) - 3k, \end{aligned}$$

and hence

$$Y_k(\sigma) + k = 2(C_k(x) - k). \quad (8.13)$$

Thus completing the square in $Y_k + k$ produces exactly the prefix error $C_k(x) - k$.

8.1.7 Logical-field correction

The main gaps supply the quadratic term in Y_k . To obtain a square centered at $C_k(x) = k$, we must also supply the matching linear term. For state jobs i, j , write

$$\alpha_{ij} = p_i p_j \min\{r_i, r_j\}.$$

Define a_v^0 as the coefficient of σ_v in the early state-job cost after substituting $c_i = \sigma_v \epsilon_i$ in every block:

$$a_v^0 = \frac{1}{4} \sum_{\{i,j\} \subseteq \mathcal{B}_v} \alpha_{ij} (\epsilon_i + \epsilon_j) + \frac{1}{4} \sum_{\substack{i \in \mathcal{B}_v \\ j \notin \mathcal{B}_v}} \alpha_{ij} \epsilon_i. \quad (8.14)$$

The first sum collects within-block pairs; the second collects pairs joining block v to another block, with every physical pair counted once. Thus a_v^0 is exactly the baseline coefficient of σ_v and is computable in polynomial time. [Section A](#) gives the indicator expansion.

The target coefficient that completes the main-gap squares is

$$a_v^* = \frac{1}{4} \sum_{k=1}^n \Delta_k k d_{v,k}. \quad (8.15)$$

Indeed,

$$\sum_v a_v^* \sigma_v = \sum_{k=1}^n \frac{\Delta_k k}{4} Y_k(\sigma).$$

Adding this to the main-gap square gives

$$\frac{\Delta_k}{8} Y_k^2 + \frac{\Delta_k k}{4} Y_k = \frac{\Delta_k}{8} (Y_k + k)^2 - \frac{\Delta_k}{8} k^2.$$

The final term is constant, and (8.13) converts the square into $(\Delta_k/2)(C_k(x) - k)^2$.

To adjust the actual field a_v^0 toward a_v^* , define the signed square-processing moment

$$D_v = \sum_{i \in B_v} \epsilon_i p_i^2.$$

For an occurrence pair, the signed square difference is

$$(P_{vh} + 1)^2 - P_{vh}^2 = 2P_{vh} + 1,$$

whereas for the controller it is

$$-(P_{v0} + 3)^2 + P_{v0}^2 = -6P_{v0} - 9.$$

The six copies of T cancel between the three occurrence pairs and the controller. Substituting the definitions of the P values gives

$$D_v = 2M \sum_{h=1}^3 \kappa(k_{vh}) + 2(Q_1 + Q_2 + Q_3) - 6Q_0 - 6 > 0. \quad (8.16)$$

Positivity follows because the M term is positive and dominates the fixed code terms. Choose an integer

$$\eta_v = \text{round} \left(\frac{4(a_v^0 - a_v^*)}{D_v} \right), \quad (8.17)$$

where either nearest integer is permitted in a tie, and define

$$e_v = a_v^0 - \frac{\eta_v D_v}{4} - a_v^*. \quad (8.18)$$

Then

$$|e_v| \leq \frac{D_v}{8}, \quad \sum_{v=1}^n D_v < 7N^2 M. \quad (8.19)$$

The first inequality is just nearest-integer rounding. The chosen integer η_v differs from the ideal real value $4(a_v^0 - a_v^*)/D_v$ by at most $1/2$; multiplying that error by $D_v/4$ gives $D_v/8$. Thus exact field cancellation is unnecessary: the remaining field error is explicitly bounded and will be absorbed into Γ .

8.1.8 Tardiness ratios, the anchor, and the load-lock scale

The tardiness ratio of every state job in block v will be

$$s_i = HL_v + \eta_v. \quad (8.20)$$

We now define H so that the HL_v layer forces the target block loads. Recall that

$$b_v = \sum_{i \in \mathcal{T} \cap \mathcal{B}_v} p_i$$

is the actual tardy processing load of block v , whereas β_v is its prescribed half-load. By (8.4), $b_v = \beta_v$ holds exactly for the two coded states.

For a state job i , let $\text{blk}(i)$ be its block index. The worked examples later retain the shorthand $b(i) = \text{blk}(i)$; here $\text{blk}(i)$ avoids confusion with the block load b_v . For state jobs i, j , define

$$\bar{\eta}_{ij} = \begin{cases} \eta_{\text{blk}(i)}, & \text{blk}(i) = \text{blk}(j), \\ \eta_{\max\{\text{blk}(i), \text{blk}(j)\}}, & \text{blk}(i) \neq \text{blk}(j). \end{cases}$$

Let $\mathcal{J}_S$ be the set of all $8n$ state jobs, and define

$$\begin{aligned} U = 1 + & \sum_{\{i,j\} \subseteq \mathcal{J}_S} p_i p_j \min\{r_i, r_j\} + \sum_{i \in \mathcal{J}_S} |\eta_{\text{blk}(i)}| p_i^2 \\ & + \sum_{\{i,j\} \subseteq \mathcal{J}_S} |\bar{\eta}_{ij}| p_i p_j. \end{aligned} \quad (8.21)$$

The quantity U is a deliberately crude upper bound on the oscillation of every assignment-dependent term that is not multiplied by H . Here the oscillation of a function means its maximum value minus its minimum value over all early–tardy assignments. Taking absolute values term by term is wasteful but safe; the added 1 makes the later inequality strict. Put

$$H = U + 2 \max_v |\eta_v| + 2. \quad (8.22)$$

Add an anchor job a with

$$p_a = H, \quad r_a = 1 + \max_{i \in \mathcal{J}_S} r_i.$$

The earliness weights of all jobs are $w_i^E = p_i r_i$.

With H chosen as in (8.22), the state-job tardiness ratios (8.20) are positive and satisfy

$$s_i > s_j \quad \text{whenever } \text{blk}(i) < \text{blk}(j).$$

To see this directly, let $\eta_{\max} = \max_v |\eta_v|$. Positivity follows from $HL_v \geq 2H > \eta_{\max}$. For consecutive blocks,

$$(HL_v + \eta_v) - (HL_{v+1} + \eta_{v+1}) \geq 2H - 2\eta_{\max} > 0.$$

Thus the dominant HL_v layer fixes the tardy block order despite the field-correction offsets. Their tardiness weights are $w_i^L = p_i s_i$.

The proof below combines these tardy terms with the anchor’s ordinary early-pair terms. The choice $L_v = 2\lambda_v$ makes the squared-processing terms cancel exactly.

Lemma I.15 (Global load lock). *Among canonical early–tardy assignments with the anchor early, every optimum satisfies*

$$b_v = \beta_v \quad (v = 1, \dots, n).$$

Consequently, every optimal block is in one of its two coded states.

Proof. The calculation has one purpose: isolate an assignment-dependent quadratic centered at the target load vector β , then show that every nonzero integer load error costs more than all remaining terms can save. Let q_v be the sum of squared processing times of the tardy jobs in block v , and let $Q_v = \sum_{i \in \mathcal{B}_v} p_i^2$. The block square-sums q_v, Q_v are unrelated to the

code constants q, Q_h . Write $b = (b_1, \dots, b_n)^\top$.

Dominant tardiness cost. Ignore the offsets η_v temporarily and retain only the HL_v part of each tardiness ratio. Within block v ,

$$\sum_{\{i,j\} \subseteq \mathcal{T} \cap \mathcal{B}_v} p_i p_j = \frac{1}{2}(b_v^2 - q_v).$$

Adding the tardy self terms q_v gives $(b_v^2 + q_v)/2$. Between two blocks u and v , all pair terms aggregate to $K_{uv}b_u b_v$. Hence the full dominant tardiness contribution is

$$\frac{H}{2} \left(b^\top K b + \sum_{v=1}^n L_v q_v \right).$$

Interaction with the early anchor. The anchor has processing time H and early ratio larger than every state-job ratio. Therefore, for every early state job i , the minimum in the early pair term is r_i . The early jobs in block v have total processing $2\beta_v - b_v$ and squared-processing sum $Q_v - q_v$. Using $r_i = f_v + (L_v/2)p_i$, the anchor interaction with that block is

$$H \left[f_v(2\beta_v - b_v) + \frac{L_v}{2}(Q_v - q_v) \right].$$

Summing these block contributions gives the full anchor–state-job interaction used by the construction.

Cancellation and centering. The q_v terms in the two displayed expressions cancel exactly. After discarding assignment-independent constants, the remaining dominant expression is

$$\frac{H}{2} b^\top K b - H f^\top b.$$

Since $f = K\beta$ and K is symmetric,

$$\frac{H}{2} b^\top K b - H \beta^\top K b = \frac{H}{2} (b - \beta)^\top K (b - \beta) - \frac{H}{2} \beta^\top K \beta.$$

The final term is constant. Thus the assignment-dependent dominant penalty is exactly

$$\frac{H}{2} (b - \beta)^\top K (b - \beta). \tag{8.23}$$

Positive definiteness of the minimum kernel. Let $L_{n+1} = 0$, and let $z^{(k)}$ have ones in positions $1, \dots, k$ and zeros afterward. Entrywise,

$$K = \sum_{k=1}^n (L_k - L_{k+1}) z^{(k)} (z^{(k)})^\top,$$

because $K_{uv} = \sum_{k \geq \max\{u,v\}} (L_k - L_{k+1})$. Consequently, for every vector y ,

$$y^\top K y = \sum_{k=1}^n (L_k - L_{k+1}) \left(\sum_{v=1}^k y_v \right)^2. \quad (8.24)$$

All coefficients are at least 2. If y is a nonzero integer vector, at least one integer prefix sum is nonzero, so $y^\top K y \geq 2$.

Dominance over all other terms. Let R_0 denote the complete assignment-dependent remainder after the H -scaled expression and all assignment-independent constants have been removed. Once the fixed tardiness block order is used, R_0 consists exactly of state–state early pair terms, tardiness-offset self terms, and tardiness-offset pair terms. The oscillation of a coefficient times a 0–1 indicator is at most the coefficient’s absolute value; hence (8.21) gives

$$\text{osc}(R_0) \leq U - 1 < U.$$

For a wrong integer load vector $b \neq \beta$, equation (8.23) is at least H . A correctly loaded coded assignment exists and has zero lock penalty. Comparing any wrong-load assignment with any such correct-load assignment therefore gives

$$F_{\text{wrong}} - F_{\text{correct}} \geq H - \text{osc}(R_0) \geq H - (U - 1) > 0.$$

Thus no optimum among the canonical anchor-early assignments has a wrong load. Every such optimum has $b = \beta$, and (8.4) together with lemma I.11 forces each block into one of its two coded states. $\square$

Intuition. The anchor is not merely a large job. Its early interactions supply the linear term $-H(K\beta)^\top b$ that shifts the minimum of the tardiness quadratic from zero load to the prescribed load β . The minimum kernel then penalizes every nonzero integer load-error vector by a visible amount.

Choose any coded assignment, put the anchor early, and compute its schedule cost C^*

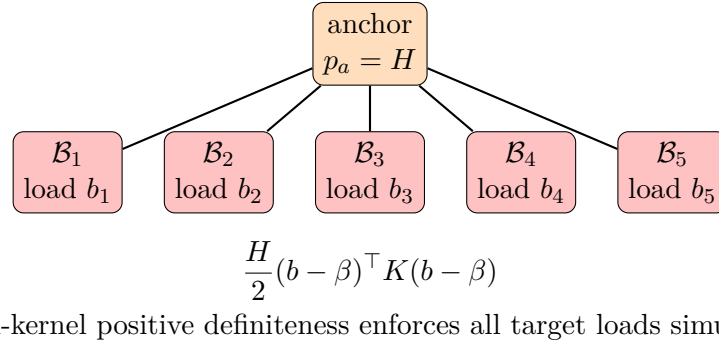


Figure 7: Schematic of the anchor-coupled global load lock. The figure shows five blocks only; the construction uses n blocks.

using the state-job ratios already fixed. This cost does not depend on the anchor's tardiness ratio. Set

$$s_a = 1 + \left\lceil \frac{C^*}{H^2} \right\rceil, \quad w_a^L = H s_a. \quad (8.25)$$

If the anchor were tardy, its tardy self-cost alone would be

$$p_a^2 s_a = H^2 \left(1 + \left\lceil C^*/H^2 \right\rceil\right) > C^*.$$

All costs are nonnegative, while the chosen coded canonical schedule with an early anchor has cost C^* . Therefore no optimal canonical assignment can have the anchor tardy. By [lemma I.3](#), at least one globally optimal schedule is canonical; choosing such a schedule, its anchor is early. This argument is independent of whether the chosen coded assignment represents an exact cover; it only supplies an explicit feasible benchmark.

Finally, set

$$d = \sum_{j \in \mathcal{J}} p_j.$$

This completes the construction of the AWET jobs.

8.1.9 Canonical objective, residual bound, and decision threshold

On the coded state space, write $c_i = \sigma_v \epsilon_i$ in block v and $x_v = (1 + \sigma_v)/2$. The dominant load-lock term is constant. The blockwise tardiness offset η_v changes the logical field of block v by $-\eta_v D_v/4$; cross-block tardy costs are constant because every block has tardy load β_v . Therefore the field coefficient is $a_v^* + e_v$.

At this point every schedule that can be optimal is represented by the n logical spins σ .

The remaining task is purely algebraic: separate the large, intended main-gap contribution from the small, unintended within-band and rounding contributions.

Expand the full canonical objective as

$$\Phi_{\text{can}}(\sigma) = C_0 + \sum_{v=1}^n (a_v^* + e_v) \sigma_v + \sum_{1 \leq u < v \leq n} J_{uv} \sigma_u \sigma_v, \quad (8.26)$$

where C_0 is the computable constant coefficient and J_{uv} is given by (8.10). Define the main-gap coefficient

$$J_{uv}^{\text{main}} = \frac{1}{4} \sum_{k=1}^n \Delta_k d_{u,k} d_{v,k}. \quad (8.27)$$

Let

$$\mathcal{P}(x) = \sum_{k=1}^n \frac{\Delta_k}{2} (C_k(x) - k)^2. \quad (8.28)$$

Define

$$A = C_0 - \frac{1}{8} \sum_{k=1}^n \Delta_k \left(k^2 + \sum_{v=1}^n d_{v,k}^2 \right), \quad (8.29)$$

$$R(\sigma) = \sum_{v=1}^n e_v \sigma_v + \sum_{1 \leq u < v \leq n} (J_{uv} - J_{uv}^{\text{main}}) \sigma_u \sigma_v. \quad (8.30)$$

How to read this step. Split every J_{uv} into its main-gap part and its within-band remainder. The main-gap quadratic terms contribute $\Delta_k Y_k^2/8$ up to a constant, and the target fields contribute $\Delta_k k Y_k/4$. Completing the square gives $\Delta_k (Y_k + k)^2/8$, while (8.13) converts this into $\Delta_k (C_k - k)^2/2$. The lemma below records this simultaneous calculation in full and identifies exactly where every constant and residual term goes.

Lemma I.16 (Exact master decomposition). *For every coded state $\sigma \in \{-1, +1\}^n$ and $x_v = (1 + \sigma_v)/2$,*

$$\Phi_{\text{can}}(\sigma) = A + \mathcal{P}(x) + R(\sigma). \quad (8.31)$$

The identity is exact: $R(\sigma)$ contains the explicitly identified within-band and rounding terms, not an omitted asymptotic error.

Proof. The purpose of the calculation is to assign every physical term to exactly one of three accounts: the intended prefix-square penalty, the explicitly bounded residual, or the state-independent constant. The calculation has three parts. The main-gap couplings provide the quadratic terms in the prefix sums, the target fields provide the linear terms needed

to complete the squares, and A supplies the state-independent correction for the omitted diagonal terms and the constants k^2 .

First split each coupling into its main-gap and residual portions. By (8.26) and the definition (8.30),

$$\Phi_{\text{can}}(\sigma) = C_0 + \sum_{v=1}^n a_v^* \sigma_v + \sum_{1 \leq u < v \leq n} J_{uv}^{\text{main}} \sigma_u \sigma_v + R(\sigma).$$

For a fixed prefix k , recall that $Y_k(\sigma) = \sum_v d_{v,k} \sigma_v$. Since $\sigma_v^2 = 1$,

$$Y_k(\sigma)^2 = \sum_{v=1}^n d_{v,k}^2 + 2 \sum_{1 \leq u < v \leq n} d_{u,k} d_{v,k} \sigma_u \sigma_v.$$

Solving for the cross-block sum and using (8.27) yields

$$\sum_{1 \leq u < v \leq n} J_{uv}^{\text{main}} \sigma_u \sigma_v = \frac{1}{8} \sum_{k=1}^n \Delta_k Y_k(\sigma)^2 - \frac{1}{8} \sum_{k=1}^n \Delta_k \sum_{v=1}^n d_{v,k}^2.$$

The second term is the diagonal correction: J_{uv}^{main} sums only over $u < v$, whereas Y_k^2 also contains $d_{v,k}^2 \sigma_v^2 = d_{v,k}^2$.

Next use (8.15) and interchange the two finite sums:

$$\sum_{v=1}^n a_v^* \sigma_v = \frac{1}{4} \sum_{k=1}^n \Delta_k k \sum_{v=1}^n d_{v,k} \sigma_v = \frac{1}{4} \sum_{k=1}^n \Delta_k k Y_k(\sigma).$$

Substitution gives

$$\begin{aligned} \Phi_{\text{can}}(\sigma) &= C_0 - \frac{1}{8} \sum_{k=1}^n \Delta_k \sum_{v=1}^n d_{v,k}^2 \\ &\quad + \sum_{k=1}^n \left[\frac{\Delta_k}{8} Y_k(\sigma)^2 + \frac{\Delta_k k}{4} Y_k(\sigma) \right] + R(\sigma). \end{aligned}$$

Complete the square separately for each prefix:

$$\frac{\Delta_k}{8} Y_k^2 + \frac{\Delta_k k}{4} Y_k = \frac{\Delta_k}{8} (Y_k + k)^2 - \frac{\Delta_k}{8} k^2.$$

Consequently, the state-independent terms combine exactly into (8.29), and

$$\Phi_{\text{can}}(\sigma) = A + \frac{1}{8} \sum_{k=1}^n \Delta_k (Y_k(\sigma) + k)^2 + R(\sigma).$$

Finally, (8.13) gives $Y_k(\sigma) + k = 2(C_k(x) - k)$. Therefore

$$\frac{1}{8} \sum_{k=1}^n \Delta_k (Y_k + k)^2 = \sum_{k=1}^n \frac{\Delta_k}{2} (C_k(x) - k)^2 = \mathcal{P}(x)$$

by (8.28). This proves (8.31). $\square$

The coefficient difference $J_{uv} - J_{uv}^{\text{main}}$ is the profile integral over the union of the element and controller bands. Since $|F_v(t)| \leq 3$, one block pair contributes at most $9W/4$ in absolute value. There are fewer than $n^2/2$ block pairs, so

$$\left| \sum_{u < v} (J_{uv} - J_{uv}^{\text{main}}) \sigma_u \sigma_v \right| \leq \frac{9}{8} n^2 W.$$

Together with (8.19), this gives

$$|R(\sigma)| \leq \Gamma, \quad \Gamma = \frac{9}{8} n^2 W + \frac{1}{8} \sum_{v=1}^n D_v. \quad (8.32)$$

By lemma I.14 and (8.19),

$$\Gamma < 18N^5 T + N^2 M.$$

Let $\Delta_{\min} = \min_{1 \leq k \leq n} \Delta_k$. By lemma I.14, $\Delta_{\min} > GM = N^{60}$. From the displayed bound on Γ ,

$$2\Gamma + 1 < 36N^5 T + 2N^2 M + 1 = 36N^{55} + 2N^{42} + 1.$$

For $N \geq 4$, this is smaller than $N^{60}/2$. Hence

$$\frac{\Delta_{\min}}{2} > 2\Gamma + 1. \quad (8.33)$$

This is the decisive scale separation: one unit of squared prefix error costs more than the largest possible favorable swing of the residual on both sides of the comparison, with an additional unit to accommodate the integer floor.

The decision bound of the constructed AWET instance is

$$B = \lfloor A + \Gamma \rfloor. \quad (8.34)$$

All quantities in (8.34) are exactly computable rationals with denominators dividing 8. Actual schedule costs are integers because processing times, weights, and canonical completion times are integers. Therefore the floor causes no loss in the yes-direction and creates a clean integer threshold.

8.2 The “if” proof

Lemma I.17. *If the RPE instance is a yes-instance, then the constructed AWET instance has a schedule of cost at most B .*

Proof. Let $x \in \{0, 1\}^n$ satisfy $C_k(x) = k$ for every k , and set $\sigma_v = 2x_v - 1$.

Build the early–tardy assignment. In block v , choose the coded orientation $c_i = \sigma_v \epsilon_i$, and put the anchor early. Every block then has tardy load β_v . By [proposition I.4](#), the resulting nonempty early set has a feasible compact canonical schedule with integral completion times.

Evaluate its prefix penalty. Every prefix equation is exact, so each square in (8.28) is zero and $\mathcal{P}(x) = 0$.

Use the residual bound and integer threshold. The master identity and $|R(\sigma)| \leq \Gamma$ give

$$\Phi_{\text{can}}(\sigma) = A + R(\sigma) \leq A + \Gamma.$$

The left side is an integer. Hence

$$\Phi_{\text{can}}(\sigma) \leq \lfloor A + \Gamma \rfloor = B.$$

Thus the constructed scheduling instance is a yes-instance. □

8.3 The “only if” proof

Lemma I.18. *If the constructed AWET instance has a schedule of cost at most B , then the RPE instance is a yes-instance.*

Proof. Assume some feasible schedule has cost at most B , and choose an optimal schedule. Apply the structural lemma to make it compact and canonical without increasing cost.

Gate 1: the anchor is early. The schedule selected above is optimal and canonical. By the choice of s_a in (8.25), a tardy anchor in such a schedule would contribute its full self term, which exceeds the explicit feasible benchmark C^* . Hence this optimal canonical schedule has the anchor early.

Gate 2: every block is coded. With an early anchor, lemma I.15 forces $b_v = \beta_v$ in every block. By lemma I.11 and (8.4), block v has a unique logical spin $\sigma_v \in \{-1, +1\}$ and selection variable $x_v = (1 + \sigma_v)/2$.

Gate 3: every prefix equation is exact. Suppose some prefix equation fails. Since $C_k(x) - k$ is an integer, at least one squared error is at least 1, and therefore

$$\mathcal{P}(x) \geq \frac{\Delta_{\min}}{2}.$$

The master identity, the lower residual bound $R(\sigma) \geq -\Gamma$, and (8.33) imply

$$\begin{aligned} \Phi_{\text{can}}(\sigma) &\geq A + \frac{\Delta_{\min}}{2} - \Gamma \\ &> A + \Gamma + 1 \\ &> \lfloor A + \Gamma \rfloor = B, \end{aligned}$$

contradicting the assumed cost. Thus $C_k(x) = k$ for all k , and the RPE instance is a yes-instance. $\square$

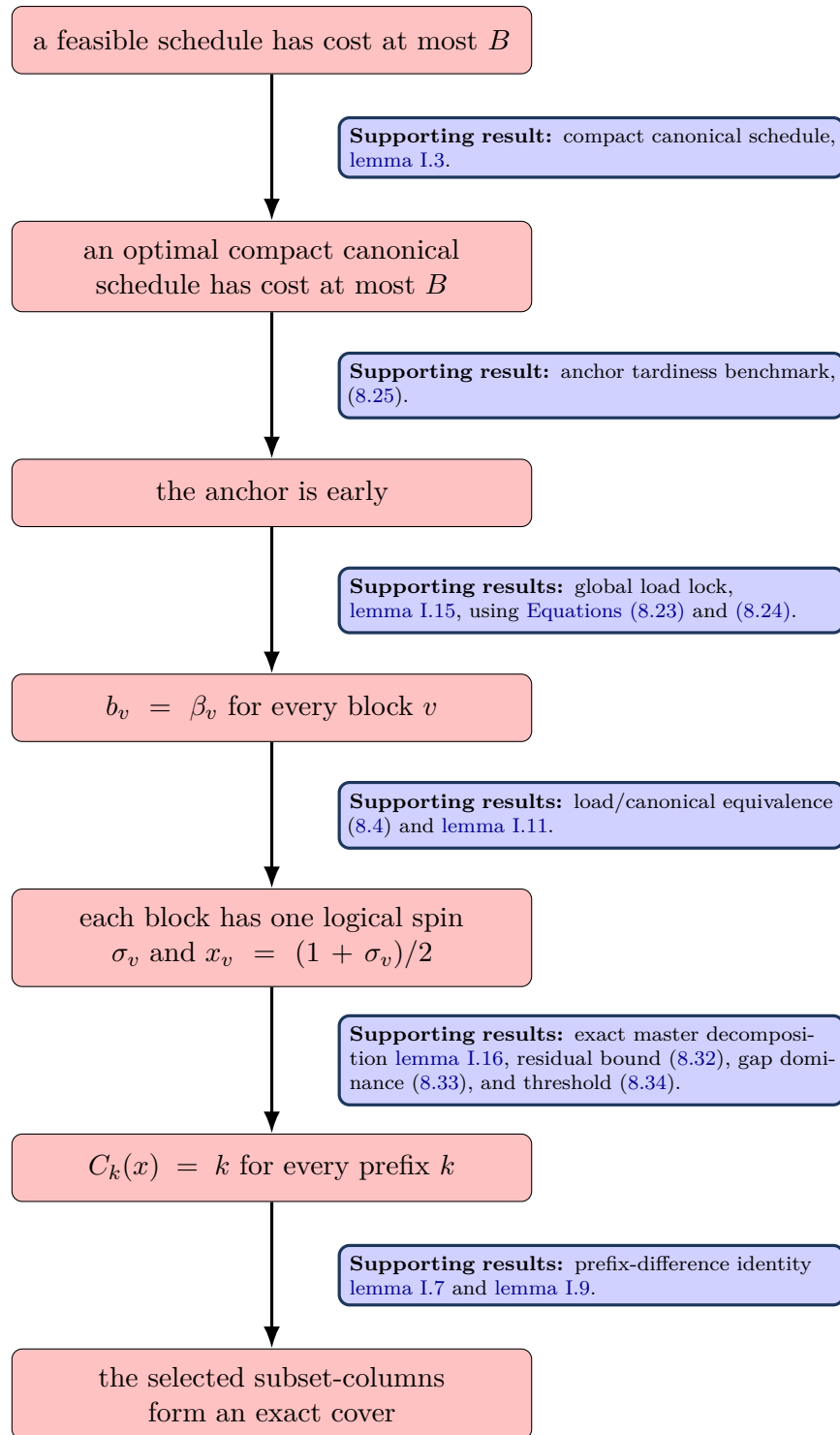


Figure 8: Expanded only-if implication chain. The enlarged vertical gaps keep every supporting-result callout visually separate from the implication nodes.

Section exit. Section 8 has established an exact reduction interface: every threshold-feasible schedule can be replaced without higher cost by a canonical schedule with an early anchor, one coded state per block, and zero prefix error, while every source solution constructs such a schedule. Section 9 uses this interface only to audit encoding length and to verify that all output integers remain polynomially bounded even in unary.

9 Polynomial and unary bounds

Lemma I.19. *The transformation in [Section 8](#) can be performed in polynomial time, and every integer in the resulting AWET instance is bounded by N^ν for a fixed constant ν independent of the source instance.*

Proof. The point is not to optimize the exponent, but to verify the dependency chain: each new scale is obtained from only polynomially many previously bounded quantities. We give explicit coarse bounds. Since the source instances used in the reduction have $N = n + 1 \geq 4$, fixed numerical constants can be absorbed into small extra powers of N .

Step 1: processing times and kernel data. The pair codes Q_h and q are fixed constants. Because $\kappa(k) \leq n < N$, $M = N^{40}$, and $T = N^{50}$, every base value P_{vh} and every state-job processing time satisfy

$$p_i < N^{51}.$$

The bound $4T < \beta_v < 5T$ from [lemma I.12](#) gives $\beta_v < N^{52}$. Also

$$\lambda_v < N^{21}, \quad L_v < N^{21}, \quad K_{uv} < N^{21}.$$

Therefore

$$f_v = \sum_{u=1}^n K_{vu} \beta_u < N \cdot N^{21} \cdot N^{52} = N^{74}.$$

This is the first useful hierarchy: processing times are of order N^{50} , whereas the additive kernel field f_v is still only a fixed polynomial.

Step 2: early ratios, effective ratios, and early weights. Using (8.5),

$$r_i = f_v + \lambda_v p_i < N^{75}.$$

The explicit formulas for the effective atoms give the same bound $|\rho_{vh}| < N^{75}$, and hence every gap $\Delta_k < N^{75}$. Each state-job earliness weight consequently satisfies

$$w_i^E = p_i r_i < N^{126}.$$

For later use, every early state-state pair coefficient obeys

$$\alpha_{ij} = p_i p_j \min\{r_i, r_j\} < N^{177}.$$

Step 3: logical-field correction. For one fixed block, (8.14) contains fewer than $64N$ physical pair terms. The preceding bound on α_{ij} therefore gives

$$|a_v^0| < N^{181}.$$

Moreover, $d_{v,k} \leq 3$, $k < n + 1 = N$, and $\Delta_k < N^{75}$, so

$$|a_v^*| < N^{78}.$$

Equation (8.16) gives the two-sided estimate

$$M = N^{40} < D_v < N^{43}.$$

Nearest-integer rounding changes a number by at most $1/2$. Hence

$$\begin{aligned} |\eta_v| &\leq \frac{4|a_v^0 - a_v^*|}{D_v} + \frac{1}{2} \\ &< N^{144}. \end{aligned}$$

Thus the field-correction offsets are polynomially bounded even though they are defined by a quotient: the denominator D_v is itself at least N^{40} .

Step 4: the load-lock scale and state-job tardiness data. There are $8n < 8N < N^3$ state jobs and fewer than N^6 unordered state-job pairs. In (8.21), the early pair sum is therefore less than N^{183} . The terms involving η_v have individual magnitude at most

$$|\eta_v|p_i^2 < N^{144}N^{102} = N^{246}.$$

After multiplying by the number of jobs or pairs, the largest contribution to U is less than N^{252} . Absorbing the three displayed sums and the constant 1 gives

$$U < N^{253}, \quad H = U + 2 \max_v |\eta_v| + 2 < N^{254}.$$

It follows that the anchor processing time is $p_a = H < N^{254}$ and that every state tardiness ratio and weight satisfy

$$s_i = HL_v + \eta_v < N^{276}, \quad w_i^L = p_i s_i < N^{327}.$$

The anchor early ratio is less than N^{76} , so $w_a^E = p_a r_a < N^{330}$.

Step 5: the anchor tardiness weight and the decision threshold. To bound C^* , use the canonical set objective rather than completion times. There are fewer than N^6 state-state pairs. A tardy state-state pair has cost coefficient at most $p_i p_j \min\{s_i, s_j\} < N^{378}$, and an anchor-state early pair has coefficient at most

$$p_a p_i r_i < N^{254} N^{51} N^{75} = N^{380}.$$

Each state self-term has coefficient $p_i^2 s_i < N^{378}$. These bounds are independent of the coded signs. There are fewer than N^6 state-state pairs, fewer than N^3 anchor-state pairs, and fewer than N^3 state self-terms. Since the anchor is early, its tardiness self-term and tardiness pair terms are absent. Consequently, every coded assignment with the anchor early satisfies the uniform bound

$$0 \leq \Phi_{\text{can}}(\sigma) < N^{384} + N^{383} + N^{381} < N^{385}.$$

In particular, the benchmark schedule satisfies $C^* < N^{385}$. Consequently,

$$s_a = 1 + \lceil C^*/H^2 \rceil < N^{386}, \quad w_a^L = H s_a < N^{640}.$$

The due date satisfies

$$d = p_a + \sum_{i \in \mathcal{I}_S} p_i < N^{255}.$$

To bound C_0 , average (8.26) over all sign vectors $\sigma \in \{-1, +1\}^n$. Every linear term averages to zero, as does every quadratic term $\sigma_u \sigma_v$ with $u \neq v$. Therefore

$$C_0 = 2^{-n} \sum_{\sigma \in \{-1, +1\}^n} \Phi_{\text{can}}(\sigma), \quad 0 \leq C_0 < N^{385}.$$

This averaging identity proves the bound; it is not an enumeration step in the reduction, since C_0 is computed directly by collecting the constant coefficients. Since $\Delta_k < N^{75}$ and $d_{v,k} \leq 3$, the correction in (8.29) is less than N^{80} , so $|A| < N^{386}$. Furthermore, (8.8) and (8.19) give $\Gamma < N^{58}$. Therefore the integer threshold $B = \lfloor A + \Gamma \rfloor$ has magnitude less than N^{387} .

We have thus obtained one completely explicit universal choice: every integer written into the AWET instance is smaller than N^{650} . The exponent is deliberately loose; only its independence from the source instance matters.

Step 6: running time and unary size. The construction creates $8n + 1 = O(N)$ jobs. All displayed quantities are computed using polynomially many additions, multiplications, comparisons, minimum operations, and exact divisions or roundings of rationals whose numerators and denominators have $O(\log N)$ bits. In particular, U and C^* can be evaluated by examining $O(n^2)$ job pairs, and $\lfloor A + \Gamma \rfloor$ can be computed exactly after multiplying by 8, since the relevant denominators divide 8.

Hence the binary running time is polynomial in the source size. Under unary encoding, the output contains $O(N)$ integers, each at most N^{650} , so its total unary length is $O(N^{651})$. This is polynomial in n , which proves the required strong reduction bound. $\square$

Proof checkpoint. The construction is numerically large but not pseudopolynomial. “Large” means a high fixed power of N , not an exponent that depends on the source data. Writing every output number in unary therefore still produces only polynomially many symbols.

Proof of theorem I.2. By Gonzalez [7], RXC3 is NP-complete. Lemmas I.8 and I.9 give a polynomial reduction from RXC3 to RPE. Lemmas I.17 and I.18 give a polynomial reduction

from RPE to AWET. By [lemma I.19](#), every numerical value in the AWET instance is polynomially bounded, so the reduction establishes strong NP-hardness in the standard sense of Garey and Johnson [6]. Membership in NP was noted after [theorem I.2](#). Hence AWET is strongly NP-complete and, in particular, unary NP-hard. $\square$

10 Concluding remarks

The reduction uses the asymmetry of the earliness and tardiness weights in two distinct ways. The early ratio order carries the exact-cover signal through independently positioned occurrence pairs, while the tardiness ratios and the early anchor jointly create a global positive-definite load lock. The controller pair closes each block’s signed profile, preventing signal below the final element band. The field-correction offsets need not be exact: the polynomially large prefix gaps dominate the rounding error and all within-band interactions.

The proof can be summarized as four reusable design lessons. First, convert schedule geometry into a set objective before introducing gadgets. Second, give every local gadget a uniquely decodable arithmetic signature. Third, separate logical signal intervals from residual intervals geometrically. Fourth, use a positive-definite global penalty to enforce local feasibility simultaneously rather than trying to enforce each block with an isolated ad hoc penalty.

The result resolves the published strong-complexity question for positive-integer asymmetric weighted earliness–tardiness scheduling at the boundary due date $d = \sum_j p_j$ [13]. It also shows that the difficulty is not caused by a large number of jobs: the construction uses only eight state jobs per source set and one global anchor. The numerical exponents are deliberately loose; smaller scales may be possible without changing the proof architecture. For complete numerical verification of both reduction directions, see the yes- and no-instances in [Sections B.1](#) and [B.2](#).

Reusable idea. A profile-and-gap compiler can turn cumulative combinatorial constraints into separated squared penalties. Local arithmetic signatures identify the permitted states, a global positive-definite penalty enforces all local loads at once, and a residual budget converts approximate field correction into an exact decision gap. This pattern is potentially useful whenever a pairwise minimum kernel can be represented by one-dimensional signed profiles.

Why Part II is next. Part I has established exact intractability for unrestricted asymmetric data. The remaining question is how much near-optimal structure survives despite that hardness. Part II uses the same canonical objective but replaces scale separation by relaxation and rounding, producing a universal constant-factor guarantee.

Part II

General Approximation

Anchored Semidefinite Relaxation and Deterministic Threshold Rounding

Part synopsis. This Part proves a polynomial-time constant-factor approximation for unrestricted AWET. One early anchor is enumerated, a locally consistent semidefinite relaxation is solved, and jobs are rounded by a common marginal threshold. Early pairs are paid by local joint consistency. Tardy pairs and tardy self-costs form a positive-semidefinite minimum-kernel matrix, allowing a global covariance payment. Threshold optimization gives $3 + 2\sqrt{2} + \varepsilon$.

Part positioning. **Established boundary.** FPTAS results are known for symmetric or otherwise structured common-due-date settings, whereas unrestricted asymmetry remained outside those schemes. **Result.** For every fixed $\varepsilon > 0$, unrestricted AWET has a polynomial-time $(3 + 2\sqrt{2} + \varepsilon)$ -approximation. **Technique.** An anchored local-consistency SDP is rounded by a deterministic marginal threshold; early pairs are paid locally, while the tardy self-cost completes a positive-semidefinite minimum kernel that pays all tardy pairs globally.

Notation reset. This Part uses a_{ij} and b_{ij} for early- and tardy-pair coefficients, c_i for the tardy self-cost, e_i, t_i for SDP marginals, and e_{ij}, t_{ij} for pseudo-joint masses. The symbols $\mathcal{A}, \mathcal{B}, \mathcal{U}$ denote the three SDP objective accounts. The threshold is $\theta = 1 - \delta$.

Proof contract.

- **Input:** an arbitrary positive-integer AWET instance and a positive rational accuracy parameter ε .
- **Relaxation:** enumerate an early anchor and solve one polynomial-size semidefinite program for each anchor.
- **Rounding invariant:** the anchor remains early, early pairs are charged to pseudo-joint early mass, and tardy pairs are charged to the completed minimum kernel.
- **Output:** the cheapest canonical schedule produced by the anchored threshold roundings.
- **Precision obligation:** additive SDP error is absorbed into the requested ε margin using only polynomially many bits.

Introduction and overview

This Part presents a fully specified approximation theorem for unrestricted AWET. The proof is constructive: it gives the SDP, the deterministic rounding rule, the schedule-recovery step, and the finite precision needed for polynomial running time. The result complements earlier FPTAS work in symmetric or otherwise structured common-due-date models: here the two weight systems may be unrelated, so the guarantee is a universal constant rather than an approximation scheme [12, 15, 16].

How to read this step. The logical chain is

canonical early–tardy partition $\longrightarrow$ anchored SDP $\longrightarrow$ threshold rounding
 $\longrightarrow$ direct early-pair payment and PSD tardy-pair self-payment $\longrightarrow$ constant ratio.

The main new inequality is [lemma II.15](#). Readers interested mainly in implementation may read [Sections 11, 14 and 20](#) first.

Algorithm at a glance. For each possible early anchor h : (1) solve the anchored SDP; (2) read the early marginals e_i ; (3) choose a rational threshold near $2 - \sqrt{2}$ and put i early when e_i reaches it; (4) construct the canonical V-shaped schedule for that partition; and (5) evaluate its cost exactly. Return the cheapest of the n candidates. Sections 14–19 prove why these steps are valid; Section 20 gives the implementation in full.

11 Problem definition and main theorem

Definition II.1 (Asymmetric weighted earliness–tardiness scheduling). An instance of AWET consists of jobs $\mathcal{J} = \{1, \dots, n\}$. Job i has positive integer processing time p_i , positive integer unit earliness weight w_i^E , and positive integer unit tardiness weight w_i^L . All jobs are available at time zero and are processed nonpreemptively on one machine. The common due date is

$$d = P := \sum_{i \in \mathcal{J}} p_i. \quad (11.1)$$

For completion time C_i , define

$$E_i = (d - C_i)^+, \quad T_i = (C_i - d)^+.$$

The objective is

$$F = \sum_{i \in \mathcal{J}} (w_i^E E_i + w_i^L T_i). \quad (11.2)$$

Positive rational data can be scaled to integers without changing approximation ratios. Define

$$r_i = \frac{w_i^E}{p_i}, \quad s_i = \frac{w_i^L}{p_i}. \quad (11.3)$$

Definition II.2 (Approximation ratio). A polynomial-time algorithm is a C -approximation for a minimization problem if it always returns a feasible solution of value at most $C \text{ OPT}$.

Theorem II.3 (Main approximation theorem). *For every $\varepsilon > 0$, unrestricted AWET has a polynomial-time*

$$(3 + 2\sqrt{2} + \varepsilon)$$

-approximation. When the accuracy parameter is supplied as a positive rational, the running time is polynomial in the input length and in $\log(1/\min\{\varepsilon, 1\})$.

Since $3 + 2\sqrt{2} \approx 5.828427$, the theorem gives a universal constant ratio for the fully asymmetric problem.

Corollary II.4 (Convenient threshold). *The same method with threshold $\theta = 7/12$ gives factor 6 in the exact-SDP model and factor $6 + \varepsilon$ in the ordinary finite-precision model.*

Intuition. The early and tardy sides are deliberately treated differently. There is no early self-cost, so early pairs are paid through a threshold greater than one half.

The tardy side does have a self-cost. Those diagonal terms complete the tardy pair coefficients into a positive-semidefinite minimum-kernel matrix, which pays globally for all rounded tardy pairs.

12 Canonical side assignment

Lemma II.5 (Compact canonical schedule). *There is an optimal schedule with no idle time between consecutive jobs, with at least one completion exactly at d , and with the following V-shaped orders: read outward from d , jobs completing no later than d are in nonincreasing r_i order, and tardy jobs are in nonincreasing s_i order.*

Proof. Choose an optimal schedule with minimum internal idle time. An idle interval before d can be shortened by shifting the block to its left toward d ; every affected early completion moves toward its due date. The symmetric operation removes an interval after d . If an interval contains d , move both adjacent blocks toward d . Thus an optimum can be chosen compact.

Fix a compact sequence. Its length is $P = d$. If the sequence begins at $z \geq 0$, its cost is a convex piecewise-linear function of z . A minimum may be chosen at $z = 0$, when the final job completes at d , or at a breakpoint, when another completion equals d .

For adjacent early jobs i, j , with j closer to d , the order-dependent contribution is $w_i^E p_j = p_i p_j r_i$; reversing them gives $p_i p_j r_j$. Thus the smaller ratio lies farther from d . For tardy jobs, with i closer to d than j , the order-dependent term is $w_j^L p_i = p_i p_j s_j$, so the larger tardiness ratio lies closer to d . $\square$

For $i \neq j$, define

$$a_{ij} = p_i p_j \min\{r_i, r_j\}, \quad (12.1)$$

$$b_{ij} = p_i p_j \min\{s_i, s_j\}, \quad (12.2)$$

and define the tardy self coefficient

$$c_i = p_i^2 s_i = p_i w_i^L. \quad (12.3)$$

All these coefficients are positive integers, since

$$a_{ij} = \min\{w_i^E p_j, p_i w_j^E\}, \quad b_{ij} = \min\{w_i^L p_j, p_i w_j^L\}.$$

Proposition II.6 (Canonical partition objective). *For every nonempty $E \subseteq \mathcal{J}$, let $T = \mathcal{J} \setminus E$. The canonical schedule that orders E by nondecreasing r_i and ends it at d , then orders T by nonincreasing s_i and starts it at d , has cost*

$$\Phi(E) = \sum_{\{i,j\} \subseteq E} a_{ij} + \sum_{\{i,j\} \subseteq T} b_{ij} + \sum_{i \in T} c_i. \quad (12.4)$$

Every nonempty E gives a feasible schedule of this form, and

$$\text{OPT} = \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E). \quad (12.5)$$

Proof. The early block starts at

$$d - \sum_{i \in E} p_i = \sum_{i \in T} p_i \geq 0,$$

so the schedule is feasible. For an early pair, the nondecreasing- r order contributes $p_i p_j \min\{r_i, r_j\} = a_{ij}$. For a tardy pair, the nonincreasing- s order contributes b_{ij} . Each tardy job contributes its own processing time to its delay, producing c_i . This proves (12.4).

By lemma II.5, some optimal schedule has this canonical form for the nonempty set of jobs completing no later than d . Conversely every nonempty set gives a feasible canonical schedule. Therefore the minimum of (12.4) equals OPT. $\square$

How to use this result. Once E has been selected, sort it by increasing r_i , begin it at $d - \sum_{i \in E} p_i$, and finish it at d . Sort T by decreasing s_i and begin it at d . Ties may be broken arbitrarily. The cost can be evaluated exactly from (12.4).

Proof checkpoint. All continuous-time scheduling choices have now disappeared. The approximation task is exactly the binary minimization of (12.4) over nonempty E .

13 Why one early anchor is enumerated

The rounding threshold will exceed $1/2$. Without an extra condition, every early marginal could lie below that threshold, producing an inadmissible empty early set. For every $h \in \mathcal{J}$, we therefore solve one relaxation with job h forced early.

Lemma II.7 (Correct-anchor lower bound). *If E^* is an optimal early set and $h \in E^*$, then the optimum of any valid relaxation with h forced early is at most OPT.*

Proof. The incidence vector of E^* is feasible for the anchored integral problem. A relaxation cannot have larger optimum value. $\square$

Intuition. The enumeration guesses only one certificate that the early side is nonempty, not the whole partition. At least one of the n guesses is compatible with the unknown optimum.

14 The anchored semidefinite relaxation

Use label $y_i = +1$ for early and $y_i = -1$ for tardy. The SDP uses unit vectors $v_0, v_1, \dots, v_n$ and sets

$$m_i = \langle v_0, v_i \rangle, \quad \rho_{ij} = \langle v_i, v_j \rangle. \quad (14.1)$$

Define pseudo-marginals

$$e_i = \frac{1 + m_i}{2}, \quad t_i = \frac{1 - m_i}{2} = 1 - e_i, \quad (14.2)$$

and pseudo-joint quantities

$$e_{ij} = \frac{1 + m_i + m_j + \rho_{ij}}{4}, \quad (14.3)$$

$$t_{ij} = \frac{1 - m_i - m_j + \rho_{ij}}{4}. \quad (14.4)$$

The two mixed local atoms are

$$e_i - e_{ij} = \frac{1 + m_i - m_j - \rho_{ij}}{4}, \quad e_j - e_{ij} = \frac{1 - m_i + m_j - \rho_{ij}}{4}. \quad (14.5)$$

For integral vectors $v_i = y_i v_0$, these four quantities are the four ordinary joint-event indicators.

Remark II.8 (Probabilistic interpretation). Suppose integral early–tardy partitions are combined with nonnegative weights that sum to one. The weights may be viewed as probabilities of a random integral partition. Then

$$m_i = \mathbb{E}[y_i], \quad \rho_{ij} = \mathbb{E}[y_i y_j],$$

and equations (14.2)–(14.4) have their ordinary probability meanings:

$$e_i = \Pr(i \in E), \quad t_i = \Pr(i \in T), \quad e_{ij} = \Pr(i, j \in E), \quad t_{ij} = \Pr(i, j \in T).$$

Indeed, for example,

$$\Pr(i \in E) = \mathbb{E}\left[\frac{1 + y_i}{2}\right] = \frac{1 + m_i}{2},$$

and

$$\Pr(i, j \in E) = \mathbb{E}\left[\frac{(1 + y_i)(1 + y_j)}{4}\right] = \frac{1 + m_i + m_j + \rho_{ij}}{4}.$$

The tardy formulas follow in the same way from $(1 - y_i)/2$. A general feasible SDP point need not arise from a global probability distribution over integral partitions; this is why e_i, t_i, e_{ij}, t_{ij} are called pseudo-marginals and pseudo-joint quantities. The probability interpretation will be used later to construct a transparent feasible point in the worked example; it does not make the rounding algorithm randomized.

Definition II.9 (Anchored SDP). For anchor h , the relaxation SDP_h minimizes

$$L_h = \sum_{i < j} a_{ij} e_{ij} + \sum_{i < j} b_{ij} t_{ij} + \sum_i c_i t_i \quad (14.6)$$

over unit-vector Gram matrices, subject to

$$e_{ij} \geq 0, \quad t_{ij} \geq 0, \quad (14.7)$$

$$e_i - e_{ij} \geq 0, \quad e_j - e_{ij} \geq 0 \quad (14.8)$$

for every $i < j$, and

$$m_h = 1. \quad (14.9)$$

Equivalently, use a matrix $X \succeq 0$ with $X_{kk} = 1$, $m_i = X_{0i}$, and $\rho_{ij} = X_{ij}$.

Because v_0 and v_h are unit vectors, $m_h = 1$ implies $v_h = v_0$ and $e_h = 1$.

Proposition II.10 (Validity). *Every integral partition with h early embeds in SDP_h with the same objective value. Therefore, if h belongs to an optimal early set,*

$$\text{SDP}_h \leq \text{OPT}. \quad (14.10)$$

Proof. Set $v_i = y_i v_0$. Then $m_i = y_i$, $\rho_{ij} = y_i y_j$, and the local quantities become the four 0–1 joint-event indicators. The SDP objective is exactly (12.4). Apply lemma II.7. $\square$

How to use this result. Use one symmetric matrix variable X of order $n + 1$. Add

$X \succeq 0$, $X_{kk} = 1$, the four pairwise local inequalities, and $X_{0h} = 1$. The objective is linear in X . The SDP has $O(n^2)$ constraints.

Remark II.11 (Why the local inequalities matter). The Gram constraint alone need not make every off-diagonal pseudo-joint quantity nonnegative. The early proof uses $t_{ij} \geq 0$, and the tardy proof treats $\sum b_{ij}t_{ij}$ as a nonnegative objective component. Local consistency is therefore essential.

15 Deterministic threshold rounding

Choose

$$\theta \in \left(\frac{1}{2}, 1\right), \quad \delta = 1 - \theta \in \left(0, \frac{1}{2}\right), \quad (15.1)$$

and round

$$E_\theta = \{i : e_i \geq \theta\}, \quad T_\theta = \mathcal{J} \setminus E_\theta. \quad (15.2)$$

The anchor satisfies $e_h = 1$, so E_θ is nonempty.

Split the SDP objective into

$$\mathcal{A} = \sum_{i < j} a_{ij}e_{ij}, \quad \mathcal{B} = \sum_{i < j} b_{ij}t_{ij}, \quad \mathcal{U} = \sum_i c_i t_i. \quad (15.3)$$

Thus $L_h = \mathcal{A} + \mathcal{B} + \mathcal{U}$.

Intuition. Two rounded-early jobs have early marginals whose sum exceeds one, forcing positive pseudo-joint early mass. A rounded-tardy job has tardy marginal at least δ . These two certificates drive the two different charging arguments.

16 Charging the early side

The local quantities satisfy

$$e_{ij} = e_i + e_j - 1 + t_{ij}. \quad (16.1)$$

Lemma II.12 (Early-pair threshold bound).

$$\sum_{\{i,j\} \subseteq E_\theta} a_{ij} \leq \frac{\mathcal{A}}{2\theta - 1} = \frac{\mathcal{A}}{1 - 2\delta}. \quad (16.2)$$

Proof. If $i, j \in E_\theta$, then by (16.1) and $t_{ij} \geq 0$,

$$e_{ij} \geq e_i + e_j - 1 \geq 2\theta - 1 > 0.$$

Thus $a_{ij} \leq a_{ij}e_{ij}/(2\theta - 1)$. Sum over rounded-early pairs and enlarge the right side to all pairs. $\square$

How to read this step. This explains why the threshold must exceed $1/2$. At exactly $1/2$, two jobs can each have marginal $1/2$ while their joint-early mass is zero.

17 Tardy minimum-kernel self-payment

17.1 The completed tardy kernel

Define the full matrix

$$K_{ij}^{\text{tar}} = p_i p_j \min\{s_i, s_j\} \quad (i, j \in \mathcal{J}). \quad (17.1)$$

Its off-diagonal entries are b_{ij} and its diagonal entries are

$$K_{ii}^{\text{tar}} = p_i^2 s_i = c_i.$$

Here the superscript tar labels the tardy kernel; throughout the tutorial, vector and matrix transposes are written with $^\top$.

Lemma II.13 (Minimum-kernel PSD). *The matrix K^{tar} is positive semidefinite and entrywise nonnegative.*

Proof. For $u \geq 0$, define $q_i(u) = p_i \mathbf{1}\{u \leq s_i\}$. Then

$$K_{ij}^{\text{tar}} = \int_0^\infty q_i(u) q_j(u) du. \quad (17.2)$$

Therefore, for every $x \in \mathbb{R}^n$,

$$x^\top K^{\text{tar}} x = \int_0^\infty \left(\sum_i x_i q_i(u) \right)^2 du \geq 0.$$

Entrywise nonnegativity is immediate. $\square$

Intuition. At level u , only jobs with $s_i \geq u$ remain active. The minimum kernel integrates the square of their weighted sum. The tardy self-cost is precisely the diagonal needed to obtain this Gram matrix.

17.2 Covariance domination

Set

$$z_i = \frac{v_0 - v_i}{2}. \quad (17.3)$$

Then

$$\langle z_i, z_j \rangle = t_{ij}, \quad \|z_i\|^2 = t_i, \quad \langle z_i, v_0 \rangle = t_i. \quad (17.4)$$

Let $T^{\text{SDP}} = (t_{ij})_{i,j}$, with $t_{ii} = t_i$, and let $t = (t_1, \dots, t_n)^\top$.

Lemma II.14 (Covariance domination).

$$T^{\text{SDP}} - tt^\top \succeq 0. \quad (17.5)$$

Proof. Let $u_i = z_i - t_i v_0$. By (17.4),

$$\langle u_i, u_j \rangle = t_{ij} - t_i t_j.$$

Thus $T^{\text{SDP}} - tt^\top$ is the Gram matrix of the u_i . □

How to read this step. The matrix $T^{\text{SDP}} - tt^\top$ behaves like a covariance matrix. The SDP joint-tardy matrix dominates the rank-one product of its tardy marginals in positive-semidefinite order, even though entrywise domination need not hold.

17.3 The payment inequality

For positive-semidefinite matrices A, B ,

$$\text{Tr}(AB) \geq 0, \quad (17.6)$$

because $\text{Tr}(AB) = \text{Tr}(A^{1/2}BA^{1/2})$.

Lemma II.15 (PSD tardy-pair self-payment).

$$\sum_{\{i,j\} \subseteq T_\theta} b_{ij} \leq \frac{\mathcal{U} + 2\mathcal{B}}{2\delta^2}. \quad (17.7)$$

Proof. By lemmas II.13 and II.14 and (17.6),

$$\text{Tr}(K^{\text{tar}}(T^{\text{SDP}} - tt^\top)) \geq 0,$$

so

$$t^\top K^{\text{tar}} t \leq \text{Tr}(K^{\text{tar}} T^{\text{SDP}}). \quad (17.8)$$

The right side equals

$$\text{Tr}(K^{\text{tar}} T^{\text{SDP}}) = \sum_i c_i t_i + 2 \sum_{i < j} b_{ij} t_{ij} = \mathcal{U} + 2\mathcal{B}. \quad (17.9)$$

If $i \in T_\theta$, then $t_i = 1 - e_i > \delta$. Since K^{tar} is entrywise nonnegative,

$$\begin{aligned} t^\top K^{\text{tar}} t &\geq 2 \sum_{\{i,j\} \subseteq T_\theta} b_{ij} t_i t_j \\ &\geq 2\delta^2 \sum_{\{i,j\} \subseteq T_\theta} b_{ij}. \end{aligned} \quad (17.10)$$

Combine (17.8)–(17.10). □

Proof checkpoint. The tardy self-cost does more than pay for individual tardy jobs. It supplies the diagonal that turns the tardy pair matrix into the PSD kernel K^{tar} . The trace inequality then pays for all rounded tardy pairs globally, with no vertex-degree factor.

Lemma II.16 (Tardy self-cost bound).

$$\sum_{i \in T_\theta} c_i \leq \frac{\mathcal{U}}{\delta}. \quad (17.11)$$

Proof. For $i \in T_\theta$, $t_i > \delta$, so $c_i \leq c_i t_i / \delta$. Sum over T_θ . □

18 Approximation ratio and threshold optimization

Combining [lemmas II.12](#), [II.15](#) and [II.16](#) gives

$$\Phi(E_\theta) \leq \frac{\mathcal{A}}{1-2\delta} + \frac{\mathcal{B}}{\delta^2} + \left(\frac{1}{\delta} + \frac{1}{2\delta^2}\right)\mathcal{U}. \quad (18.1)$$

Because $0 < \delta < 1/2$,

$$\frac{1}{\delta} + \frac{1}{2\delta^2} \leq \frac{1}{\delta^2}. \quad (18.2)$$

Define

$$C(\delta) = \max \left\{ \frac{1}{1-2\delta}, \frac{1}{\delta^2} \right\}. \quad (18.3)$$

Then

$$\Phi(E_\theta) \leq C(\delta)L_h. \quad (18.4)$$

Theorem II.17 (Exact-SDP rounding theorem). *For every anchor h , every feasible point of SDP_h with objective value L_h , and every $\delta \in (0, 1/2)$, threshold rounding with $\theta = 1 - \delta$ returns a nonempty early set satisfying*

$$\Phi(E_\theta) \leq C(\delta)L_h.$$

In particular, rounding an optimal feasible point gives

$$\Phi(E_\theta) \leq C(\delta) \text{SDP}_h.$$

For an anchor in an optimal early set, this further implies

$$\Phi(E_\theta) \leq C(\delta) \text{OPT}.$$

Proof. The anchor makes the rounded early set nonempty, and the first inequality is [\(18.4\)](#). The feasible region is compact, so an optimal feasible point exists; substituting $L_h = \text{SDP}_h$ gives the second inequality. The final inequality uses [\(14.10\)](#). $\square$

The two terms in [\(18.3\)](#) are equal when

$$\delta^2 = 1 - 2\delta. \quad (18.5)$$

The root in $(0, 1/2)$ is

$$\delta^* = \sqrt{2} - 1, \quad \theta^* = 2 - \sqrt{2}. \quad (18.6)$$

At this value,

$$C(\delta^*) = \frac{1}{(\sqrt{2} - 1)^2} = 3 + 2\sqrt{2}. \quad (18.7)$$

Intuition. A smaller δ means a larger early threshold. This improves the early denominator $1 - 2\delta$ but worsens the tardy denominator δ^2 . The optimal threshold balances the two losses exactly.

Corollary II.18 (Optimized exact-SDP factor). *Enumerating all anchors and using $\theta^* = 2 - \sqrt{2}$ gives factor $3 + 2\sqrt{2}$ in the exact-SDP model.*

Proof. Choose an anchor from an optimal early set and apply [theorem II.17](#). Returning the cheapest result over all anchors cannot be worse. $\square$

For the rational threshold $\theta = 7/12$, $\delta = 5/12$, and

$$\frac{1}{1 - 2\delta} = 6, \quad \frac{1}{\delta^2} = \frac{144}{25} < 6, \quad \frac{1}{\delta} + \frac{1}{2\delta^2} = \frac{132}{25} < 6.$$

This proves [corollary II.4](#) in the exact-SDP model.

How to use this result. Use $\theta = 7/12$ for the simplest proof and implementation. Use a rational approximation to $2 - \sqrt{2}$ when the best constant is desired.

19 Finite precision and polynomial running time

Precision convention. The Turing-model algorithm receives positive rational accuracy parameters. We may assume $0 < \varepsilon \leq 1$ and $0 < \alpha \leq 1$: if a requested tolerance exceeds 1, replacing it by 1 only strengthens the resulting guarantee. The usual statement for an arbitrary real $\varepsilon > 0$ follows by supplying any smaller positive rational. Thus all logarithmic precision bounds below are nonnegative.

A numerical SDP solver normally returns a point that is feasible only up to tolerance, whereas the threshold proof uses exact local inequalities and an exact Gram constraint. The goal of this section is not to demand unrealistic numerical precision. It is to convert a certified approximate point into an exactly feasible rational point while spending only the additive

objective tolerance already reserved in the approximation analysis. Exact facial reduction handles the anchor, an explicit rational interior point supplies uniform slack, and a small convex mixture repairs all residuals simultaneously.

Weak-optimization interface used below. We use the standard rational weak-optimization theorem for a well-bounded rational spectrahedron in affine coordinates; see Grötschel et al. [19], together with the semidefinite-programming formulations in Alizadeh [1], Vandenberghe and Boyd [20]. Suppose the feasible region lies in a known ball of radius R and contains a known relative ball of radius $r > 0$. Given positive rational feasibility and objective tolerances η_f and η_o , weak optimization returns a rational point, in time polynomial in the rational encoding length, $\log(R/r)$, $\log(1/\eta_f)$, and $\log(1/\eta_o)$, such that the affine matrix has least eigenvalue at least $-\eta_f$, every specified affine slack is at least $-\eta_f$, and the objective is at most the true optimum plus η_o . If the quoted weak-membership theorem is formulated with a Euclidean distance tolerance, divide that tolerance by a rational upper bound on the operator norms of the affine matrix map, the slack functionals, and the objective. Those norms have polynomial encoding length here, so this rescaling preserves polynomial time. The next lemma verifies the required inner and outer bounds for the reduced anchored SDP and then repairs the weak output to exact feasibility.

Lemma II.19 (Positive integral floor). *If $n \geq 2$, then $\text{OPT} \geq 1$.*

Proof. If a feasible canonical partition has a tardy job, its objective contains a positive integer self-term c_i . If all jobs are early, then $n \geq 2$ and the objective contains a positive integer early-pair term a_{ij} . $\square$

Lemma II.20 (Exact rational realization of an anchored SDP point). *Fix an anchor h and a rational tolerance $0 < \alpha \leq 1$. For $n \geq 2$, one can compute, in time polynomial in the binary input length and $\log(1/\alpha)$, a rational Gram matrix $X^{(h)}$ that satisfies every constraint of SDP_h exactly and whose objective value is at most*

$$\text{SDP}_h + \alpha.$$

Proof. The proof first removes the anchor degeneracy, then obtains a near-optimal rational point, and finally repairs every feasibility residual without exceeding the reserved objective tolerance. We give the feasibility repair explicitly.

Step 1: reduce the anchor face exactly. The Gram matrix X is indexed by $\{0\} \cup \mathcal{J}$ and satisfies

$$X \succeq 0, \quad X_{00} = X_{hh} = 1, \quad X_{0h} = 1.$$

Writing $\mathbf{e}_k$ for the k th standard coordinate vector, consequently

$$(\mathbf{e}_0 - \mathbf{e}_h)^\top X (\mathbf{e}_0 - \mathbf{e}_h) = 1 + 1 - 2 = 0.$$

For a positive-semidefinite matrix, $z^\top X z = 0$ implies $Xz = 0$. Hence rows and columns 0 and h are identical. Delete the duplicate row and column h and optimize over a reduced matrix

$$Y \in \mathbb{S}^n, \quad \text{indexed by } \{0\} \cup (\mathcal{J} \setminus \{h\}).$$

Conversely, duplicating row and column 0 reconstructs the original anchored matrix. We impose the unit diagonal identically by using only the off-diagonal entries of Y as variables.

For an anchor pair h, i , duplication gives

$$e_{hi} = e_i, \quad t_{hi} = 0, \quad e_i - e_{hi} = 0, \quad e_h - e_{hi} = t_i.$$

The two zero relations are identities, and $e_i, t_i \geq 0$ follow from $Y \succeq 0$ and unit diagonal. Thus only the local inequalities for pairs $i, j \neq h$ need be retained in the reduced formulation.

Step 2: exhibit a rational relative-interior point. Take

$$Y^\circ = I_n.$$

For every pair $i, j \neq h$, one has $m_i = m_j = \rho_{ij} = 0$, so all four local atoms equal $1/4$. For an anchor pair, the two nontrivial marginals equal $1/2$. Also $Y^\circ \succ 0$ with smallest eigenvalue one.

For explicit boundedness, add the redundant rational inequalities

$$-1 \leq Y_{k\ell} \leq 1 \quad (k \neq \ell),$$

which every positive-semidefinite unit-diagonal matrix already satisfies. The point I_n has unit slack in these inequalities. Moreover, if E is symmetric with zero diagonal and $\|E\|_F \leq 1/8$,

then

$$\lambda_{\min}(I_n + E) \geq 1 - \|E\|_2 \geq \frac{7}{8}.$$

Each nonanchor local atom changes by at most $3\|E\|_F/4 \leq 3/32$, so it remains at least $5/32$; each anchor marginal changes by at most $\|E\|_F/2 \leq 1/16$. Hence, in the affine space of unit-diagonal symmetric matrices with the Frobenius norm, the reduced spectrahedron contains the relative ball of radius $1/8$ about I_n . The box inequalities also give

$$\|Y - I_n\|_F \leq \sqrt{n(n-1)} < n,$$

so the same feasible region lies in the explicitly known outer ball of radius n about I_n .

Step 3: obtain a certified weak solution. Put

$$W := \sum_{i < j} (a_{ij} + b_{ij}) + \sum_i c_i.$$

Since $n \geq 2$ and all coefficients are positive integers, $W \geq 1$. Define

$$\gamma := \frac{\alpha}{6W}, \quad \tau := \frac{\gamma}{8}. \tag{19.1}$$

Because $0 < \alpha \leq 1$, one has $0 < \gamma \leq 1/6$. Invoke the weak-optimization interface above on the reduced, explicitly bounded SDP, with the diagonal equations built into the parameterization, feasibility tolerance $\eta_f = \tau$, and objective tolerance $\eta_o = \alpha/2$. It returns a rational matrix $\tilde{Y}$ such that

$$\tilde{Y} + \tau I_n \succeq 0, \tag{19.2}$$

$$s_\nu(\tilde{Y}) \geq -\tau \quad \text{for every retained local or box slack } s_\nu, \tag{19.3}$$

$$\ell_h(\tilde{Y}) \leq \text{SDP}_h + \frac{\alpha}{2}, \tag{19.4}$$

where ℓ_h denotes the reduced anchored objective. The relative inner radius $1/8$, outer radius n , and polynomial-bit coefficient norms established above are exactly the conditioning data required by the stated weak-optimization interface.

Step 4: mix with the interior point. Set

$$Y^r := (1 - \gamma)\tilde{Y} + \gamma I_n.$$

This matrix is rational and keeps the unit diagonal exactly. From (19.2),

$$Y^r \succeq (\gamma - (1 - \gamma)\tau)I_n \succeq (\gamma - \tau)I_n = \frac{7\gamma}{8}I_n \succ 0.$$

Every retained local inequality has slack at least $1/4$ at I_n , and therefore

$$s_\nu(Y^r) \geq -(1 - \gamma)\tau + \frac{\gamma}{4} \geq \frac{\gamma}{4} - \tau = \frac{\gamma}{8} > 0.$$

The redundant box constraints are repaired by the same calculation. Duplicating row and column 0 now produces a rational matrix $X^{(h)}$ satisfying the original anchor equation, positive-semidefinite constraint, unit-diagonal equations, and all local inequalities exactly.

It remains to control the objective. The approximate box constraints and $\tau \leq 1$ imply $|\tilde{Y}_{k\ell}| \leq 2$. Hence every pair pseudo-atom has absolute value at most $7/4$, every unary marginal t_i has absolute value at most $3/2$, and

$$|\ell_h(\tilde{Y})| \leq 2W, \quad 0 \leq \ell_h(I_n) \leq W.$$

Using (19.4),

$$\begin{aligned} \ell_h(Y^r) &\leq \ell_h(\tilde{Y}) + \gamma |\ell_h(I_n) - \ell_h(\tilde{Y})| \\ &\leq \text{SDP}_h + \frac{\alpha}{2} + 3\gamma W \\ &\leq \text{SDP}_h + \alpha. \end{aligned}$$

By (19.1), the last added term is exactly $3\gamma W = \alpha/2$.

Step 5: bit complexity and exact thresholding. The binary lengths of W , γ , and τ are polynomial in the input length and $\log(1/\alpha)$. Thus the weak optimization and the rational convex combination use polynomially many bit operations. The entries of $X^{(h)}$, and hence the marginals $e_i = (1 + X_{0i}^{(h)})/2$, are exact rationals. Comparison with a rational threshold θ is therefore exact, including the boundary case $e_i = \theta$. $\square$

How to read this step. The weak solver is used only to approach the optimum. Exact feasibility is restored afterward by an explicitly quantified convex mixture with the rational interior point I_n . The mixture consumes only the prescribed additive tolerance, so every rounding inequality proved for feasible SDP points remains valid without an appeal to floating-point continuity.

A rational near-optimal threshold. Let $0 < \varepsilon \leq 1$, put

$$\eta := \frac{\varepsilon}{512},$$

and recall $\delta^* = \sqrt{2} - 1$, the unique zero in $(0, 1/2)$ of $g(x) = x^2 + 2x - 1$. Exact rational evaluation gives

$$g\left(\frac{53}{128}\right) = -\frac{7}{16384} < 0, \quad g\left(\frac{107}{256}\right) = \frac{697}{65536} > 0.$$

Thus δ^* lies in the dyadic interval $[53/128, 107/256] \subset [0.4, 0.42]$. Bisect this interval using the sign of g and exact rational arithmetic until its width is at most 2η , and take the midpoint as δ . This requires $O((\log(1/\varepsilon)))$ steps, keeps δ dyadic, and guarantees

$$|\delta - \delta^*| \leq \eta.$$

The numerator and denominator of δ consequently use $O((\log(1/\varepsilon)))$ bits. On $[0.4, 0.42]$, for

$$f_1(x) = \frac{1}{1-2x}, \quad f_2(x) = \frac{1}{x^2},$$

one has

$$|f'_1(x)| \leq \frac{625}{8}, \quad |f'_2(x)| \leq \frac{125}{4}.$$

Since $f_1(\delta^*) = f_2(\delta^*) = 3 + 2\sqrt{2}$, the mean-value theorem gives

$$\begin{aligned} C(\delta) &\leq C(\delta^*) + \frac{625}{8}|\delta - \delta^*| \\ &\leq 3 + 2\sqrt{2} + \frac{625}{4096}\varepsilon \\ &< 3 + 2\sqrt{2} + \frac{\varepsilon}{2}. \end{aligned}$$

Hence both δ and $\theta = 1 - \delta$ are exact rationals of the required precision.

Theorem II.21 (Bit-model guarantee). *For every positive rational ε , rational threshold selection and polynomial-precision SDP optimization give a schedule of value at most*

$$(3 + 2\sqrt{2} + \varepsilon) \text{OPT}.$$

After the precision normalization above, the running time is polynomial in the input length and $\log(1/\varepsilon)$.

Proof. Replace ε by $\min\{\varepsilon, 1\}$ if necessary; the stronger guarantee obtained for the smaller parameter also satisfies the original request. The case $n = 1$ is solved exactly by completing the sole job at d . Assume $n \geq 2$. Use the dyadic bisection construction above to obtain a rational $\delta \in (0, 1/2)$ satisfying

$$C(\delta) < 3 + 2\sqrt{2} + \frac{\varepsilon}{2},$$

and write $\bar{C} = C(\delta)$. For each anchor, apply [lemma II.20](#) with

$$\alpha = \frac{\varepsilon}{2\bar{C}}.$$

For an anchor in an optimal early set, [proposition II.10](#) and the realization lemma give an exactly feasible rational point of value at most $\text{OPT} + \alpha$. Since the rounding inequalities apply to every feasible SDP point,

$$\Phi(E_\theta) \leq \bar{C}(\text{OPT} + \alpha) = \bar{C} \text{OPT} + \frac{\varepsilon}{2}.$$

By [lemma II.19](#), this is at most

$$\left(\bar{C} + \frac{\varepsilon}{2}\right) \text{OPT} \leq (3 + 2\sqrt{2} + \varepsilon) \text{OPT}.$$

The bisection construction gives δ and θ with $O(\log(1/\varepsilon))$ bits, and $\alpha = \varepsilon/(2\bar{C})$ has the same polynomial precision. The realization lemma then uses only polynomially many additional bits and operations. $\square$

Remark II.22 (Formal versus floating-point implementation). The theorem is a Turing-model statement: a rational weak-optimization routine produces $\tilde{Y}$, and the displayed convex

mixture produces an exactly feasible rational Gram matrix before thresholding. A practical floating-point solver may implement the same plan with certified residual bounds or rational reconstruction, but unverified solver tolerances are not used in the proof.

Section exit. Section 19 has converted weak numerical optimization into an exactly feasible rational SDP point with polynomial bit length and controlled additive objective loss. Section 20 uses that result only through Lemma II.20: threshold comparisons are exact rational comparisons, and every returned schedule is evaluated from the original integer objective.

20 Algorithm and implementation details

Algorithm: Anchored-SDP threshold approximation for AWET

Input: Jobs (p_i, w_i^E, w_i^L) and a positive rational accuracy parameter ε .

1. If $n = 1$, complete the unique job at d and stop.
2. Replace ε by $\min\{\varepsilon, 1\}$, and compute r_i, s_i and a_{ij}, b_{ij}, c_i .
3. Use the dyadic bisection construction of Section 19 to choose rational δ with $C(\delta) < 3 + 2\sqrt{2} + \varepsilon/2$, write $\bar{C} = C(\delta)$, and set $\theta = 1 - \delta$.
4. For every anchor $h \in \mathcal{J}$:
 - (a) use lemma II.20 with $\alpha = \varepsilon/(2\bar{C})$ to compute an exactly feasible rational point of SDP_h (14.6-14.9) whose value is at most $\text{SDP}_h + \alpha$;
 - (b) set $E_h = \{i : e_i \geq \theta\}$ and $T_h = \mathcal{J} \setminus E_h$;
 - (c) schedule E_h by increasing r_i ending at d and T_h by decreasing s_i beginning at d ;
 - (d) evaluate the exact cost from (12.4).
5. Return the cheapest of the n schedules.

Theorem II.23 (Algorithmic guarantee). *The algorithm is feasible, runs in polynomial time, and satisfies theorem II.3.*

Proof. Each rounded early set contains its anchor. By [proposition II.6](#), the two ratio sorts produce a feasible canonical schedule of the exact partition cost. Let E^* be optimal and choose $h \in E^*$. This anchor is enumerated; its SDP optimum is at most OPT by [proposition II.10](#); and [theorem II.21](#) gives the desired ratio. Returning the cheapest candidate cannot be worse.

Coefficient construction takes $O(n^2)$ arithmetic operations. There are n SDPs, each with a matrix of order $n + 1$ and $O(n^2)$ linear constraints. Standard weak SDP optimization, the two sorts, and exact candidate evaluation are polynomial. $\square$

How to use this result. After the rational feasibility repair, only the first Gram-matrix row is needed: $e_i = (1 + X_{0i})/2$. Compare these exact rationals with θ , sort the two sides, and evaluate the resulting schedule exactly. Thus neither threshold membership nor the final candidate comparison depends on floating-point tie handling.

21 Worked example

This example follows every line of the construction: instance data, the coefficients a_{ij} , b_{ij} , and c_i , a feasible anchored SDP moment point, all marginal and joint quantities, the three objective accounts $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{U}$, threshold rounding, schedule recovery, and comparison with an optimal schedule. The actual algorithm solves the anchored SDP. For hand verification, this example instead chooses an explicit convex mixture of integral partitions, so every marginal and joint mass can be reproduced without a numerical solver. The displayed point is pedagogical and is not asserted to be SDP-optimal.

Instance data and due date

Take the following six jobs.

job i	p_i	w_i^E	w_i^L	$r_i = w_i^E/p_i$	$s_i = w_i^L/p_i$
1	1	1	6	1	6
2	2	6	4	3	2
3	3	6	15	2	5
4	4	20	4	5	1
5	2	8	8	4	4
6	5	30	15	6	3

The ratios in the last two columns use (11.3). For example, $r_4 = w_4^E/p_4 = 20/4 = 5$ and $s_4 = w_4^L/p_4 = 4/4 = 1$. The processing times are

$$(p_1, \dots, p_6) = (1, 2, 3, 4, 2, 5),$$

so five different processing-time values occur. Equation (11.1) gives

$$d = P = 1 + 2 + 3 + 4 + 2 + 5 = 17.$$

The pair coefficients a_{ij} and b_{ij}

For every pair, the second column below substitutes the data into (12.1), and the third column substitutes them into (12.2).

pair	$a_{ij} = p_i p_j \min\{r_i, r_j\}$, from (12.1)	$b_{ij} = p_i p_j \min\{s_i, s_j\}$, from (12.2)
$\{1, 2\}$	$1 \cdot 2 \min\{1, 3\} = 2$	$1 \cdot 2 \min\{6, 2\} = 4$
$\{1, 3\}$	$1 \cdot 3 \min\{1, 2\} = 3$	$1 \cdot 3 \min\{6, 5\} = 15$
$\{1, 4\}$	$1 \cdot 4 \min\{1, 5\} = 4$	$1 \cdot 4 \min\{6, 1\} = 4$
$\{1, 5\}$	$1 \cdot 2 \min\{1, 4\} = 2$	$1 \cdot 2 \min\{6, 4\} = 8$
$\{1, 6\}$	$1 \cdot 5 \min\{1, 6\} = 5$	$1 \cdot 5 \min\{6, 3\} = 15$
$\{2, 3\}$	$2 \cdot 3 \min\{3, 2\} = 12$	$2 \cdot 3 \min\{2, 5\} = 12$
$\{2, 4\}$	$2 \cdot 4 \min\{3, 5\} = 24$	$2 \cdot 4 \min\{2, 1\} = 8$
$\{2, 5\}$	$2 \cdot 2 \min\{3, 4\} = 12$	$2 \cdot 2 \min\{2, 4\} = 8$
$\{2, 6\}$	$2 \cdot 5 \min\{3, 6\} = 30$	$2 \cdot 5 \min\{2, 3\} = 20$
$\{3, 4\}$	$3 \cdot 4 \min\{2, 5\} = 24$	$3 \cdot 4 \min\{5, 1\} = 12$
$\{3, 5\}$	$3 \cdot 2 \min\{2, 4\} = 12$	$3 \cdot 2 \min\{5, 4\} = 24$
$\{3, 6\}$	$3 \cdot 5 \min\{2, 6\} = 30$	$3 \cdot 5 \min\{5, 3\} = 45$
$\{4, 5\}$	$4 \cdot 2 \min\{5, 4\} = 32$	$4 \cdot 2 \min\{1, 4\} = 8$
$\{4, 6\}$	$4 \cdot 5 \min\{5, 6\} = 100$	$4 \cdot 5 \min\{1, 3\} = 20$
$\{5, 6\}$	$2 \cdot 5 \min\{4, 6\} = 40$	$2 \cdot 5 \min\{4, 3\} = 30$

Equation (12.3) supplies one tardy self coefficient per job:

job	substitution into $c_i = p_i^2 s_i = p_i w_i^L$	c_i
1	$1^2 \cdot 6 = 1 \cdot 6$	6
2	$2^2 \cdot 2 = 2 \cdot 4$	8
3	$3^2 \cdot 5 = 3 \cdot 15$	45
4	$4^2 \cdot 1 = 4 \cdot 4$	16
5	$2^2 \cdot 4 = 2 \cdot 8$	16
6	$5^2 \cdot 3 = 5 \cdot 15$	75

Thus

$$c = (6, 8, 45, 16, 16, 75).$$

A feasible anchored SDP moment point

Force job 1 early. Consider the following distribution over five integral early sets. The numbers in the probability column are chosen convex-combination weights, not quantities calculated from the scheduling data. Equivalently, for interpretation, one may imagine selecting one of the five integral partitions according to these probabilities. The final column is obtained from the canonical partition formula (12.4).

scenario	probability	early set $E^{(q)}$	tardy set $T^{(q)}$	$\Phi(E^{(q)})$
S_1	3/10	$\{1, 2, 3, 5\}$	$\{4, 6\}$	154
S_2	3/10	$\{1, 2, 5, 6\}$	$\{3, 4\}$	164
S_3	1/5	$\{1, 2, 6\}$	$\{3, 4, 5\}$	158
S_4	1/10	$\{1, 3, 5\}$	$\{2, 4, 6\}$	164
S_5	1/10	$\{1, 3, 4, 5\}$	$\{2, 6\}$	180

The probabilities sum to one, and job 1 belongs to every early set. Each row therefore gives an integral Gram point with $v_i = y_i v_0$ and anchor $m_1 = 1$. The probability-weighted average of those Gram matrices is positive semidefinite and satisfies all local constraints. Hence the distribution defines a feasible point of the SDP anchored at job 1, as in [proposition II.10](#).

For a distribution over integral points, equations (14.2)–(14.4) have their ordinary probability meanings:

$$e_i = \Pr(i \in E), \quad t_i = \Pr(i \in T), \quad e_{ij} = \Pr(i, j \in E), \quad t_{ij} = \Pr(i, j \in T).$$

The marginal quantities e_i and t_i

The early marginals are obtained by adding the probabilities of the scenarios in which the job is early. The tardy marginal is then $t_i = 1 - e_i$, exactly as in (14.2).

job	calculation of $e_i = \Pr(i \in E)$	e_i	$t_i = 1 - e_i$
1	$3/10 + 3/10 + 1/5 + 1/10 + 1/10$	1	0
2	$3/10 + 3/10 + 1/5$	$4/5$	$1/5$
3	$3/10 + 1/10 + 1/10$	$1/2$	$1/2$
4	$1/10$	$1/10$	$9/10$
5	$3/10 + 3/10 + 1/10 + 1/10$	$4/5$	$1/5$
6	$3/10 + 1/5$	$1/2$	$1/2$

Therefore

$$e = \left(1, \frac{4}{5}, \frac{1}{2}, \frac{1}{10}, \frac{4}{5}, \frac{1}{2}\right), \quad t = \left(0, \frac{1}{5}, \frac{1}{2}, \frac{9}{10}, \frac{1}{5}, \frac{1}{2}\right).$$

The joint quantities e_{ij} and t_{ij}

The next table performs the same probability addition for every pair. By (14.3), the middle column is e_{ij} ; by (14.4), the last column is t_{ij} . For example, jobs 2 and 6 are both early in S_2 and S_3 , so $e_{26} = 3/10 + 1/5 = 1/2$; they are both tardy in S_4 and S_5 , so $t_{26} = 1/10 + 1/10 = 1/5$.

pair	$e_{ij} = \Pr(i, j \in E)$, from (14.3)	$t_{ij} = \Pr(i, j \in T)$, from (14.4)
$\{1, 2\}$	$3/10 + 3/10 + 1/5 = 4/5$	0
$\{1, 3\}$	$3/10 + 1/10 + 1/10 = 1/2$	0
$\{1, 4\}$	$1/10$	0
$\{1, 5\}$	$3/10 + 3/10 + 1/10 + 1/10 = 4/5$	0
$\{1, 6\}$	$3/10 + 1/5 = 1/2$	0
$\{2, 3\}$	$3/10$	0
$\{2, 4\}$	0	$1/10$
$\{2, 5\}$	$3/10 + 3/10 = 3/5$	0
$\{2, 6\}$	$3/10 + 1/5 = 1/2$	$1/10 + 1/10 = 1/5$
$\{3, 4\}$	$1/10$	$3/10 + 1/5 = 1/2$
$\{3, 5\}$	$3/10 + 1/10 + 1/10 = 1/2$	$1/5$
$\{3, 6\}$	0	0
$\{4, 5\}$	$1/10$	$1/5$
$\{4, 6\}$	0	$3/10 + 1/10 = 2/5$
$\{5, 6\}$	$3/10$	0

The SDP accounts $\mathcal{A}$, $\mathcal{B}$, $\mathcal{U}$, and L

Equation (15.3) defines the early-pair account. Substituting every nonzero e_{ij} from the preceding table gives

$$\begin{aligned}\mathcal{A} &= 2 \left(\frac{4}{5} \right) + 3 \left(\frac{1}{2} \right) + 4 \left(\frac{1}{10} \right) + 2 \left(\frac{4}{5} \right) + 5 \left(\frac{1}{2} \right) \\ &\quad + 12 \left(\frac{3}{10} \right) + 12 \left(\frac{3}{5} \right) + 30 \left(\frac{1}{2} \right) \\ &\quad + 24 \left(\frac{1}{10} \right) + 12 \left(\frac{1}{2} \right) + 32 \left(\frac{1}{10} \right) + 40 \left(\frac{3}{10} \right) \\ &= 57.\end{aligned}$$

The omitted pair terms have $e_{ij} = 0$.

The tardy-pair account in the same equation uses every nonzero t_{ij} :

$$\begin{aligned}\mathcal{B} &= 8 \left(\frac{1}{10} \right) + 20 \left(\frac{1}{5} \right) + 12 \left(\frac{1}{2} \right) + 24 \left(\frac{1}{5} \right) \\ &\quad + 8 \left(\frac{1}{5} \right) + 20 \left(\frac{2}{5} \right) \\ &= \frac{126}{5} = 25.2.\end{aligned}$$

These six terms correspond respectively to pairs $\{2, 4\}$, $\{2, 6\}$, $\{3, 4\}$, $\{3, 5\}$, $\{4, 5\}$, and $\{4, 6\}$.

Finally, the unary account in (15.3) is

$$\begin{aligned}\mathcal{U} &= 6(0) + 8 \left(\frac{1}{5} \right) + 45 \left(\frac{1}{2} \right) + 16 \left(\frac{9}{10} \right) + 16 \left(\frac{1}{5} \right) + 75 \left(\frac{1}{2} \right) \\ &= \frac{396}{5} = 79.2.\end{aligned}$$

For this feasible point, write $L = L_1$. Equation (14.6), or equivalently the account split immediately following (15.3), gives

$$L = \mathcal{A} + \mathcal{B} + \mathcal{U} = 57 + \frac{126}{5} + \frac{396}{5} = \frac{807}{5} = 161.4.$$

There is also a useful independent check. Because this point is a probability mixture of

integral partitions, its SDP objective is their expected canonical cost:

$$\begin{aligned} L &= \frac{3}{10}(154) + \frac{3}{10}(164) + \frac{1}{5}(158) + \frac{1}{10}(164) + \frac{1}{10}(180) \\ &= \frac{807}{5}. \end{aligned}$$

Pedagogical point versus algorithmic output. The threshold rule below is the algorithmic rule, but it is being applied to the auditable convex-mixture point just constructed, not to an SDP optimum returned by a solver. The comparison with the true scheduling optimum therefore illustrates the rounding mechanics rather than the worst-case certificate of the implemented algorithm.

Threshold rounding

Equation (18.6) gives

$$\theta^* = 2 - \sqrt{2} \approx 0.585786.$$

Applying the rule (15.2) to the six early marginals gives

i	1	2	3	4	5	6
e_i	1	0.8	0.5	0.1	0.8	0.5
$e_i \geq \theta^*$	yes	yes	no	no	yes	no

Hence

$$E_{\theta^*} = \{1, 2, 5\}, \quad T_{\theta^*} = \{3, 4, 6\}.$$

Exact cost of the rounded canonical schedule

First evaluate the partition directly from (12.4). The early-pair part is

$$a_{12} + a_{15} + a_{25} = 2 + 2 + 12 = 16.$$

The tardy-pair part is

$$b_{34} + b_{36} + b_{46} = 12 + 45 + 20 = 77.$$

The tardy self part is

$$c_3 + c_4 + c_6 = 45 + 16 + 75 = 136.$$

Therefore the exact canonical cost is

$$\Phi(E_{\theta^*}) = 16 + 77 + 136 = 229.$$

Now recover the actual schedule, as prescribed after (12.5). The early jobs are sorted by increasing ratio,

$$1 \ (r_1 = 1), \quad 2 \ (r_2 = 3), \quad 5 \ (r_5 = 4),$$

and their block starts at

$$d - (p_1 + p_2 + p_5) = 17 - (1 + 2 + 2) = 12.$$

The tardy jobs are sorted by decreasing ratio,

$$3 \ (s_3 = 5), \quad 6 \ (s_6 = 3), \quad 4 \ (s_4 = 1),$$

and their block starts at $d = 17$. The next table evaluates every job from the original objective (11.2).

position	job	interval	C_i	deviation	applicable weight	cost
1	1	[12, 13]	13	$E_1 = 17 - 13 = 4$	$w_1^E = 1$	$1(4) = 4$
2	2	[13, 15]	15	$E_2 = 17 - 15 = 2$	$w_2^E = 6$	$6(2) = 12$
3	5	[15, 17]	17	$E_5 = 0$	$w_5^E = 8$	$8(0) = 0$
4	3	[17, 20]	20	$T_3 = 20 - 17 = 3$	$w_3^L = 15$	$15(3) = 45$
5	6	[20, 25]	25	$T_6 = 25 - 17 = 8$	$w_6^L = 15$	$15(8) = 120$
6	4	[25, 29]	29	$T_4 = 29 - 17 = 12$	$w_4^L = 4$	$4(12) = 48$
				total	$4 + 12 + 0 + 45 + 120 + 48 = 229$	

Thus the job-by-job cost from (11.2) agrees exactly with the pair-and-self calculation from (12.4).

The optimal solution and its cost

Equation (12.5) permits exact enumeration because there are only $2^6 - 1 = 63$ nonempty early sets. Evaluating (12.4) for every set and grouping the results by $|E|$ gives the following minimum in each cardinality class.

$ E $	minimizing early set within this class	minimum $\Phi(E)$
1	$\{6\}$	194
2	$\{3, 6\}$	116
3	$\{1, 3, 6\}$	102
4	$\{1, 2, 3, 6\}$	122
5	$\{1, 2, 3, 5, 6\}$	164
6	$\{1, 2, 3, 4, 5, 6\}$	332

Since these six classes partition all nonempty early sets, the unique optimum is

$$E^* = \{1, 3, 6\}, \quad T^* = \{2, 4, 5\}, \quad \text{OPT} = 102.$$

Its cost from (12.4) is

$$\begin{aligned}
\Phi(E^*) &= \underbrace{(a_{13} + a_{16} + a_{36})}_{\text{early pairs}} + \underbrace{(b_{24} + b_{25} + b_{45})}_{\text{tardy pairs}} + \underbrace{(c_2 + c_4 + c_5)}_{\text{tardy self terms}} \\
&= (3 + 5 + 30) + (8 + 8 + 8) + (8 + 16 + 16) \\
&= 38 + 24 + 40 = 102.
\end{aligned}$$

For completeness, the optimal canonical schedule can also be checked job by job. The early order is 1, 3, 6 because $1 < 2 < 6$ are their r -values, and its start time is

$$17 - (p_1 + p_3 + p_6) = 17 - (1 + 3 + 5) = 8.$$

The tardy order is 5, 2, 4 because $4 > 2 > 1$ are their s -values.

position	job	interval	C_i	deviation	applicable weight	cost
1	1	$[8, 9]$	9	$E_1 = 17 - 9 = 8$	$w_1^E = 1$	$1(8) = 8$
2	3	$[9, 12]$	12	$E_3 = 17 - 12 = 5$	$w_3^E = 6$	$6(5) = 30$
3	6	$[12, 17]$	17	$E_6 = 0$	$w_6^E = 30$	$30(0) = 0$
4	5	$[17, 19]$	19	$T_5 = 19 - 17 = 2$	$w_5^L = 8$	$8(2) = 16$
5	2	$[19, 21]$	21	$T_2 = 21 - 17 = 4$	$w_2^L = 4$	$4(4) = 16$
6	4	$[21, 25]$	25	$T_4 = 25 - 17 = 8$	$w_4^L = 4$	$4(8) = 32$
total						$8 + 30 + 0 + 16 + 16 + 32 = 102$

The rounded schedule therefore has additive gap $229 - 102 = 127$ and comparison ratio

$$\frac{\Phi(E_{\theta^*})}{\text{OPT}} = \frac{229}{102} \approx 2.2451 < 3 + 2\sqrt{2}.$$

How to read this step. The example makes the global tardy-side argument visible. Jobs 3 and 6 are both rounded tardy even though their pseudo-joint quantity is $t_{36} = 0$, while their actual canonical pair cost is $b_{36} = 45$. That cost cannot be paid by an entrywise comparison with t_{36} . It is covered only by the global PSD inequality, which combines the pair account $\mathcal{B}$ with the diagonal self account $\mathcal{U}$.

22 Why the proof works

22.1 Early pairs use local consistency

The early estimate needs no PSD property of the early kernel. It uses only $e_{ij} = e_i + e_j - 1 + t_{ij}$ and $t_{ij} \geq 0$. The threshold replaces the missing early diagonal.

22.2 Tardy pairs use the kernel diagonal

The tardy self-costs are exactly the diagonal of K^{tar} . They complete the off-diagonal tardy pair matrix into a minimum-kernel Gram matrix. This structure is stronger than a generic weighted binary constraint system.

22.3 The rounding is deterministic

No random hyperplane is needed. A common marginal threshold avoids the linear-versus-quadratic mismatch that can make root-relative hyperplane rounding poor on unary tardiness terms.

22.4 The anchor is a feasibility device

The proof uses the anchor only to ensure $E_\theta \neq \emptyset$. Enumerating all jobs guarantees one anchor in the optimal early set.

Proof checkpoint.

local joint consistency pays for early pairs + minimum-kernel PSD with its diagonal pays for tardy pairs
--

Neither ingredient alone is sufficient for unrestricted asymmetric AWET.

23 Concluding remarks

The canonical side-assignment formulation of AWET has two unrelated minimum kernels and an asymmetric diagonal term. The correct approximation proof uses that asymmetry rather than trying to remove it. A threshold above one half pays for early pairs through local pseudo-probability consistency. The tardy self-cost completes the tardy minimum kernel into a positive-semidefinite matrix, and the SDP covariance matrix pays globally for all rounded tardy pairs. Balancing the two threshold losses gives $3 + 2\sqrt{2}$.

The resulting algorithm is direct: enumerate an early anchor, solve a polynomial-size SDP, threshold the early marginals, and build the canonical V-shaped schedule. This establishes a polynomial-time constant-factor approximation for unrestricted AWET.

Reusable idea. When pair costs form a positive-semidefinite kernel only after diagonal completion, a unary or self-cost term can be an asset rather than a nuisance. Combine that diagonal with a covariance domination inequality to pay all rounded pairs globally, and use a separate local-consistency argument on any side lacking a natural diagonal.

Why Part III is next. Part II has established a universal approximation guarantee without restricting either ratio order. It does not classify exact solvability on highly structured order patterns. Part III moves to the maximally opposed strict order, where a completed load square gives weak hardness and a small two-resource state space gives pseudopolynomial solvability.

Part III

Antithetical Ordering

Weak NP-Completeness at Exact Reversal

Part synopsis. This Part classifies the strict antithetical ratio-order class. A positional tagging lemma converts PARTITION into an exact half-load question. Affine opposite ratio systems and one anchor then produce a dominant completed square in the selected processing load, while an explicitly bounded cubic residual cannot erase a unit load gap. The same reversed order yields a direct two-resource pseudopolynomial recurrence. Exact reversal is also a separable permutation, so the weak hardness and pseudopolynomial algorithm proved here are compatible with the separating-tree FPTAS in Part IV rather than in tension with it.

Part positioning. **Starting boundary.** The unrestricted problem is strongly NP-complete, but that construction does not classify the exact-reversal subclass. **Result.** Strict antithetical AWET is weakly NP-complete and has an exact pseudopolynomial algorithm. **Technique.** Binary positional tags create an exact half-load target; an anchor converts the dominant scheduling contribution into a completed load square, and a two-resource recurrence exploits the reversed order.

24 Introduction to the antithetical theorem

The antithetical class orders the two Smith-ratio systems in opposite directions. Unlike the agreeable class, it does not contain the symmetric Hall–Posner hard core, so a separate reduction is required. The construction below starts from PARTITION, encodes the source choice as an exact half-load condition, and then makes the dominant scheduling cost a convex square in that selected load. A single anchor supplies the required linear field and is itself forced to the early side.

Exact reversal is simultaneously hard and highly structured. Its ratio permutation is the reverse permutation, which is separable after the fixed strict refinements used here. Consequently, the weak hardness and pseudopolynomial recurrence in this Part and the separating-tree FPTAS in Part IV describe complementary exact and approximation properties of the same order extreme.

The canonical side-assignment formula is the only scheduling structural fact needed, and it is recorded first for completeness.

An AWET instance consists of jobs $j \in \mathcal{J} = \{1, \dots, n\}$ with positive integer processing time p_j , earliness weight w_j^E , and tardiness weight w_j^L . All jobs have the common due date

$$d = P := \sum_{j \in \mathcal{J}} p_j.$$

For completion time C_j , the objective is

$$F = \sum_{j \in \mathcal{J}} \left(w_j^E (d - C_j)^+ + w_j^L (C_j - d)^+ \right).$$

Put

$$r_j = \frac{w_j^E}{p_j}, \quad s_j = \frac{w_j^L}{p_j}.$$

Definition III.1 (Strict antithetical ratio order). An instance is *strict antithetical* if its jobs can be indexed so that

$$r_1 < r_2 < \dots < r_n \quad \text{and} \quad s_1 > s_2 > \dots > s_n.$$

Thus the two strict Smith-ratio orders are exact reversals.

The standard canonical-schedule theorem converts the problem into a binary side assignment. The following form is the only scheduling fact used in the reductions.

Proposition III.2 (Canonical early-set objective). *For every nonempty $E \subseteq \mathcal{J}$, put $T = \mathcal{J} \setminus E$. The compact canonical schedule that orders E chronologically by nondecreasing r_i , ends its early block at d , orders T chronologically by nonincreasing s_i , and begins its tardy block at d has cost*

$$\Phi(E) = \sum_{\{i,j\} \subseteq E} p_i p_j \min\{r_i, r_j\} + \sum_{\{i,j\} \subseteq T} p_i p_j \min\{s_i, s_j\} + \sum_{i \in T} p_i^2 s_i. \quad (24.1)$$

Every nonempty E gives a feasible canonical schedule of this form, and

$$\text{OPT} = \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E).$$

Proof. Fix a nonempty early set E and let $T = \mathcal{J} \setminus E$. Its canonical early block starts at

$$d - \sum_{i \in E} p_i = \sum_{i \in T} p_i \geq 0,$$

ends at d , and is followed immediately by the canonical tardy block. Thus the displayed construction is feasible.

Read outward from the due date, the adjacent-exchange argument orders the early jobs by nonincreasing $r_j = w_j^E/p_j$; equivalently, their chronological order before d is nondecreasing in r_j . The tardy jobs are read both chronologically and outward from d , and are ordered by nonincreasing $s_j = w_j^L/p_j$.

To verify the pair coefficients, let early job i lie farther from d than early job j . The processing of j contributes $w_j^E p_j = p_i p_j r_i$ to the earliness of i , and the exchange order places the smaller of r_i, r_j farther from d . Hence the pair contributes $p_i p_j \min\{r_i, r_j\}$. Similarly, if tardy job i is closer to d than tardy job j , the processing of i contributes $p_i w_j^L = p_i p_j s_j$ to the tardiness of j , and the exchange order places the larger tardiness ratio closer to d . The tardy pair accordingly contributes $p_i p_j \min\{s_i, s_j\}$. Each tardy job also contributes the self term $p_i^2 s_i$. Summing these terms proves (24.1).

Because every nonempty E supplies a feasible canonical schedule, $\text{OPT} \leq \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E)$. Conversely, by lemma II.5, a global optimum can be chosen compact, with a nonempty early/on-time arm and the two canonical ratio orders. If E^* is its early set, the preceding

calculation gives its cost as $\Phi(E^*)$. Therefore $\text{OPT} \geq \min_{\emptyset \neq E \subseteq \mathcal{J}} \Phi(E)$, proving equality. $\square$

All quantities in (24.1) are integers when the weights and processing times are integers, since

$$p_i p_j \min\{r_i, r_j\} = \min\{w_i^E p_j, p_i w_j^E\},$$

and similarly on the tardy side. Hence whenever one canonical objective value is strictly larger than another, it exceeds it by at least one. This unit integrality fact is used in the separation below.

Proof checkpoint. From this point onward, optimization and the reduction threshold can be analyzed through canonical binary side assignments: every assignment gives a feasible canonical schedule, and some global optimum is canonical. The reduction therefore needs only to control the algebra of the canonical objective.

Notation reset. The source instance consists of positive integers $b_1, \dots, b_m$ with total $2B$. Binary positional tags produce strictly increasing values $a_1, \dots, a_N$ with total $Q = \sum_{i=1}^N a_i$. For an early-side vector $x \in \{0, 1\}^N$, the selected tagged load is

$$S(x) = \sum_{i=1}^N a_i x_i.$$

The later construction sets the residual budget $U = 3Q^3$, the dominant scale $M = 8Q^3$, and the common-due-date scale $D = M + Q$. The symbols C and K denote, respectively, the all-variable-tardy baseline with the anchor early and the decision threshold. These quantities are previewed here so that the proof architecture can be read before their formal construction.

25 Antithetical hardness: proof architecture

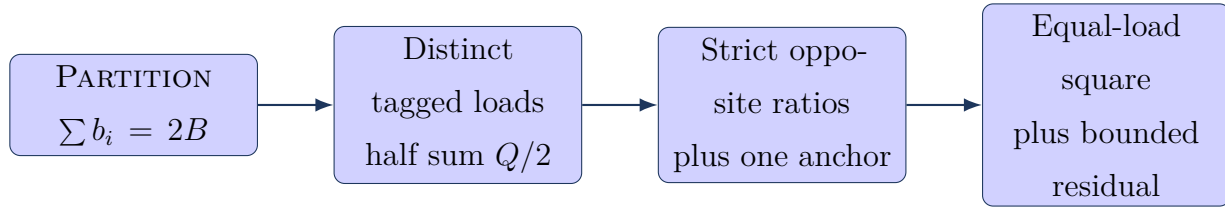
Proof contract.

- **Input:** a binary PARTITION instance with total sum $2B$.
- **Output:** a positive-integer AWET instance with strict reversed ratio orders and threshold K .
- **Forward invariant:** a subset of sum B becomes a tagged load $Q/2$ and eliminates the dominant square term.

- **Reverse invariant:** if any schedule has cost at most K , a global optimum can be chosen canonical; that canonical optimum keeps the anchor early and has tagged load exactly $Q/2$.
- **Size obligation:** the tag, scale, weights, and threshold have polynomial binary encoding length.

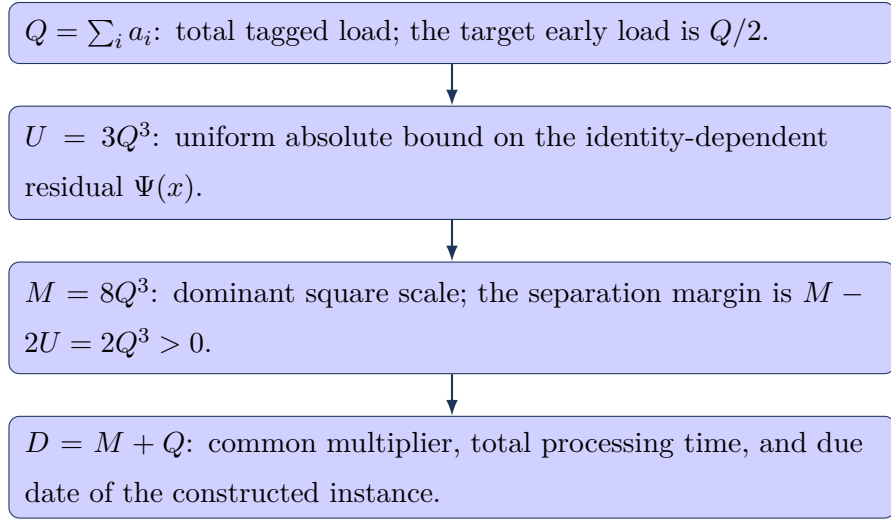
26 Reduction roadmap

The antithetical case does not contain the symmetric hard core, so a separate reduction is needed. The construction has four layers.



- Step 1.** Binary positional tags force every half-sum subset to select exactly one number from each source pair, while the low-order part carries the original partition equation.
- Step 2.** Each tagged number becomes a processing time. Early ratios increase affinely with processing time and tardiness ratios decrease affinely, producing a strict antithetical chain.
- Step 3.** A single final anchor supplies the linear field needed to make the dominant assignment cost $DM(S^2 - QS)$.
- Step 4.** The remaining cubic expression has absolute value at most $3Q^3$. Taking $M = 8Q^3$ leaves a strict gap between $S = Q/2$ and every other integer load.

Scale ledger. The four scales should be remembered by role rather than by magnitude alone.



The proof later uses the ledger only through $|\Psi| \leq U$, the unit load gap, and $M > 2U$.

The reduction is developed below from source equivalence through the cost identity, anchor proof, gap separation, and encoding-length audit.

27 A distinct half-sum compiler

Begin with the standard PARTITION problem on positive integers $b_1, \dots, b_m$ [6]. If the total sum is odd, output the following fixed strict-antithetical no-instance of AWET: two jobs

$$(p_1, w_1^E, w_1^L) = (1, 1, 2), \quad (p_2, w_2^E, w_2^L) = (2, 4, 2),$$

with due date $d = 3$ and decision bound $K = 1$. Its ratios satisfy $r_1 = 1 < 2 = r_2$ and $s_1 = 2 > 1 = s_2$. The three nonempty early sets have canonical costs 4, 2, and 2, respectively, so this fixed target instance is a no-instance.

It therefore remains only to describe an even-total source instance. Write

$$\sum_{i=1}^m b_i = 2B,$$

and ask whether some subcollection sums to B . Put

$$\kappa = 2B + 1, \quad L_i = \kappa 2^{i-1},$$

and create the two numbers

$$a_{2i-1} = L_i, \quad a_{2i} = L_i + b_i \quad (i = 1, \dots, m). \quad (27.1)$$

Let $N = 2m$ and $Q = \sum_{j=1}^N a_j$.

Arithmetic guide. For any candidate subset of the tagged numbers, let $m_i \in \{0, 1, 2\}$ denote the number selected from the pair $\{a_{2i-1}, a_{2i}\}$. The positional argument uses the strict superincreasing inequality

$$2^{k-1} > \sum_{i=1}^{k-1} 2^{i-1}.$$

It does *not* appeal to uniqueness of an arbitrary base-two representation with digits in $\{0, 1, 2\}$. What matters is more specific: the target coefficient at every position is 1, while each deviation $m_i - 1$ lies in $\{-1, 0, 1\}$. Consequently, the largest nonzero deviation cannot be cancelled by all lower positions together.

Lemma III.3 (Binary pair tagging). *The numbers in (27.1) are strictly increasing, Q is even, and the following are equivalent:*

- (i) *some subset of $\{b_1, \dots, b_m\}$ sums to B ;*
- (ii) *some subset of $\{a_1, \dots, a_N\}$ sums to $Q/2$.*

Moreover, every half-sum subset chooses exactly one member of each pair $\{a_{2i-1}, a_{2i}\}$.

Proof. The total tagged sum is

$$\begin{aligned} Q &= \sum_{i=1}^m (L_i + (L_i + b_i)) \\ &= 2 \sum_{i=1}^m L_i + \sum_{i=1}^m b_i \\ &= 2 \sum_{i=1}^m L_i + 2B. \end{aligned}$$

Hence Q is even and

$$\frac{Q}{2} = \sum_{i=1}^m L_i + B. \quad (27.2)$$

Consider an arbitrary subset of the $2m$ tagged numbers. For pair i , let $m_i \in \{0, 1, 2\}$ be the number of selected tags and let $z_i \in \{0, 1\}$ indicate whether the second tag $L_i + b_i$ is selected. Its sum is therefore

$$\sum_{i=1}^m m_i L_i + \sum_{i=1}^m z_i b_i = \kappa \sum_{i=1}^m m_i 2^{i-1} + \sum_{i=1}^m z_i b_i. \quad (27.3)$$

Suppose this sum equals $Q/2$. Substituting (27.2) and $L_i = \kappa 2^{i-1}$ gives

$$\kappa \sum_{i=1}^m m_i 2^{i-1} + \sum_{i=1}^m z_i b_i = \kappa \sum_{i=1}^m 2^{i-1} + B. \quad (27.4)$$

Both B and $\sum_i z_i b_i$ lie in $[0, 2B] = [0, \kappa - 1]$. Reducing (27.4) modulo κ therefore gives not merely a congruence but the exact equality

$$\sum_{i=1}^m z_i b_i = B. \quad (27.5)$$

Subtract (27.5) from (27.4) and divide by κ :

$$\begin{aligned} \sum_{i=1}^m m_i 2^{i-1} &= \sum_{i=1}^m 2^{i-1}, \\ \sum_{i=1}^m (m_i - 1) 2^{i-1} &= 0. \end{aligned}$$

Suppose, for contradiction, that some $m_i \neq 1$, and let k be the largest such index. Then $m_k - 1 \in \{-1, 1\}$, and the preceding equality can be written line by line as

$$\begin{aligned} 0 &= \sum_{i=1}^m (m_i - 1) 2^{i-1} \\ &= (m_k - 1) 2^{k-1} + \sum_{i=1}^{k-1} (m_i - 1) 2^{i-1}. \end{aligned}$$

Moving the lower-position sum to the other side and taking absolute values gives

$$\begin{aligned} 2^{k-1} &= |(m_k - 1) 2^{k-1}| \\ &= \left| \sum_{i=1}^{k-1} (m_i - 1) 2^{i-1} \right| \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{i=1}^{k-1} |m_i - 1| 2^{i-1} \\
&\leq \sum_{i=1}^{k-1} 2^{i-1} \\
&= 2^{k-1} - 1,
\end{aligned}$$

a contradiction. Hence $m_i = 1$ for every i . Every half-sum subset therefore chooses exactly one tag from each pair, and (27.5) decodes a source subset of sum B .

Conversely, suppose $I \subseteq \{1, \dots, m\}$ satisfies $\sum_{i \in I} b_i = B$. Select $L_i + b_i$ when $i \in I$ and select L_i otherwise. Exactly one tag is chosen from each pair, and its sum is

$$\sum_{i=1}^m L_i + \sum_{i \in I} b_i = \sum_{i=1}^m L_i + B = \frac{Q}{2}.$$

This proves the equivalence.

Finally, $L_i < L_i + b_i$ because $b_i > 0$. For $i = 1, \dots, m-1$, one also has $b_i < \kappa$, so

$$L_i + b_i < \kappa 2^{i-1} + \kappa \leq \kappa 2^i = L_{i+1}.$$

The within-pair inequalities and these cross-pair inequalities together show that the tagged sequence is strictly increasing. $\square$

Example III.4 (A small yes-instance). For $(b_1, b_2, b_3) = (1, 2, 3)$, one has $B = 3$, $\kappa = 7$, and

$$(a_1, \dots, a_6) = (7, 8, 14, 16, 28, 31).$$

The total is $Q = 104$ and the half-total is 52. The source solution $1 + 2 = 3$ becomes

$$8 + 16 + 28 = 52,$$

whereas the source solution $3 = 3$ becomes

$$7 + 14 + 31 = 52.$$

Each tagged solution visibly selects one number from every pair.

28 The strict antithetical scheduling instance

Start with the strictly increasing tagged values

$$0 < a_1 < a_2 < \cdots < a_N, \quad Q = \sum_{i=1}^N a_i,$$

where Q is even. Define

$$U = 3Q^3, \quad M = 8Q^3, \quad D = M + Q. \quad (28.1)$$

For every variable job $i = 1, \dots, N$, set

$$p_i = a_i, \quad (28.2)$$

$$r_i = MQ + Da_i, \quad (28.3)$$

$$s_i = M(Q + 2M) - Da_i, \quad (28.4)$$

$$w_i^E = p_i r_i, \quad w_i^L = p_i s_i. \quad (28.5)$$

Add one anchor job g with

$$p_g = M, \quad (28.6)$$

$$r_g = MQ + D(Q + 1), \quad (28.7)$$

$$s_g = DM, \quad (28.8)$$

$$w_g^E = p_g r_g, \quad w_g^L = p_g s_g. \quad (28.9)$$

The common due date is especially simple:

$$d = \sum_{i=1}^N p_i + p_g = Q + M = D.$$

Lemma III.5 (Strict antithetical order and positive data). *All constructed values are positive integers, all processing times are pairwise distinct, and*

$$r_1 < r_2 < \cdots < r_N < r_g, \quad s_1 > s_2 > \cdots > s_N > s_g.$$

Table 5: The two job types in the antithetical reduction.

Job	Processing time	Early ratio	Tardiness ratio	Role
Variable i	a_i	$MQ + Da_i$	$M(Q + 2M) - Da_i$	carries the tagged subset load
Anchor g	M	$MQ + D(Q + 1)$	DM	forces the side and creates the dominant linear field

Proof. The tagged a_i are strictly increasing, and $M > Q > a_N$. Therefore

$$r_{i+1} - r_i = D(a_{i+1} - a_i) > 0, \quad s_i - s_{i+1} = D(a_{i+1} - a_i) > 0.$$

Also $r_g > MQ + DQ \geq r_N$. For positivity,

$$s_N \geq M(Q + 2M) - DQ = 2M^2 - Q^2 > 0.$$

Finally,

$$s_N - s_g \geq M(Q + 2M) - DQ - DM = M^2 - MQ - Q^2 > 0,$$

because $M > 2Q$. The anchor processing time M exceeds every a_i , so all processing times are distinct as well. $\square$

29 The exact equal-load square

Assume temporarily that the anchor is early. Let $x_i = 1$ if variable job i is early and $x_i = 0$ if it is tardy, and define

$$S(x) = \sum_{i=1}^N a_i x_i, \quad P_i = \sum_{h=1}^i a_h.$$

Let C be the cost with the anchor early and every variable tardy:

$$C = \sum_{1 \leq i < j \leq N} a_i a_j s_j + \sum_{i=1}^N a_i^2 s_i. \quad (29.1)$$

Define the residual

$$\Psi(x) := \sum_{1 \leq i < j \leq N} a_i a_j (a_i - a_j) x_i x_j \quad (29.2)$$

$$+ \sum_{i=1}^N a_i \left(a_i P_i + \sum_{j>i} a_j^2 \right) x_i. \quad (29.3)$$

Lemma III.6 (Exact cost identity). *For every $x \in \{0, 1\}^N$ with the anchor early,*

$$\Phi(x) = C + DM(S(x)^2 - QS(x)) + D\Psi(x). \quad (29.4)$$

Proof of lemma III.6. The calculation has three explicit purposes: write the canonical objective as a quadratic polynomial in the side bits, isolate its linear and pair coefficients by finite differences, and regroup the dominant terms into a function of the single aggregate load $S(x)$. The residual $\Psi(x)$ will then contain every dependence on job identity and position.

Assume that the anchor job g is early. Recall that

$$Q = \sum_{i=1}^N a_i, \quad D = M + Q,$$

and that variable job i has

$$p_i = a_i, \quad r_i = MQ + Da_i, \quad s_i = M(Q + 2M) - Da_i.$$

The anchor has processing time $p_g = M$ and early ratio

$$r_g = MQ + D(Q + 1) > r_i \quad (i = 1, \dots, N).$$

Moreover, for $i < j$,

$$r_i < r_j \quad \text{and} \quad s_i > s_j.$$

Let $x_i = 1$ when variable job i is early and $x_i = 0$ when it is tardy, and put

$$S(x) = \sum_{i=1}^N a_i x_i, \quad P_i = \sum_{h=1}^i a_h.$$

The all-variable-tardy baseline, with the anchor still early, is

$$C = \sum_{1 \leq i < j \leq N} a_i a_j s_j + \sum_{i=1}^N a_i^2 s_i. \quad (29.5)$$

The residual appearing in the statement of the lemma is

$$\Psi(x) = \sum_{1 \leq i < j \leq N} a_i a_j (a_i - a_j) x_i x_j + \sum_{i=1}^N a_i \left(a_i P_i + \sum_{j>i} a_j^2 \right) x_i. \quad (29.6)$$

We first write the full canonical cost as a function of the side vector x . If variable job i is early, its early pair with the anchor contributes

$$p_g p_i \min\{r_g, r_i\} = M a_i r_i.$$

For $i < j$, two early variable jobs contribute $a_i a_j r_i$, because $r_i < r_j$, whereas two tardy variable jobs contribute $a_i a_j s_j$, because $s_i > s_j$. A tardy variable job i also contributes its self term $a_i^2 s_i$. Hence

$$\begin{aligned} \Phi(x) &= \sum_{i=1}^N M a_i r_i x_i + \sum_{1 \leq i < j \leq N} a_i a_j r_i x_i x_j \\ &\quad + \sum_{1 \leq i < j \leq N} a_i a_j s_j (1 - x_i)(1 - x_j) + \sum_{i=1}^N a_i^2 s_i (1 - x_i). \end{aligned} \quad (29.7)$$

Setting $x = \mathbf{0}$ in (29.7) gives exactly (29.5), so

$$\Phi(\mathbf{0}) = C.$$

We next define the coefficients that will be used in the quadratic expansion. Let $\mathbf{e}_i$ denote the binary vector whose only nonzero coordinate is coordinate i . Define

$$h_i := \Phi(\mathbf{e}_i) - \Phi(\mathbf{0}), \quad (29.8)$$

and, for $i < j$, define the two-variable interaction coefficient

$$q_{ij} := \Phi(\mathbf{e}_i + \mathbf{e}_j) - \Phi(\mathbf{e}_i) - \Phi(\mathbf{e}_j) + \Phi(\mathbf{0}). \quad (29.9)$$

We now verify the coefficient representation directly. Subtracting (29.5) from (29.7) and expanding $(1 - x_i)(1 - x_j) = 1 - x_i - x_j + x_i x_j$ gives

$$\begin{aligned} \Phi(x) - C &= \sum_{i=1}^N M a_i r_i x_i + \sum_{1 \leq i < j \leq N} a_i a_j r_i x_i x_j \\ &\quad + \sum_{1 \leq i < j \leq N} a_i a_j s_j - \sum_{1 \leq i < j \leq N} a_i a_j s_j x_i - \sum_{1 \leq i < j \leq N} a_i a_j s_j x_j \\ &\quad + \sum_{1 \leq i < j \leq N} a_i a_j s_j x_i x_j + \sum_{i=1}^N a_i^2 s_i - \sum_{i=1}^N a_i^2 s_i x_i \\ &\quad - \sum_{1 \leq i < j \leq N} a_i a_j s_j - \sum_{i=1}^N a_i^2 s_i. \end{aligned}$$

The two displayed pairs of constant sums cancel. Hence

$$\begin{aligned} \Phi(x) - C &= \sum_{i=1}^N M a_i r_i x_i - \sum_{i=1}^N a_i^2 s_i x_i \\ &\quad - \sum_{1 \leq i < j \leq N} a_i a_j s_j x_i - \sum_{1 \leq i < j \leq N} a_i a_j s_j x_j \\ &\quad + \sum_{1 \leq i < j \leq N} a_i a_j (r_i + s_j) x_i x_j. \end{aligned}$$

Grouping all linear terms that multiply a fixed x_i yields

$$\begin{aligned} \Phi(x) - C &= \sum_{i=1}^N \left(M a_i r_i - a_i^2 s_i - a_i \sum_{j=i+1}^N a_j s_j - a_i s_i \sum_{j=1}^{i-1} a_j \right) x_i \\ &\quad + \sum_{1 \leq i < j \leq N} a_i a_j (r_i + s_j) x_i x_j. \end{aligned}$$

For a quadratic polynomial with zero constant term, evaluation at $\mathbf{e}_i$ isolates the coefficient of x_i , while the two-coordinate finite difference in (29.9) isolates the coefficient of $x_i x_j$. Therefore the definitions (29.8)–(29.9) give the exact expansion

$$\Phi(x) - C = \sum_{i=1}^N h_i x_i + \sum_{1 \leq i < j \leq N} q_{ij} x_i x_j. \quad (29.10)$$

The following calculations recover these two displayed coefficients term by term and then simplify them using the constructed ratios.

Calculation of q_{ij} . Fix $i < j$. In the four values appearing in (29.9), every term not involving

both i and j cancels. The pair $\{i, j\}$ itself has the following costs:

side vector for jobs i, j	cost of pair $\{i, j\}$
$(0, 0)$	$a_i a_j s_j$
$(1, 0)$	0
$(0, 1)$	0
$(1, 1)$	$a_i a_j r_i$.

The mixed cases cost zero because the two jobs lie on opposite sides of the due date. Substituting these four pair costs into (29.9) gives

$$q_{ij} = a_i a_j (r_i + s_j). \quad (29.11)$$

Using the definitions of r_i and s_j and then $D = M + Q$,

$$\begin{aligned}
 q_{ij} &= a_i a_j (MQ + Da_i + M(Q + 2M) - Da_j) \\
 &= a_i a_j (2MQ + 2M^2 + D(a_i - a_j)) \\
 &= a_i a_j (2M(M + Q) + D(a_i - a_j)) \\
 &= D a_i a_j (2M + a_i - a_j) \\
 &= 2DM a_i a_j + D a_i a_j (a_i - a_j). \quad (29.12)
 \end{aligned}$$

Calculation of h_i . Fix i . By definition (29.8), h_i is the exact change in cost when job i alone is changed from tardy to early, while the anchor remains early and every other variable job remains tardy. The affected terms are as follows.

affected term	cost at $\mathbf{0}$	cost at $\mathbf{e}_i$	change
anchor- i pair	0	$Ma_i r_i$	$+Ma_i r_i$
tardy self term of i	$a_i^2 s_i$	0	$-a_i^2 s_i$
pair $\{j, i\}$, $j < i$	$a_j a_i s_i$	0	$-a_j a_i s_i$
pair $\{i, j\}$, $j > i$	$a_i a_j s_j$	0	$-a_i a_j s_j$.

For $j < i$, the all-tardy pair cost is $a_j a_i \min\{s_j, s_i\} = a_j a_i s_i$, since $s_j > s_i$. For $j > i$, it is $a_i a_j \min\{s_i, s_j\} = a_i a_j s_j$, since $s_i > s_j$. After job i becomes early, each such variable pair is

split across the two sides and therefore has zero pair cost. Every term not listed in the table is unchanged and cancels in (29.8). Summing the displayed changes gives

$$h_i = Ma_i r_i - a_i^2 s_i - \sum_{j < i} a_j a_i s_i - \sum_{j > i} a_i a_j s_j. \quad (29.13)$$

Because

$$a_i^2 s_i + \sum_{j < i} a_j a_i s_i = a_i s_i \left(a_i + \sum_{j < i} a_j \right) = a_i P_i s_i,$$

dividing (29.13) by $a_i > 0$ yields

$$\frac{h_i}{a_i} = Mr_i - P_i s_i - \sum_{j > i} a_j s_j. \quad (29.14)$$

We now substitute the constructed ratios into every term of (29.14):

$$\begin{aligned} \frac{h_i}{a_i} &= M(MQ + Da_i) - P_i(M(Q + 2M) - Da_i) \\ &\quad - \sum_{j > i} a_j(M(Q + 2M) - Da_j) \\ &= M^2Q + MDa_i - M(Q + 2M)P_i + Da_iP_i \\ &\quad - M(Q + 2M) \sum_{j > i} a_j + D \sum_{j > i} a_j^2 \\ &= M^2Q + MDa_i - M(Q + 2M) \left(P_i + \sum_{j > i} a_j \right) \\ &\quad + D \left(a_i P_i + \sum_{j > i} a_j^2 \right). \end{aligned} \quad (29.15)$$

Since P_i contains precisely $a_1, \dots, a_i$ and $Q = \sum_{j=1}^N a_j$,

$$P_i + \sum_{j > i} a_j = Q. \quad (29.16)$$

Applying (29.16) to (29.15) gives

$$\frac{h_i}{a_i} = M^2Q + MDa_i - MQ(Q + 2M) + D \left(a_i P_i + \sum_{j > i} a_j^2 \right). \quad (29.17)$$

The scalar part in (29.17) simplifies line by line as

$$M^2Q - MQ(Q + 2M) = M^2Q - MQ^2 - 2M^2Q = -MQ(M + Q) = -MQD.$$

Therefore

$$\begin{aligned} \frac{h_i}{a_i} &= MDa_i - MQD + D \left(a_i P_i + \sum_{j>i} a_j^2 \right) \\ &= DM(a_i - Q) + D \left(a_i P_i + \sum_{j>i} a_j^2 \right). \end{aligned} \quad (29.18)$$

Multiplying (29.18) by a_i gives the required explicit linear coefficient:

$$h_i = DM(a_i^2 - Qa_i) + D a_i \left(a_i P_i + \sum_{j>i} a_j^2 \right). \quad (29.19)$$

Finally, substitute (29.12) and (29.19) into the exact coefficient expansion (29.10). Grouping the terms multiplied by DM separately from those multiplied only by D gives

$$\begin{aligned} \Phi(x) - C &= DM \left[\sum_{i=1}^N (a_i^2 - Qa_i)x_i + 2 \sum_{1 \leq i < j \leq N} a_i a_j x_i x_j \right] \\ &\quad + D \left[\sum_{1 \leq i < j \leq N} a_i a_j (a_i - a_j)x_i x_j + \sum_{i=1}^N a_i \left(a_i P_i + \sum_{j>i} a_j^2 \right) x_i \right]. \end{aligned} \quad (29.20)$$

Because $x_i \in \{0, 1\}$, one has $x_i^2 = x_i$, and hence

$$\begin{aligned} S(x)^2 &= \left(\sum_{i=1}^N a_i x_i \right)^2 \\ &= \sum_{i=1}^N a_i^2 x_i^2 + 2 \sum_{1 \leq i < j \leq N} a_i a_j x_i x_j \\ &= \sum_{i=1}^N a_i^2 x_i + 2 \sum_{1 \leq i < j \leq N} a_i a_j x_i x_j. \end{aligned} \quad (29.21)$$

Since $S(x) = \sum_i a_i x_i$, the first bracket in (29.20) is exactly

$$S(x)^2 - QS(x).$$

By definition (29.6), the second bracket in (29.20) is exactly $\Psi(x)$. Consequently,

$$\Phi(x) = C + DM\left(S(x)^2 - QS(x)\right) + D\Psi(x),$$

which proves the lemma. $\square$

How to read the identity. The scheduling construction does not merely approximate a partition objective. The first term after C is an exact convex quadratic in the total selected processing load. All dependence on the identities and order positions of the selected jobs is isolated in the explicitly displayed residual Ψ . This separation makes it possible to compare a unit integer load gap with one uniform worst-case residual bound.

Lemma III.7 (Uniform residual bound). *For every $x \in \{0, 1\}^N$,*

$$|\Psi(x)| \leq 3Q^3 = U.$$

Proof. Because $a_i < a_j$ for $i < j$,

$$\begin{aligned} \left| \sum_{i < j} a_i a_j (a_i - a_j) x_i x_j \right| &\leq \sum_{i < j} a_i a_j (a_j - a_i) \\ &\leq \sum_{i < j} a_i a_j^2 \leq Q \sum_j a_j^2 \leq Q^3. \end{aligned}$$

For the two positive parts of (29.3),

$$\sum_i a_i^2 P_i x_i \leq Q \sum_i a_i^2 \leq Q^3,$$

and

$$\sum_i a_i x_i \sum_{j > i} a_j^2 \leq Q \sum_j a_j^2 \leq Q^3.$$

The triangle inequality gives the claim. $\square$

Completing the square uses the exact algebraic identity

$$S^2 - QS = \left(S - \frac{Q}{2}\right)^2 - \frac{Q^2}{4}.$$

Multiplying by DM gives the completed-square identity:

$$DM(S^2 - QS) = DM\left(S - \frac{Q}{2}\right)^2 - \frac{DMQ^2}{4}. \quad (29.22)$$

Since Q is even and S is an integer, every non-half load has squared distance at least one.

Section exit. We have established the exact decomposition $\Phi = C - DMQ^2/4 + DM(S - Q/2)^2 + D\Psi$, together with the uniform bound $|\Psi| \leq U$. The next section uses this result only through the unit integer load gap and the inequality $M > 2U$ to force the anchor, separate yes- and no-instances, and define the decision threshold.

30 Forcing the anchor and separating yes from no

Define the integer decision threshold

$$K = C - \frac{DMQ^2}{4} + DU. \quad (30.1)$$

The next lemma rules out the only case not covered by [lemma III.6](#).

Lemma III.8 (The anchor is forced early). *Every canonical assignment of cost at most K places the anchor in the early set.*

Proof. We first show that the threshold is positive. By [lemma III.5](#), every variable tardiness ratio exceeds $s_g = DM$. Hence each term in the baseline C is strictly larger than the same processing product multiplied by DM , and

$$\begin{aligned} C &> DM \left(\sum_{i < j} a_i a_j + \sum_i a_i^2 \right) \\ &= \frac{DM}{2} \left(Q^2 + \sum_i a_i^2 \right) \\ &\geq \frac{DMQ^2}{2}. \end{aligned}$$

Consequently,

$$K = C - \frac{DMQ^2}{4} + DU$$

$$\begin{aligned}
&> \frac{DMQ^2}{4} + DU \\
&> 0.
\end{aligned}$$

We next derive an upper bound on K . Since $Q < M$ and $Da_i > 0$,

$$s_i = M(Q + 2M) - Da_i < M(Q + 2M) < 3M^2.$$

Also

$$\sum_{i < j} a_i a_j + \sum_i a_i^2 = \frac{1}{2} \left(Q^2 + \sum_i a_i^2 \right) \leq Q^2.$$

Thus

$$C < 3M^2Q^2. \tag{30.2}$$

Because $D = M + Q < 2M$ and $U = 3Q^3$,

$$\begin{aligned}
K &< C + DU \\
&< 3M^2Q^2 + 6MQ^3.
\end{aligned}$$

Now $M = 8Q^3$ and $Q \geq 1$, so

$$Q^2 \leq Q^3 = \frac{M}{8}, \quad 6MQ^3 = \frac{3}{4}M^2.$$

Substitution gives

$$\begin{aligned}
K &< \frac{3}{8}M^3 + \frac{3}{4}M^2 \\
&\leq \frac{3}{8}M^3 + \frac{3}{32}M^3 \quad (M \geq 8) \\
&= \frac{15}{32}M^3 \\
&< M^3.
\end{aligned}$$

However, if the anchor is tardy, its tardy self-cost alone is

$$\begin{aligned}
p_g^2 s_g &= M^2(DM) \\
&= DM^3
\end{aligned}$$

$$\begin{aligned}
&> M^3 \\
&> K.
\end{aligned}$$

Therefore every canonical assignment of cost at most K places the anchor early. $\square$

Theorem III.9 (Correctness of the antithetical reduction). *The source PARTITION instance is a yes-instance if and only if the constructed strict-antithetical AWET instance has a schedule of cost at most K .*

Proof. Suppose first that the source is feasible. By [lemma III.3](#), there is a tagged side vector x with

$$S(x) = \frac{Q}{2}.$$

Place the anchor early and use the canonical schedule for this side assignment. Combining the exact identity with the completed square gives

$$\begin{aligned}
\Phi(x) &= C + DM \left(S(x) - \frac{Q}{2} \right)^2 - \frac{DMQ^2}{4} + D\Psi(x) \\
&= C - \frac{DMQ^2}{4} + D\Psi(x).
\end{aligned}$$

By [lemma III.7](#), $\Psi(x) \leq U$, so

$$\begin{aligned}
\Phi(x) &\leq C - \frac{DMQ^2}{4} + DU \\
&= K.
\end{aligned}$$

Thus a source partition produces a threshold-feasible schedule.

Conversely, suppose the constructed instance has a schedule of cost at most K . Then its global optimum is at most K . By [lemma II.5](#), choose an optimal schedule in compact canonical form; by [proposition III.2](#), its cost is $\Phi(x) \leq K$ for the corresponding variable-side vector x . The anchor-forcing lemma [lemma III.8](#) places the anchor early. The identity and the residual lower bound $\Psi(x) \geq -U$ therefore give

$$\begin{aligned}
\Phi(x) &= C - \frac{DMQ^2}{4} + DM \left(S(x) - \frac{Q}{2} \right)^2 + D\Psi(x) \\
&\geq C - \frac{DMQ^2}{4} + DM \left(S(x) - \frac{Q}{2} \right)^2 - DU.
\end{aligned}$$

If $S(x) \neq Q/2$, then $S(x)$ and $Q/2$ are integers, so

$$\left|S(x) - \frac{Q}{2}\right| \geq 1 \quad \text{and} \quad \left(S(x) - \frac{Q}{2}\right)^2 \geq 1.$$

Hence

$$\begin{aligned} \Phi(x) &\geq C - \frac{DMQ^2}{4} + DM - DU \\ &= \left(C - \frac{DMQ^2}{4} + DU\right) + DM - 2DU \\ &= K + D(M - 2U). \end{aligned}$$

Finally,

$$M - 2U = 8Q^3 - 6Q^3 = 2Q^3 > 0,$$

so the last bound is strictly larger than K , a contradiction. Therefore the chosen canonical optimum satisfies $S(x) = Q/2$. The binary tagging lemma then selects exactly one tag from each pair and decodes a source subset of sum B . $\square$

Lemma III.10 (Binary encoding length of the antithetical reduction). *Let L_{src} be the binary input length of the source PARTITION instance. The constructed AWET instance and threshold have size polynomial in L_{src} and can be computed in polynomial time.*

Proof. There are $N = 2m$ variable jobs and one anchor. From (27.1),

$$a_{2i-1} = \kappa 2^{i-1}, \quad a_{2i} = \kappa 2^{i-1} + b_i,$$

so every tagged number, and hence $Q = \sum_i a_i$, has $O(m + \log(B + 1)) = O(L_{\text{src}})$ bits. For $Q \geq 1$,

$$M = 8Q^3, \quad U = 3Q^3, \quad D = M + Q \leq 9Q^3.$$

A variable job satisfies

$$p_i \leq Q, \quad r_i = O(Q^4), \quad s_i = O(Q^6),$$

and therefore

$$w_i^E = O(Q^5), \quad w_i^L = O(Q^7).$$

For the anchor,

$$p_g = O(Q^3), \quad w_g^E = O(Q^7), \quad w_g^L = O(Q^9).$$

The baseline C is a sum of $O(m^2)$ products of integers having polynomially many bits. The threshold

$$K = C - \frac{DMQ^2}{4} + DU$$

has the same property; it is integral because Q is even. All displayed quantities are obtained by polynomially many integer operations on polynomial-bit operands. Hence the instance map is a polynomial-time binary reduction. $\square$

Theorem III.11 (Pseudopolynomial algorithm for strict antithetical order). *Let jobs be indexed so that $r_1 < \dots < r_n$ and $s_1 > \dots > s_n$. Put*

$$W_E = \sum_{i=1}^n w_i^E, \quad P = \sum_{i=1}^n p_i.$$

Then the optimal value of AWET can be computed exactly in

$$O(n(W_E + 1)(P + 1))$$

arithmetic operations and $O((W_E + 1)(P + 1))$ rolling memory. An optimal early set and canonical schedule can be recovered by storing one predecessor per reachable state, using $O(n(W_E + 1)(P + 1))$ memory, or by a standard recomputation scheme.

Proof. For $i < j$, the early ratio minimum is r_i and the tardiness ratio minimum is s_j . Thus the canonical objective for side bits $x_i \in \{0, 1\}$ is

$$\begin{aligned} \Phi(x) = & \sum_{i < j} w_i^E p_j x_i x_j + \sum_{i < j} p_i w_j^L (1 - x_i)(1 - x_j) \\ & + \sum_i p_i w_i^L (1 - x_i). \end{aligned}$$

Process jobs in the index order. After jobs $1, \dots, j$ have been assigned, retain

$$A = \sum_{\substack{i \leq j \\ x_i = 1}} w_i^E, \quad T = \sum_{\substack{i \leq j \\ x_i = 0}} p_i,$$

and let $G_j(A, T)$ be the least cost of all terms whose later endpoint is at most j . Initialize

$$G_0(0, 0) = 0$$

and set all other entries to $+\infty$.

Suppose a state (A, T) is present before job j is inserted. If job j is early, every earlier early job i contributes $w_i^E p_j$, so the exact increment is

$$p_j A.$$

The transition is

$$G_j(A + w_j^E, T) \leftarrow \min\{G_j(A + w_j^E, T), G_{j-1}(A, T) + p_j A\}.$$

If job j is tardy, every earlier tardy job contributes $p_i w_j^L$, totaling $T w_j^L$, and job j also contributes its self-cost $p_j w_j^L$. The tardy transition is

$$G_j(A, T + p_j) \leftarrow \min\{G_j(A, T + p_j), G_{j-1}(A, T) + w_j^L(T + p_j)\}.$$

Each early pair, tardy pair, and tardy self-term is charged exactly once, when its later-indexed job is inserted. Conversely, every side assignment follows one recurrence path. Because all earliness weights are positive, the nonempty-early condition is exactly $A > 0$ at the terminal layer. Therefore

$$\text{OPT} = \min_{A > 0, T} G_n(A, T).$$

There are at most $(W_E + 1)(P + 1)$ states per layer and two constant-time transitions per state, proving the stated value-computation bounds. Storing the minimizing predecessor for each reachable state permits direct reconstruction of an optimal side vector; alternatively, rolling tables can be recomputed during a backtracking pass. $\square$

Corollary III.12 (Strict-antithetical NP-completeness and pseudopolynomial solvability). *The decision version of positive-integer AWET is NP-complete and pseudopolynomially solvable even when the early- and tardiness-ratio orders are both strict and are exact reversals of one another. In the standard terminology for numerical scheduling problems, the restriction is therefore weakly NP-complete.*

Proof. The correctness theorem and [lemma III.10](#) give a polynomial-time binary reduction from PARTITION, so the restricted problem is NP-hard. It belongs to NP because an early set and its canonical orders form a polynomially checkable certificate whose cost is computable by exact integer arithmetic. The value algorithm in [theorem III.11](#) is pseudopolynomial. Thus the restricted decision problem is NP-complete and pseudopolynomially solvable, which is the usual weak-NP-completeness classification. In particular, it is not strongly NP-complete unless $P = NP$. $\square$

Corollary III.13 (FPTAS for strict antithetical order). *Positive-integer AWET restricted to strict antithetical ratio orders admits an FPTAS.*

Proof. Under the strict inequalities $r_1 < \dots < r_n$ and $s_1 > \dots > s_n$, the two ratio orders are unique, and the induced ratio permutation is $n(n-1)\dots 1$. This is the iterated skew sum of singleton permutations, hence is separable. Therefore the separating-tree FPTAS of [theorem IV.11](#) applies. $\square$

Section exit. The reduction is now complete: the constructed instance has a schedule of cost at most K exactly when a compact canonical optimum has the anchor early and tagged load $Q/2$; binary tagging decodes that load to a source partition, and the same strict order admits the two-resource pseudopolynomial recurrence. The worked examples use only the tag equivalence, the completed-square identity, and the certified margin $D(M - 2U) > 0$ to instantiate both directions numerically.

31 Two worked examples

How to read this step. **Conceptual route.** Compare the tagged half-load calculation and the closest non-half loads in the two examples. **Audit route.** Verify every job parameter, threshold component, and finite-state cost table. The first route explains the square-gap mechanism; the second reproduces the full numerical certificate.

The two source instances have four numbers each. The first is a yes-instance of PARTITION; the second changes only the final source number and is a no-instance. Each source instance produces eight tagged variable jobs and one anchor job. The comparison table predicts the distinction that the detailed calculations will verify.

case	source data	half-load target	attainable tagged load	least square displacement	threshold outcome
yes	(1, 2, 5, 8), $B = 8$	$Q/2 = 263$	263	$(S - Q/2)^2 = 0$	two canonical schedules have cost below K
no	(1, 2, 5, 10), $B = 9$	$Q/2 = 294$	closest are 293, 295	$(S - Q/2)^2 \geq 1$	every schedule has cost above K

All arithmetic below is exact.

31.1 A “yes” instance

The source instance is

$$(b_1, b_2, b_3, b_4) = (1, 2, 5, 8), \quad \sum_i b_i = 16 = 2B, \quad B = 8.$$

It has two complementary-looking witnesses: $b_4 = 8$ and $b_1 + b_2 + b_3 = 1 + 2 + 5 = 8$.

How to read this step. The purpose of the example is to verify both directions of the reduction, not merely to exhibit one cheap schedule. The “if” calculation starts with the source subset $\{1, 2, 3\}$, carries it through the binary tags, constructs the corresponding early set, and evaluates its exact canonical cost. The “only if” calculation starts hypothetically from an arbitrary schedule of cost at most K , passes to a compact canonical global optimum, forces that optimum’s anchor early and its selected tagged load to $Q/2$, and then decodes the original partition witness pair by pair.

The source PARTITION instance The exhaustive source-level subset audit is small enough to display. It makes clear that the half-total 8 is attained exactly by $\{4\}$ and $\{1, 2, 3\}$.

Subset	Sum	Subset	Sum
$\emptyset$	0	$\{4\}$	8
$\{1\}$	1	$\{1, 4\}$	9
$\{2\}$	2	$\{2, 4\}$	10
$\{1, 2\}$	3	$\{1, 2, 4\}$	11
$\{3\}$	5	$\{3, 4\}$	13
$\{1, 3\}$	6	$\{1, 3, 4\}$	14
$\{2, 3\}$	7	$\{2, 3, 4\}$	15
$\{1, 2, 3\}$	8	$\{1, 2, 3, 4\}$	16

Binary tagging and the exact half load Here $\kappa = 2B + 1 = 17$ and

$$(L_1, L_2, L_3, L_4) = (17, 34, 68, 136).$$

The four tagged pairs are therefore

$$\{17, 18\}, \quad \{34, 36\}, \quad \{68, 73\}, \quad \{136, 144\}.$$

Thus

$$(a_1, \dots, a_8) = (17, 18, 34, 36, 68, 73, 136, 144), \quad Q = 526, \quad \frac{Q}{2} = 263.$$

The source witness $\{1, 2, 3\}$ says “take the second member” of pairs 1, 2, and 3, and “take the first member” of pair 4. Hence

$$x = (0, 1, 0, 1, 0, 1, 1, 0), \quad S(x) = 18 + 36 + 73 + 136 = 263 = Q/2.$$

The other source witness $\{4\}$ gives the complementary tagged selection

$$x' = (1, 0, 1, 0, 1, 0, 0, 1), \quad S(x') = 17 + 34 + 68 + 144 = 263.$$

This concretely illustrates the pair-tagging lemma: every source choice is recorded only in the low-order increments b_i , whereas the positional terms L_i force exactly one tag from each pair.

Numerical scales and the nine-job instance The scale values are

$$U = 3Q^3 = 436594728, \quad M = 8Q^3 = 1164252608, \quad D = M + Q = 1164253134.$$

The common due date is $d = D = 1164253134$. The table below lists every job parameter needed to reproduce the instance. The displayed ratios are integers; the actual weights are $w_i^E = p_i r_i$ and $w_i^L = p_i s_i$.

Purpose. The table lists the nine job parameters needed to reproduce the antithetical yes-instance. **Verification rule.** For each variable job, substitute $p_i = a_i$ into the affine ratio formulas and multiply $p_i r_i$ and $p_i s_i$ to recover the two weights; for the anchor, use the four displayed anchor definitions.

Table 6: Nine-job yes-instance: processing times, ratios, and physical weights.

Job	p	$r = w^E/p$	$s = w^L/p$	w^E	w^L
1	17	632189175086	2710968863074171858	10747215976462	46086470672260921586
2	18	633353428220	2710968861909918724	11400361707960	48797439514378537032
3	34	651981478364	2710968843281868580	22167370264376	92172940671583531720
4	36	654309984632	2710968840953362312	23555159446752	97594878274321043232
5	68	691566084920	2710968803697262024	47026493774560	184345878651413817632
6	73	697387350590	2710968797875996354	50909276593070	197900722244947733842
7	136	770735298032	2710968724528048912	104820000532352	368691746535814652032
8	144	780049323104	2710968715214023840	112327102526976	390379494990819432960
g	1164252608	1225958273426	1355484747631673472	1427325117135397595008	1578126652534397663180414976

The strict antithetical inequalities can now be inspected numerically:

$$r_1 < r_2 < \cdots < r_8 < r_g, \quad s_1 > s_2 > \cdots > s_8 > s_g.$$

For example, $r_2 - r_1 = D(18 - 17) = 1164253134 > 0$, while $s_1 - s_2$ is the same positive quantity. The same affine difference calculation applies to every adjacent pair.

Baseline, square term, residual allowance, and threshold For this instance the all-variable-tardy baseline with the anchor early is

$$C = 445854051730378609479774.$$

Also

$$DM = 1355484747631673472, \quad \frac{DMQ^2}{4} = 93757524508935222384768,$$

and

$$DU = 508306780361877552.$$

Consequently

$$K = C - \frac{DMQ^2}{4} + DU = 352097035528223748972558.$$

For the witness $x = (0, 1, 0, 1, 0, 1, 1, 0)$, direct substitution in the residual formula gives

$$\Psi(x) = 15322886.$$

Because $S(x) = Q/2$, the completed-square term is at its exact minimum, so

$$\begin{aligned} \Phi(x) &= C - \frac{DMQ^2}{4} + D\Psi(x) \\ &= 352096545061161434519730. \end{aligned}$$

Thus

$$K - \Phi(x) = 490467062314452828 > 0.$$

The witness schedule is therefore not merely feasible; it lies strictly below the decision threshold.

The “if” case: source witness $\Rightarrow$ threshold-feasible schedule We now trace the forward implication without suppressing any logical step.

1. The source subset $\{1, 2, 3\}$ has sum $1 + 2 + 5 = 8 = B$.
2. For each selected source item choose $L_i + b_i$, and for each unselected item choose L_i . This gives 18, 36, 73, 136.
3. Their tagged sum is $263 = Q/2$. The associated binary side vector is $x = (0, 1, 0, 1, 0, 1, 1, 0)$.
4. Put the anchor g on the early arm and put precisely the four tagged jobs selected by x on the early arm. Put the remaining four variable jobs on the tardy arm. Proposition III.2 supplies the compact canonical schedule for this side assignment.
5. The exact identity gives $\Phi = C + DM(S^2 - QS) + D\Psi$. Since $S = 263$, the dominant term equals $-DMQ^2/4$.
6. The exact residual is $15322886 < U = 436594728$. The inequality $D\Psi < DU$ and the numerical evaluation above gives $\Phi = 352096545061161434519730 < K$.

This is the theorem’s forward implication instantiated completely.

The “only if” case: threshold-feasible schedule $\Rightarrow$ source witness Now begin at the target end. Suppose some schedule for this nine-job instance has cost at most K .

1. Its global optimum is at most K . Choose a compact canonical optimal schedule, whose cost is therefore also at most K .
2. The anchor-forcing lemma applies to this canonical optimum because the anchor self-cost when tardy already exceeds K . Hence g must be early.
3. With the anchor early, the exact identity applies. If $S \neq 263$, integrality gives $|S - 263| \geq 1$, so the completed-square term increases by at least DM .
4. The residual can improve by at most DU relative to zero and by at most $2DU$ when compared against the threshold allowance. Every non-half load therefore satisfies

$$\Phi \geq K + D(M - 2U).$$

Here

$$D(M - 2U) = 338871186907918368 > 0,$$

contradicting $\Phi \leq K$. Hence $S = 263$.

5. The tagging lemma now forces one selected number from each pair. Write $z_i = 1$ when the second tag $L_i + b_i$ is selected. Reducing the equality $S = Q/2$ modulo $\kappa = 17$ gives

$$z_1(1) + z_2(2) + z_3(5) + z_4(8) = 8.$$

6. The source audit above shows that this equation has exactly two 0–1 solutions: $(z_1, z_2, z_3, z_4) = (1, 1, 1, 0)$ and $(0, 0, 0, 1)$. The chosen canonical optimum therefore decodes to the source partition $\{1, 2, 3\}$ or $\{4\}$.

The reverse implication therefore recovers a genuine source witness rather than merely a numerically balanced tagged load.

An exhaustive half-load audit for the yes-instance There are $2^8 = 256$ variable-side vectors. The binary-tagging lemma says that any vector with $S = 263$ must choose exactly one tag from each pair, reducing the search to the $2^4 = 16$ source patterns already listed. Exactly two of those patterns have low-order sum 8. Hence there are exactly two half-load vectors, the two vectors x, x' displayed above. Direct evaluation gives

$$\Phi(x) = 352096545061161434519730,$$

while for $x' = (1, 0, 1, 0, 1, 0, 0, 1)$,

$$\Psi(x') = 15680566, \quad \Phi(x') = 352096545477591495488850 < K.$$

Every other variable-side vector has $|S - 263| \geq 1$ and is excluded by the certified square-gap argument. This completes both the constructive and the decoding audit.

31.2 A similar “no” instance

Change only the last source number from 8 to 10:

$$(b_1, b_2, b_3, b_4) = (1, 2, 5, 10), \quad \sum_i b_i = 18 = 2B, \quad B = 9.$$

The first three numbers sum to only 8, while the fourth number alone already exceeds 9. Thus the instance is intuitively close to the preceding one but crosses the partition boundary.

How to read this step. The forward direction is now demonstrated by showing precisely where the construction of a threshold-feasible schedule fails: there is no source subset of sum B , hence no tagged half load. The reverse direction begins with a hypothetical target schedule of cost at most K , passes to a compact canonical global optimum, and derives the impossible source equation $z_1 + 2z_2 + 5z_3 + 10z_4 = 9$.

The modified source instance and exhaustive subset audit The sixteen source subset sums are

Subset	Sum	Subset	Sum
$\emptyset$	0	$\{4\}$	10
$\{1\}$	1	$\{1, 4\}$	11
$\{2\}$	2	$\{2, 4\}$	12
$\{1, 2\}$	3	$\{1, 2, 4\}$	13
$\{3\}$	5	$\{3, 4\}$	15
$\{1, 3\}$	6	$\{1, 3, 4\}$	16
$\{2, 3\}$	7	$\{2, 3, 4\}$	17
$\{1, 2, 3\}$	8	$\{1, 2, 3, 4\}$	18

No entry equals $B = 9$. Hence this is a no-instance of PARTITION.

Binary tagging and the missing half load Now $\kappa = 19$ and

$$(L_1, L_2, L_3, L_4) = (19, 38, 76, 152).$$

The tagged pairs are

$$\{19, 20\}, \quad \{38, 40\}, \quad \{76, 81\}, \quad \{152, 162\},$$

so

$$(a_1, \dots, a_8) = (19, 20, 38, 40, 76, 81, 152, 162), \quad Q = 588, \quad Q/2 = 294.$$

If a tagged subset had sum 294, the tagging lemma would force exactly one number per pair and, modulo 19, would give

$$z_1 + 2z_2 + 5z_3 + 10z_4 = 9.$$

The source table proves that no such binary vector z exists. Therefore no tagged subset has load 294.

The two closest pair-respecting choices come from source sums 8 and 10. They have tagged loads

$$293 = 294 - 1 \quad \text{and} \quad 295 = 294 + 1,$$

respectively. Explicitly,

$$x^- = (0, 1, 0, 1, 0, 1, 1, 0), \quad S(x^-) = 20 + 40 + 81 + 152 = 293,$$

and

$$x^+ = (1, 0, 1, 0, 1, 0, 0, 1), \quad S(x^+) = 19 + 38 + 76 + 162 = 295.$$

Thus the dominant square is forced at least one unit away from its minimum.

Numerical scales and the nine-job no-instance The new scales are

$$U = 609892416, \quad M = 1626379776, \quad D = 1626380364,$$

and the common due date is $d = D = 1626380364$.

Purpose. The table lists the nine job parameters needed to reproduce the antithetical no-instance. **Verification rule.** Recompute each ratio from the hatted scale formulas, multiply by the displayed processing time to recover each weight, and use adjacent differences to check strict antithetical order.

Table 7: Nine-job no-instance: processing times, ratios, and physical weights.

Job	p	$r = w^E/p$	$s = w^L/p$	w^E	w^L
1	19	987212535204	5290223276973701724	18757038168876	100514242262500332756
2	20	988838915568	5290223275347321360	19776778311360	105804465506946427200
3	38	1018113762120	5290223246072474808	38688322960560	201028483350754042704
4	40	1021366522848	5290223242819714080	40854660913920	211608929712788563200
5	76	1079916215952	5290223184270020976	82073632412352	402056962004521594176
6	81	1088048117772	5290223176138119156	88131897539532	428508077267187651636
7	152	1203521123616	5290223060665113312	182935210789632	804113905221097223424
8	162	1219784927256	5290223044401309672	197605158215472	857016133193012166864
g	1626379776	1914249342684	2645112132093118464	3113296417162551158784	4301956876888488418621784064

Again the data visibly satisfy strict antithetical order. The baseline and threshold components are

$$C = 1087759799531923395684660,$$

$$\begin{aligned} DM &= 2645112132093118464, \\ \frac{DMQ^2}{4} &= 228632912249600787554304, \\ DU &= 991917049534919424. \end{aligned}$$

and therefore

$$K = 859127879199372143049780.$$

The universal certified separation is

$$D(M - 2U) = 661278033023279616 > 0.$$

The “if” route: exactly where the forward construction fails The forward proof of the theorem begins with a source subset summing to $B = 9$. The exhaustive source table shows that there is none. Equivalently, the pair-respecting tagged loads nearest $Q/2 = 294$ are 293 and 295, not 294. Therefore the forward recipe cannot produce a vector x with $S = Q/2$, and the dominant square cannot attain its minimum.

This failure is quantitative. For the closest lower-load vector x^- ,

$$\Psi(x^-) = 21387022,$$

so the exact identity gives

$$\Phi(x^-) = 859129567177887326484828 = K + 1687978515183435048 > K.$$

For the closest upper-load vector x^+ ,

$$\Psi(x^+) = 22087552,$$

and

$$\Phi(x^+) = 859129568317215562877748 = K + 1689117843419827968 > K.$$

Thus even the two load-nearest canonical candidates miss the threshold by more than 1.68×10^{18} .

The “only if” route: a hypothetical target witness is impossible Suppose, for contradiction, that some schedule for the no-instance has cost at most K .

1. Its global optimum is at most K . Choose a compact canonical optimal schedule, whose cost is therefore also at most K .
2. The anchor-forcing lemma implies that g is early in this canonical optimum.
3. Since the anchor is early, the exact identity and completed-square equation apply.
4. If $S \neq 294$, integrality gives $|S - 294| \geq 1$. The theorem’s gap calculation then yields

$$\Phi \geq K + D(M - 2U) = K + 661278033023279616 > K,$$

a contradiction. Hence the hypothetical canonical optimum would have to satisfy $S = 294$.

5. By binary tagging, a half-load subset must choose exactly one member of each tag pair. Let z_i indicate use of the second member. Reduction modulo $\kappa = 19$ yields

$$z_1 + 2z_2 + 5z_3 + 10z_4 = 9.$$

6. But the exhaustive sixteen-row source audit contains no subset sum 9. This is impossible.

Therefore the target instance has no schedule of cost at most K .

Finite-state audit of the closest canonical assignments For completeness, all $2^8 = 256$ variable-side vectors were evaluated from the exact identity. The table below lists the closest load levels and, for each displayed level, the best exact canonical cost found there. The first row is the global minimum among anchor-early assignments.

S	$S - Q/2$	best Ψ shown		$\Phi - K$
293	-1	21387022	1687978515183435048	
295	+1	22087552	1689117843419827968	
292	-2	21341770	9623241314498558712	
296	+2	22130722	9624524450539497240	
291	-3	21273742	22848691335560748840	
297	+3	22196980	22850192871715247472	
290	-4	21228412	41364402536390677968	
298	+4	22240072	41366047880349722208	

The numerical audit is stronger than the proof requires: the theoretical lower gap already excludes every non-half load uniformly, whereas the enumeration shows that the actual best no-instance cost exceeds K by substantially more.

Side-by-side conclusion The two examples differ only in replacing the last source item 8 by 10. In the yes-instance, the low-order equation has solutions $1 + 2 + 5 = 8$ and $8 = 8$, producing two tagged half-load assignments and two schedules below threshold. In the no-instance, the low-order target becomes 9; the first three source items stop at 8 and the last starts at 10. Consequently the nearest tagged loads are $Q/2 \pm 1$, the square term pays at least one full unit of DM , and the residual is too small to erase the gap. This is precisely the finite-arithmetic “if and only if” mechanism of the antithetical reduction.

Reusable idea. A hard binary choice can sometimes be isolated by making one aggregate quantity pay a dominant convex square while proving a uniform bound on every identity-dependent residual. The reduction then needs only an integral unit gap and a scale choice larger than twice the residual budget; the detailed form of the residual may remain complicated.

Why Part IV is next. Part III has classified exact reversal as weakly NP-complete and pseudopolynomially solvable. It leaves open how far the algorithmic structure extends beyond one reverse permutation. Part IV replaces the single load state by a four-coordinate separating-tree interface and uses recursive direct/skew order compatibility to obtain an FPTAS for a larger class.

Part IV

Separating-Tree Analysis

An FPTAS for Separable Ratio Permutations

Part synopsis. For fixed total refinements of the early- and tardiness-ratio orders, suppose the induced ratio permutation is separable. A separating tree then recursively divides the jobs by direct and skew sums. Every tree node determines a precise block of jobs, and a side assignment on that block is summarized by four nonnegative coordinates: early processing, tardy processing, early weight, and tardy weight. The cost of pairs crossing two child blocks is an exact nonnegative bilinear form. This gives an exact pseudopolynomial dynamic program for the original scheduling problem. Keeping one minimum-cost candidate per coordinated geometric box converts that recurrence into an FPTAS.

Part positioning. **Starting boundary.** The canonical side-assignment objective is quadratic, but a separable ratio permutation fixes the orientation of every pair crossing a direct or skew decomposition node. **Result.** For fixed total ratio-order refinements whose ratio permutation is separable, AWET admits an FPTAS. **Technique.** An ordered separating tree compresses every future cross interaction into a four-coordinate nonnegative additive interface, and coordinated geometric trimming preserves one cheapest representative per box.

Notation reset. The symbols $\prec_E$ and $\prec_L$ denote fixed total orders refining nondecreasing early and tardiness ratios, and π is their ratio permutation. For a node M of an ordered separating tree, $J(M)$ denotes exactly the set of leaves below M ; this set is called the *separating-tree block* of M . For a side assignment on $J(M)$, the signature is $z_M = (e_M, t_M, u_M, v_M)$, where the four entries are defined explicitly in [definition IV.4](#). The root height is H , the trimming ratio is $\lambda = 1 + \varepsilon/(4H)$, and zero is always stored in a singleton bucket.

32 Introduction and proof map

The argument has two layers. First, an exact separating-tree dynamic program enumerates every side assignment through a four-coordinate interface and returns an optimal schedule. Its number of numerical states is pseudopolynomial. Second, coordinated geometric trimming stores only one representative per four-dimensional box. Nonnegativity and bilinearity ensure that the resulting multiplicative loss can be charged by separating-tree height.

Proof contract.

- **Input class:** positive-integer AWET with $d = \sum_j p_j$, together with two fixed total ratio-order refinements whose ratio permutation is separable.
- **Exact layer:** define a local canonical cost and a four-coordinate table at every separating-tree node; prove that the root recurrence equals the optimum of the original scheduling problem.
- **Approximation layer:** replace the exact table by one least-cost candidate per geometric box while preserving zero coordinates exactly.
- **Guarantee:** coordinate inflation is at most λ per level and cost inflation at most λ^2 per level, giving $\lambda^{2H} \leq 1 + \varepsilon$.
- **Scope:** no distinct-ratio, balanced-tree, bounded-degree, or bounded-height assumption is imposed beyond $H \leq n - 1$ for a binary separating tree.

33 Ratio-order costs and separable permutations

Let

$$r_j = \frac{w_j^E}{p_j}, \quad s_j = \frac{w_j^L}{p_j}.$$

Ratio comparisons are made exactly by cross multiplication. Ties do not change any pair coefficient in the canonical objective, but a permutation requires total orders. We therefore fix one total order $\prec_E$ refining nondecreasing r_j and one total order $\prec_L$ refining nondecreasing s_j . A deterministic job-index tie break is sufficient.

Definition IV.1 (Ratio permutation and direct/skew sums). List the jobs as

$$j_1 \prec_E j_2 \prec_E \cdots \prec_E j_n.$$

For each position k , let $\pi(k)$ be the position of job j_k in the order $\prec_L$. The permutation π is the *ratio permutation* of the chosen refinements.

Write S_m for the set of permutations of $\{1, \dots, m\}$. For $\alpha \in S_p$ and $\beta \in S_q$, their *direct sum* $\alpha \oplus \beta \in S_{p+q}$ is defined by

$$(\alpha \oplus \beta)(k) = \begin{cases} \alpha(k), & 1 \leq k \leq p, \\ p + \beta(k - p), & p < k \leq p + q, \end{cases}$$

and their *skew sum* $\alpha \ominus \beta \in S_{p+q}$ is defined by

$$(\alpha \ominus \beta)(k) = \begin{cases} q + \alpha(k), & 1 \leq k \leq p, \\ \beta(k - p), & p < k \leq p + q. \end{cases}$$

Thus, in a direct sum, every entry of the left permutation is smaller than every entry of the right permutation; in a skew sum, every entry of the left permutation is larger than every entry of the right permutation.

The ratio permutation depends on the chosen refinements when ratios tie. This causes no objective ambiguity: interchanging equal-ratio jobs leaves the relevant minimum coefficient unchanged. The results below apply to the fixed refinements actually used by the algorithm; they do not require every possible tie refinement to have the same decomposition.

Definition IV.2 (Separable permutation, separable instance, and separating tree). A *separating tree* for a permutation $\pi \in S_n$ is a rooted plane binary tree whose leaves, from left to right, are labeled

$$\pi(1), \pi(2), \dots, \pi(n).$$

For every tree node, the leaves below that node occupy consecutive leaf positions and their labels form a consecutive set of integers. Every internal node has one of two labels:

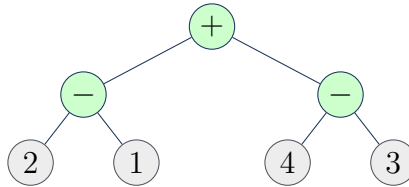
- it is *positive* or *direct*, written $M = A \oplus B$, when every label below its left child A is smaller than every label below its right child B ;
- it is *negative* or *skew*, written $M = A \ominus B$, when every label below its left child A is larger than every label below its right child B .

A permutation is *separable* when it has a separating tree. Equivalently, it can be constructed recursively from the singleton permutation 1 by direct and skew sums. An AWET instance with fixed refinements $(\prec_E, \prec_L)$ is *separable* when its ratio permutation π is separable.

The leaves are the jobs in $\prec_E$ order, while their labels are the corresponding ranks in $\prec_L$. For example,

$$2143 = (1 \ominus 1) \oplus (1 \ominus 1).$$

Its separating tree has a positive root, two negative children, and leaves 2, 1, 4, 3 from left to right:



A separating tree need not be unique: associativity permits rotations among adjacent internal nodes with the same sign. Such rotations do not change the direct/skew relations used by the algorithm.

The strict antithetical ordering studied in Part III has ratio permutation $n(n-1)\cdots 1$, an iterated skew sum of singleton permutations. It is therefore separable, and the FPTAS in [theorem IV.11](#) applies to that class in addition to the exact pseudopolynomial algorithm of Part III.

Separable permutations can be recognized in linear time by a stack scan, and a separating tree can be recovered from the same computation [4]. This preprocessing is therefore no bottleneck for the dynamic programs below.

For comparison with the graph terminology, the *inversion graph* $G_\times$ joins two jobs when their relative orders in $\prec_E$ and $\prec_L$ disagree. A permutation is separable if and only if its inversion graph is a cograph [4, 5]. At a positive node, the two child graphs are combined

by disjoint union; at a negative node, they are combined by join. Consequently, a suitably $\prec_E$ -ordered binary refinement of the graph-theoretic cotree is equivalent to a separating tree: the children are ordered by their left-to-right intervals in $\prec_E$, disjoint-union nodes become positive/direct nodes, and join nodes become negative/skew nodes. The scheduling development below uses only the permutation and its separating tree.

Definition IV.3 (Separating-tree block). For a separating-tree node M , let $J(M)$ be the set of leaf jobs below M . We call $J(M)$ the *separating-tree block* of M . It is a contiguous interval in $\prec_E$, and the consecutive label condition in the separating tree makes it a contiguous interval in $\prec_L$ as well.

Exact assumptions for the FPTAS. The algorithm assumes only the standard positive-integer AWET model with $d = \sum_j p_j$, fixed total refinements of the two ratio orders, and separability of the resulting ratio permutation. A separating tree need not be supplied: it can be recognized and recovered in linear time. Equal ratios are allowed, and the binary separating tree may be completely unbalanced. Separability is a sufficient structural hypothesis for this algorithm, not a proved necessary condition for the existence of an FPTAS by some other method.

Put

$$P = \sum_{j \in \mathcal{J}} p_j, \quad W_E = \sum_{j \in \mathcal{J}} w_j^E, \quad W_L = \sum_{j \in \mathcal{J}} w_j^L.$$

34 Local side assignments and the four signature coordinates

Fix a separating-tree node M . A local side assignment is a subset $E_M \subseteq J(M)$ of jobs assigned to the early/on-time side; its tardy complement is

$$T_M = J(M) \setminus E_M.$$

Definition IV.4 (Four-coordinate signature). For the local side assignment (E_M, T_M) , define

$$e_M = \sum_{j \in E_M} p_j, \quad \text{early processing load,} \quad (34.1)$$

$$t_M = \sum_{j \in T_M} p_j, \quad \text{tardy processing load,} \quad (34.2)$$

$$u_M = \sum_{j \in E_M} w_j^E, \quad \text{earliness weight carried by early jobs,} \quad (34.3)$$

$$v_M = \sum_{j \in T_M} w_j^L, \quad \text{tardiness weight carried by tardy jobs.} \quad (34.4)$$

The signature is

$$z_M = z_M(E_M) := (e_M, t_M, u_M, v_M) \in \mathbb{Z}_{\geq 0}^4. \quad (34.5)$$

Although $e_M + t_M = \sum_{j \in J(M)} p_j$, both processing coordinates are retained. This makes every coordinate nonnegative and additive, represents zero tardy load exactly, and avoids subtraction in both the exact recurrence and the trimming proof.

The entries u_M and v_M are not the total early and tardy weights of all jobs in the block. They are side-specific sums: u_M includes w_j^E only for jobs currently assigned early, whereas v_M includes w_j^L only for jobs currently assigned tardy.

For a local assignment E_M , define its *internal canonical cost* by restricting the global canonical objective to jobs and pairs lying wholly in $J(M)$:

$$\phi_M(E_M) := \sum_{\{i,j\} \subseteq E_M} p_i p_j \min\{r_i, r_j\} \quad (34.6)$$

$$\begin{aligned} &+ \sum_{\{i,j\} \subseteq T_M} p_i p_j \min\{s_i, s_j\} \\ &+ \sum_{i \in T_M} p_i w_i^L. \end{aligned} \quad (34.7)$$

At a leaf containing job j , the two possible states are therefore

$$z_j^E = (p_j, 0, w_j^E, 0), \quad \phi_j(\{j\}) = 0, \quad (34.8)$$

$$z_j^{\text{tar}} = (0, p_j, 0, w_j^L), \quad \phi_j(\emptyset) = p_j w_j^L. \quad (34.9)$$

35 Cross costs and exact bilinear convolution

Let M have left child A and right child B . A *cross pair for the split* (A, B) is an unordered two-element set $\{i, j\}$ with $i \in J(A)$ and $j \in J(B)$. The *cross cost* is the total canonical-objective contribution of those cross pairs $\{i, j\}$ for which i and j are assigned to the same

side. For generic signatures

$$z_A = (e_A, t_A, u_A, v_A), \quad z_B = (e_B, t_B, u_B, v_B),$$

define

$$g_{\oplus}(z_A, z_B) = u_A e_B + v_A t_B, \quad (35.1)$$

$$g_{\ominus}(z_A, z_B) = u_A e_B + t_A v_B. \quad (35.2)$$

The first formula is used at a direct node and the second at a skew node. The two cases are summarized below.

node	early cross relation	early term	tardy cross relation	tardy term and total
direct $A \oplus B$	$A \prec_E B$	$u_A e_B$	$A \prec_L B$	$v_A t_B$; hence $g_{\oplus} = u_A e_B + v_A t_B$
skew $A \ominus B$	$A \prec_E B$	$u_A e_B$	$B \prec_L A$	$t_A v_B$; hence $g_{\ominus} = u_A e_B + t_A v_B$

Lemma IV.5 (Exact nonnegative bilinear convolution). *Let $M = A \oplus B$ or $M = A \ominus B$. For every side assignment on $J(M)$,*

$$z_M = z_A + z_B \quad (35.3)$$

and

$$\phi_M(E_M) = \phi_A(E_A) + \phi_B(E_B) + g_{\circ}(z_A, z_B), \quad (35.4)$$

where $\circ = \oplus$ at a direct node and $\circ = \ominus$ at a skew node. Every term in $g_{\circ}$ is nonnegative and is the product of one coordinate from each child.

Proof. Because $J(A)$ and $J(B)$ are disjoint, each signature coordinate splits into its two child sums. For example,

$$\begin{aligned} e_M &= \sum_{j \in E_M} p_j \\ &= \sum_{j \in E_A} p_j + \sum_{j \in E_B} p_j \end{aligned}$$

$$= e_A + e_B.$$

The same calculation gives $t_M = t_A + t_B$, $u_M = u_A + u_B$, and $v_M = v_A + v_B$, proving (35.3).

At either node type, every job of A precedes every job of B in the nondecreasing early-ratio order. Thus $r_i \leq r_j$ for $i \in J(A)$ and $j \in J(B)$. The cross contribution of pairs that are both early assigned is

$$\begin{aligned} & \sum_{i \in E_A} \sum_{j \in E_B} p_i p_j \min\{r_i, r_j\} \\ &= \sum_{i \in E_A} \sum_{j \in E_B} p_i p_j r_i \\ &= \sum_{i \in E_A} \sum_{j \in E_B} w_i^E p_j \\ &= \left(\sum_{i \in E_A} w_i^E \right) \left(\sum_{j \in E_B} p_j \right) \\ &= u_A e_B. \end{aligned}$$

Suppose first that $M = A \oplus B$. Then every job of A also precedes every job of B in the nondecreasing tardiness-ratio order, so $s_i \leq s_j$ for $i \in J(A)$ and $j \in J(B)$. Hence the cross contribution of pairs that are both tardy is

$$\begin{aligned} & \sum_{i \in T_A} \sum_{j \in T_B} p_i p_j \min\{s_i, s_j\} \\ &= \sum_{i \in T_A} \sum_{j \in T_B} p_i p_j s_i \\ &= \sum_{i \in T_A} \sum_{j \in T_B} w_i^L p_j \\ &= \left(\sum_{i \in T_A} w_i^L \right) \left(\sum_{j \in T_B} p_j \right) \\ &= v_A t_B. \end{aligned}$$

Adding the early and tardy cross contributions gives $g_{\oplus}(z_A, z_B)$.

Now suppose that $M = A \ominus B$. Every job of B precedes every job of A in the nondecreasing tardiness-ratio order, so $s_j \leq s_i$ for $i \in J(A)$ and $j \in J(B)$. The tardy cross contribution is

then

$$\begin{aligned}
& \sum_{i \in T_A} \sum_{j \in T_B} p_i p_j \min\{s_i, s_j\} \\
&= \sum_{i \in T_A} \sum_{j \in T_B} p_i p_j s_j \\
&= \sum_{i \in T_A} \sum_{j \in T_B} p_i w_j^L \\
&= \left(\sum_{i \in T_A} p_i \right) \left(\sum_{j \in T_B} w_j^L \right) \\
&= t_A v_B.
\end{aligned}$$

Adding the early cross term gives $g_{\ominus}(z_A, z_B)$.

Every tardy self-term belongs to a leaf, every within-child pair belongs to the corresponding child cost, and every cross pair is charged at the unique lowest separating-tree node at which its two jobs lie in different child blocks. Thus no term is omitted or counted twice, and (35.4) follows. $\square$

36 The exact pseudopolynomial dynamic program

The exact recurrence is stated before trimming so that both its state space and its connection to the original scheduling problem are explicit.

Definition IV.6 (Exact separating-tree table). For each node M and integer vector $z \in \mathbb{Z}_{\geq 0}^4$, define

$$\mathcal{D}_M(z) := \min\{\phi_M(E_M) : E_M \subseteq J(M), z_M(E_M) = z\}. \quad (36.1)$$

If no assignment has signature z , set $\mathcal{D}_M(z) = +\infty$. With every finite entry, store a backpointer to an assignment attaining it.

At a leaf j , the only finite entries are

$$\mathcal{D}_j(p_j, 0, w_j^E, 0) = 0, \quad (36.2)$$

$$\mathcal{D}_j(0, p_j, 0, w_j^L) = p_j w_j^L. \quad (36.3)$$

At an internal node $M = A \circ B$, where $\circ \in \{\oplus, \ominus\}$, use

$$\mathcal{D}_M(z) = \min\left\{\mathcal{D}_A(z_A) + \mathcal{D}_B(z_B) + g_\circ(z_A, z_B) : \right. \quad (36.4)$$

$$\left. z_A, z_B \in \mathbb{Z}_{\geq 0}^4, \quad z_A + z_B = z, \quad \mathcal{D}_A(z_A), \mathcal{D}_B(z_B) < +\infty\right\}. \quad (36.5)$$

Theorem IV.7 (Exact separating-tree dynamic program). *For every separating-tree node M and every signature z ,*

$$\mathcal{D}_M(z) = \min\left\{\phi_M(E_M) : E_M \subseteq J(M), \quad z_M(E_M) = z\right\}, \quad (36.6)$$

where the minimum of the empty set is $+\infty$. Whenever $\mathcal{D}_M(z) < +\infty$, the stored backpointers reconstruct an assignment attaining this value. If R is the root, then the optimum value of the original scheduling instance is

$$\text{OPT} = \min\left\{\mathcal{D}_R(z) : z = (e, t, u, v) \in \mathbb{Z}_{\geq 0}^4, \quad e > 0\right\}. \quad (36.7)$$

Backtracking from a minimizing root state gives an optimal early set and hence an optimal canonical schedule. The recurrence is pseudopolynomial.

Proof. We prove the following exact induction assertion $\mathcal{I}(M)$ for every separating-tree node M : for every signature z , equality (36.6) holds, and every finite table entry has a stored assignment attaining it.

At a leaf, (36.2) and (36.3) list the only two side assignments, so $\mathcal{I}(M)$ holds.

Let $M = A \circ B$ be internal and assume $\mathcal{I}(A)$ and $\mathcal{I}(B)$. Fix an arbitrary assignment E_M . It restricts uniquely to

$$E_A = E_M \cap J(A), \quad E_B = E_M \cap J(B).$$

Write $z_A = z_A(E_A)$ and $z_B = z_B(E_B)$. By signature additivity,

$$z_M(E_M) = z_A + z_B.$$

By the child induction assertions and lemma IV.5,

$$\phi_M(E_M) = \phi_A(E_A) + \phi_B(E_B) + g_\circ(z_A, z_B)$$

$$\begin{aligned}
&\geq \mathcal{D}_A(z_A) + \mathcal{D}_B(z_B) + g_o(z_A, z_B) \\
&\geq \mathcal{D}_M(z_M(E_M)).
\end{aligned}$$

The first inequality can be strict: the restrictions of an arbitrary parent assignment need not be minimum-cost child assignments among all assignments having signatures z_A and z_B . Thus the recurrence value is no larger than the cost of any assignment with the requested parent signature.

Conversely, let z be any signature with $\mathcal{D}_M(z) < +\infty$, and take a pair (z_A, z_B) attaining the minimum in (36.5). By $\mathcal{I}(A)$ and $\mathcal{I}(B)$, the stored child backpointers yield assignments E_A and E_B of costs $\mathcal{D}_A(z_A)$ and $\mathcal{D}_B(z_B)$. Their union

$$E_M = E_A \cup E_B$$

has signature $z_A + z_B = z$, and lemma IV.5 gives

$$\begin{aligned}
\phi_M(E_M) &= \mathcal{D}_A(z_A) + \mathcal{D}_B(z_B) + g_o(z_A, z_B) \\
&= \mathcal{D}_M(z).
\end{aligned}$$

Therefore the recurrence value is attainable and is exactly optimal for that signature. For any signature z' with $\mathcal{D}_M(z') = +\infty$, the first half shows that no assignment can have signature z' , because any such assignment would generate a finite recurrence candidate. Storing the two child backpointers proves the attainability part of $\mathcal{I}(M)$.

At the root, positive processing times imply

$$e_R > 0 \quad \Longleftrightarrow \quad E_R \neq \emptyset.$$

By proposition I.4, every nonempty E_R defines a feasible due-date-aligned canonical schedule of cost $\phi_R(E_R)$, and the global optimum is the minimum of this expression over all nonempty early sets. Hence minimizing the exact root table over $e_R > 0$ proves (36.7). Backpointers recover the side assignment. Scheduling the early jobs in nondecreasing r_j order so that the early block ends at d , and scheduling the tardy jobs in reverse $\prec_L$ order (equivalently, nonincreasing s_j order) starting at d , then recovers an optimal schedule for the original problem.

It remains to bound the exact state space. Every reachable coordinate satisfies

$$0 \leq e, t \leq P, \quad 0 \leq u \leq W_E, \quad 0 \leq v \leq W_L.$$

Consequently, every node has at most

$$S := (P + 1)^2(W_E + 1)(W_L + 1) \quad (36.8)$$

finite signatures. A direct implementation scans at most S^2 child-state pairs at each of $O(n)$ internal nodes, for at most $O(nS^2)$ exact-arithmetic operations. All table costs and backpointers have polynomial binary encoding length. The running time is therefore polynomial in n, P, W_E, W_L and hence pseudopolynomial in the original binary input. $\square$

How to use this result. The exact recurrence is already a self-contained algorithm for numerically small instances. Construct the ordered separating tree, fill the two leaf states, combine child tables by (36.5), choose the least root entry with $e > 0$, and follow backpointers to recover the early/tardy assignment and its canonical schedule.

37 Geometric trimming of the exact recurrence

A one-job instance is solved directly. Hence assume the binary separating tree has root height $H \geq 1$, where leaves have height 0. Fix a rational accuracy parameter $0 < \varepsilon \leq 1$ and put

$$\lambda = 1 + \frac{\varepsilon}{4H}. \quad (37.1)$$

Definition IV.8 (Geometric buckets and boxes). Zero has its own bucket, denoted $\perp$. For an integer $x > 0$, define

$$\beta_\lambda(x) = \lfloor \log_\lambda x \rfloor.$$

For a signature $z = (e, t, u, v)$, its box is

$$\text{box}_\lambda(z) = (\beta_\lambda(e), \beta_\lambda(t), \beta_\lambda(u), \beta_\lambda(v)), \quad (37.2)$$

where $\beta_\lambda(0) = \perp$.

The bucket index need not be computed with floating-point logarithms. Because ε is

rational, write $\lambda = a/b > 1$ in lowest terms; the integers a and b have polynomial binary length. For an integer coordinate $x > 0$, the index $q = \beta_\lambda(x)$ is characterized exactly by

$$a^q \leq xb^q, \quad a^{q+1} > xb^{q+1}.$$

A doubling search followed by binary search finds q using exact integer comparisons, with powers computed by repeated squaring. For a coordinate bounded by U , the running-time estimate below gives $q = O((H/\varepsilon) \log U)$. The integers compared have bit length $O(q(\log a + \log b) + \log U)$, which is polynomial in the binary input length and $1/\varepsilon$. Thus box membership is computable in the ordinary Turing model without real-number arithmetic.

Trimmed recurrence. At a leaf, retain both exact states. At an internal node $M = A \circ B$, generate one parent candidate from every pair of retained child candidates. Compute its signature and exact cost by

$$\begin{aligned} z_M &= z_A + z_B, \\ c_M &= c_A + c_B + g_\circ(z_A, z_B). \end{aligned}$$

Group the generated candidates by $\text{box}_\lambda(z_M)$. In every nonempty box, retain a candidate of minimum exact cost, together with its child backpointers. At the root, return the minimum-cost retained candidate with $e > 0$ and reconstruct its schedule.

Lemma IV.9 (Same-box replacement). *Let candidate (z', c') be discarded in favor of retained candidate $(\hat{z}, \hat{c})$ from the same box. Then*

$$\hat{c} \leq c', \tag{37.3}$$

$$\hat{z}_\ell = 0 \iff z'_\ell = 0, \tag{37.4}$$

$$\hat{z}_\ell \leq \lambda z'_\ell \quad \text{for every coordinate } \ell. \tag{37.5}$$

Proof. The retained candidate minimizes exact cost in the box, proving (37.3). The zero bucket contains only zero, proving (37.4). If a coordinate is positive and has bucket index q , both coordinate values lie in

$$[\lambda^q, \lambda^{q+1}),$$

so the retained value is strictly smaller than λ times the discarded value. This proves

(37.5). □

Lemma IV.10 (Representative invariant). *Let M have height $h(M)$. For every exact state (z, c) represented in $\mathcal{D}_M$, the trimmed table contains a retained state $(\hat{z}, \hat{c})$ such that*

$$\hat{z}_\ell = 0 \iff z_\ell = 0, \quad (37.6)$$

$$\hat{z}_\ell \leq \lambda^{h(M)} z_\ell \quad \text{for every coordinate } \ell, \quad (37.7)$$

$$\hat{c} \leq \lambda^{2h(M)} c. \quad (37.8)$$

Proof. The induction controls two quantities in the order needed by the recurrence. First it bounds coordinate inflation before the parent is trimmed; then nonnegative bilinearity converts those coordinate bounds into a cost bound. The singleton zero bucket is tracked separately so that root feasibility is never lost.

We use induction on $h(M)$. A leaf has height 0, and both leaf states are retained exactly, so all three claims hold with equality.

Let $M = A \circ B$ have height $h \geq 1$, and fix a finite exact entry (z, c) of $\mathcal{D}_M$, so $c = \mathcal{D}_M(z)$. Choose child signatures z_A, z_B attaining the minimum in (36.5), and put

$$c_A = \mathcal{D}_A(z_A), \quad c_B = \mathcal{D}_B(z_B).$$

Then

$$z = z_A + z_B, \quad c = c_A + c_B + g_\circ(z_A, z_B).$$

By induction, choose retained child representatives $(\hat{z}_A, \hat{c}_A)$ and $(\hat{z}_B, \hat{c}_B)$. Before trimming at M , combine them into

$$\tilde{z} = \hat{z}_A + \hat{z}_B, \quad (37.9)$$

$$\tilde{c} = \hat{c}_A + \hat{c}_B + g_\circ(\hat{z}_A, \hat{z}_B). \quad (37.10)$$

Because $h(A), h(B) \leq h - 1$, coordinate additivity gives, for every coordinate ℓ ,

$$\begin{aligned} \tilde{z}_\ell &= \hat{z}_{A,\ell} + \hat{z}_{B,\ell} \\ &\leq \lambda^{h(A)} z_{A,\ell} + \lambda^{h(B)} z_{B,\ell} \\ &\leq \lambda^{h-1} (z_{A,\ell} + z_{B,\ell}) \end{aligned}$$

$$= \lambda^{h-1} z_\ell.$$

Every term in g_o is a product of one nonnegative child coordinate from A and one from B . For any such term,

$$\begin{aligned} \widehat{z}_{A,q} \widehat{z}_{B,r} &\leq \lambda^{h(A)} z_{A,q} \lambda^{h(B)} z_{B,r} \\ &= \lambda^{h(A)+h(B)} z_{A,q} z_{B,r} \\ &\leq \lambda^{2(h-1)} z_{A,q} z_{B,r}. \end{aligned}$$

All three quantities c_A , c_B , and $g_o(z_A, z_B)$ are nonnegative. Summing the termwise bounds and using the child cost bounds therefore yields

$$\begin{aligned} \widetilde{c} &= \widehat{c}_A + \widehat{c}_B + g_o(\widehat{z}_A, \widehat{z}_B) \\ &\leq \lambda^{2(h-1)} c_A + \lambda^{2(h-1)} c_B + \lambda^{2(h-1)} g_o(z_A, z_B) \\ &= \lambda^{2(h-1)} (c_A + c_B + g_o(z_A, z_B)) \\ &= \lambda^{2(h-1)} c. \end{aligned}$$

Let $(\widehat{z}, \widehat{c})$ be the state retained in the box containing $(\widetilde{z}, \widetilde{c})$. By [lemma IV.9](#),

$$\begin{aligned} \widehat{z}_\ell &\leq \lambda \widetilde{z}_\ell \\ &\leq \lambda^h z_\ell, \end{aligned}$$

and

$$\begin{aligned} \widehat{c} &\leq \widetilde{c} \\ &\leq \lambda^{2(h-1)} c \\ &\leq \lambda^{2h} c. \end{aligned}$$

A coordinate sum is zero exactly when both child coordinates are zero. The induction hypothesis preserves those child zero patterns, and the singleton zero bucket preserves the parent zero pattern. This proves [\(37.6\)](#)–[\(37.8\)](#). $\square$

Theorem IV.11 (FPTAS for separable AWET). *Fix total orders $\prec_E$ and $\prec_L$ refining*

nondecreasing early and tardiness ratios. If the resulting ratio permutation π is separable, equivalently if a separating tree exists, then for every rational $\varepsilon > 0$ the trimmed separating-tree algorithm returns a feasible schedule of cost at most

$$(1 + \varepsilon) \text{OPT}.$$

Its running time is polynomial in the binary input length and $1/\varepsilon$.

Proof. If $\varepsilon > 1$, run the construction with accuracy parameter 1. Its factor 2 is at most $1 + \varepsilon$. Hence it remains to consider $0 < \varepsilon \leq 1$.

Let R be the root, of height H , and let (z^*, OPT) be an optimal exact root state from [theorem IV.7](#). Because the early set is nonempty, the first coordinate of z^* is positive. By [lemma IV.10](#), the trimmed root table contains a representative $(\hat{z}, \hat{c})$ with the same zero pattern. Its first coordinate is therefore also positive, so the representative is feasible. Its cost satisfies

$$\begin{aligned} \hat{c} &\leq \lambda^{2H} \text{OPT} \\ &= \left(1 + \frac{\varepsilon}{4H}\right)^{2H} \text{OPT} \\ &\leq \exp(\varepsilon/2) \text{OPT} \\ &\leq (1 + \varepsilon) \text{OPT}. \end{aligned}$$

For completeness, the last inequality follows because

$$f(\varepsilon) = \log(1 + \varepsilon) - \frac{\varepsilon}{2}$$

has $f(0) = 0$ and

$$f'(\varepsilon) = \frac{1}{1 + \varepsilon} - \frac{1}{2} = \frac{1 - \varepsilon}{2(1 + \varepsilon)} \geq 0 \quad (0 \leq \varepsilon \leq 1).$$

The algorithm returns the cheapest feasible retained root state, so its cost is no larger than $\hat{c}$.

We next prove the fully polynomial bound. For a positive coordinate bounded by U , the number of positive buckets is at most $1 + \lceil \log_\lambda U \rceil$; one additional bucket stores zero. Define

$$B_P = 2 + \lceil \log_\lambda \max\{1, P\} \rceil,$$

$$B_E = 2 + \lceil \log_\lambda \max\{1, W_E\} \rceil ,$$

$$B_L = 2 + \lceil \log_\lambda \max\{1, W_L\} \rceil .$$

Every node retains at most

$$B := B_P^2 B_E B_L \tag{37.11}$$

boxes. Put $x = \varepsilon/(4H)$. Since $0 < x \leq 1/4$,

$$\log(1+x) \geq \frac{x}{2}.$$

Consequently, for $U \geq 1$,

$$\begin{aligned} \log_\lambda U &= \frac{\log U}{\log(1+x)} \\ &\leq \frac{2 \log U}{x} \\ &= \frac{8H}{\varepsilon} \log U. \end{aligned}$$

Because $H \leq n-1$, the bound B is polynomial in n , $1/\varepsilon$, and the binary lengths of P, W_E, W_L .

The ratio orders are obtained by exact sorting. From the resulting permutation, separability can be recognized and an ordered separating tree recovered in linear time [4]. After binarization there are $O(n)$ internal nodes. Each node examines at most B^2 pairs of retained child states and performs constant-dimensional exact arithmetic on polynomial-bit integers and rationals. Exact bucket comparisons are polynomial as noted after [definition IV.8](#). Therefore the total running time is polynomial in the input length and $1/\varepsilon$. Backpointers and the fixed canonical orders reconstruct the returned schedule. The one-job case was handled directly. $\square$

Section exit. The exact separating-tree recurrence has now been converted into an FPTAS: every exact state has a retained representative with the same zero pattern, coordinate inflation at most λ^h , and cost inflation at most λ^{2h} ; the root choice then gives $\lambda^{2H} \leq 1 + \varepsilon$. The worked example uses only the two cross forms, the same-box replacement rule, and root backtracking to show where one locally cheaper representative loses globally.

38 Worked example

This example uses six jobs, exactly four distinct processing times, an ordered separating tree containing both direct and skew nodes, and genuine geometric trimming at an internal node. Most importantly, the trimmed dynamic program does *not* return an exact optimum for the displayed accuracy parameter. The example therefore shows both why trimming is useful and how its controlled information loss can affect the final schedule.

38.1 Instance and ratio orders

Let the six jobs have the following data.

job j	p_j	w_j^E	w_j^L	$r_j = w_j^E/p_j$	$s_j = w_j^L/p_j$
1	1	140	315	140	315
2	1	141	321	141	321
3	1	145	322	145	322
4	2	292	200	146	100
5	3	441	600	147	200
6	4	592	1200	148	300

Thus

$$(p_1, \dots, p_6) = (1, 1, 1, 2, 3, 4),$$

so the instance has exactly four distinct processing-time values, namely 1, 2, 3, 4. The nonrestrictive common due date is

$$d = P = 1 + 1 + 1 + 2 + 3 + 4 = 12.$$

The nondecreasing early-ratio order is

$$1 \prec_E 2 \prec_E 3 \prec_E 4 \prec_E 5 \prec_E 6,$$

whereas the nondecreasing tardiness-ratio order is

$$4 \prec_L 5 \prec_L 6 \prec_L 1 \prec_L 2 \prec_L 3.$$

Therefore the ratio permutation, written in the $\prec_E$ indexing, is

$$\pi = (4, 5, 6, 1, 2, 3) = 123 \ominus 123.$$

The displayed skew-sum decomposition proves that the permutation is separable. Equivalently, its inversion graph is the cograph $K_{3,3}$. Use the ordered binary separating tree

$$A = (1 \oplus 2) \oplus 3, \quad B = (4 \oplus 5) \oplus 6, \quad R = A \ominus B.$$

Here and in the figure, the leaf labels $1, \dots, 6$ are job identifiers. Under the formal convention of [definition IV.2](#), their permutation labels—the corresponding $\prec_L$ ranks read from left to right in $\prec_E$ order—are 4, 5, 6, 1, 2, 3. The separating-tree height is $H = 3$.

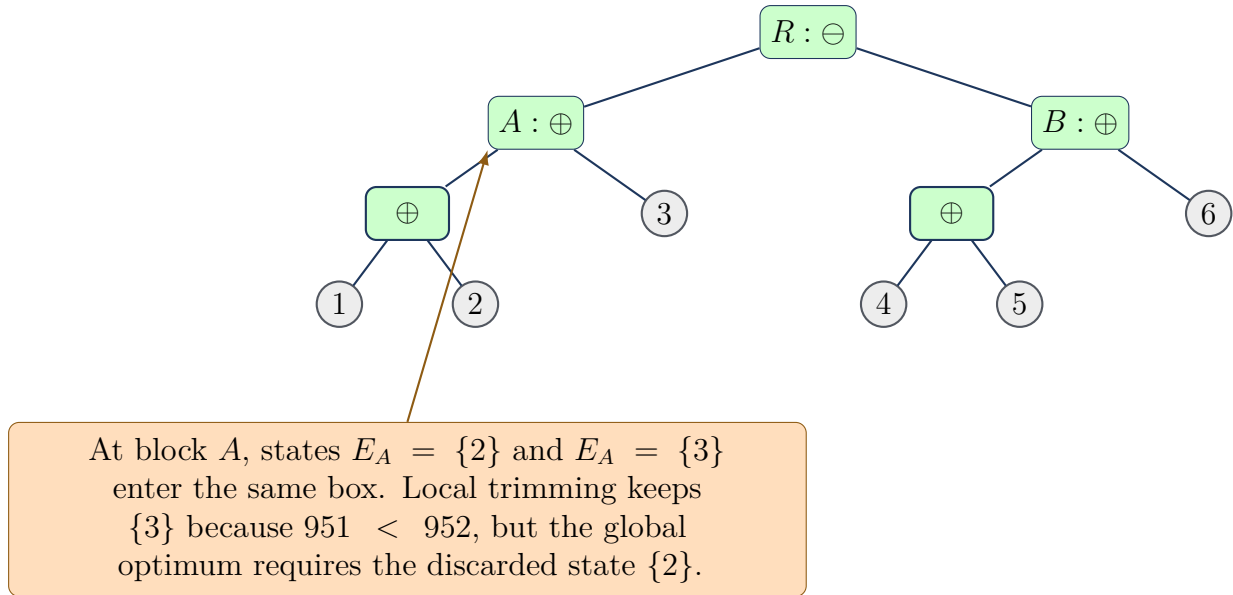


Figure 9: Job-labeled ordered separating tree for the six-job example. Direct nodes form the two three-job blocks, the root is skew, and the annotation marks the left-block trimming collision that removes the globally optimal path.

38.2 Trimming parameter

All calculations below use the four-coordinate signature from [definition IV.4](#) and the two cross-cost formulas in (35.1)–(35.2). Every required coordinate and cross term is instantiated where it is used.

Choose

$$\varepsilon = 0.50.$$

Since $H = 3$,

$$\lambda = 1 + \frac{\varepsilon}{4H} = 1 + \frac{0.50}{12} = \frac{25}{24} \approx 1.0416667.$$

For a positive integer x , put

$$\beta(x) = \left\lfloor \log_{25/24} x \right\rfloor,$$

and give zero its own bucket $\perp$. A signature is assigned the four-dimensional box

$$\text{box}(e, t, u, v) = (\beta(e), \beta(t), \beta(u), \beta(v)).$$

The bucket values used in the important left-subtree calculation are

x	1	2	3	140	141	145	281	285	286	315	321	322	426	636	637	643
$\beta(x)$	0	16	26	121	121	121	138	138	138	140	141	141	148	158	158	158

and $\beta(958) = 168$.

38.3 The left block A: trimming actually deletes states

First combine jobs 1 and 2 at the direct node $1 \oplus 2$. The four exact candidates are

early set in $\{1, 2\}$	signature (e, t, u, v)	exact cost	box
$\emptyset$	$(0, 2, 0, 636)$	951	$(\perp, 16, \perp, 158)$
$\{1\}$	$(1, 1, 140, 321)$	321	$(0, 0, 121, 141)$
$\{2\}$	$(1, 1, 141, 315)$	315	$(0, 0, 121, 140)$
$\{1, 2\}$	$(2, 0, 281, 0)$	140	$(16, \perp, 138, \perp)$

All four boxes are distinct, so no trimming occurs yet.

Now combine this table with job 3 at the direct node $A = (1 \oplus 2) \oplus 3$. Before trimming there are eight exact candidates:

early set E_A	signature (e_A, t_A, u_A, v_A)	cost c_A	box
$\emptyset$	$(0, 3, 0, 958)$	1909	$(\perp, 26, \perp, 168)$
$\{1\}$	$(1, 2, 140, 643)$	964	$(0, 16, 121, 158)$
$\{2\}$	$(1, 2, 141, 637)$	952	$(0, 16, 121, 158)$
$\{3\}$	$(1, 2, 145, 636)$	951	$(0, 16, 121, 158)$
$\{1, 2\}$	$(2, 1, 281, 322)$	462	$(16, 0, 138, 141)$
$\{1, 3\}$	$(2, 1, 285, 321)$	461	$(16, 0, 138, 141)$
$\{2, 3\}$	$(2, 1, 286, 315)$	456	$(16, 0, 138, 140)$
$\{1, 2, 3\}$	$(3, 0, 426, 0)$	421	$(26, \perp, 148, \perp)$

This is the first genuinely important trimming step. The three single-early candidates

$$E_A = \{1\}, \quad E_A = \{2\}, \quad E_A = \{3\}$$

all lie in the same box $(0, 16, 121, 158)$. Their exact local costs are 964, 952, 951, respectively. The trimmed recurrence therefore keeps only

$$E_A = \{3\}, \quad z_A = (1, 2, 145, 636), \quad c_A = 951,$$

and discards the candidate

$$E_A = \{2\}, \quad z_A = (1, 2, 141, 637), \quad c_A = 952.$$

Notice the tradeoff: the retained state is locally cheaper by only 1, but its early-weight coordinate is larger by 4.

A second collision occurs between $E_A = \{1, 2\}$ and $E_A = \{1, 3\}$. Both lie in box $(16, 0, 138, 141)$, so the latter, with cost 461, replaces the former, with cost 462. Thus the eight exact candidates at A are compressed to five retained candidates.

The first collision is the one that will matter globally: the discarded state $E_A = \{2\}$ is part of the exact optimum of the six-job instance.

38.4 The right block B

For $B = (4 \oplus 5) \oplus 6$, all eight exact side assignments survive trimming for this value of λ . The state that will be paired with the best left-side candidates is

$$E_B = \{5, 6\}, \quad T_B = \{4\}.$$

Its signature is

$$z_B = (7, 2, 1033, 200),$$

because

$$e_B = 3 + 4 = 7, \quad t_B = 2, \quad u_B = 441 + 592 = 1033, \quad v_B = 200.$$

Its exact internal cost is

$$\begin{aligned} c_B &= p_5 p_6 \min\{r_5, r_6\} + p_4 w_4^L \\ &= (3)(4)(147) + (2)(200) \\ &= 1764 + 400 \\ &= 2164. \end{aligned}$$

38.5 Root convolution and the returned solution

All jobs of $A = \{1, 2, 3\}$ precede all jobs of $B = \{4, 5, 6\}$ in $\prec_E$, whereas all jobs of B precede all jobs of A in $\prec_L$. Thus the root relation is $R = A \ominus B$, so the root is skew and

$$g_\ominus(z_A, z_B) = u_A e_B + t_A v_B.$$

Using the retained left state $E_A = \{3\}$ and the right state $E_B = \{5, 6\}$ gives

$$\begin{aligned} g_\ominus &= (145)(7) + (2)(200) \\ &= 1015 + 400 \\ &= 1415. \end{aligned}$$

Hence the corresponding root cost is

$$\begin{aligned} c_R &= c_A + c_B + g_\ominus(z_A, z_B) \\ &= 951 + 2164 + 1415 \\ &= 4530. \end{aligned}$$

The resulting early set is therefore

$$\hat{E} = \{3, 5, 6\}.$$

The five retained states at A and the eight retained states at B generate $5 \cdot 8 = 40$ root candidates. Root trimming compresses these to 35 boxes. The least-cost retained feasible root candidate is the one above, of cost 4530, so the trimmed dynamic program returns $\hat{E} = \{3, 5, 6\}$.

38.6 What was lost by trimming

Now reconsider the discarded left state

$$E_A^* = \{2\}, \quad z_A^* = (1, 2, 141, 637), \quad c_A^* = 952.$$

Combine it with the *same* right state $E_B = \{5, 6\}$. The skew cross cost would have been

$$\begin{aligned} g_\ominus(z_A^*, z_B) &= (141)(7) + (2)(200) \\ &= 987 + 400 \\ &= 1387. \end{aligned}$$

Thus this discarded path has total cost

$$\begin{aligned} c_R^* &= 952 + 2164 + 1387 \\ &= 4503. \end{aligned}$$

This calculation isolates the trimming effect exactly. Replacing $E_A^* = \{2\}$ by $\{3\}$ saves

$$952 - 951 = 1$$

in local cost, but increases the root cross term by

$$(145 - 141)e_B = 4 \cdot 7 = 28.$$

The net loss is therefore

$$28 - 1 = 27,$$

which is exactly

$$4530 - 4503 = 27.$$

38.7 Exact optimum for comparison

A direct enumeration of all $2^6 - 1 = 63$ nonempty early sets gives the five smallest exact objective values

early set	exact cost
$\{2, 5, 6\}$	4503
$\{1, 5, 6\}$	4508
$\{3, 5, 6\}$	4530
$\{5, 6\}$	4673
$\{1, 2, 5, 6\}$	4793

Hence the exact optimum is unique and equals

$$\text{OPT} = 4503, \quad E^* = \{2, 5, 6\}.$$

The trimmed algorithm returns

$$\hat{\Phi} = 4530 > 4503 = \text{OPT}.$$

Thus this six-job example is genuinely approximate rather than accidentally exact.

38.8 Canonical schedules and direct cost checks

The common due date is $d = 12$.

For the exact optimum $E^* = \{2, 5, 6\}$, the early processing load is $1 + 3 + 4 = 8$, so the early block starts at time $12 - 8 = 4$. The early jobs are sequenced in nondecreasing r_j order:

$$2, 5, 6.$$

Their completion times are 5, 8, 12. The tardy jobs are $\{1, 3, 4\}$ and are sequenced in nonincreasing s_j order:

$$3, 1, 4,$$

with completion times 13, 14, 16. Therefore

$$\begin{aligned}\Phi(E^*) &= 141(12 - 5) + 441(12 - 8) + 592(12 - 12) \\ &\quad + 322(13 - 12) + 315(14 - 12) + 200(16 - 12) \\ &= 987 + 1764 + 0 + 322 + 630 + 800 \\ &= 4503.\end{aligned}$$

For the returned set $\hat{E} = \{3, 5, 6\}$, the early block also has load 8 and starts at time 4. The canonical early order is

$$3, 5, 6,$$

with completion times 5, 8, 12. The tardy jobs $\{1, 2, 4\}$ are ordered

$$2, 1, 4,$$

with completion times 13, 14, 16. Thus

$$\begin{aligned}\Phi(\hat{E}) &= 145(12 - 5) + 441(12 - 8) + 592(12 - 12) \\ &\quad + 321(13 - 12) + 315(14 - 12) + 200(16 - 12) \\ &= 1015 + 1764 + 0 + 321 + 630 + 800 \\ &= 4530.\end{aligned}$$

This independently confirms the root-DP calculations.

38.9 Approximation-factor audit and effect of tighter trimming

For the displayed choice $\varepsilon = 0.50$ and $H = 3$,

$$\lambda^{2H} = \left(\frac{25}{24}\right)^6 \approx 1.2775344 < 1.50 = 1 + \varepsilon.$$

The theorem therefore guarantees a returned cost at most 1.5 OPT. The actual ratio in this example is much better:

$$\frac{4530}{4503} \approx 1.005996.$$

The example also illustrates how the accuracy parameter controls information loss. If the same instance is run with

$$\varepsilon = 0.40, \quad \lambda = 1 + \frac{0.40}{12} = \frac{31}{30},$$

the important single-early states in block A no longer collapse in the same way; the exact-optimal state $E_A = \{2\}$ survives, and the trimmed recurrence returns the exact optimum 4503. Thus the $\varepsilon = 0.50$ run visibly demonstrates trimming loss, whereas a modestly tighter geometric grid restores the optimum on this instance.

38.10 What this example demonstrates

This six-job instance separates the essential ideas of the separating-tree FPTAS. The exact recurrence first generates distinct interface states. Geometric trimming then identifies states whose four coordinates are close enough to occupy one box and keeps only the cheapest local representative. Here that local choice is not globally exact: the state $E_A = \{3\}$ is cheaper inside block A than $E_A = \{2\}$ by 1, but its larger early-weight coordinate creates 28 additional units of skew-root interaction. The final error is therefore 27, while remaining far inside the proved approximation bound.

The example also shows why the four-coordinate signature matters. The trimmed recurrence is not merely remembering the number or processing load of early jobs. It must retain enough information about both processing and side-specific weights to price future direct or skew interactions. Geometric trimming compresses that information rather than deleting it indiscriminately, and the approximation theorem controls precisely the resulting loss.

Concluding synthesis

The separating-tree result combines an exact decomposition with a controlled loss of numerical resolution. The decomposition is structural: direct and skew nodes determine the orientation of every cross pair. The approximation is numerical: one minimum-cost representative is retained in each four-dimensional geometric box. Because the interface is additive, nonnegative, and bilinear, coordinate inflation compounds by at most λ per level and cost inflation by at most λ^2 per level.

Reusable idea. When a recursive decomposition makes every cross interaction a nonnegative bilinear form of a small additive signature, approximate the interface rather than the underlying assignments. Keeping one cheapest representative per coordinated geometric box preserves reconstructibility and supports a height-based multiplicative induction.

Why Part V is next. Part IV has extended the structured algorithmic regime from exact reversal to every fixed refinement pair whose ratio permutation is separable. What remains is to compare exact hardness, unrestricted approximation, weak hardness, and structured approximation without conflating their meanings. Part V synthesizes the four certificates, identifies their transferable lessons, and states the next technical milestones.

Part V

Global Synthesis

Exact Complexity, Approximation, and Order Structure

Part synopsis. This concluding Part integrates the four theorem chains. It separates exact from approximation classifications, explains why weak NP-completeness and an FPTAS coexist at exact reversal, compares the certificates and interfaces used in the proofs, and converts the remaining boundaries into named technical problems with concrete first milestones.

Part positioning. Established in Parts I–IV. General positive-integer AWET at the boundary due date is strongly NP-complete but has a universal constant-factor approximation; strict antithetical instances are weakly NP-complete and pseudopolynomially solvable; and fixed refinement pairs with separable ratio permutation admit an FPTAS. **Result of this Part.** The four classifications are placed in one landscape without treating exact and approximation complexity as interchangeable. **Technique.** The synthesis compares each structural obstacle with its certificate or interface, its loss or gap control, and its algorithmic consequence.

39 What the four technical Parts establish

Part I settles the published strong-complexity question for positive-integer asymmetric weighted earliness–tardiness scheduling at $d = \sum_j p_j$. Its theorem is strong NP-completeness. Its reusable technique is a compiler that separates a prefix-square signal, an anchor-coupled minimum-kernel load lock, and a bounded residual while keeping every output number polynomially bounded.

Part II shows that exact intractability does not preclude a universal performance guarantee. Its theorem is a polynomial-time $(3 + 2\sqrt{2} + \varepsilon)$ -approximation for unrestricted AWET. Its reusable technique is asymmetric payment: local pseudo-probability consistency pays the early pairs, while the tardy self-cost completes a positive-semidefinite minimum kernel whose covariance inequality pays the tardy pairs globally.

Part III classifies the exact-reversal extreme. Its theorem is weak NP-completeness together with an exact pseudopolynomial recurrence. Its reusable technique is a dominant completed square in one selected load, with every identity-dependent interaction confined to a uniformly bounded residual. The same reversed order is separable, so the FPTAS of Part IV also applies after the fixed strict refinements.

For fixed total ratio-order refinements, Part IV treats instances whose ratio permutation is separable. Its theorem is an FPTAS. Its reusable technique is an ordered separating tree whose direct and skew nodes expose a four-coordinate nonnegative additive interface; the exact pseudopolynomial recurrence is then compressed by coordinated geometric trimming.

40 Combined landscape

The conclusions are summarized in Figure 10. The horizontal direction changes the order promise; the vertical direction changes the required quality of solution. No arrow should be read as an automatic transfer between exact and approximation complexity.

The antithetical subclass illustrates why the categories must remain distinct. Its decision problem is NP-complete, so a polynomial exact algorithm is unlikely. Its pseudopolynomial recurrence makes that hardness weak, and its reverse permutation is separable, so an FPTAS is available. Weak NP-hardness, pseudopolynomial solvability, and an FPTAS are therefore mutually consistent, as in other knapsack-type regimes.

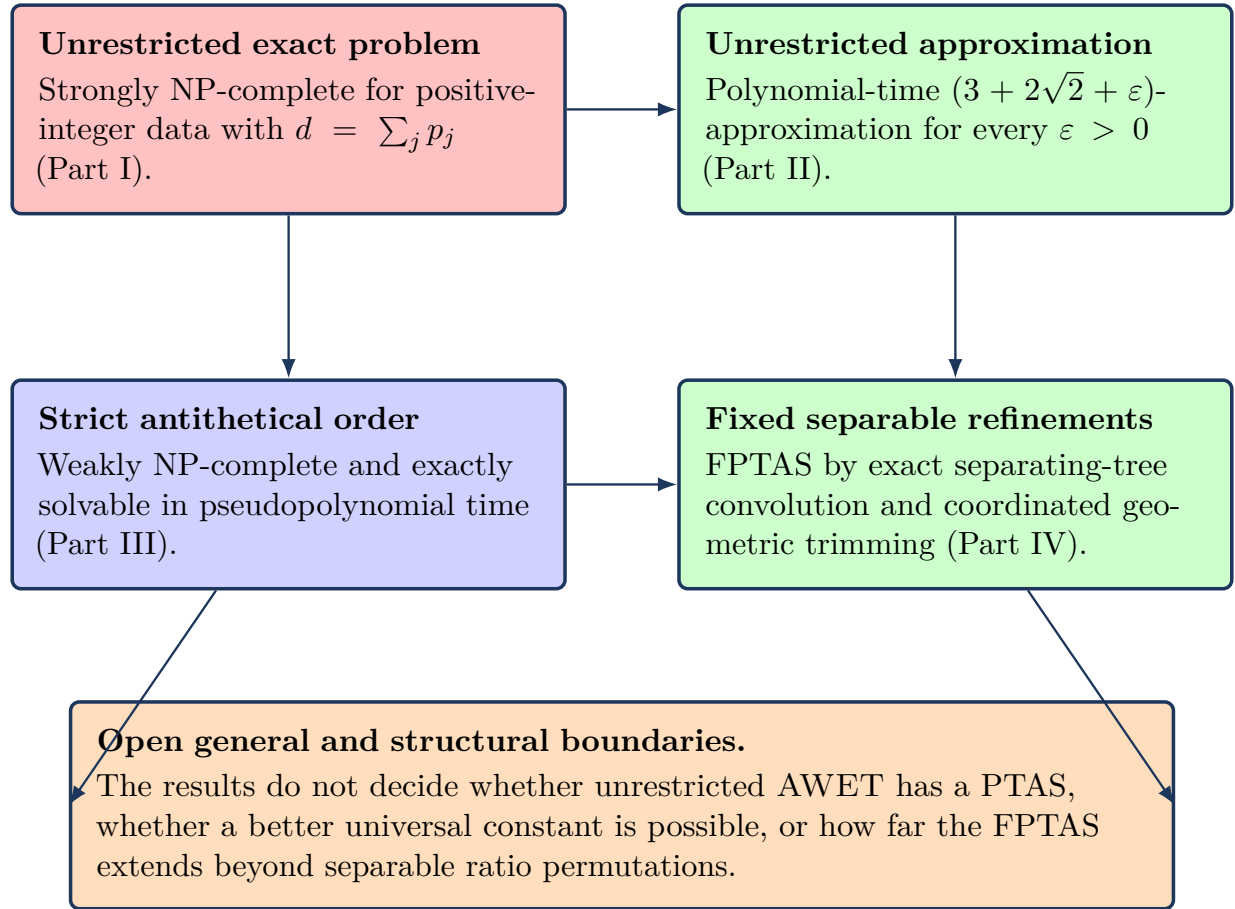


Figure 10: Combined exact-complexity and approximation landscape. Red marks exact intractability, green marks approximation algorithms, blue marks a mixed weak-hardness/pseudopolynomial regime, and amber marks unresolved boundaries.

41 Synthesis by obstacle, certificate, and control

The following table compares the proof architectures rather than merely restating the theorems.

Part	structural obstacle	certificate or interface	loss or gap control	consequence
I	many local exact-cover conditions must be encoded by nonnegative scheduling costs	effective-atom prefix profiles plus an anchor-coupled minimum-kernel load lock	main gaps dominate two residual swings and an integer unit: $\Delta_{\min}/2 > 2\Gamma + 1$	strong NP-completeness with polynomially bounded integer data
II	rounded tardy pairs may have zero pseudo-joint mass, so entrywise payment fails	anchored local SDP and the diagonal-completed tardy minimum kernel	threshold denominators are balanced; finite-precision loss is absorbed in the requested ε	universal $(3 + 2\sqrt{2} + \varepsilon)$ -approximation
III	exact reversal destroys the Part I compiler but leaves a global load choice	tagged half-load square plus residual Ψ	the unit load gap pays DM , while both residual swings cost at most $2DU$ and $M > 2U$	weak NP-completeness and a two-resource pseudopolynomial algorithm
IV	future cross-block cost must be summarized without retaining every side assignment	four-coordinate nonnegative additive signature on an ordered separating tree	one factor λ per coordinate level and at most λ^2 per cost level, with $\lambda^{2H} \leq 1 + \varepsilon$	exact pseudopolynomial recurrence and FPTAS

Two broad design choices recur. A *global certificate* is appropriate when one aggregate condition can dominate all unwanted interactions: the load lock of Part I, the covariance inequality of Part II, and the completed square of Part III have this character. A *bounded constructive interface* is appropriate when the instance decomposes recursively: Part IV stores exactly the information needed to price interactions with the unprocessed part of the tree.

This distinction is transferable beyond the present model. In another two-sided sequencing or quadratic partition problem, the first audit should ask whether a global norm, kernel, or

convex aggregate can certify feasibility or cost. If not, the next question is whether an order or graph decomposition makes every cross interaction factor through a small nonnegative signature. Only after those structural questions should one choose between scale-separated hardness, relaxation and rounding, pseudopolynomial enumeration, or interface trimming.

42 Methodological, modeling, and implementation implications

Methodological implication. Canonicalization is the common entry point. It removes continuous placement and local sequencing choices before any gadget, relaxation, or dynamic program is introduced. This prevents proof machinery from paying repeatedly for decisions that an exchange argument already determines.

Model-structure implication. The ratio orders are informative data objects in their own right. Two instances with similar processing times and weights can have different algorithmic structure if their early and tardy ratios induce different inversion graphs. This observation is structural rather than managerial: it identifies which mathematical representation should be inspected before an algorithm is selected.

Implementation implication. A solver can use the following diagnostic workflow.

1. Compute the two ratio orders exactly by cross multiplication and fix deterministic tie refinements.
2. Construct the ratio permutation and test whether it is separable.
3. If it is, use the separating-tree exact recurrence for small numerical ranges or the separating-tree FPTAS for general binary data.
4. Otherwise, use the unrestricted anchored-SDP approximation, while exact enumeration or the antithetical recurrence remains available when the order pattern or numerical range permits it.

This workflow does not claim that nonseparable instances lack stronger algorithms. It uses only the guarantees established in the tutorial and leaves room for future structural tests.

Hardness-design implication. Parts I and III show two different uses of magnitude. The strong reduction uses a hierarchy of polynomial scales to isolate many simultaneous signals; the antithetical reduction uses one dominant convex square and one residual budget. The first architecture supports strong hardness, whereas the second remains compatible with a pseudopolynomial target algorithm. The numerical form of a reduction should therefore be aligned with the exact complexity classification it seeks to prove.

43 Named open problems and first milestones

Open Problem A: the unrestricted PTAS boundary. Does unrestricted AWET admit a PTAS, or can a no-PTAS theorem be proved by an approximation-preserving reduction whose relative gap survives the positive objective baseline? A concrete first milestone is to prove or refute a bounded-core decomposition after conditioning on one early anchor, with a residual class that can be spliced at $O(\varepsilon)$ relative cost.

Open Problem B: a better universal approximation ratio. Can the factor $3 + 2\sqrt{2}$ be improved? The current proof pays the early and tardy sides separately. A first milestone is a relaxation and rounding inequality that couples the early local-consistency account with the tardy minimum-kernel account, thereby replacing the maximum of two threshold losses by one joint bound.

Open Problem C: structured classes beyond separable permutations. Which nonseparable ratio permutations still admit an FPTAS? A first milestone is to handle one bounded-size prime node in a substitution decomposition while retaining separating-tree trimming around it. A successful result would clarify whether the natural parameter is prime-node size, distance to separability, or another interface-width measure.

Open Problem D: practical acceleration with certificates. Can the n anchored SDPs and the theoretically possible separating-tree boxes be reduced substantially in practice without weakening the guarantees? A first milestone is to derive valid anchor-screening lower bounds and combine them with output-sensitive dictionaries that store only reachable signature boxes. Warm starts and sparse Gram representations can then be assessed on top of those certified reductions.

44 Final perspective

This tutorial gives a deliberately balanced picture. General positive-integer AWET is strongly NP-complete, yet it has a universal constant-factor approximation. Exact reversal remains NP-complete, but only weakly, and its order structure supports both a pseudopolynomial exact algorithm and an FPTAS through the broader separating-tree theorem. The decisive theme is not simply whether the two ratio orders agree. It is how much of the canonical quadratic interaction can be controlled by a dominant certificate or compressed into a bounded constructive interface.

The four theorem chains therefore support one final methodological claim: expose the canonical objective first, then match the proof device to the structure that remains. Hardness requires a gap that dominates every residual; unrestricted approximation requires a global relaxation certificate; weakly hard order patterns may expose small pseudopolynomial states; and recursively decomposable orders may permit approximation of interfaces rather than assignments.

A Algebra of the canonical objective

This appendix certifies the two coefficient formulas consumed by the Part I master decomposition. The early indicator expansion creates the baseline fields and couplings, while a common tardiness offset changes only one block's field on the coded manifold. Keeping the expansions separate identifies the destination of every constant, linear, and quadratic term.

A.1 Baseline early field

For an early pair i, j in the same block v ,

$$\mathbf{1}\{i, j \in E\} = \frac{1}{4}(1 + \sigma_v \epsilon_i)(1 + \sigma_v \epsilon_j) = \frac{1}{4}(1 + \epsilon_i \epsilon_j + \sigma_v(\epsilon_i + \epsilon_j)).$$

For $i \in \mathcal{B}_u$ and $j \in \mathcal{B}_v$, $u \neq v$,

$$\mathbf{1}\{i, j \in E\} = \frac{1}{4}(1 + \sigma_u \epsilon_i + \sigma_v \epsilon_j + \sigma_u \sigma_v \epsilon_i \epsilon_j).$$

Collecting the coefficients of σ_v gives (8.14); collecting the coefficients of $\sigma_u \sigma_v$ and applying lemma I.13 gives (8.10). The constant terms are retained inside C_0 . Hence every term in the physical early objective has a declared destination: constant, field, or coupling.

A.2 Effect of a common tardiness offset

On the coded state space, the tardy load of block v is β_v . Therefore every cross-block tardy contribution is independent of σ . Within block v , the contribution of the offset η_v is

$$\eta_v \left(\sum_{\{i,j\} \subseteq T \cap \mathcal{B}_v} p_i p_j + \sum_{i \in T \cap \mathcal{B}_v} p_i^2 \right) = \frac{\eta_v}{2}(\beta_v^2 + q_v),$$

where

$$q_v = \sum_{i \in \mathcal{B}_v} p_i^2 \frac{1 - \sigma_v \epsilon_i}{2} = \frac{1}{2}(Q_v - \sigma_v D_v).$$

Thus the offset changes the logical field by $-\eta_v D_v/4$, as used in (8.18).

B Two worked examples

This appendix provides reproducibility certificates for the Part I reduction, not merely illustrative examples. The yes-instance verifies the no-carry arithmetic, pair synchronization, strict ratio separation, load lock, master identity, decision threshold, and forward decoding chain. A degree-preserving exchange then produces a no-instance and verifies the corresponding reverse contradiction. Printing the full integer data makes every claimed inequality and schedule cost independently auditable without hidden computation.

How to read this step. Theorem-verification lane. Read the source and prefix instances, the scale and gap summaries through Section B.1.7, the master-identity and threshold summary in Section B.1.9, and the forward and reverse conclusions. The landscape job catalogues and full schedule tables may be skipped without breaking the proof narrative. **Full arithmetic-audit lane.** Follow every defining recurrence and use each “Purpose and verification” statement to reproduce the displayed integer tables and all 64 coded-state evaluations.

Arithmetic guide. Long integers are never meant to be compared visually as undifferentiated strings. The certificate map and each table lead-in name a defining expression or subtraction. Verify one row from that expression, then use only the property consumed later: equality, sign, strict order, stated lower bound, or final difference. Full digit strings are retained as exact evidence; the defining expressions are the audit interface.

B.1 A “yes” instance

This first example specializes the reduction to a six-element yes-instance. No scale is compressed and no floating-point approximation is used. The resulting scheduling instance has 48 state jobs and one anchor job. Because the proof uses deliberately separated powers of N , its integers are long; the line-breaking typewriter format preserves every digit.

How to read this step. Theorem-verification lane. Track the exact cover, prefix equations, two-state block property, band-gap certificate, load-lock slack, master threshold, and final schedule. **Full arithmetic-audit lane.** Reproduce every row from the formulas named in Table 8. In either lane, do not compare long integers visually;

Table 8: Certificate map for Appendix B. The defining-expression column is the preferred audit route for the long integers.

certificate	defining expression	claim verified	where consumed
Tables 10 and 22	$P_{vh} = T + M\kappa(k_{vh}) + Q_h$ and $\beta_v = \sum_h P_{vh} + 3$	integral bases and equal balanced loads	two-state synchrono- nization and the global load lock
Tables 11 and 23	$f = K\beta$ and $f_v - f_{v+1} =$ $2 \sum_{u \leq v} \beta_u$	minimum-kernel fields and block order	early ratios and load- lock centering
Tables 12, 13 and 24	$\rho = f_v + \lambda_v(2P + c)$; sub- tract adjacent endpoints	effective atoms, strict bands, main gaps, and to- tal width	pair compression, scale separation, and Γ
Tables 14, 15, 25 and 26	profile integrals and $4e_v = 4a_v^0 - \eta_v D_v - 4a_v^*$	main/residual couplings and bounded field correc- tion	exact master decom- position and residual bound
Tables 16, 17, 27 and 28	$w_j^E = p_j r_j$, $w_j^L = p_j s_j$	the physical positive- integer AWET instances	feasibility and exact input reproducibility
Tables 18, 19 and 29	canonical completion times and $A + \mathcal{P} + R - B$	yes schedule below threshold and every no state above threshold	forward and reverse correctness certifi- cates

audit the defining recurrence or subtraction and retain only the equality, sign, or bound used later.

B.1.1 The restricted exact-cover instance

Let $X = \{e_1, e_2, e_3, e_4, e_5, e_6\}$. The six sets are

$$\begin{aligned} S_1 &= \{e_1, e_2, e_3\}, & S_2 &= \{e_1, e_2, e_5\}, & S_3 &= \{e_2, e_4, e_6\}, \\ S_4 &= \{e_1, e_3, e_6\}, & S_5 &= \{e_4, e_5, e_6\}, & S_6 &= \{e_3, e_4, e_5\}. \end{aligned}$$

With elements in the rows and subsets in the columns, its incidence matrix is

$$A^{\text{inc}} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}. \quad (\text{B.1})$$

Every element row and every subset column contains three ones, so this is a valid three-uniform, three-regular RXC3 instance. Since an exact cover of six elements by 3-sets must contain exactly two subset-columns, it is enough to inspect pairwise intersections. The intersection-cardinality matrix is

$$\begin{pmatrix} 3 & 2 & 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 1 & 2 & 1 \\ 2 & 1 & 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \end{pmatrix}.$$

Its only off-diagonal zero is the pair (S_1, S_5) . Moreover, the corresponding incidence columns satisfy

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus the subset-columns S_1 and S_5 form not merely an exact cover, but the unique exact cover. The uniqueness is not required by the reduction, but it makes the final 64-state audit especially transparent: exactly one binary state should have zero prefix penalty.

B.1.2 The associated prefix-exactness instance

For each set S_v , the prefix datum is $d_{v,k} = |S_v \cap \{e_1, \dots, e_k\}|$. The full prefix matrix is shown in [Table 9](#). The exact-cover vector and spin vector are

$$x = (1, 0, 0, 0, 1, 0), \quad \sigma = 2x - \mathbf{1} = (+1, -1, -1, -1, +1, -1). \quad (\text{B.2})$$

	S_1	S_2	S_3	S_4	S_5	S_6
e_1	1	1	0	1	0	0
e_2	1	1	1	0	0	0
e_3	1	0	0	1	0	1
e_4	0	0	1	0	1	1
e_5	0	1	0	0	1	1
e_6	0	0	1	1	1	0

selected exact-cover columns: S_1 and S_5

Figure 11: The incidence matrix with elements in rows and subsets in columns. The highlighted subset-columns sum rowwise to the all-ones vector.

Table 9: Prefix data with subset-columns preserved. Row k gives the cumulative counts through element row e_k ; the last column evaluates $C_k(x) = \sum_v d_{v,k} x_v$ at $x = (1, 0, 0, 0, 1, 0)$.

Prefix	S_1	S_2	S_3	S_4	S_5	S_6	$C_k(x)$
$k = 1$	1	1	0	1	0	0	1
$k = 2$	2	2	1	1	0	0	2
$k = 3$	3	2	1	2	0	1	3
$k = 4$	3	2	2	2	1	2	4
$k = 5$	3	3	2	2	2	3	5
$k = 6$	3	3	3	3	3	3	6

The six equations are, explicitly,

$$C_1(x) = 1 \cdot x_1 + 1 \cdot x_2 + 0 \cdot x_3 + 1 \cdot x_4 + 0 \cdot x_5 + 0 \cdot x_6 = 1,$$

$$C_2(x) = 2x_1 + 2x_2 + x_3 + x_4 = 2,$$

$$C_3(x) = 3x_1 + 2x_2 + x_3 + 2x_4 + x_6 = 3,$$

$$C_4(x) = 3x_1 + 2x_2 + 2x_3 + 2x_4 + x_5 + 2x_6 = 4,$$

$$C_5(x) = 3x_1 + 3x_2 + 2x_3 + 2x_4 + 2x_5 + 3x_6 = 5,$$

$$C_6(x) = 3(x_1 + x_2 + x_3 + x_4 + x_5 + x_6) = 6.$$

Substitution of (B.2) gives $(C_1, \dots, C_6) = (1, 2, 3, 4, 5, 6)$. Conversely, successive differences

recover the element-cover equations:

$$\begin{aligned} C_1 - C_0 &= x_1 + x_2 + x_4 = 1, & C_2 - C_1 &= x_1 + x_2 + x_3 = 1, & C_3 - C_2 &= x_1 + x_4 + x_6 = 1, \\ C_4 - C_3 &= x_3 + x_5 + x_6 = 1, & C_5 - C_4 &= x_2 + x_5 + x_6 = 1, & C_6 - C_5 &= x_3 + x_4 + x_5 = 1. \end{aligned}$$

These are exactly the six element-row equations of (B.1). For instance, the S_1 column of the prefix table is $(1, 2, 3, 3, 3, 3)^\top$ because the cumulative sum down incidence column S_1 increases in rows e_1, e_2, e_3 and then remains constant. This illustrates concretely why successive prefix differences recover the original incidence rows.

B.1.3 Numerical scales and base processing values

Here $n = 6$ and $N = n + 1 = 7$. Therefore

$$Q_0 = 10, \quad Q_1 = 100, \quad Q_2 = 1000, \quad Q_3 = 10000, \quad q = 11110.$$

$$G = 7^{20} = 79792266297612001,$$

$$M = 7^{40} = 6366805760909027985741435139224001,$$

$$T = 7^{50} = 1798465042647412146620280340569649349251249.$$

The level map is

$$(\kappa(1), \dots, \kappa(6)) = (6, 5, 4, 3, 2, 1).$$

The scale relationships are especially easy to see symbolically:

$$M = G^2 = 7^{40}, \quad T = 7^{10}M = 7^{50}, \quad GM = 7^{60}.$$

Thus T dominates all block-level M terms, while the ratio-space gap GM dominates block-field variation on the order of $N^2T = 7^{52}$. The common controller base is

$$P_{v0} = T + 10 = 1798465042647412146620280340569649349251259 \quad (v = 1, \dots, 6).$$

All other base processing values and block loads are displayed in Table 10. For example, $S_1 = \{e_1, e_2, e_3\}$, so

$$P_{11} = T + 6M + 100,$$

$$\begin{aligned}
P_{12} &= T + 5M + 1000, \\
P_{13} &= T + 4M + 10000, \\
\beta_1 &= P_{10} + P_{11} + P_{12} + P_{13} + 3.
\end{aligned}$$

Substitution gives the first row of [Table 10](#). The three coefficients 6, 5, 4 are not arbitrary: they are the reversed element levels of e_1, e_2, e_3 . The pair codes 100, 1000, 10000 identify occurrence positions $h = 1, 2, 3$ even when two sets contain the same element.

Purpose. The table certifies the integral base values and target load of every coded block. **Verification rule.** Form each $P_{vh} = T + M\kappa(k_{vh}) + Q_h$, then add $P_{v0} + P_{v1} + P_{v2} + P_{v3} + 3$; the two orientations must both sum to the displayed β_v .

Table 10: Exact base processing values and balanced block loads.

v	S_v	$(\kappa_1, \kappa_2, \kappa_3)$	P_{v0}	P_{v1}	P_{v2}	P_{v3}	β_v
1	$\{1, 2, 3\}$	$(6, 5, 4)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650744814409 51165420269276825 045372254	17984650681146351 90256392283535389 906157253	7193860266091735000 1165411484001244853 76124
2	$\{1, 2, 5\}$	$(6, 5, 2)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650744814409 51165420269276825 045372254	17984650553810236 68438336312052519 627709251	7193860253358123478 2984851769172542069 28122
3	$\{2, 4, 6\}$	$(5, 3, 1)$	17984650426474121 46620280340569649 349251259	17984650744814409 51165420269276825 045371354	17984650617478294 29347364297793954 766924252	17984650490142179 07529308326311084 488485250	7193860227890900434 6623732339515136500 32118
4	$\{1, 3, 6\}$	$(6, 4, 1)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650681146351 90256392283535389 906148253	17984650490142179 07529308326311084 488485250	7193860240624511956 4804292054343839284 80120
5	$\{4, 5, 6\}$	$(3, 2, 1)$	17984650426474121 46620280340569649 349251259	17984650617478294 29347364297793954 766923352	17984650553810236 68438336312052519 627700251	17984650490142179 07529308326311084 488485250	7193860208790483151 9352892767272082323 60115
6	$\{3, 4, 5\}$	$(4, 3, 2)$	17984650426474121 46620280340569649 349251259	17984650681146351 90256392283535389 906147353	17984650617478294 29347364297793954 766924252	17984650553810236 68438336312052519 627709251	7193860227890900434 6623732339515136500 32118

Arithmetic guide. To audit a row of Table 10, first form each P_{vh} from $T + M\kappa(k_{vh}) + Q_h$ and then add the four bases plus 3 to obtain β_v . Rows 3 and 6 have the same β because their three reversed levels have the same sum $5 + 3 + 1 = 4 + 3 + 2 = 9$; the fixed pair-code total is identical in every block.

B.1.4 The 48 state jobs and arithmetic synchronization

Block v contains the eight jobs specified in Table 2. To make the arithmetic visible, consider block 1. In the $\sigma_1 = +1$ orientation its signed processing sum is

$$\begin{aligned} & -(P_{10} + 3) + P_{10} + (P_{11} + 1) - P_{11} + (P_{12} + 1) - P_{12} + (P_{13} + 1) - P_{13} \\ & = -3 + 1 + 1 + 1 = 0. \end{aligned}$$

The complementary $\sigma_1 = -1$ orientation also has sum zero. For an arbitrary orientation, the decomposition (8.3) becomes

$$2TU + 2MZ + 20u_0 + 200u_1 + 2000u_2 + 20000u_3 + R, \quad |R| \leq 6.$$

The symbols U and Z have exactly the meanings used in the general proof: U is the total pair imbalance and Z is the reversed-element-level imbalance. The exact inequalities are

$$2T > 6nM + 2q + 6, \quad 2M > 2q + 6, \quad 20 > 6.$$

Thus $U = Z = 0$, then $(u_0, u_1, u_2, u_3) = (0, 0, 0, 0)$, and finally the residual equation forces the displayed state or its complement. A direct exhaustive enumeration of all $2^8 = 256$ sign vectors in each of the six blocks gives the same conclusion. In the job order

$$(A_{v0}, B_{v0}, A_{v1}, B_{v1}, A_{v2}, B_{v2}, A_{v3}, B_{v3}),$$

the only zero-load vectors are

$$(-, +, +, -, +, -, +, -) \quad \text{and} \quad (+, -, -, +, -, +, -, +).$$

Hence all six blocks have exactly the required two balanced states.

B.1.5 Kernel data, early ratios, and effective atoms

The six values of $\lambda_v = G + 7 - v$, the loads β_v , and the fields $f_v = \sum_u \min\{L_v, L_u\}\beta_u$ are in [Table 11](#). As one sample calculation, since L_1 is the largest kernel level,

$$f_1 = \sum_{u=1}^6 L_u \beta_u.$$

The displayed exact value agrees with direct multiplication and summation. A useful local check is

$$f_1 - f_2 = 2\beta_1 = \frac{14387720532183470000233082296}{800248970752248},$$

which is the $v = 1$ case of [lemma I.12](#). Every state-job early ratio is then $r_i = f_v + \lambda_v p_i$.

Purpose. The table certifies the block loads, kernel levels, and affine fields used by the load lock. **Verification rule.** Compute one row from $f = K\beta$, then audit successive rows by $f_v - f_{v+1} = 2 \sum_{u \leq v} \beta_u$.

Table 11: Block loads and minimum-kernel data.

v	β_v	λ_v	$L_v = 2\lambda_v$	f_v
1	719386026609173500011 654114840012448537612 4	79792266297612 007	15958453259522 4014	688817294128443960104362198021598 7825344444649279254568462502
2	719386025335812347829 848517691725420692812 2	79792266297612 006	15958453259522 4012	688817294128443958665590144803251 7825111362352479005597710254
3	719386022789090043466 237323395151365003211 8	79792266297612 005	15958453259522 4010	688817294128443955788046040913280 0868281309701844248213101762
4	719386024062451195648 042920543438392848012 0	79792266297612 004	15958453259522 4008	688817294128443951471729891445128 3042126510583306463528429034
5	719386020879048315193 528927672720823236011 5	79792266297612 003	15958453259522 4006	688817294128443945716641693852074 1303010853053899910986796066
6	719386022789090043466 237323395151365003211 8	79792266297612 002	15958453259522 4004	688817294128443938522781454500923 3260024616971038941980442868

Arithmetic guide. Each row of Table 11 is generated from the previous one by subtracting twice the cumulative block load: $f_v - f_{v+1} = 2 \sum_{u \leq v} \beta_u$. This recurrence is much easier to audit than recomputing every long dot product independently.

The 24 compressed atoms are shown in Table 12. For example,

$$\rho_{11} = f_1 + \lambda_1(2P_{11} + 1) = \frac{71751801506004388200718894869745}{48885544592536491609034929479}.$$

The controller atom in every block has mass -3 ; every occurrence atom has mass $+1$. For an occurrence row, one checks the effective ratio by $f_v + \lambda_v(2P_{vh} + 1)$; for a controller row, by $f_v + \lambda_v(2P_{v0} + 3)$. The expression $2P$ plus the small constant is the difference-of-squares calculation from the theory, not an independently chosen coordinate.

Purpose. The table fixes every effective-atom mass, element label, and ratio used in the profile calculation. **Verification rule.** Use mass +1 and $\rho = f_v + \lambda_v(2P_{vh} + 1)$ for occurrences, and mass -3 with $\rho = f_v + \lambda_v(2P_{v0} + 3)$ for controllers.

Table 12: All effective atoms in the worked instance.

block pair	kind	element	mass	effective ratio ρ
1	0	controller	–	-3 7175180144504176491196408644698275728267965338815663737832149
1	1	occurrence	e_1	1 7175180150600438820071889486974548885544592536491609034929479
1	2	occurrence	e_2	1 7175180149584395098592642679928503359331967330128854016182065
1	3	occurrence	e_3	1 7175180148568351377113395872882457833120634758480120311948051
2	0	controller	–	-3 7175180144504176473211758027219981434794322360876116068577380
2	1	occurrence	e_1	1 7175180150600438802087238793094585461162613729654839694986520
2	2	occurrence	e_2	1 7175180149584395080607991998782151456768044494774954954685308
2	3	occurrence	e_5	1 7175180146536263916170251615844849443585342172690650645057272
3	0	controller	–	-3 7175180144504176440839386903025440184723709029102059985466367
3	1	occurrence	e_2	1 7175180149584395048235620810919552597607007679507211777723307
3	2	occurrence	e_4	1 7175180147552307605277127247761907632453837526633847152404287
3	3	occurrence	e_6	1 7175180145520220162318633684604262667301960008474503841566267
4	0	controller	–	-3 7175180144504176394079295323049098065328349229424976602291118

Continued on next page

Table 12 (continued)

block pair	kind	element	mass	effective ratio ρ
4	1	occurrence e_1	1	7175180150600438722954775936120363829879968940409256887323878
4	2	occurrence e_3	1	7175180148568351279996282398429941908362910730501632818899062
4	3	occurrence e_6	1	7175180145520220115558542091894309026088544237314550179923038
5	0	controller $-$	-3	7175180144504176332931483261823732032972131018879125362155629
5	1	occurrence e_4	1	7175180147552307497369223530158530349793780061434355156796181
5	2	occurrence e_5	1	7175180146536263775889976774046930910853378741003081251833575
5	3	occurrence e_6	1	7175180145520220054410730017935331471914270055285828661319569
6	0	controller $-$	-3	7175180144504176257395950783017399696745334254878857657299910
6	1	occurrence e_3	1	7175180148568351143312937756529351365235304358013215944716282
6	2	occurrence e_4	1	7175180147552307421833691013151363448112959009064812318199878
6	3	occurrence e_5	1	7175180146536263700354444269773375530991906294830430006115874

How to read this step. In Table 12, compare only rows carrying the same element to form one band. For example, e_1 occurs in blocks 1, 2, and 4; the maximum of their three ratios is the upper endpoint of $\mathcal{R}_1$, and the minimum is its lower endpoint. The controller rows are grouped in $\mathcal{R}_0$.

Taking minima and maxima within each element level gives Table 13. The bands are strictly ordered

$$\mathcal{R}_1 > \mathcal{R}_2 > \mathcal{R}_3 > \mathcal{R}_4 > \mathcal{R}_5 > \mathcal{R}_6 > \mathcal{R}_0.$$

The narrowest main gap is Δ_4 .

Purpose. The table certifies the within-band widths and separating main gaps used in the scale-separation lemma. **Verification rule.** Take minima and maxima over atoms with the same element label, subtract endpoints to obtain widths and the “gap below,” and use only the sign and stated lower bound later.

Table 13: Effective bands, widths, and exact gaps below the bands.

band	lower endpoint	upper endpoint	width	gap below
$\mathcal{R}_1$	71751801506004387229547759 36120363829879968940409256 887323878	71751801506004388200718894 86974548885544592536491609 034929479	9711711355085418505 5664623596082352147 605601	1016043624362133256 1918604705480016102 80402871141813
$\mathcal{R}_2$	71751801495843950482356208 10919552597607007679507211 777723307	71751801495843950985926426 79928503359331967330128854 016182065	5035702186900895076 1724959650621642238 458758	1016043671122224938 0370947644863729210 27091465775256
$\mathcal{R}_3$	71751801485683511433129377 56529351365235304358013215 944716282	71751801485683513771133958 72882457833120634758480120 311948051	2338004581163531064 6788533040046690436 7231769	1016043538035810508 7674437327814668313 79368792311995
$\mathcal{R}_4$	71751801475523074218336910 13151363448112959009064812 318199878	71751801475523076052771272 47761907632453837526633847 152404287	1834434362346105441 8434087851756903483 4204409	1016043505663439397 3065140045276168363 74161673142606
$\mathcal{R}_5$	71751801465362637003544442 69773375530991906294830430 006115874	71751801465362639161702516 15844849443585342172690650 645057272	2158158073460714739 1259343587786022063 8941398	1016043538035810585 1691128636899462863 55926164549607
$\mathcal{R}_6$	71751801455202200544107300 17935331471914270055285828 661319569	71751801455202201623186336 84604262667301960008474503 841566267	1079079036666689311 9538768995318867518 0246698	1016043563214321373 2370557436463047164 70164923487420
$\mathcal{R}_0$	71751801445041762573959507 83017399696745334254878857 657299910	71751801445041764911964086 44698275728267965338815663 737832149	2338004578616808760 – 3152263108393680608 0532239	

The total within-band width and minimum gap are

$$W = \frac{11222421986452480676091195490}{79725635487220872}, \quad \Delta_{\min} = \Delta_4 = \frac{10160435056634393973065140045}{27616836374161673142606}.$$

For example, the first gap is computed as

$$\Delta_1 = \rho_1 - \bar{\rho}_2.$$

The table’s “gap below” column performs this subtraction for every band. The important comparison is not the number of digits but that every gap is roughly six orders of magnitude longer than the sum of all within-band widths in this particular instance.

B.1.6 Cross-block couplings and logical-field correction

For every block pair $u < v$, direct exact evaluation uses the 16 atom-pair terms in (8.10) and the main-gap coefficient in (8.27). To avoid repeated quarters, Table 14 lists $4J_{uv}$, $4J_{uv}^{\text{main}}$, and their difference. These are exact integers.

Purpose. The table certifies every cross-block coupling and its split into main-gap signal and within-band residual. **Verification rule.** Evaluate the 16 atom-pair terms, multiply by four to avoid fractions, compute the main-gap sum from $d_{u,k}d_{v,k}\Delta_k$, and subtract the two columns.

Table 14: Exact compressed cross-block coefficients, scaled by four.

$u \ v$	$4J_{uv}$	$4J_{uv}^{\text{main}}$	$4(J_{uv} - J_{uv}^{\text{main}})$
1 2	3556153014386573426160237417661 6278035024822539117778	355615244822977200704395034183 74254326344771319303686	566156801419116287075824202370 8680051219814092
1 3	264171363556042424895355923731 90354673478623208816986	264171322874762736563637836294 39267520804300646026555	406812796883317180874375108715 2674322562790431
1 4	3048131096455761015761959627127 5247883125806571430953	304813065259458384388579752983 31669625222809894104353	4438611771718761620972943578257 902996677326600
1 5	1828878556224211316413789664327 2598523828821569358634	182887838141340740620677208885 39270675489526318112240	174810803910207017575473332784 8339295251246394
1 6	274331783757355413160777894786 08820936384401166606761	274331755593392555357969326915 36360537817896208124864	281639628578028085678707246039 8566504958481897
2 3	2336900536741607815788557443604 9353535201886057872113	233690017381135843529833118876 02567016676608507429348	362930249380490226254844678651 8525277550442765
2 4	264171362836656399941551921556 61250705627384692494548	2641713243854733862671005982371 3502289715748963195151	384511830136744513233194774841 5911635729299397
2 5	172727421429049968071288643783 26947436541046080912646	1727274030847063466476120688401 1653839115364644969634	1834434362142367657494315293597 425681435943012
2 6	243850478047912713353914221936 94028716143603217098066	243850450099765662324164609496 99660033690204069527657	279481470510297496124399436868 2453399147570409
3 4	203208742425445692225358005212 33764930250657255135288	203208709909197382446079913957 36301522937664712554878	3251624830977927809125497463407 312992542580410
3 5	152406550668333759041964806091 27666525762826615153450	152406532323990134944229811566 31761266403512315870420	183443436240977349945249590525 9359314299283030
3 6	203208735123677544967245632952 91371255942519401253571	2032087085783332396814167862584 8354343243056783566841	265453443052858288466944301691 2699462617686730
4 5	152406553941540169385636524012 59073180536669740444233	152406532323990134944229811566 31761266403512315870420	2161755003444140671244627311914 133157424573813

Continued on next page

Table 14 (continued)

$u \ v \ 4J_{uv}$	$4J_{uv}^{\text{main}}$	$4(J_{uv} - J_{uv}^{\text{main}})$
4 6 213369173309641144471492251789	213369143958691344769091223586	293509497997024010282036100779
90828970935777657213120	29821174622425575878836	6313352081334284
5 6 172727430385405973283429128005	1727274030847063466476120688401	273006996266358170591657928629
90940138111547715453003	1653839115364644969634	8996183070483369

How to read this step. The first two coefficient columns in Table 14 are close because they differ only on intervals lying inside bands. Their difference is the precise within-band residual for that block pair. Summing the absolute values of such differences is unnecessary; the uniform profile bound $|F_v| \leq 3$ yields the simpler global estimate used in Γ .

The field-correction arithmetic is summarized in Table 15. The columns are scaled so that

$$4e_v = 4a_v^0 - \eta_v D_v - 4a_v^* \quad (\text{B.3})$$

can be checked by integer subtraction. For block 1, for example,

$$\eta_1 = \text{round}\left(\frac{4a_1^0 - 4a_1^*}{D_1}\right) = \frac{12160865322039464013179252192182}{-70129329906859157558367453001536} = 73135,$$

and the remainder in (B.3) is

$$4e_1 = \frac{91993098399800992075465492602}{897143}.$$

The bound $|e_v| \leq D_v/8$ is verified in every row. Because the table displays $4e_v$, the convenient integer check is $|4e_v| \leq D_v/2$. For block 1, the displayed residual 91993098399800992075465492602897143 is smaller than $D_1/2$, as required.

Purpose. The table certifies the integer field correction and the residual error in each block. **Verification rule.** Check $4e_v = 4a_v^0 - \eta_v D_v - 4a_v^*$ by integer subtraction and verify $|4e_v| \leq D_v/2$.

Table 15: Exact logical-field correction. All displayed field columns are multiplied by four.

v	$4a_v^0$	$4a_v^*$	D_v	η_v	$4e_v$
1	-232277602169999044004 9663451156042800224093 5718792495585799850159 1171215910783929350760 75583630791887268	599465694275295438 507290975188407760 05094889033779729 742164	191004172827270 839572243054176 7925219218270129329906 92602897143	-121608653220394640131 7925219218270129329906 8591575583674530015367 3135	919930983998 009920754654 92602897143
2	-206468978841734728773 1865554410681094766054 1066093452746135549401 2734604590001168019885 89378943123704295	528342647907683547 352007103023859081 65460135964273320 846160	165536949783634 727629277313619 5473990760683176220511 26642419105	-124726823293289057041 9845305870995880671213 5473990760683176220511 4467	445001472244 464564652214 26642419105
3	-160013456587725036655 3771259612880364985318 2454880499211205269725 8218711497023502424470 40798952925932636	416577855198752871 507867647860710287 87207812961896975 054152	114602503696362 503743345832506 3053209901139119678537 828761470939	-139624747650957194248 9276711699727977310940 3053209901139119678537 8386	-22622512333 360885622529 828761470939
4	-196145528355563217097 0000779723572354534104 4315144813932488116709 6184154249642299817112 47984553556902256	457219597583448519 332809564549634308 91626322209974773 950156	140069726739998 615686311573062 3444187969097337811885 640297051854	-140034205049642231928 3444187969097337811885 7348725914766676146632 277130310353	-67239674239 640297051854 277130310353
5	-134204832528533012398 5813501584149115687795 8675776002961694190276 4505471100182360256730 74055215771459374	325133935408696481 591841880406434151 05518876960840054 10146	764016691309083 358288972216707 7100520313483974937642 87676810674	-175656937937551388015 5340717775192073871514 7100520313483974937642 6987	364836129403 523952449957 87676810674

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Table 15 (continued)

v	$4a_v^0$	$4a_v^*$	D_v	η_v	$4e_v$
6	-185822076736453798625	447059158678098902	114602503696362	-162144866598016402741	298915986900
	7857583723117565677020	005581395755480143	503743345832506	6841667157158991027305	642672396921
	0985970200493102699693	76933260853094498	054152	4411249072660304013938	62084116139
	6275517993528400205381			9225	
	31195353118101563				

Arithmetic guide. The quotient defining η_v is not expected to be integral. Rounding chooses the nearest integer multiplier of D_v ; equation (B.3) then records the exact leftover. The proof needs only the half-step bound, not a small decimal relative error.

B.1.7 The load-lock scale, anchor, and due date

The four terms in (8.21) evaluate to

$$\begin{aligned}
 1 &= 1, \\
 \sum_{i < j} p_i p_j \min\{r_i, r_j\} &= \frac{256549985336378567112187831773905469666007855288234150446}{938428348374058837795295352207450157651340141314012679426} \\
 &\quad 31440865121723230297964517549394530, \\
 \sum_i |\eta_{b(i)}| p_i^2 &= \frac{223514294183632542249314160825811265093651729277674819612}{326492630727111950644933978364173031968927116195290618081} \\
 &\quad 21836053790729275879854346002172010721524, \\
 \sum_{i < j} |\bar{\eta}_{ij}| p_i p_j &= \frac{562093418052650571895673292558307306876005080517367319774}{850385852645843472540618343347088295348930077885807518366} \\
 &\quad 628326198584559884180249290425125667473428.
 \end{aligned}$$

Therefore

$$\begin{aligned}
 U &= \frac{584444873126012359758461419859671610775}{917220045920330559498079809561389505010} \\
 &\quad 995520713040819290838554639350711576018 \\
 &\quad 104883816154281783333934391815227589483
 \end{aligned}$$

and

$$\begin{aligned}
 \max_v |\eta_v| &= \frac{17565693793755138801553407177751}{92073871514710052031348397493764} \\
 &\quad 26987.
 \end{aligned}$$

The load-lock scale is

$$\begin{aligned}
 H &= U + 2 \max_v |\eta_v| + 2 = \frac{584444873126012359758461419859671610775}{917220045920330559498079809561389505010} \\
 &\quad 995520713392133166713657415381779719573 \\
 &\quad 143298590457223793740204071313980443459.
 \end{aligned}$$

In particular,

$$\begin{aligned}
 H - U &= \frac{35131387587510277603106814355503}{84147743029420104062696794987528} > 0. \\
 &\quad 53976
 \end{aligned}$$

This positive difference is the numerical certificate for load-lock dominance. A wrong integer load vector pays at least H , while every non-dominant term together can improve by less than U . Every state job in block v receives the common tardiness ratio $s_i = HL_v + \eta_v$. All six values are positive and decrease strictly with v , as visible in the full job table below.

The anchor data are

$$\begin{aligned}
 p_a &= H, \\
 r_a &= 1 + \max_i r_i = \frac{70316765459424392105577557335952}{68355444518632781564950501995}, \\
 C^* &= \frac{267886806177807976135173003247823899187}{721758959732956168790713347663219908907} \\
 &\quad \frac{798227605566471005179125919095260487893}{414629014523117639278783931399668030172} \\
 &\quad \frac{330838659432497804801216895928228214103}{907831843093389083896975601641119383754} \\
 &\quad \frac{333813593128421093636747252}{s_a = 1 + \lceil C^*/H^2 \rceil = 2.}
 \end{aligned}$$

The computed ratio satisfies $0 < C^*/H^2 \leq 1$, so the ceiling is 1 and $s_a = 2$. Its tardy self-cost is

$$\begin{aligned}
 p_a^2 s_a &= \frac{683151619446561368413615410444933623585}{099609900914566805231553229996594117852} \\
 &\quad \frac{535253587426979235841601201985188673749}{202537372133835586671076633705422428293} \\
 &\quad \frac{801024314335090374585925621334531619631}{706967943361532152450437040923960956249} \\
 &\quad \frac{613431193297658282439578766783389222785}{684354764168571996115924105420591769362},
 \end{aligned}$$

which exceeds C^* by

$$\begin{aligned}
 &\frac{683151619446561368413615410444933623585}{099609900914298918425375422020458944849} \\
 &\frac{287429688239257476881868245816397960401}{539317463226037359065510162700243302374} \\
 &\frac{705763826441675745571402503695252835700}{307299913189201313791004543119159739353} \\
 &\frac{685202979193750450596485377699492247184}{043235380414238182522795684326955022110}.
 \end{aligned}$$

Thus no optimal canonical assignment can place the anchor tardy. The common due date is the total processing time of all 49 jobs:

$$\begin{aligned}
 d &= \frac{584444873126012359758461419859671610775}{917220045920330559498079809561389505010} \\
 &\quad \frac{995520713392133166713657415381779805899}{466147883766136104722754835310286860893}.
 \end{aligned}$$

B.1.8 The constructed 49-job instance

The decision instance consists of the 49 triples (p_j, w_j^E, w_j^L) , the due date d , and the threshold B . Tables 16 and 17 lists every job. The first table gives the intermediate ratios; the second

gives the actual integer weights. The equalities $w_j^E = p_j r_j$ and $w_j^L = p_j s_j$ hold row by row.

How to read this step. The ratios are construction aids; the actual scheduling input contains only processing times and weights. To audit one row, multiply the displayed p_j by r_j and s_j . Jobs in the same block share a tardiness ratio, while their early ratios differ according to their processing values. The anchor is recognizable by $p_a = H$ and $s_a = 2$.

Purpose. The table records the ratio-level construction data for all 49 jobs of the yes-instance. **Verification rule.** Compute $r_j = f_v + \lambda_v p_j$ and $s_j = HL_v + \eta_v$ within each block; the anchor is checked from $p_a = H$, $r_a = 1 + \max r_j$, and $s_a = 2$.

Table 16: All jobs: metadata, processing times, and early/tardiness ratios.

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{1,0}$	1	0	—	-1	17984650426474121466202803	703167654289430804612001531245713177680	93268361905489682802055503317404676472373959319
					40569649349251262	620511373585859565336	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$B_{1,0}$	1	0	—	1	17984650426474121466202803	703167654289430804612001531245713177680	93268361905489682802055503317404676472373959319
					40569649349251259	6204874359059706729315	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$A_{1,1}$	1	1	e_1	1	17984650808482467120744482	703167654594243921055775573359526835544	93268361905489682802055503317404676472373959319
					55018260184595356	4518632781564950501994	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$B_{1,1}$	1	1	e_1	-1	17984650808482467120744482	703167654594243921055775573359526835544	93268361905489682802055503317404676472373959319
					55018260184595355	4518552989298652889987	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$A_{1,2}$	1	2	e_2	1	17984650744814409511654202	703167654543441734981813233007224559233	93268361905489682802055503317404676472373959319
					69276825045372255	8206029600187441128287	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$B_{1,2}$	1	2	e_2	-1	17984650744814409511654202	703167654543441734981813233007224559233	93268361905489682802055503317404676472373959319
					69276825045372254	8205949807921143516280	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$A_{1,3}$	1	3	e_3	1	17984650681146351902563922	703167654492639548907850892654922282923	93268361905489682802055503317404676472373959319
					83535389906157254	2539743775820589011280	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$B_{1,3}$	1	3	e_3	-1	17984650681146351902563922	703167654492639548907850892654922282923	93268361905489682802055503317404676472373959319
					83535389906157253	2539663983554291399273	65631670907676270113881635896637721561726085767
							89370860560342566200591849400908225745875287729
							05120802618877369924761012351291
$A_{2,0}$	2	0	—	-1	17984650426474121466202803	703167654289430802993382973762624962995	93268361905489681633165757065379956955451119600
					40569649349251262	2842476365960279561826	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
							90075686207245191695671000023041

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Table 16 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$B_{2,0}$	2	0	–	1	17984650426474121466202803	703167654289430802993382973762624962995	93268361905489681633165757065379956955451119600
					40569649349251259	2842236989161386725808	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$A_{2,1}$	2	1	e_1	1	17984650808482467120744482	703167654594243919437157012056355164313	93268361905489681633165757065379956955451119600
					55018260184595356	6988080963055795154390	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$B_{2,1}$	2	1	e_1	–1	17984650808482467120744482	703167654594243919437157012056355164313	93268361905489681633165757065379956955451119600
					55018260184595355	6988001170789497542384	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$A_{2,2}$	2	2	e_2	1	17984650744814409511654202	703167654543441733363194672340733464093	93268361905489681633165757065379956955451119600
					69276825045372255	9703463523113425003784	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$B_{2,2}$	2	2	e_2	–1	17984650744814409511654202	703167654543441733363194672340733464093	93268361905489681633165757065379956955451119600
					69276825045372254	9703383730847127391778	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$A_{2,3}$	2	3	e_5	1	17984650553810236684383363	703167654391035175141307653193868363434	93268361905489681633165757065379956955451119600
					12052519627709252	8352302480961270189766	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$B_{2,3}$	2	3	e_5	–1	17984650553810236684383363	703167654391035175141307653193868363434	93268361905489681633165757065379956955451119600
					12052519627709251	8352222688694972577760	31309515724232260929815523997021759649448184765
							69460446292468721165589756924170711267759696005
$A_{3,0}$	3	0	–	–1	17984650426474121466202803	703167654289430799935992365607912052650	93268361905489680464276010813355237438528279880
					40569649349251262	2509485161553545702072	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
$B_{3,0}$	3	0	–	1	17984650426474121466202803	703167654289430799935992365607912052650	93268361905489680464276010813355237438528279880
					40569649349251259	2509245784754652866057	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
$A_{3,1}$	3	1	e_2	1	17984650744814409511654202	703167654543441730305804061002617673294	93268361905489680464276010813355237438528279880
					69276825045371355	4158730571863144218537	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
$B_{3,1}$	3	1	e_2	–1	17984650744814409511654202	703167654543441730305804061002617673294	93268361905489680464276010813355237438528279880
					69276825045371354	4158650779596846606532	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
							45188562827573378993608458872204

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Table 16 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{3,2}$	3	2	e_4	1	17984650617478294293473642	703167654441837358157879382844735425036	93268361905489680464276010813355237438528279880
					97793954766924253	7573654135180831559027	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
							45188562827573378993608458872204
$B_{3,2}$	3	2	e_4	-1	17984650617478294293473642	703167654441837358157879382844735425036	93268361905489680464276010813355237438528279880
					97793954766924252	7573574342914533947022	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
							45188562827573378993608458872204
$A_{3,3}$	3	3	e_6	1	17984650490142179075293083	703167654340232986009954704686853176779	93268361905489680464276010813355237438528279880
					26311084488485251	1634895055509176140017	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
							45188562827573378993608458872204
$B_{3,3}$	3	3	e_6	-1	17984650490142179075293083	703167654340232986009954704686853176779	93268361905489680464276010813355237438528279880
					26311084488485250	1634815263242878528012	96987360540788251745749412097405797737170283763
							49550032024477078587739927244465685470468097937
							45188562827573378993608458872204
$A_{4,0}$	4	0	—	-1	17984650426474121466202803	703167654289430795439829711875019055372	93268361905489679295386264561330517921605440161
					40569649349251262	7430026054119511778082	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$B_{4,0}$	4	0	—	1	17984650426474121466202803	703167654289430795439829711875019055372	93268361905489679295386264561330517921605440161
					40569649349251259	7429786677320618942070	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$A_{4,1}$	4	1	e_1	1	17984650808482467120744482	703167654594243911883603742528582343600	93268361905489679295386264561330517921605440161
					55018260184595356	3239801753993356682458	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$B_{4,1}$	4	1	e_1	-1	17984650808482467120744482	703167654594243911883603742528582343600	93268361905489679295386264561330517921605440161
					55018260184595355	3239721961727059070454	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$A_{4,2}$	4	2	e_3	1	17984650681146351902563922	703167654492639539735679065644061247524	93268361905489679295386264561330517921605440161
					83535389906148254	4710696800181322470050	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$B_{4,2}$	4	2	e_3	-1	17984650681146351902563922	703167654492639539735679065644061247524	93268361905489679295386264561330517921605440161
					83535389906148253	4710617007915024858046	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001
$A_{4,3}$	4	3	e_6	1	17984650490142179075293083	703167654340232981513792050317279603410	93268361905489679295386264561330517921605440161
					26311084488485251	7527450206640002982038	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
							14179572284932834575415817036001

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Table 16 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$B_{4,3}$	4	3	e_6	-1	17984650490142179075293083	703167654340232981513792050317279603410	93268361905489679295386264561330517921605440161
					26311084488485250	7527370414373705370034	62665205357344242561683300197789835824892382761
							29639617756630320679479928560035923602735862605
$A_{5,0}$	5	0	—	-1	17984650426474121466202803	703167654289430789504895010017223666799	14179572284932834575415817036001
					40569649349251262	1492156077917620893852	93268361905489678126496518309305798404682600442
							28343050173900233377617188298173873912614481759
$B_{5,0}$	5	0	—	1	17984650426474121466202803	703167654289430789504895010017223666799	09729203488431430016327688691528432681466901897
					40569649349251259	1491916701118728057843	27302227179508439259799946049767
							93268361905489678126496518309305798404682600442
$A_{5,1}$	5	1	e_4	1	17984650617478294293473642	703167654441837347726782023433963582640	28343050173900233377617188298173873912614481759
					97793954766923353	2316597563266220602125	09729203488431430016327688691528432681466901897
							27302227179508439259799946049767
$B_{5,1}$	5	1	e_4	-1	17984650617478294293473642	703167654441837347726782023433963582640	93268361905489678126496518309305798404682600442
					97793954766923352	2316517770999922990122	28343050173900233377617188298173873912614481759
							09729203488431430016327688691528432681466901897
$A_{5,2}$	5	2	e_5	1	17984650553810236684383363	703167654391035161652819685628383610693	27302227179508439259799946049767
					12052519627700252	2115937347629268120822	93268361905489678126496518309305798404682600442
							28343050173900233377617188298173873912614481759
$B_{5,2}$	5	2	e_5	-1	17984650553810236684383363	703167654391035161652819685628383610693	09729203488431430016327688691528432681466901897
					12052519627700251	2115857555362970508819	27302227179508439259799946049767
							93268361905489678126496518309305798404682600442
$A_{5,3}$	5	3	e_6	1	17984650490142179075293083	703167654340232975578857347822803638746	28343050173900233377617188298173873912614481759
					26311084488485251	2561594489002972863819	09729203488431430016327688691528432681466901897
							27302227179508439259799946049767
$B_{5,3}$	5	3	e_6	-1	17984650490142179075293083	703167654340232975578857347822803638746	93268361905489678126496518309305798404682600442
					26311084488485250	2561514696736675251816	28343050173900233377617188298173873912614481759
							09729203488431430016327688691528432681466901897
$A_{6,0}$	6	0	—	-1	17984650426474121466202803	703167654289430782131188266401331647838	27302227179508439259799946049767
					40569649349251262	4975732647299265289392	93268361905489676957606772057281078887759760722
							94020894990456224193551076398557912000336580756
$B_{6,0}$	6	0	—	1	17984650426474121466202803	703167654289430782131188266401331647838	89818789220723887395449890236631337340622069008
					40569649349251259	4975493270500372453386	30808706484740439353881222200611
							93268361905489676957606772057281078887759760722

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Table 16 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{6,1}$	6	1	e_3	1	17984650681146351902563922	703167654492639526427037615076929231262	93268361905489676957606772057281078887759760722
					83535389906147354	9960704422212111385576	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
$B_{6,1}$	6	1	e_3	-1	17984650681146351902563922	703167654492639526427037615076929231262	93268361905489676957606772057281078887759760722
					83535389906147353	9960624629945813773574	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
$A_{6,2}$	6	2	e_4	1	17984650617478294293473642	703167654441837340353075277908029835406	93268361905489676957606772057281078887759760722
					97793954766924253	8788029948010298127374	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
$B_{6,2}$	6	2	e_4	-1	17984650617478294293473642	703167654441837340353075277908029835406	93268361905489676957606772057281078887759760722
					97793954766924252	8787950155744000515372	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
$A_{6,3}$	6	3	e_5	1	17984650553810236684383363	703167654391035154279112940739130439550	93268361905489676957606772057281078887759760722
					12052519627709252	8261672830819142085372	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
$B_{6,3}$	6	3	e_5	-1	17984650553810236684383363	703167654391035154279112940739130439550	93268361905489676957606772057281078887759760722
					12052519627709251	8261593038552844473370	94020894990456224193551076398557912000336580756
							89818789220723887395449890236631337340622069008
							30808706484740439353881222200611
a	-	-	-	-	58444487312601235975846141	703167654594243921055775573359526835544	2
					98596716107759172200459203	4518632781564950501995	
					30559498079809561389505010		
					99552071339213316671365741		
					53817797195731432985904572		
					23793740204071313980443459		

Purpose. The table records the physical positive-integer input triples for the 49-job yes-instance. **Verification rule.** Multiply each displayed p_j by its ratio in Table 16 to recover w_j^E and w_j^L ; no visual comparison of full digit strings is required.

Table 17: The 49 actual AWET job triples. The processing-time column is repeated for completeness.

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$A_{1,0}$	17984650426474121466202803405696493 49251262	12646224453599219326700847020084580656881203566	16773988847201077280415568447091583739686005811
		97702825281084050064254977600534998478018463881	30427864201215420861304644721859353565318578303
		0249454032	63261530701460745989355193681531979728334732921
			84732310090776362623007575902184517656382779933
$B_{1,0}$	17984650426474121466202803405696493 49251259	12646224453599219326700847020084580656881182041	16773988847201077280415568447091583739685977830
		43659474028136691141661568117816518088416819181	79570699510731359196205122581565182346538988614
		9035957585	13134300413357329540278294549885127945761696110
			58915499813790344867525548655412141397750908397
$A_{1,1}$	17984650808482467120744482550182601 84595356	12646224727697089120677578060354606619030675665	16773989203494003614667471991721827370517301345
		89256686619235452860299136572780947026288868547	26567869853120773826218599295692316771925091278
		7961139864	72212092091791937430192924315246415225397215089
			50822692889447927541729991476298112330069140157
$B_{1,1}$	17984650808482467120744482550182601 84595355	12646224727697089120677578060354606619030668490	16773989203494003614667471991721827370517292018
		71241626575353445671350439117892392567035219386	42948814956292753271185425248927593032331894715
		8926210385	55503015329090798613833957938030797964539536152
			42217089463785921623235982394040653577191849645
$A_{1,2}$	17984650744814409511654202692768250 45372255	12646224682014110805509073672185811119202619787	16773989144111849225625488067616786765387283035
		68403343121070718465830129758200200499679367544	77440081209832682669076134974187470052067795559
		5125477185	70622544144045042886387882236178597240377135849
			61066229095184080171129896118768119182883135169
$B_{1,2}$	17984650744814409511654202692768250 45372254	12646224682014110805509073672185811119202612612	16773989144111849225625488067616786765387273708
		50388384681560859201562136907864267302946354659	93821026313004662114042960927422746312474598996
		1109295120	53913467381343904070028915858962979979519456912
			52460625669522074252635887036510660430005844657
$A_{1,3}$	17984650681146351902563922835353899 06157254	12646224636331132496809522303266532088556565572	16773989084729694836583504143511746160332812099
		61015037018981540904061266830774126704357672304	42656956873511087280643449446944914037102661494
		4059825120	12554774075522560850210495603610592202556447006
			76697513163168172602003367046654024341750278032
			1002664892723895704935914914

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{1,3}$	17984650681146351902563922835353899 06157253	12646224636331132496809522303266532088556558397 43000180183843829564473978584990814640881824292 3747877069	16773989084729694836583504143511746160332802772 59037901976683066725610275400180190297509464930 95845697312821422033851529226394974941698768069 68091909737506166683509357964396565588872987520 0200046015353970943923563623
$A_{2,0}$	17984650426474121466202803405696493 49251262	12646224453599219297590558089947430905148042864 87127554870142291438486786885432874760975609791 4537524412	16773988847201077070194833712764543784236109306 53589102710720962315065376734522711021760084178 03245743330527184574410135219686266931674428637 19871873949050749266351023112076983632811010727 9041911951081977458098327742
$B_{2,0}$	17984650426474121466202803405696493 49251259	12646224453599219297590558089947430905148021339 33084203617200327911078620891002436464267346956 6331792272	16773988847201077070194833712764543784236081326 02731938020271967342353415335814047488172074785 17774016362699290108690224566896783486131457799 06483099887415781573643297990738950841930833705 1983290215506890445098258619
$A_{2,1}$	17984650808482467120744482550182601 84595356	12646224727697089091567288443188792452212284420 19165562314551012139237266489821876304107651578 7697012840	16773989203494003404446732792138405232137672357 28288022169183825479657129362687992636212153260 59077831081552562085992258405780645242500539583 21125470693375644946666219973785348532343530906 8091255054507636789261597596
$B_{2,1}$	17984650808482467120744482550182601 84595355	12646224727697089091567288443188792452212277245 01150502270670803415357957031275760042734686094 8002026320	16773989203494003404446732792138405232137663030 44668967272367493822086475563118438125016150129 63920588758943263930752288188184150760652882637 16662546006163989049096978266672670935383471899 2405047809315941118261574555
$A_{2,2}$	17984650744814409511654202692768250 45372255	12646224682014110776398784169524774557579511618 30424943829964783220111966837967232052846921899 5063612920	16773989144111849015404749612242761657495942794 99400414547444704410183179264279638607747672190 53596794203019711826887996903025201587353074407 69671651805582756643280101935896396401888749799 7567258139274333975422127455
$B_{2,2}$	17984650744814409511654202692768250 45372254	12646224682014110776398784169524774557579504443 12409985390456722420912088622821555248397444404 0108927612	16773989144111849015404749612242761657495933468 15781359650628372752612525464710084096551669059 58439551880410413671648026685428707105505417461 65208727118371100745710860228783718804928690792 1881050894082638304422104414
$A_{2,3}$	17984650553810236684383363120525196 27709252	12646224544965175869706989464029819202035105905 42004772848036261228454180679123439633403147234 1913915032	16773988965965385848278800072555830933629513176 12783441625116783896880266257247997057173953975 27780316005999539061387583252674106262149438962 31735724575636246419344502632098400858896154160 9100912101257152834048875332

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{2,3}$	17984650553810236684383363120525196 27709251	12646224544965175869706989464029819202035098730 23990119221644644203292596194179081099185878169 1268857760	16773988965965385848278800072555830933629503849 29164386728300452239309612457678442545977950844 32623073683390240906147613035077611780301782016 27272799888424590521775260924985723261936095153 3414704856065457163048852291
$A_{3,0}$	17984650426474121466202803405696493 49251262	12646224453599219242604456785099799770310546667 12049380552042200908158865524147643426296932975 1322014864	16773988847201076859974098978437503828786212801 76750341220226503768826108747186068478201590052 43229955959572437683172935152998396558705079550 27707769278449759326325727690052204812485498735 2531789779496331388343721448
$B_{3,0}$	17984650426474121466202803405696493 49251259	12646224453599219242604456785099799770310525141 58006029299109949092087957892092226313588202357 1365615763	16773988847201076859974098978437503828786184821 25893176529812575488501708090062912629805160956 22413732312020065200810012979066281450192174685 26747034966092127128508393719488090769547375178 6843307059359350562967104836
$A_{3,1}$	17984650744814409511654202692768250 45371355	12646224682014110721412681834134147570295808288 76945374039094077907254731399668014135010523278 0439807635	16773989144111848805184011156868736549596208401 64211340760878242053970026417425052644498819958 91703949996261450581844320748183165159188504106 22769661225547527764603706968477178512900876840 2860650798725990793667316420
$B_{3,1}$	17984650744814409511654202692768250 45371354	12646224682014110721412681834134147570295801113 58930415599589254345173639444408253434242572556 8662084328	16773989144111848805184011156868736549596199074 80592285864073599293861892865050667361700010260 18098542113743993087723346690205793456350869151 22449416454761650365331262311622473831921502321 4297823225346997185208444216
$A_{3,2}$	17984650617478294293473642977939547 66924253	12646224590648154110482532756771859529371352922 5171685557419442219021243624960330990255498608 3607381831	16773989025347540027100046285496243461280932618 89767080436319455993038957529854575451426758254 28091804111912093924310705530547274212751344202 84237814085660964188882758347627006684645775632 6685453015068949662275163612
$B_{3,2}$	17984650617478294293473642977939547 66924252	12646224590648154110482532756771859529371345747 33702100326658914506296467434197085606502835223 6454977544	16773989025347540027100046285496243461280923292 06148025539514813232930823977480190168627948555 54486396229394636430189731472569902509913709247 83917569314875086789610313690772302003666401113 8122625441689956053816291408
$A_{3,3}$	17984650490142179075293083263110844 88485251	12646224499282197525428195756407636716421097776 86205250964024478951005316064866596798094188798 7415389267	16773988906583231249016081414123750373041204209 29667484229367026807989662874804888928713256311 6828350661896843964666959929622176251157319825 39688610311381334719963530249884750789324277160 9413599601260137047747863204

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{3,3}$	17984650490142179075293083263110844 88485250	12646224499282197525428195756407636716421090601 68190698942008247087636855638599866602093341348 3573823000	16773988906583231249016081414123750373041194882 46048429332562384047881529322430503645914446612 94678098736450982152545985871644804548319684870 39368365540595457320691085593030046108344902642 0850772027881143439288991000
$A_{4,0}$	17984650426474121466202803405696493 49251262	12646224453599219161742543197145508006937069848 44569673162900640116541286924575844936687242053 7002439484	16773988847201076649753364244110463873336316296 99911579729732045222586840759849425934643095926 83214168588643747793283018248174741373861834494 76900650428831211599166512819371015204478966471 6681597980289208233448683262
$B_{4,0}$	17984650426474121466202803405696493 49251259	12646224453599219161742543197145508006937048322 90526321909982416327944372195156246431918414560 7195566130	16773988847201076649753364244110463873336288316 4905441503935318363465000084431177771438247127 27053448261366897294277084553099994602378995602 88367951519210827201309711741662933128600812217 7964743181785481985997575259
$A_{4,1}$	17984650808482467120744482550182601 84595356	12646224727697088955719270527460750116389911770 95504351002964072273090705220599386405163978151 3313465048	16773989203494002984005254392971560955378414381 31728326801309928786534189496679344364786277224 32809309061057497446007405179127485986815476429 04512846831357192032328335810151497181864887478 0698673985323972051469411356
$B_{4,1}$	17984650808482467120744482550182601 84595355	12646224727697088955719270527460750116389904595 77489290959091776795497093184216398408269937225 6426141170	16773989203494002984005254392971560955378405054 48109271904516974923888576191500128310384660957 80755735618631880613005427280769237062987863465 08335280528150397233042735450915469823238836060 1126389052489396635652375355
$A_{4,2}$	17984650681146351902563922835353899 06148254	12646224636331132331851216007024271176285942779 93619309945057338805996428400352823728775983909 6774792700	16773989084729694205921291010017861928039716092 97764428530369090617208371530954158111973933811 58065181397792824364937403076409237659625910292 76327065810696498690692328663239428472546698230 8536446231244470544761292254
$B_{4,2}$	17984650681146351902563922835353899 06148253	12646224636331132331851216007024271176285935604 75604453109929339177756585406161987437702933746 3955893638	16773989084729694205921291010017861928039706766 14145373633576136754562758225774942057572317545 06011607955367207531935425178050988735798297328 80149499507489703891406728304003401113920646812 8964161298409895128944256253
$A_{4,3}$	17984650490142179075293083263110844 88485251	12646224499282197444566281870740924460484297829 51494847816803320668653847515771344411659603251 3580921538	16773988906583231038795345935587313387103018957 32588541590054882602161429203369194905119857574 55785824458864587208463322362936132450080735395 92430613811685673213081066098480618897862761732 8272840784627149285024521251

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{4,3}$	17984650490142179075293083263110844 88485250	12646224499282197444566281870740924460484290654	16773988906583231038795345935587313387103009630
		33480295794791764814444658084868735557235871796	48969486693261928739515815898189978850718241308
		3400998500	03732251016438970375461344464577883526253122431
			96253047508478878413795465739244591539236710314
$A_{5,0}$	17984650426474121466202803405696493 49251262	12646224453599219055004817280282645237743434971	8700555851792573869207485250
		48637747284659178241999014382769209060794132716	16773988847201076439532629509783423917886419792
		3379041224	23072818239237586676347572772512783391084601801
			23198381217651728058388618816417428028778871552
$B_{5,0}$	17984650426474121466202803405696493 49251259	12646224453599219055004817280282645237743413445	44598236166896435800018738585187028377523443590
		94594396031759298797020467263159317421488468978	7641862283560747196039555954
		7292574337	16773988847201076439532629509783423917886391811
			72215653548893791780798293598560642913071333298
$A_{5,1}$	17984650617478294293473642977939547 66923353	12646224590648153922882891190868716615887963591	31693164210650399542739673600200049594326098633
		29119937714251945198116971397224650701482408806	68493583223995945969357992732206584370466525400
		9383925125	0960323758242967796201406653
			16773989025347539606658572351585781367443012973
$B_{5,1}$	17984650617478294293473642977939547 66923352	12646224590648153922882891190868716615887956416	29119937714251945198116971397224650701482408806
		11105182483502208275763955544189671323476265371	57499064136680942026216240719391962154273575213
		4227128944	13513230736891847482353354349245926474691074529
			78865501901141045289185789473575920515532182607
$A_{5,2}$	17984650553810236684383363120525196 27700252	12646224544965175627121246868090510011678964258	3678576257408442424412508751
		66106588563673969752381958584811719374400465263	16773989025347539606658572351585781367443003646
		3535847144	73880009239899677061033147661407915328269152378
			83011491734558071310470372610506800329873483556
$B_{5,2}$	17984650553810236684383363120525196 27700251	12646224544965175627121246868090510011678957083	86830617586840882012298874189249105846513209877
		48091934937296380754704553891720634036526364955	1451396748969182624466458984
		2284013569	16773988965965385217616591404318328884266149652
			89331999579588731219747833290273086199987538089
$A_{5,3}$	17984650490142179075293083263110844 88485251	12646224499282197337828555564561819552362241811	96357727066001130077616138373197657576534884371
		98579824785089504592669076902507504858819620892	3186223607934458077970337674516605544494593555
		4813033569	4791828109011318098150441284
			16773988965965385217616591404318328884266140326

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{5,1}$	17984650490142179075293083263110844 88485250	12646224499282197337828555564561819552362234636 80565272763084063519667283369360313431814092309 6151714000	16773988906583230828574610457050876401164824378 51890544053961473431150102473949454055522036003 12786403296363628753148024445332035556047983894 75387031774570495777347851779628194958117146577 3831918810179313534945436750
$A_{6,0}$	17984650426474121466202803405696493 49251262	12646224453599218922391279149015987405940085629 64380316462463892338619639658598411376998615740 0951212704	16773988847201076229311894775456383962436523287 46234056748743128130108304785176140847526107675 63182593846748075551274601958984307466368018031 33809961341190547077224827102594326367795236176 8097810410577307204808921082
$B_{6,0}$	17984650426474121466202803405696493 49251259	12646224453599218922391279149015987405940064104 10336965209586673553384734438689387776722152082 7979312974	16773988847201076229311894775456383962436495306 95376892058434399926946586352809508054704419469 36332880160022269018982645221624297368945311085 70133339624568683580517728162474107705724986934 1978356189259245561142319249
$A_{6,1}$	17984650681146351902563922835353899 06147354	12646224636331132092499948477949527526974189281 16203350670331324614118940092741528017622370785 6966165904	16773989084729693785479815587688605773225218953 25016050177112628758780403798646253874561032478 54618589801712931729422685419820152470905724117 86281700418659501860667732074422055974318167196 7360549274615050048714833294
$B_{6,1}$	17984650681146351902563922835353899 06147353	12646224636331132092499948477949527526974182105 98188493835216993320343287157605004487186569464 1021449622	16773989084729693785479815587688605773225209626 41396995280343052691059830987857376276953803076 45668685239470996218658699840700149105098155135 98389493179785547361765365761048649753628084115 8654064534175696167492632683
$A_{6,2}$	17984650617478294293473642977939547 66924253	12646224590648153790269351616841868401079688785 64464116097174977451760615431133473742492582443 9603801622	16773989025347539396437835384630550320536644379 77089166641569388039980510592623870611667810004 33571597200036258609280368503254294494349898888 59778839891078311152788746761745659048944035004 4923795243689892824607318583
$B_{6,2}$	17984650617478294293473642977939547 66924252	12646224590648153790269351616841868401079681610 46449360866432794082659300294788662446591675962 7285601744	16773989025347539396437835384630550320536635052 93470111744799811972259937781834993014060580602 24621692637794323098516382924134291128542329906 71886632652204356653886380448372252828253951923 6217310503250538943385117972
$A_{6,3}$	17984650553810236684383363120525196 27709252	12646224544965175494507707774983725339005033981 80157095836053327318370527486369821505345187775 8578261744	16773988965965385007395855181572494867923617179 43506946939592295857820384776510027967332744455 06751558758037222677421242458715699559102826878 60154614242528561554076899773659649713242857864 5013438771531173500324752972

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Table 17 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{6,3}$	17984650553810236684383363120525196	12646224544965175494507707774983725339005026806	16773988965965385007395855181572494867923607852
	27709251	62142442209683291873943550148816722314715704732	59887892042822719790099811965721150369725515052
		8572145870	97801654195795287166657256879595696193295257896
			72262407003654607055174533460286243492552774783
a			6306954031091819619102552361
	58444487312601235975846141985967161	41096273067564857043370818290520427918012556640	11688897462520247195169228397193432215518344400
	07759172200459203305594980798095613	04894352468585871745373565093807405946198875657	91840661118996159619122779010021991041426784266
	89505010995520713392133166713657415	87396827810054656640124590248570663723634992646	33342731483076355943914628659718091444758748040
	38177971957314329859045722379374020	79421120835429148137023477008457916018721182774	8142627960886918
	4071313980443459	35471069330460272774764200705	

B.1.9 The master identity and decision threshold

For this instance,

$$\sum_{v=1}^6 D_v = \frac{80221752587453752620342082754}{2356930}$$

and

$$\Gamma = \frac{9}{8}(6^2)W + \frac{1}{8} \sum_v D_v = \frac{18180323618093129571561463571 \cdot 4017263362700959729}{4}.$$

The exact constant coefficient and shifted constant are

$$C_0 = \frac{267886806177807976135173003247823899187 \cdot 721758959732956168790713347663219908907 \cdot 798227605566471005179125919095260487893 \cdot 414629014523117639278783931399668030172 \cdot 330838659432497804801216895928228214103 \cdot 907831843093234899276852755333078435488}{1}, \quad A = \frac{535773612355615952270346006495647798375 \cdot 443517919465912337581426695326439817815 \cdot 596455211132942010358251838190520975786 \cdot 829258029046235278557567862799336060344 \cdot 661677318864995609602433791856456428207 \cdot 815663686186408327918875608758123919221}{2}.$$

The gap-dominance comparison is

$$\frac{\Delta_{\min}}{2} = \frac{50802175283171969865325700226}{3808418187080836571303} > \frac{18180323618093129571561463571 \cdot 4017263362700959731/2}{4017263362700959731/2} = 2\Gamma + 1.$$

The left side is the least possible main penalty for one nonzero integer prefix error. The right side allows the residual to favor the yes-state by Γ , favor the no-state by another Γ , and still leaves one full unit for the integer floor. The positive margin is

$$\frac{\Delta_{\min}}{2} - (2\Gamma + 1) = \frac{10158617024272584660107983898}{91902819110798972182875/2}.$$

Finally,

$$B = \lfloor A + \Gamma \rfloor = \frac{267886806177807976135173003247823899187 \cdot 721758959732956168790713347663219908907 \cdot 798227605566471005179125919095260487893 \cdot 414629014523117639278783931399668030172 \cdot 330838659432497804801216895928228214103 \cdot 907831843093204164004888613424294783539}{481975740781438304901918017}.$$

B.1.10 The “if” case: constructing and evaluating the schedule

Select S_1 and S_5 , so $x = (1, 0, 0, 0, 1, 0)$ and $\sigma = (+, -, -, -, +, -)$. In a block with $\sigma_v = +1$, the early jobs are

$$B_{v0}, A_{v1}, A_{v2}, A_{v3},$$

and the tardy jobs are

$$A_{v0}, B_{v1}, B_{v2}, B_{v3}.$$

For $\sigma_v = -1$, the two lists are interchanged. Thus the exact-cover assignment contains 24 early state jobs, 24 tardy state jobs, and the early anchor. Every block has tardy load β_v ; for example, in selected block 1,

$$(P_{10} + 3) + P_{11} + P_{12} + P_{13} = \beta_1,$$

whereas in unselected block 2,

$$P_{20} + (P_{21} + 1) + (P_{22} + 1) + (P_{23} + 1) = \beta_2.$$

Because $C_k(x) = k$ for all k , every squared term in $\mathcal{P}(x)$ is zero, so the prefix penalty is exactly zero. No cancellation among different prefixes is involved. The residual is

$$R(\sigma) = \frac{-2343399975092575105438479690 + 358125499641790995}{2}.$$

Consequently, the master identity gives the integer cost

$$\Phi_{\text{cover}} = A + R(\sigma) = \begin{array}{r} 267886806177807976135173003247823899187 \\ 721758959732956168790713347663219908907 \\ 798227605566471005179125919095260487893 \\ 414629014523117639278783931399668030172 \\ 330838659432497804801216895928228214103 \\ 907831843093204163958266104391515672057 \\ 859076967098059714405782588. \end{array}$$

This satisfies

$$B - \Phi_{\text{cover}} = \frac{46622509032779111481622898773}{683378590496135429} > 0.$$

Hence the constructed AWET instance is a yes-instance.

For completeness, [Table 18](#) gives the entire compact canonical schedule. The early jobs

are listed in increasing r_j , which is chronological order before d : reading in the opposite direction, outward from d , gives the nonincreasing order of [lemma I.3](#). The anchor is last and completes at d . The tardy jobs are listed chronologically in decreasing s_j . The early block starts at

43163161424646654456155491275

381998153208717,

which is precisely the total tardy processing load. The final completion time exceeds d by

43163161424646654456155491275

381998153208717.

The equality of the initial start offset and final tardiness span is forced by $d = \sum_j p_j$: both equal the total processing load assigned to the tardy side.

Purpose. The table certifies feasibility and the job-by-job objective value of the canonical schedule induced by the exact cover.

Verification rule. Accumulate each completion time from the previous row, subtract from or add to d according to the side, and multiply the deviation by the applicable physical weight.

Table 18: Canonical schedule induced by the exact cover. Deviation means earliness on the early side and tardiness on the tardy side.

pos.	job	side	start	completion	deviation	cost contribution	
1	A _{6,0}	early	4316316142464665445615549	4496162646729406660277577161	5844448731260123597584614	73910210463068704814293649704631232284572444	
			1275381998153208717	5951647502459979	1985967161077591722004592	97618896383901032528237088047629893164689181	
					0330559498079809561389505	89786075260895904745088321575844352894846984	
					0109955207133921331667136	00713267770148646949636646873014856818043214	
					5741538177976093783968058	66959178132008213979305692781739749787876687	
			9699533328951138883662784	494823254429411827840326437762426011456	400914		
	2	B _{5,0}	early	4496162646729406660277577	4676009150994147874939605195	5844448731260123597584614	73910210463068705589346674383828672699396114
				1615951647502459979	6521296851711238	1985967161077591722004592	19281542455547013701249375542744933875977893
						0330559498079809561389505	94656558460855089272242778530758111980280510
						0109955207133921331667136	12525818922892033276907807435419203761917436
					5741538177975913937463794	36320184664671355893492628719973144171478097	
			2287386708670798314013435	105588215938053021839793671487807403735	149655		
	3	A _{4,0}	early	4676009150994147874939605	4855855655258889089601633229	5844448731260123597584614	73910210463068706213169841196228643775332006
				1956521296851711238	7090946200962500	1985967161077591722004592	16381788211454711174880266143032127674642432
						0330559498079809561389505	65237811689676080630328988419230019666931644
						0109955207133921331667136	68375828986723602978606933629513557940933342
					5741538177975734090959529	98058091392947021566113011008942959529078423	
			4875240088390457744364085	578249687296422919248200774764755349212	898393		
	4	A _{3,0}	early	4855855655258889089601633	5035702159523630304263661263	5844448731260123597584614	73910210463068706685763149472613853027858744
				2297090946200962500	7660595550213762	1985967161077591722004592	09007470570126930879010041375473654876513311
						0330559498079809561389505	55850148319944118516093990464610802100803768
						0109955207133921331667136	33266580996099811463492469665645928473102595
					5741538177975554244455264	45652304164140316417170144267338569494101236	
			7463093468110117174714736	194745688271039291555897254602704955184	647131		
	5	A _{2,0}	early	5035702159523630304263661	5215548663788371518925689297	5844448731260123597584614	73910210463068707007126599480671217450784853
				2637660595550213762	8230244899465024	1985967161077591722004592	53025105237546730499221037905822415601719315
						0330559498079809561389505	06451381846810056693413186186593048029477328
						0109955207133921331667136	67882509577933824415492717834876416175207140
					5741538177975374397951000	20040833012273473733257070418109784656581990	
			0050946847829776605065387	128731451775565616568571698782695454028	395869		

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
6	$B_{1,0}$	early	5215548663788371518925689 2978230244899465024	5395395168053112733587717331 8799894248716283	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177975194551446735 2638800227549436035416038 144610	73910210463068707177260190685026903056493409 17193408434826816885303628944066410108805749 73546730689664714925174101210912510365440977 24690175831507789801772362502230637528009574 68561325359916564513346920005836362681667039 569283831785299700500876147538056366850	
7	$A_{5,3}$	early	5395395168053112733587717 3318799894248716283	557524167295453452430468164 5110978737201534	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177975014704941833 8420892698241109724331549 659359	73910210730060530054579651355368122130015413 68453165248289528134948887054647388631706090 63693368787843681815021469646636333721465787 01682834067043411234172154141588087742456214 99671759999799720253193198667211430914284051 234383646955813374701281878565382022271	
8	$B_{4,3}$	early	5575241672954534524304648 1645110978737201534	5755088177855956315093578997 1422063225686784	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177974834858436932 4202985168932783413247061 174109	73910210730060530678402820443106910633770416 05378189443252299236593918593421500124467341 49754262562228645923615215606185357931768860 58825361661551215857989699708667511864038014 89215735241233856329589774669750196463692349 672208673265149034997720480192947836500	
9	$B_{3,3}$	early	5755088177855956315093578 9971422063225686784	5934934682757378105846509829 7733147714172034	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177974655011932030 9985077639624457102162572 688859	73910210730060531150996130459457098114439288 64069952361337247320885811691314271951235630 95338467992062548033470753794092983346449921 00458371894964978220507421361607552352930550 71058642256603038644887999700228489524171247 730883196974394524662788433936137957000	
10	$B_{6,3}$	early	5934934682757378105846509 8297733147714172034	6114781188295480472690343460 9785667341881285	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177974475165426492 8961409201288145049642944 979608	73910210997052353782567062292219002691825112 04686891197014854918428918464748849027802372 98701240393876869030882910607925559076568533 98384017568445357116199120936492316760688779 48054311462436809626045337283215821313635027 795209144561763319588606832143951418960	
11	$A_{5,2}$	early	6114781188295480472690343 4609785667341881285	6294627693833582839534177092 1838186969581537	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177974295318920954 7937740762951832997123317 279356	73910210997052354557620092592841752096851368 77936668523518837647768825860452619755827144 89424822710585737758665204522044921379367725 81931043319587318020694595208946057339122981 00245648749332010518114447253230466619383431 256787288808649120980891724296762759264	

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
12	$B_{2,3}$	early	6294627693833582839534177 0921838186969581537	6474474199371685206378010723 3890706597290788	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177974115472415416 6914072324615520944603689 570105	73910210997052355975400028129474169534235836 52885587763766033190341473808458830443304362 95557063452003819484077805209347603823646383 68967562185870816177762452554697375789247790 69611579336829840135998823422057482767255125 479982824322894700308067013108793264800	
13	$B_{6,2}$	early	6474474199371685206378010 7233890706597290788	6654320705546468149312747153 1684661364215040	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973935625909241 9084642977251223150648922 645853	73910211264044178323414964984914157989648691 09276321209639868207470830852155951644671871 93986819575425898176037665192030022771638725 04422801513172458235485894762277785920091329 86887236309081854252499220964306321485546370 484335511299773384367556416207711167632	
14	$A_{5,1}$	early	6654320705546468149312747 1531684661364215040	6834167211721251092247483582 9478616131138393	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973755779403067 1255213629886925356694155 722500	73910211264044179098467998096249562601040039 81724858060048248331470459781920840299065451 89412866511727393458971419327472445486591433 93921494967402619464326608139488595410347621 57815429405514830579319839677059353107471112 823054813705752181374798175030277812500	
15	$B_{3,2}$	early	6834167211721251092247483 5829478616131138393	7014013717896034035182220012 7272570898062645	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973575932896892 3425784282522627562739388 798248	73910211264044180194884485230946132740366487 64808064191955892421602702602998293621200001 73861108932103329413686313133350678268966255 86846123878341957279328938175536819995198935 78586126066263506088251269747246644398849097 386134594874971344745556710015986542912	
16	$B_{6,1}$	early	7014013717896034035182220 0127272570898062645	7193860224707497554207859241 0807960804209998	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973396086390080 8790594026130344027349482 650895	73910211531036002902070331943543493351001318 44811778422105867731537121772327506420177369 39977293006214206928452467986654886325610387 69579504384564310210990249873213970827918254 75410465990875696285634248551721598138215926 018607102679536876663879418452255711690	
17	$B_{4,2}$	early	7193860224707497554207859 2410807960804209998	7373706731518961073233498469 4343350710358251	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973216239883269 4155403769738060491959576 502642	73910211531036004300946543779346797272472079 05694334089073135211949953958658924040707907 07302766066658232479761637195945423262405712 47841651021272797839556426174606047814318230 81334568672828918437421034307414751213398952 650210039109503427899238409295777991596	

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
18	$A_{1,3}$	early	7373706731518961073233498 4694343350710358251	7553553238330424592259137697 7878740616515505	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177973036393376457 9520213513345776956569670 345388	73910211531036005265036907723238737215006317 23030047651241072183250472462635047806572143 13604522617744080375713362493603442354281910 01740124505214147861643528118657604293347576 49571350873551684690926270705378236399173821 679167605326000235272714037845278546560	
19	$B_{3,1}$	early	7553553238330424592259137 6977878740616515505	7733399745778568687375679724 7155565661886859	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177972856546869009 8079262347925507679744624 974034	73910211798027829390002697066171893305274871 1956930200136239291887669190 35565067421734241003266102171140942278496124 77795471505189707548384922518549339743778523 35534398011043119418968821471535279341283123 158614098374267783848169138857318339152	
20	$B_{2,2}$	early	7733399745778568687375679 7247155565661886859	7913246253226712782492221751 6432390707259113	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177972676700361561 6638311182505238402919579 601780	73910211798027829711366153097184962347037031 16910456455061504203998688146862169460554062 00076491298737983179195116033483284344308860 62812204730078272011890037608340535392447001 07178952363797674925873353300697971263869367 625096653647124958720818025265806349360	
21	$A_{1,2}$	early	7913246253226712782492221 7516432390707259113	8093092760674856877608763778 5709215752631368	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177972496853854113 5197360017084969126094534 229525	73910211798027829881499747647627151834921669 41555899242481158400534987117851040919205402 57664843061216239444860714154682952004268551 04820940928439522749077500896687784638356660 80329564717289145443936281774104406562681769 016448173641775105340238712631940887125	
22	$B_{4,1}$	early	8093092760674856877608763 7785709215752631368	8272939268759681548816208604 0727475937226723	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177972317007346028 6950647942636714107834349 634170	73910212065019653571679680666510960086003940 91605454151922182971674432887840108749949564 84341874730535900270786053061259407501475718 72654745649100695541596413646643356059482987 28298347845565925329099075607640393280606696 759651254522841894750283506814275778900	
23	$B_{2,1}$	early	8272939268759681548816208 6040727475937226723	8452785776844506220023653429 5745736121822078	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177972137160837943 8703935868188459089574165 038815	73910212065019654365636456618290257201693583 95821264638279285868030663111172727654074693 78637290623583271000936676354531869100850569 70269939918180777708333186679558678336659053 67225800303227826972929895782190318855790328 109206863652055014099213618721451610800	

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
24	$A_{1,1}$	early	8452785776844506220023653 4295745736121822078	8632632284929330891231098255 0763996306417434	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177971957314329859 0457223793740204071313980 443459	73910212065019654535770051837949748887484408 07854743021715277149965867363668916755089050 18093758962535219122979162391709709118624218 61832211241867061138674235320034558985149120 46023195597542284438381594036679975903913966 430719007648663435856998409667842949576	
25	a	early	8632632284929330891231098 2550763996306417434	5844448731260123597584614198 0 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 0589946614788376613610472275 4835310286860893	0	0	
26	$A_{1,0}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177980589946614788 3766136104722754835310286 860893	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 0769793119053117828272500309 5404959636112155	1798465042647412146620280 340569649349251262 46395651161932441133563283184266876434841859 53726310169668853631440011998993005418422316 56425554444610018152722310159683205357171897 7228868891193533982289329809852340062791677 72637423296867168165935405534246503404		
27	$B_{1,1}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177980769793119053 1178282725003095404959636 112155	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 0949639627137942499479945135 0423219820707510	3596930123495658858694728 595587909533846617 91084960232820901196352714709736952787312404 10344571712948442670706726231195521302208811 37323192672489235902827444097975985968961391 01639335226599014027490436363669519185		
28	$B_{1,2}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177980949639627137 9424994799451350423219820 707510	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 1129486134586086594596487161 9700044866079764	5395395197977099809860148 864864734579218871 54275189954673398544116947116157191441485387 32493115261654915097754440964585548420328002 72712811213262663149737141387434770484978636 27690260035250545254209196615597257094		
29	$B_{1,3}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177981129486134586 0865945964871619700044866 079764	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 1309332641397550113622126390 3235434772237017	7193860266091735000116541 148400124485376124 95881183123311671244748056229148045275579144 99767033010560095380854121893156114964893637 74937517058535822022101379726524774182897910 50923484704178678823051093427607928975179532 660115789252552996835945329915599137252		

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
30	$A_{2,1}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177981309332641397 5501136221263903235434772 237017	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 1489179149482374784829571215 8253694956832373	8992325346939981712190989 403418384669971480	15083716828387672166459010828640294437303565 95214286491548756884644859138989585592858666 86198205294590230442790140088404763215332931 34743082533954349007576321452348659240694101 29907918709523160028370072058965910845234965 906210709120724086790440645512556562080	
31	$A_{2,2}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177981489179149482 3747848295712158253694956 832373	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 1669025656930518879946113242 7530520002204628	1079079042142142266335640 9672695209715343735	18100460138530906808825097171626435948168369 37758769236853880977233642158703396588154629 20923446659501084232022293654927589577970983 39509350834002792284887628651474119562673610 28670456933340486250679882901876209470627671 072094940290679150242181725950305744425	
32	$A_{2,3}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177981669025656930 5188799461132427530520002 204628	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 1848872162468621246789946873 9583039629913880	1258925547680244633179474 5984747729343052987	21117203245760353733220125711398704451605986 19775460672716195143372984308990084241385350 82856431419694700338661184031273758117271729 34279247578599639077646486591351157363974026 23486575652453154573353796984314308423770165 706514389559413986933027103014633216684	
33	$B_{2,0}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177981848872162468 6212467899468739583039629 913880	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 2028718666733362461451974908 0152688979165139	1438772051944985847841502 6325317378692304246	24133946352989801338149511037986248207515013 64559067373394062096608522444301208580710905 16589958731549286753358897235720586056485494 34736496408886179163838821034501924804516261 90393079842504502031335735079350698135074244 308166572149541902837697335311139796274	
34	$A_{3,1}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 5741538177982028718666733 3624614519749080152688979 165139	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 2208565174181506556568516934 9429514024536494	1618618559393129942958044 6594594203737675601	27150690143718321443276367348212207320068140 69397084040473123643143482439134879487967992 63043745745683063169621959712583389184679492 00739757383324343804906030256669029781932450 52395142569300078822740114028242025286534412 619689963951874004066937717894180668420	
35	$A_{3,2}$	tardy	5844448731260123597584614 1985967161077591722004592 0330559498079809561389505 0109955207133921331667136 574153817798208565174181 5065565685169349429514024 536494	5844448731260123597584614198 5967161077591722004592033055 9498079809561389505010995520 7133921331667136574153817798 2388411680356289499503253364 7223468791460747	1798465065567912885892781 0892388158504599854	30167433272307114737747546281721095367038445 94958162006046971077721936317285334083765409 97978037065554097120302369066491905678539209 75438505252595479334524241319377907978459035 82737867691819838804741192164888591701258601 426717718362086956864378703562441312648	

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
36	$A_{3,3}$	tardy	5844448731260123597584614	5844448731260123597584614198	1978311570469334676645711	33184176336817870207444064697207663006424171	
			1985967161077591722004592	5967161077591722004592033055	9218699242993085105	50304638452851659491386183734785657661033727	
			0330559498079809561389505	9498079809561389505010995520		47104838270055980171830933338599651822941990	
			0109955207133921331667136	7133921331667136574153817798		65055126893304077922937369137787950436601963	
			5741538177982388411680356	2568258185257711290256184197		46302753904543803746501708525989719919273598	
			2894995032533647223468791	3534553279945998		288251651019347906098211568111469976420	
			460747				
37	$B_{3,0}$	tardy	5844448731260123597584614	5844448731260123597584614198	2158158074734075891307739	36200919476086337140642889081890810683485252	
			1985967161077591722004592	5967161077591722004592033055	9559268892342336364	86599497143928431043310928582202472500917373	
			0330559498079809561389505	9498079809561389505010995520		82547147758271393222197350554068026413361613	
			0109955207133921331667136	7133921331667136574153817798		25848586451516000447049001812137637952421370	
			5741538177982568258185257	2748104689522452504918212231		50533184631101234505563593265095344815219238	
			7112902561841973534553279	4104202629197257		116415018489842002759205073433163056304	
			945998				
38	$A_{4,1}$	tardy	5844448731260123597584614	5844448731260123597584614198	2338004582818900562515184	39217663629923738580490832851564405324900974	
			1985967161077591722004592	5967161077591722004592033055	7814287152526931720	38502439514879674490660872557897837637068495	
			0330559498079809561389505	9498079809561389505010995520		94779551425241336025880018860264830693075823	
			0109955207133921331667136	7133921331667136574153817798		04641156998365484581627225786505857060183757	
			5741538177982748104689522	2927951197607277176125657056		59271767832839726585924334151867509601289111	
			4525049182122314104202629	9122462813792613		094157944437130814270506403565204612320	
			197257				
39	$A_{4,2}$	tardy	5844448731260123597584614	5844448731260123597584614198	2517851089630364081540824	42234406694434494050187353676233260971352169	
			1985967161077591722004592	5967161077591722004592033055	0097822542433079974	09670132106723124714062723107136476431924089	
			0330559498079809561389505	9498079809561389505010995520		89528941974264525750028716769121717534452981	
			0109955207133921331667136	7133921331667136574153817798		09568940177818445589494144845073010222537598	
			5741538177982927951197607	3107797704418740695151296285		72964585643453154679742692138968335654190178	
			2771761256570569122462813	2657852719940867		761025729140972200642660175953568721396	
			792613				
40	$A_{4,3}$	tardy	5844448731260123597584614	5844448731260123597584614198	2697697594531785872293754	45251149523992443456637455915271674175489470	
			1985967161077591722004592	5967161077591722004592033055	8424133626921565225	45154115008495708384378056696035784799467888	
			0330559498079809561389505	9498079809561389505010995520		46690639721299821151608110583972585310405358	
			0109955207133921331667136	7133921331667136574153817798		98552672295048512834846890442556669418238123	
			5741538177983107797704418	3287644209320162485904227117		83590471045702484328703941818674234035435827	
			7406951512962852657852719	8968937208426118		663464638134446685617062898183195096475	
			940867				
41	$B_{4,0}$	tardy	5844448731260123597584614	5844448731260123597584614198	2877544098796527086955782	48267892620542218405833270586434026380614788	
			1985967161077591722004592	5967161077591722004592033055	8764703276270816484	95690743577304369393858612301937238985951935	
			0330559498079809561389505	9498079809561389505010995520		58105136409591336974724434594973156279725192	
			0109955207133921331667136	7133921331667136574153817798		24417608434509219666250486029269717735931780	
			5741538177983287644209320	3467490713584903700566255151		47859778549276155201998142090668581023364074	
			1624859042271178968937208	9538586557677377		665069288513868026454307049048167769356	
			426118				

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
42	$A_{5,0}$	tardy	5844448731260123597584614	5844448731260123597584614198	3057390603061268301617810	51284635877287087765735871056777577838239209	
			1985967161077591722004592	5967161077591722004592033055	9105272925620067746	75742650294721940745276609350752883434231612	
			0330559498079809561389505	9498079809561389505010995520		96779482557432789578099030333009578186372630	
			0109955207133921331667136	7133921331667136574153817798		83234696115152665695993040483601326537544733	
			5741538177983467490713584	3647337217849644915228283186		24862215717729581597266924858745590575773744	
			9037005662551519538586557	0108235906928639		772010109880457968824902813037237659684	
			677377				
43	$B_{5,1}$	tardy	5844448731260123597584614	5844448731260123597584614198	3237237109236051244552547	54301379742773317822066451188295270506851154	
			1985967161077591722004592	5967161077591722004592033055	3403066880386991098	65601470351085012841296556459266920646512936	
			0330559498079809561389505	9498079809561389505010995520		75117593351763203929002813121100185191321887	
			0109955207133921331667136	7133921331667136574153817798		57367753998682106418406650663652562685484751	
			5741538177983647337217849	3827183724024427858163019615		22330829476977518867151853177021610010063127	
			6449152282831860108235906	7902190673851991		471635619057592821071333646546390124432	
			928639				
44	$B_{5,2}$	tardy	5844448731260123597584614	5844448731260123597584614198	3417083614774153611396380	57318122850002765653840627731913236800280808	
			1985967161077591722004592	5967161077591722004592033055	9715119400014691349	53940991809399705622310975488536807328552483	
			0330559498079809561389505	9498079809561389505010995520		44528515574062454668971688451036693341765701	
			0109955207133921331667136	7133921331667136574153817798		33009898019798546771666966491717270003365590	
			5741538177983827183724024	4007030229562530225006853246		03708118757033602648215184288617047099749031	
			4278581630196157902190673	9954710301552242		233107417814434919500109396589108886433	
			851991				
45	$B_{5,3}$	tardy	5844448731260123597584614	5844448731260123597584614198	3596930119675575402149311	60334865925193194648842403351449161579106872	
			1985967161077591722004592	5967161077591722004592033055	8041430484503176599	49023503014492300852804753671885057286377960	
			0330559498079809561389505	9498079809561389505010995520		26828343227826105183677108722368111037453704	
			0109955207133921331667136	7133921331667136574153817798		02713035548695628740214911449789903710695048	
			5741538177984007030229562	4186876734463952015759784079		54236998717012066932124670695291028235304033	
			530225068532469954710301	6265794790037492		608447220086719980422812399514434613250	
			552242				
46	$A_{6,1}$	tardy	5844448731260123597584614	5844448731260123597584614198	3776776626487038921174951	63351609908155826564764780053773222184580033	
			1985967161077591722004592	5967161077591722004592033055	0324965874409323953	68215778136786720755973500139328972084064023	
			0330559498079809561389505	9498079809561389505010995520		37144580535915623515587183728304993726952709	
			0109955207133921331667136	7133921331667136574153817798		89675535599690655648556421242420021253847977	
			5741538177984186876734463	4366723241275415534785423307		72778457213867621669662437588193947848071367	
			9520157597840796265794790	9801184696184846		470067426203707259157381807089636091182	
			037492				
47	$A_{6,2}$	tardy	5844448731260123597584614	5844448731260123597584614198	3956623132661821864109687	66368353004705601400538205942436835542269062	
			1985967161077591722004592	5967161077591722004592033055	4622759829176248206	32386897915647649076314672361247387146666927	
			0330559498079809561389505	9498079809561389505010995520		59012207765647575122132538803286516298582454	
			0109955207133921331667136	7133921331667136574153817798		06281842252992497388954808054997296679547952	
			5741538177984366723241275	4546569747450198477720159737		42663134208775723729034492809007645945592866	
			4155347854233079801184696	7595139463109099		817009696617975866685566914861724212098	
			184846				

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Table 18 (continued)

pos.	job	side	start	completion	deviation	cost	contribution
48	$A_{6,3}$	tardy	58444448731260123597584614	58444448731260123597584614198	4136469638199924230953521	69385096069216357286116838510746245055658074	
			1985967161077591722004592	5967161077591722004592033055	0934812348803957458	52863475002405104811584125489249121161461447	
			0330559498079809561389505	9498079809561389505010995520		08586825469487752321359747945013194208522711	
			0109955207133921331667136	7133921331667136574153817798		78275536987171833470644697142961535390286354	
			5741538177984546569747450	4726416252988300844563993368		92915845105923019791258594836250958939163703	
			1984777201597377595139463	9647659090818351		355678085169018950223353348829847065176	
			109099				
49	$B_{6,0}$	tardy	58444448731260123597584614	58444448731260123597584614198	4316316142464665445615549	72401838834696269850786364262710105709816551	
			1985967161077591722004592	5967161077591722004592033055	1275381998153208717	52665526210733578008154721198867395821347811	
			0330559498079809561389505	9498079809561389505010995520		82558574934219877453425765201250172512914622	
			0109955207133921331667136	7133921331667136574153817798		67594581323837929734959114270034985505449809	
			5741538177984726416252988	4906262757253042059226021403		58485762112315098513170040550430919892418197	
			3008445639933689647659090	0217308440069610		491576103380031767488954039288543693533	
			818351				

How to read this step. For each row of [Table 18](#), completion equals start plus processing time. On the early side, deviation is $d - C_j$; on the tardy side it is $C_j - d$. Multiplying the deviation by the relevant weight gives the final column. Consecutive rows also verify compactness because each start equals the preceding completion.

B.1.11 The “only if” case: numerical enforcement and exhaustive audit

Suppose a schedule has cost at most B . Choose an optimal schedule and replace it, using [lemma I.3](#), by an optimal canonical schedule. The three mechanical enforcement steps specialize as follows.

Anchor enforcement. If the anchor in this canonical schedule were tardy, its full self-cost would exceed C^* , as calculated in [Section B.1.7](#). Therefore the selected optimal canonical schedule keeps it early.

Load enforcement. For a nonzero integer load-error vector $y = b - \beta$, the kernel identity gives $y^\top Ky \geq 2$. The dominant loss is therefore at least H , whereas all remaining assignment-dependent oscillation is less than U . Numerically,

$$\begin{aligned} & 35131387587510277603106814355503 \\ H - U = & 84147743029420104062696794987528 > 0. \\ & 53976 \end{aligned}$$

Thus every optimal block has load β_v , and the arithmetic synchronization calculation restricts it to one of the two coded states.

Prefix enforcement. If a coded state is not prefix-exact, one residual $C_k(x) - k$ is a nonzero integer and

$$\mathcal{P}(x) \geq \frac{\Delta_{\min}}{2}.$$

Consequently,

$$\Phi_{\text{can}}(\sigma) \geq A + \frac{\Delta_{\min}}{2} - \Gamma = B + \frac{20317234048545169320215967797}{83805638221597944365757/4} > B.$$

Thus a schedule meeting the bound must be prefix-exact. Successive differences then give exactly one selected subset-column in every element row, so the selected subset-columns form an exact cover.

The logical space has only $2^6 = 64$ states. [Table 19](#) evaluates every one using exact

arithmetic. The column $\mathcal{P}$ is the weighted prefix penalty, R is the bounded residual, and the last column is $\Phi_{\text{can}} - B$. Exactly one row is prefix-exact: $x = 100010$, corresponding to S_1, S_5 . The least expensive non-prefix-exact row is $x = 010001$, with

$$(C_1, \dots, C_6) = (1, 2, 3, 4, 6, 6), \quad (C_k - k)_{k=1}^6 = (0, 0, 0, 0, 1, 0),$$

and it still exceeds B by

$$50797486594832285376920111793$$

$$4529122335558302158437.$$

This exhaustive audit is not needed by the general proof, but it independently demonstrates the full “only if” mechanism on the instance. It also shows why the residual cannot create a false positive: the closest violating state has only one unit of prefix error, yet its main-gap penalty still leaves an enormous positive margin over B .

Purpose. The table exhausts the 64 coded block states and certifies the master identity and decision comparison. **Verification rule.** From each bit vector compute the prefix totals, $\mathcal{P}(x)$, $R(2x - 1)$, and finally $A + \mathcal{P} + R - B$; only its sign determines the threshold outcome.

Table 19: Exact audit of all 64 coded states.

x	$C(x)$	$C(x)$ (1, . . . , 6)	– RPE?	$\mathcal{P}(x)$	$R(\sigma)$	$\Phi_{\text{can}} - B$
000000	0,0,0,0,0,0	-1,-2,-3,-4,-5,-6	no	92459962968399192009913285 55948419111061376599461978 3/2	11600998877981094437295 69693445457571586359871 6	46229947634389428753227151171780101 505566758185668676
000001	0,0,1,2,3,3	-1,-2,-2,-2,-2,-3	no	26417132704720167332446023 62530635555905807956564644 9/2	46355436959732099809220 03998029907345967878441 9/2	13208525537094734406609063830998247 305120545075461734
000010	0,0,0,1,2,3	-1,-2,-3,-3,-3,-3	no	41657785613395469788661376 63030150532960536271566748 9/2	64465980131183693888344 97569617946793961239244 9/2	20828853802486702779876148245989393 778433634643833057
000011	0,0,1,3,5,6	-1,-2,-2,-1,0,0	no	10160435966657714440716528 46414697745628040557663342 3/2	22112127937740666383773 83919492274607136002338 3/2	50801747437326057616009737057984797 16098969249079118
000100	1,1,2,2,2,3	0,-1,-1,-2,-3,-3	no	12192522571531490330843055 99070228306166896337149545 9	65697928684164338638563 76462065723020141772575 /2	12192480405618879306236059015231585 590214632767141815
000101	1,1,3,4,5,6	0,-1,0,0,0,0	no	50802183556111246901854738 2243186460513545732887628 7/2	-1490927535658854843250 28739519098701548699940 7/2	50797563928829940676719685344056036 0704197314147993
000110	1,1,2,3,4,6	0,-1,-1,-1,-1,0	no	20320871264286427146400826 82742701437568274047889732 2	58450114557926510942323 2356220731491186201257/ 2	20320419678701702714487084906999510 43618178965750429
000111	1,1,3,5,7,9	0,-1,0,1,2,3	no	76203266989288995175767809 58295258675528221054251535 7/2	-2016079333168860185850 27292843798617830160398 7/2	76202802400801877003227591294998659 52219291228209610
001000	0,1,1,2,2,3	-1,-1,-2,-2,-3,-3	no	14224609690766272722090151 82514848411387821799553435 8	77639736693852080977183 98724036050233392943127 /2	14224568121944062181870271780688917 627587494016765990
001001	0,1,2,4,5,6	-1,-1,-1,0,0,0	no	15240654167600843514981994 83907920681343431564614532 /2	-8578678335733950456901 01114987887581163123963 /2	15240195370171223319767477351984346 83083800307812619
001010	0,1,1,3,4,6	-1,-1,-2,-1,-1,0	no	40641742456634251058871785 1718890248977528671928631 /2	11240406644793049997009 91804881511018186790243 /2	40641293568747121127157494640258764 26217197090083821
001011	0,1,2,5,7,9	-1,-1,-1,1,2,3	no	86363702801278714000564330 59959992896358106885978439 9/2	-2037660913152134731874 34946154443443022831010 9/2	86363238104883695911651382191263336 19825051096583453

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Table 19 (continued)

x	$C(x)$	$C(x)$ (1, . . . , 6)	— RPE? $\mathcal{P}(x)$	$R(\sigma)$	$\Phi_{\text{can}} - B$
001100	1,2,3,4,4,6	0,0,0,0,-1,0	no	10160435380358105851691128 63689946286355926164549607 /2	-1182490778102345647382 81187799637609655366045 5
001101	1,2,4,6,7,9	0,0,1,2,2,3	no	18288783781761702797803452 89846846177053120445446762 7/2	-2558316549611407116266 70078971203165175035971 6
001110	1,2,3,5,6,9	0,0,0,1,1,3	no	11176479112628142341609128 56103430557096157214907899 3/2	-2340702277111156525146 88802565951254673216308 4
001111	1,2,4,7,9,12	0,0,1,3,4,6	no	62994699973295303883765882 36383612873336757972893628 1/2	-9864580859566362881141 97651076171918858378976 786196030330849233
010000	1,2,2,2,3,3	0,0,-1,-2,-2,-3	no	18288783781761702797803452 89846846177053120445446762 7/2	27183799396989180687194 47052225688406643273431 06638168195267313
010001	1,2,3,4,6,6	0,0,0,0,1,0	no	10160435380358105851691128 63689946286355926164549607 /2	-1452260537205991426410 25151551652656410487643 5
010010	1,2,2,3,5,6	0,0,-1,-1,0,0	no	20320870436992499060739577 37309083667753530465454601 /2	-6015865630214393222474 37019013514833823143029 16046090734344340
010011	1,2,3,5,8,9	0,0,0,1,3,3	no	19304827416914627022962031 47055387586180898146547584 9/2	-2042157077262767111460 55630045673362150080952 6
010100	2,3,4,4,5,6	1,1,1,0,0,0	no	15240654167600843514981994 83907920681343431564614532 1/2	-3505208405413028608632 22780556022832684027920 96913427469235000
010101	2,3,5,6,8,9	1,1,2,2,3,3	no	14224609690766272722090151 82514848411387821799553435 8	-5976299399282367393250 90292545203156417391036 5/2
010110	2,3,4,5,7,9	1,1,1,1,2,3	no	86363702801278714000564330 59959992896358106885978439 5/2	-5821631403965462048076 74132421036849292396449 86858019748756260
010111	2,3,5,7,10,12	1,1,2,3,5,6	no	38609654862604694435409592 28521221016800825796186789 9	-2832582472507637420862 25787150417936411662892 1/2
011000	1,3,3,4,5,6	0,1,0,0,0,0	no	50802183556111246901854738 2243186460513545732887628 3/2	-2742659219569340114337 20746738467382994682191 3860790084236740
011001	1,3,4,6,8,9	0,1,1,2,3,3	no	12192522571531490330843055 99070228306166896337149545 9	-5774871312321993335257 71856904370484620774818 5/2

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Table 19 (continued)

x	$C(x)$	$C(x)$ (1, . . . , 6)	– RPE? $\mathcal{P}(x)$	$R(\sigma)$	$\Phi_{\text{can}} - B$
011010	1,3,3,5,7,9	0,1,0,1,2,3	no	76203266989288995175767809 58295258675528221054251535 3/2	-5713723500190507897365 76202783912581041894989033716444304 98379934436168228108877 99031539238467217
011011	1,3,4,7,10,12	0,1,1,3,5,6	no	36577567743369912044162496 45076600911579900333782900 0	-3285795667615997706453 36577520649663033003339714320438916 33632840540033240104830 408782996462064915
011100	2,4,5,6,7,9	1,2,2,2,2,3	no	26417132704720167332446023 62530635555905807956564644 9/2	-2590688921212174621078 13208518310862117221224461830327023 66722529082649249424357 984386706613339715
011101	2,4,6,8,10,12	1,2,3,4,5,6	no	92459962968399192009913285 55948419111061376599461978 3/2	-1171699987618261128718 46229934861690563153871585157521398 56176832402864854337243 726962393778697530
011110	2,4,5,7,9,12	1,2,2,3,4,6	no	71123048896253768418408453 09067402252189445484011510 3/2	-1914466058078617841348 35561477082851780897762456293250003 42807935653726123680319 400094125508014425
011111	2,4,6,9,12,15	1,2,3,5,7,9	no	17171135977687406242539812 87928751657516808705590277 05/2	22345928381788773569282 85855636672220824158752492421035318 56663909256765784551322 280781360388825243
100000	1,2,3,3,3,3	0,0,0,-1,-2,-3	no	71123048633677870485582335 76052072215014675321363907 /2	44691856764287346399744 71122616471515800301016246596193355 52383618453753660435161 19925711476341556
100001	1,2,4,5,6,6	0,0,1,1,1,0	no	15240652908675302456215353 00499514977054728315002104 5/2	-3828932116026472559198 15240179255924269995613267974463022 78856843374997621404159 55863899532741375
100010	1,2,3,4,5,6	0,0,0,0,0,0	yes	0	-2343399975092575105438 -4662250903277911148162289877368337 47969035812549964179099 8590496135429
100011	1,2,4,6,8,9	0,0,1,2,3,3	no	11684500735970377861824508 60845909660115541763860783 1	-5181377842220618892778 11684452694472411518691133315519133 56206981233686337530101 190671145275717393
100100	2,3,5,5,5,6	1,1,2,1,0,0	no	71123049532910396266052441 70687858693199131179307655 /2	-1642897833839748564407 35561053829386407407301287743460674 33893335458433228516824 87699392629245648
100101	2,3,6,7,8,9	1,1,3,3,3,3	no	38609654600028794974550092 33684238656652408831834172 1/2	-2856861750097112061173 19304778992343602157339056091163863 59840158437284514107234 194573358342858587
100110	2,3,5,6,7,9	1,1,2,2,2,3	no	23369001691353492518334739 33184723679597680516832068 1/2	-2887435656156262765869 11684452507432044870080674893087031 17765842874080151170751 464931760397212896
100111	2,3,6,8,10,12	1,1,3,4,5,6	no	89411831955032517195802001 26602507234753249159729401 5/2	-1371329609750044556718 44705869155377603615032515010495767 85784035953313129712823 309917273826278838

Continued on next page

Table 19 (continued)

x	$C(x)$	$C(x)$ (1, . . . , 6)	– RPE? $\mathcal{P}(x)$	$R(\sigma)$	$\Phi_{\text{can}} - B$
101000	1,3,4,5,5,6	0,1,1,1,0,0	no	30481307148214748441110525 01795456588780621931229857 /2	-1416291236240951259705 52029347464939977411914 15425070516255856
101001	1,3,5,7,8,9	0,1,2,3,3,3	no	32513393285487609174521013 20238705079934684148563993 3/2	-2910815701939971974622 69775258805180209367078 307305777973909246
101010	1,3,4,6,7,9	0,1,1,2,2,3	no	19304827452883927735840547 66295483469155829592024288 3/2	-2988149699591832632959 49042520357971212594918 37883595158932321
101011	1,3,5,8,10,12	0,1,2,4,5,6	no	83315570640491331395772922 13156973658035524476459222 7/2	-1752604202627271641960 08859801798593137501746 767875850332038714
101100	2,4,6,7,7,9	1,2,3,3,2,3	no	18288783961608208431407906 15592588694891286594645972 7	-4084314154634278669949 17591046229523269384911 213449408924295236
101101	2,4,7,9,10,12	1,2,4,5,5,6	no	54358329642810410073522008 86485150522881840129954360 1	-1203173126091839894142 54358283590414801794778132889788170 7/2
101110	2,4,6,8,9,12	1,2,3,4,4,6	no	41657785563038448371695634 89313733726670521525683666 0	-2904521074447760073967 86155009188954254836158 716444603307415934
101111	2,4,7,10,12,15	1,2,4,6,7,9	no	95000071552711284678570944 48607460938572611525489016 8	54367598794218421136716 95000028820282179156668072418247740 /2
110000	2,4,5,5,6,6	1,2,2,1,1,0	no	55882397523467625129428206 63918461871318165870591515 5/2	-1972916171753292355661 55881933150796314034727139294778666 56021944025057971
110001	2,4,6,7,9,9	1,2,3,3,4,3	no	24385045189823071942422583 33806556466704842293375736 9	-4681404554002549632346 81554810925769391992316 108103735298555856
110010	2,4,5,6,8,9	1,2,2,2,3,3	no	15748675197449610129145793 97187804349541885519419724 2	-5116633098989866786085 15748627188324015401388472025286704 7/2
110011	2,4,6,8,11,12	1,2,3,4,6,6	no	51818220943396554223386763 53003680013026447690233273 0	-2364981555911960650937 95846229299321820929655 479452027122444520
110100	3,5,7,7,8,9	2,3,4,3,3,3	no	56898439608343667808794537 99384764258523761020415317 3/2	-1018830456563316929893 28449173334532332108256410199865849 329423009870456611
110101	3,5,8,9,11,12	2,3,5,5,6,6	no	13716587927503455562157230 80594014905259429754822425 53/2	56202033228229453458433 68582894748728564860256759710377673 752440578801167616

Continued on next page

Table 19 (continued)

x	$C(x)$	$C(x)$ (1, . . . , 6)	– RPE?	$\mathcal{P}(x)$	$R(\sigma)$	$\Phi_{\text{can}} - B$
110110	3,5,7,8,10,12	2,3,4,4,5,6	no	10770061696334739003004644 69230303283662460134831054 67/2	–4289339167374634738609 22474935610742346976297 743196423719336505	53850262601930733044735820696933760
110111	3,5,8,10,13,15	2,3,5,6,8,9	no	22251353724697954717234663 07566074839851821080511341 15/2	38819868347717296965333 91711816050327015173387 5304325540365500513	11125672705466756312507908300803652
111000	2,5,6,7,8,9	1,3,3,3,3,3	no	46738003969006594478846850 45273336993474082004454376 9/2	–1008039666527062494975 08136547633138022418756 9	23368955525654585479537001347626390
111001	2,5,7,9,11,12	1,3,4,5,6,6	no	12497335655962586127408973 30527242842126874477380091 59/2	29225057287689175135173 5544092719616603831192 694747499797595840	62486633121254458281112688974438757
111010	2,5,6,8,10,12	1,3,3,4,5,6	no	97540181324010316700098759 3819160557157492232349606 3/2	–7454637677355762107347 36203607836011140074605 745722759846433495	48770044465732345381649240052562895
111011	2,5,7,10,13,15	1,3,4,6,8,9	no	21032101453157085282486405 57499302776719265803069007 21/2	32848964343319597415086 59992260211168758427525 2591858618236637954	10516046509987281551156784047996620
111100	3,6,8,9,10,12	2,4,5,5,5,6	no	66550853027248106875485864 95510834591160871396631183 4	44224255875283891241755 97442366164077864612819 /2	66550809787651855406856498139248138
111101	3,6,9,11,13,15	2,4,6,7,8,9	no	12700544371842687475001642 86137336242252044533889233 65	12893196026276777822469 95327709689277568331890 9/2	12700540471421584265558141094505133
111110	3,6,8,10,12,15	2,4,5,6,7,9	no	10719259493714898148053480 05763314500685164279216584 01	92710873919996430351455 42389931421130881986373 /2	10719255412188363224753238924544371
111111	3,6,9,12,15,18	2,4,6,8,10,12	no	18491992593679838401982657 11189683822212275319892395 66	23201997756075195145273 05679726014219484165920 1/2	18491989208698821682460021485183785

B.1.12 How to reproduce the numerical audit

The numerical appendix is self-contained and can be regenerated with exact integer and rational arithmetic from the following sequence.

1. Enter the incidence matrix with elements in rows and subsets in columns, then compute $d_{v,k}$ by cumulative sums down subset-column v .
2. Form G, M, T , then P_{vh} and β_v from (8.1)–(8.2).
3. Form λ_v, L_v, K , and $f = K\beta$; verify $f_v - f_{v+1} = 2 \sum_{u \leq v} \beta_u$.
4. Generate the 24 atoms from the two displayed effective-ratio formulas, group them into bands, and subtract endpoints to obtain Δ_k and W .
5. Expand the early objective to obtain a_v^0 and J_{uv} , form a_v^*, D_v, η_v, e_v , and verify (B.3).
6. Form U, H , all tardiness ratios, the anchor data, C_0, A, Γ , and B .
7. Enumerate the $2^6 = 64$ logical vectors x ; for each, compute $C_k(x)$, $\mathcal{P}(x)$, $R(2x - \mathbf{1})$, and $A + \mathcal{P} + R - B$.
8. For the exact-cover state, sort the early jobs by increasing r_j and the tardy jobs by decreasing s_j , then accumulate start times, completion times, deviations, and costs.

All divisions occur in rational quantities with small fixed denominators; no floating-point approximation is necessary.

B.2 A “no” instance

How to read this step. **Theorem-verification lane.** Read the modified incidence matrix, the exhaustive two-column source audit, the single failed prefix of the closest vector, the no-instance gap summary, and the three reverse gates. The full 49-job catalogues may be skipped. **Full arithmetic-audit lane.** Recompute every hatted quantity from the same formulas used for the yes-instance and verify all 64 coded-state differences in [Table 29](#).

The second example is deliberately as close as possible to the first one. Starting from the yes-instance in [Section B.1.1](#), exchange e_3 and e_4 between subset-columns S_1 and S_3 :

$$S_1 = \{e_1, e_2, e_3\}, \quad S_3 = \{e_2, e_4, e_6\} \quad \longrightarrow \quad \hat{S}_1 = \{e_1, e_2, e_4\}, \quad \hat{S}_3 = \{e_2, e_3, e_6\}.$$

All other subset-columns are unchanged. This is a degree-preserving 2-switch: each of the two changed columns still contains three elements, and the exchanged elements still occur in three columns overall. The example therefore remains a valid three-uniform, three-regular RXC3 instance, but the unique cover of the first example is destroyed.

How to read this step. The phrases “failure of the if route” and “failure of the only-if supposition” below do not mean that either proved implication is false. In a no-instance, the forward proof has no source witness from which to build a low-cost schedule, while a hypothetical target witness in the reverse proof is driven backward to an impossible exact cover. The example traces both phenomena numerically.

B.2.1 The modified restricted exact-cover instance

Let $\widehat{X} = \{e_1, \dots, e_6\}$, and let

$$\begin{aligned} \hat{S}_1 &= \{e_1, e_2, e_4\}, & \hat{S}_2 &= \{e_1, e_2, e_5\}, & \hat{S}_3 &= \{e_2, e_3, e_6\}, \\ \hat{S}_4 &= \{e_1, e_3, e_6\}, & \hat{S}_5 &= \{e_4, e_5, e_6\}, & \hat{S}_6 &= \{e_3, e_4, e_5\}. \end{aligned}$$

With elements in rows and subset-columns in columns, the incidence matrix is

$$\hat{A}^{\text{inc}} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}. \quad (\text{B.4})$$

Every row and every column contains three ones. Its intersection-cardinality matrix is

$$\begin{pmatrix} 3 & 2 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 1 & 1 & 1 \\ 1 & 1 & 3 & 2 & 1 & 1 \\ 1 & 1 & 2 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \end{pmatrix}.$$

Every off-diagonal entry is positive. Hence no two subset-columns are disjoint. An exact cover of six elements by 3-sets would contain exactly two columns: if x were a cover, then summing the six row equations would give

$$6 = \mathbf{1}^\top \hat{A}^{\text{inc}} x = 3 \sum_{v=1}^6 x_v, \quad \text{so} \quad \sum_v x_v = 2.$$

Those two selected columns would have to be disjoint, contrary to the displayed matrix. Thus this is a no-instance of RXC3.

B.2.2 Prefix data and an exhaustive source-level audit

Define

$$\hat{d}_{v,k} = |\hat{S}_v \cap \{e_1, \dots, e_k\}|, \quad \hat{C}_k(x) = \sum_{v=1}^6 \hat{d}_{v,k} x_v.$$

The full prefix matrix is as follows.

	$\hat{S}_1$	$\hat{S}_2$	$\hat{S}_3$	$\hat{S}_4$	$\hat{S}_5$	$\hat{S}_6$	
e_1	1	1	0	1	0	0	
e_2	1	1	1	0	0	0	
e_3	0	0	1	1	0	1	row e_3 : selected sum 0 (uncovered)
e_4	1	0	0	0	1	1	row e_4 : selected sum 2 (double-covered)
e_5	0	1	0	0	1	1	
e_6	0	0	1	1	1	0	

the former cover columns $\hat{S}_1, \hat{S}_5$ now fail

Figure 12: The no-instance incidence matrix. The highlighted pair was the exact cover in Figure 11; after the 2-switch it leaves e_3 uncovered and covers e_4 twice.

Table 20: Prefix data for the no-instance. Each column is the cumulative sum down the corresponding column of (B.4).

Prefix	$\hat{S}_1$	$\hat{S}_2$	$\hat{S}_3$	$\hat{S}_4$	$\hat{S}_5$	$\hat{S}_6$
$k = 1$	1	1	0	1	0	0
$k = 2$	2	2	1	1	0	0
$k = 3$	2	2	2	2	0	1
$k = 4$	3	2	2	2	1	2
$k = 5$	3	3	2	2	2	3
$k = 6$	3	3	3	3	3	3

The six prefix equations are

$$\hat{C}_1(x) = x_1 + x_2 + x_4 = 1,$$

$$\hat{C}_2(x) = 2x_1 + 2x_2 + x_3 + x_4 = 2,$$

$$\hat{C}_3(x) = 2x_1 + 2x_2 + 2x_3 + 2x_4 + x_6 = 3,$$

$$\hat{C}_4(x) = 3x_1 + 2x_2 + 2x_3 + 2x_4 + x_5 + 2x_6 = 4,$$

$$\hat{C}_5(x) = 3x_1 + 3x_2 + 2x_3 + 2x_4 + 2x_5 + 3x_6 = 5,$$

$$\hat{C}_6(x) = 3(x_1 + x_2 + x_3 + x_4 + x_5 + x_6) = 6.$$

Purpose. The table exhausts the only possible two-column exact-cover candidates in the modified source instance. **Verification rule.** Intersect each pair, identify uncovered and doubled rows, or equivalently subtract $(1, \dots, 6)$ from its prefix-total vector; no row may have zero error.

Table 21: Exhaustive audit of the fifteen two-column candidates.

selected columns	intersection	uncovered	doubled	$\widehat{C}(x)$	$\widehat{C}(x) - (1, \dots, 6)$
$\widehat{S}_1, \widehat{S}_2$	e_1, e_2	e_3, e_6	e_1, e_2	2, 4, 4, 5, 6, 6	1, 2, 1, 1, 1, 0
$\widehat{S}_1, \widehat{S}_3$	e_2	e_5	e_2	1, 3, 4, 5, 5, 6	0, 1, 1, 1, 0, 0
$\widehat{S}_1, \widehat{S}_4$	e_1	e_5	e_1	2, 3, 4, 5, 5, 6	1, 1, 1, 1, 0, 0
$\widehat{S}_1, \widehat{S}_5$	e_4	e_3	e_4	1, 2, 2, 4, 5, 6	0, 0, -1, 0, 0, 0
$\widehat{S}_1, \widehat{S}_6$	e_4	e_6	e_4	1, 2, 3, 5, 6, 6	0, 0, 0, 1, 1, 0
$\widehat{S}_2, \widehat{S}_3$	e_2	e_4	e_2	1, 3, 4, 4, 5, 6	0, 1, 1, 0, 0, 0
$\widehat{S}_2, \widehat{S}_4$	e_1	e_4	e_1	2, 3, 4, 4, 5, 6	1, 1, 1, 0, 0, 0
$\widehat{S}_2, \widehat{S}_5$	e_5	e_3	e_5	1, 2, 2, 3, 5, 6	0, 0, -1, -1, 0, 0
$\widehat{S}_2, \widehat{S}_6$	e_5	e_6	e_5	1, 2, 3, 4, 6, 6	0, 0, 0, 0, 1, 0
$\widehat{S}_3, \widehat{S}_4$	e_3, e_6	e_4, e_5	e_3, e_6	1, 2, 4, 4, 4, 6	0, 0, 1, 0, -1, 0
$\widehat{S}_3, \widehat{S}_5$	e_6	e_1	e_6	0, 1, 2, 3, 4, 6	-1, -1, -1, -1, -1, 0
$\widehat{S}_3, \widehat{S}_6$	e_3	e_1	e_3	0, 1, 3, 4, 5, 6	-1, -1, 0, 0, 0, 0
$\widehat{S}_4, \widehat{S}_5$	e_6	e_2	e_6	1, 1, 2, 3, 4, 6	0, -1, -1, -1, -1, 0
$\widehat{S}_4, \widehat{S}_6$	e_3	e_2	e_3	1, 1, 3, 4, 5, 6	0, -1, 0, 0, 0, 0
$\widehat{S}_5, \widehat{S}_6$	e_4, e_5	e_1, e_2	e_4, e_5	0, 0, 1, 3, 5, 6	-1, -2, -2, -1, 0, 0

No row of [Table 21](#) has a zero error vector, so the associated RPE instance is also a no-instance. The last prefix equation forces any candidate to select exactly two columns; the table therefore exhausts every possible prefix-exact candidate, while each of the other binary vectors fails already at $k = 6$. The closest visual continuation of the yes-example is the former choice

$$x^\dagger = (1, 0, 0, 0, 1, 0).$$

It gives

$$\widehat{C}(x^\dagger) = (1, 2, 2, 4, 5, 6), \quad \widehat{C}(x^\dagger) - (1, \dots, 6) = (0, 0, -1, 0, 0, 0).$$

Thus the entire failure is localized at the third prefix. Successive differences make the same defect pointwise:

$$\begin{aligned} \widehat{C}_1 - \widehat{C}_0 &= x_1 + x_2 + x_4 = 1, & \widehat{C}_2 - \widehat{C}_1 &= x_1 + x_2 + x_3 = 1, & \widehat{C}_3 - \widehat{C}_2 &= x_3 + x_4 + x_6 = 0, \\ \widehat{C}_4 - \widehat{C}_3 &= x_1 + x_5 + x_6 = 2, & \widehat{C}_5 - \widehat{C}_4 &= x_2 + x_5 + x_6 = 1, & \widehat{C}_6 - \widehat{C}_5 &= x_3 + x_4 + x_5 = 1. \end{aligned}$$

The zero in the e_3 row and the two in the e_4 row are exactly the uncovered and double-covered elements in [Figure 12](#).

B.2.3 The numerical compiler for the no-instance

The dimensions and global scales are unchanged:

$$\begin{aligned} n &= 6, & N &= 7, \\ G &= 7^{20}, & M &= 7^{40}, & T &= 7^{50}, \\ (Q_0, Q_1, Q_2, Q_3) &= (10, 100, 1000, 10000). \end{aligned}$$

Only two incidence positions have changed. Consequently,

$$\widehat{P}_{1,3} = P_{1,3} - M, \quad \widehat{P}_{3,2} = P_{3,2} + M,$$

and

$$\widehat{\beta}_1 = \beta_1 - M, \quad \widehat{\beta}_3 = \beta_3 + M.$$

All base values and balanced loads are displayed in [Table 22](#); the unchanged rows are repeated so that this subsection independently specifies the no-instance.

Purpose. The table certifies integral bases and balanced target loads for every no-instance block. **Verification rule.** Reuse $P_{vh} = T + M\kappa(k_{vh}) + Q_h$ with the modified incidence data and verify both coded orientations have load $\widehat{\beta}_v$.

Table 22: Exact base processing values and balanced block loads for the no-instance.

v	$\widehat{S}_v$	$(\kappa_1, \kappa_2, \kappa_3)$	$\widehat{P}_{v0}$	$\widehat{P}_{v1}$	$\widehat{P}_{v2}$	$\widehat{P}_{v3}$	$\widehat{\beta}_v$
1	$\{1, 2, 4\}$	$(6, 5, 3)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650744814409 51165420269276825 045372254	17984650617478294 29347364297793954 766933252	7193860259724929239 2075131626586893461 52123
2	$\{1, 2, 5\}$	$(6, 5, 2)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650744814409 51165420269276825 045372254	17984650553810236 68438336312052519 627709251	7193860253358123478 2984851769172542069 28122
3	$\{2, 3, 6\}$	$(5, 4, 1)$	17984650426474121 46620280340569649 349251259	17984650744814409 51165420269276825 045371354	17984650681146351 90256392283535389 906148253	17984650490142179 07529308326311084 488485250	7193860234257706195 5714012196929487892 56119
4	$\{1, 3, 6\}$	$(6, 4, 1)$	17984650426474121 46620280340569649 349251259	17984650808482467 12074448255018260 184595355	17984650681146351 90256392283535389 906148253	17984650490142179 07529308326311084 488485250	7193860240624511956 4804292054343839284 80120
5	$\{4, 5, 6\}$	$(3, 2, 1)$	17984650426474121 46620280340569649 349251259	17984650617478294 29347364297793954 766923352	17984650553810236 68438336312052519 627700251	17984650490142179 07529308326311084 488485250	7193860208790483151 9352892767272082323 60115
6	$\{3, 4, 5\}$	$(4, 3, 2)$	17984650426474121 46620280340569649 349251259	17984650681146351 90256392283535389 906147353	17984650617478294 29347364297793954 766924252	17984650553810236 68438336312052519 627709251	7193860227890900434 6623732339515136500 32118

Purpose. The table certifies the hatted block loads, kernel levels, and affine fields. **Verification rule.** Compute one row from $\widehat{f} = \widehat{K}\widehat{\beta}$ and verify the recurrence $\widehat{f}_v - \widehat{f}_{v+1} = 2 \sum_{u \leq v} \widehat{\beta}_u$.

Table 23: Block loads and minimum-kernel data for the no-instance.

v	$\widehat{\beta}_v$	λ_v	$L_v = 2\lambda_v$	$\widehat{f}_v$
1	719386025972492923920 751316265868934615212 3	007	4014	688817294128443960104362195474876 4781708332706313514011566498
2	719386025335812347829 848517691725420692812 2	006	4012	688817294128443958665590143529890 6303293306380996135319262252
3	719386023425770619557 140121969294878925611 9	005	4010	688817294128443955788046040913280 0868281309701844248213101762
4	719386024062451195648 042920543438392848012 0	004	4008	688817294128443951471729891445128 3042126510583306463528429034
5	719386020879048315193 528927672720823236011 5	003	4006	688817294128443945716641693852074 1303010853053899910986796066
6	719386022789090043466 237323395151365003211 8	002	4004	688817294128443938522781454500923 3260024616971038941980442868

The minimum-kernel data in [Table 23](#) show that the transfer of one M -unit from $\hat{\beta}_1$ to $\hat{\beta}_3$ changes $\hat{f}_1$ and $\hat{f}_2$, but cancels from $\hat{f}_3, \dots, \hat{f}_6$, because rows 3, $\dots$, 6 of the minimum kernel assign the same coefficient to blocks 1 and 3.

The 24 effective atoms are obtained from the same two formulas

$$\hat{\rho}_{vh} = \hat{f}_v + \lambda_v(2\hat{P}_{vh} + 1) \quad (h = 1, 2, 3), \quad \hat{\rho}_{v0} = \hat{f}_v + \lambda_v(2\hat{P}_{v0} + 3).$$

Taking exact minima and maxima within each element band gives [Table 24](#). The narrowest main gap is $\hat{\Delta}_3$; this is precisely the gap charged to the single failed prefix of $x^\dagger$.

Purpose. The table certifies the no-instance band endpoints, widths, and separating gaps. **Verification rule.** Group hatted atoms by element, take endpoint differences, and use the resulting signs and minimum gap in the hatted residual-dominance calculation.

Table 24: Effective bands, widths, and gaps for the no-instance.

band	lower endpoint	upper endpoint	width	gap below
$\widehat{\mathcal{R}}_1$	71751801506004387229547759 36120363829879968940409256 887323878	71751801506004388200718894 61507325841908480593525868 478033475	9711711352538696201 2028511653116611590 709597	10160436243621332816590835 14184113553246143428037817
$\widehat{\mathcal{R}}_2$	71751801495843950482356208 10919552597607007679507211 777723307	71751801495843950985926426 54461280315695855387163113 459286061	5035702184354172771 8088847707655901681 562754	10160437214792467815788224 82576513263397014461855010
$\widehat{\mathcal{R}}_3$	71751801485683511433129377 56529351365235304358013215 944716282	71751801485683513267563740 29340730115030494416110197 315868297	1834434362728113787 4979519005809698137 1152015	10160434876787887161601621 01963550374940926910024249
$\widehat{\mathcal{R}}_4$	71751801475523074218336910 13151363448112959009064812 318199878	71751801475523076556341490 40369189263271753983072289 034692033	2338004580272178258 1515879497400747671 6492155	10160435056634394100401255 26345672807857031951590608
$\widehat{\mathcal{R}}_5$	71751801465362637003544442 69773375530991906294830430 006115874	71751801465362639161702516 03111237921767286201207780 366609270	2158158073333378623 9077537990637735036 0493396	10160435380358105851691128 63689946286355926164549607
$\widehat{\mathcal{R}}_6$	71751801455202200544107300 17935331471914270055285828 661319569	71751801455202201623186336 84604262667301960008474503 841566267	1079079036666689311 9538768995318867518 0246698	10160435632143213987042787 87282416659435905480383424
$\widehat{\mathcal{R}}_0$	71751801445041762573959507 83017399696745334254878857 657299910	71751801445041764911964086 19231052684631853395849923 180936145	2338004578362136529 – 8788651914097106552 3636235	

Thus

$$\widehat{W} = \begin{array}{c} 11222421985051783408691209333 \\ 93414062424292850, \end{array} \quad \widehat{\Delta}_{\min} = \widehat{\Delta}_3 = \begin{array}{c} 10160434876787887161601621019 \\ 63550374940926910024249. \end{array}$$

The exact field-correction data are in [Table 25](#). As before, the columns scaled by four satisfy

$$4\widehat{e}_v = 4\widehat{a}_v^0 - \widehat{\eta}_v\widehat{D}_v - 4\widehat{a}_v^*, \quad |4\widehat{e}_v| \leq \widehat{D}_v/2.$$

Purpose. The table certifies every no-instance logical-field correction and residual field error. **Verification rule.** Check $4\widehat{e}_v = 4\widehat{a}_v^0 - \widehat{\eta}_v\widehat{D}_v - 4\widehat{a}_v^*$ and $|4\widehat{e}_v| \leq \widehat{D}_v/2$.

Table 25: Exact logical-field correction for the no-instance. All displayed field columns are multiplied by four.

v	$4\widehat{a}_v^0$	$4\widehat{a}_v^*$	$\widehat{D}_v$	$\widehat{\eta}_v$	$4\widehat{e}_v$
1	-216792428487339362989 8340524601709232926765 9591390358448222236329 0192345197526623722884 14841668717901842	568984387127080695 796305634988522367 15949337269836384	178270561305452 783600760183898 294162	-121608653105591756666 6611516804924119730854 3552672881550201404915 8185	-10584813658 782193594642 372387722256
2	-206468978841734728773 1865581891827726498716 5404731754651778919036 0410646456694883458328 58134583187226111	528342646900543119 394700613934695454 84521209463473952	165536949783634 727629277313619 846160	-124726823293289057041 9845322472211903832162 6280895652186955986625 2237	-29942504744 365429559070 048954840143
3	-175498630270384710554 6609156086684761947661 1577537027532384916321 0627222771739361210411 74115963617454932	447059159325546321 688077685956826739 72701406288978080	127336115218180 559714828702784 502154	-137823138368624970100 1574636490446751088264 2955379957511953607156 7838	-61452303140 095695548705 426838309960
4	-196145528355563215415 0722686924704875710079 9992943056357209653512 9155443416997763473066 90099041956617518	457219595569167654 504668521098667875 25947549717015897	140069726739998 615686311573062 950156	-140034205049642230727 5654578998640297067851 7127529771453769235787 0883	-82020207973 900133716096 52760925667
5	-134204832528533012398 5813547386059525400945 7826536166160026962231 0014512700520676374241 24752151885357758	325133935408696486 685286489133656539 64833688098760134	764016691309083 358288972216707 10146	-175656937937551388015 5340777724026871230624 6701215626056446721476 0027	-14450833146 919828679376 961919983950

Continued on next page

Table 25 (continued)

v	$4\widehat{a}_v^0$	$4\widehat{a}_v^\star$	$\widehat{D}_v$	$\widehat{\eta}_v$	$4\widehat{e}_v$
6	-185822076736453796943	447059157167388253	114602503696362	-162144866598016401274	442437567175
	8579403900619218784535	830152016430887277	503743345832506	0654291369186469353397	024347845852
	5764894663592240233032	52864227457943348	054152	6133636015857072268289	2739796437
	5180937881954072596735			3316	
	17254903737108247				

For completeness, [Table 26](#) gives all fifteen cross-block coefficients. The third column is the main-gap coefficient

$$\hat{J}_{uv}^{\text{main}} = \frac{1}{4} \sum_{k=1}^6 \hat{\Delta}_k \hat{d}_{u,k} \hat{d}_{v,k},$$

and the last column is its exact within-band remainder.

Purpose. The table certifies the hatted cross-block couplings and their main-gap/residual decomposition. **Verification rule.** Recompute the scaled atom-pair sum and main-gap sum for one block pair, then subtract; the global proof later uses only the aggregate residual bound.

Table 26: Exact cross-block couplings for the no-instance.

$u \ v$	$\hat{J}_{uv}$	$\hat{J}_{uv}^{\text{main}}$	difference
1 2	1676472135045362036195597612754 9538099518931384542881/2	838235935155652489692907496729 2417866466646357373945	264734057056809782619296470236 6585638669794991/2
1 3	1371659006372025562099612889601 8855810076799576420385/2	137165878623990204936961163993 81346263379317292049561/2	110066061756365000624831875477 3348741142185412
1 4	1422461180935811508539167970271 2052952007277982567969/2	2844921934916017426905131631294 6806080004778012136939/4	4269556055901732043092477299824 009777952998999/4
1 5	914439278112105669667145201799 8803005260243290711335/2	914439190706703716473678142335 9223038314901082760141/2	4370270097659673352973197899834 72671103975597
1 6	660428372201418820728065869234 2569512033784817161637	264171319205894017474216022207 52404109150530333989425/4	296746735108170103254861787393 8984608934657123/4
2 3	254010928103745718101785000539 29059114534678333936147/4	127005443567355810836559908730 35673455522285340458953/2	409690340964286651830785771220 3490107653018241/4
2 4	660428407091641001763921532164 2396633631151590795640	264171323378332954489710652602 55460464290714108955723/4	394583234462158579602631412607 0233892254226837/4
2 5	863637107145249845449887827643 5697604202004154248331/2	863637015423531745971671866018 6386634386385106964837/2	458608590497391079808124655484 907809523641747
2 6	609626195119781785613167571160 5105129130923791558520	243850449092625229273413511680 61058493436466430808209/4	289552874849718535167835936202 3087228735425871/4
3 4	2235296163874297115478265727971 9495023594793085094073/4	111764789579601471076974667680 57470537467851944981841/2	372282267693938772374360455394 8659089195130391/4
3 5	762032753341668795209824030456 3833262881413307576725/2	381016330809975343727380289824 8220174015229471207615	917217181077550634508067392914 850954365161495/2
3 6	106684585252017825027460035140 36419043660944096782783/2	533422857378877279295850314424 7804908592171984289847	137762423691682899722554080922 6476600128203089/2
4 5	152406553941540169385636524012 59073180536669740444233/4	381016330809975343727380289824 8220174015229471207615	2161755003189468440808266192484 475751855613773/4

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Table 26 (continued)		
$u \ v \ \widehat{J}_{uv}$	$\widehat{J}_{uv}^{\text{main}}$	difference
4 6 5334229332741028611787306294747	533422857378877279295850314424	7589522558188288031504999023341
707242733944414303280	7804908592171984289847	41772430013433
5 6 172727430385405973283429128005	863637015423531745971671866018	273006996240890947548021816686
90940138111547715453003/4	6386634386385106964837/2	9338777501523329/4

B.2.4 Load lock, anchor, and the 49-job no-instance

The exact load-lock quantities are

$$\begin{aligned} \widehat{U} = & \begin{array}{l} 583489201983731785266607839650588849977 \\ 614462575543787905298044664841507417338 \\ 080675476531688081516158715736685814354 \\ 106598477105310962383846749076147397863, \\ \\ 17565693793755138801553407777240 \end{array} \\ \max_v |\widehat{\eta}_v| = & 26871230624670121562605644672147 \\ & 60027, \end{aligned}$$

and

$$\begin{aligned} \widehat{H} = \widehat{U} + 2 \max_v |\widehat{\eta}_v| + 2 = & \begin{array}{l} 583489201983731785266607839650588849977 \\ 614462575543787905298044664841507417338 \\ 080675476883001957391261491767753969898 \\ 911972723230244986696367878010576917919. \end{array} \end{aligned}$$

The strict dominance certificate is

$$\begin{aligned} \widehat{H} - \widehat{U} = & \begin{array}{l} 35131387587510277603106815554480 \\ 53742461249340243125211289344295 > 0. \\ 20056 \end{array} \end{aligned}$$

Choose the all-rejected coded state $\sigma = (-1, \dots, -1)$ as the explicit benchmark used in the anchor construction. Its cost is

$$\begin{aligned} \widehat{C}^* = & \begin{array}{l} 267448763683410661369552775798154563648 \\ 899529608119085375293826634448762597664 \\ 705230994596377703116027184387031023196 \\ 308427965856403257033045344487944571498 \\ 492190534308029665037069211634968042718 \\ 586903984048367505303673231949601766248 \\ 866470467640636208882322899. \end{array} \end{aligned}$$

The anchor data and common due date are

$$\begin{aligned} \widehat{p}_a = & \widehat{H}, \\ \widehat{r}_a = 1 + \max_i \widehat{r}_i = & \begin{array}{l} 70316765459424392105577557081280 \\ 45311808406689815824393605991, \end{array} \\ \widehat{s}_a = 1 + \lceil \widehat{C}^* / \widehat{H}^2 \rceil = & 2, \\ \widehat{d} = & \begin{array}{l} 583489201983731785266607839650588849977 \\ 614462575543787905298044664841507417338 \\ 080675476883001957391261491767754056225 \\ 234822016539157297678918642006883335353. \end{array} \end{aligned}$$

Its tardy self-cost is

$$\hat{p}_a^2 \hat{s}_a = \begin{array}{l} 680919297663224297469522772039887703131 \\ 818185701605521353164494680402051534719 \\ 634615271115978950656748767054982151116 \\ 098783494343969724943804661695902292754 \\ 150322500412307014293134310459733809136 \\ 889835337744566351718894261991040961104 \\ 909767454545739800961024637863943069853 \\ 588098070640688084360565913330526581122. \end{array}$$

This value exceeds $\hat{C}^*$ by

$$\begin{array}{l} 680919297663224297469522772039887703131 \\ 818185701605253904400811269740681981943 \\ 836460707467079421048629681679688324481 \\ 650020896679264493949208283992786265569 \\ 763291477215998586327277907202700763792 \\ 401890766246074161184586232326003891893 \\ 274799411827152896976976270358639396621 \\ 638496304391821613892925277121644258223 \end{array} > 0.$$

Therefore no optimal canonical assignment can place the anchor tardy.

The following two tables list every parameter of the no-instance. The first gives the construction ratios; the second gives the actual integer input triples (p_j, w_j^E, w_j^L) .

Purpose. The table records the ratio-level construction data for all 49 jobs of the no-instance. **Verification rule.** Compute each block ratio from the hatted affine formulas and check the anchor from $\hat{p}_a = \hat{H}$, $\hat{r}_a = 1 + \max \hat{r}_j$, and $\hat{s}_a = 2$.

Table 27: All jobs of the no-instance: metadata, processing times, and early/tardiness ratios.

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{1,0}$	1	0	-	-1	17984650426474121466202803	703167654289430804612001528698990873317	93115851572934093495709703243946643628787667233
					40569649349251262	0093170770118042669332	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$B_{1,0}$	1	0	-	1	17984650426474121466202803	703167654289430804612001528698990873317	93115851572934093495709703243946643628787667233
					40569649349251259	0092931393319149833311	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$A_{1,1}$	1	1	e_1	1	17984650808482467120744482	703167654594243921055775570812804531180	93115851572934093495709703243946643628787667233
					55018260184595356	8406689815824393605990	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$B_{1,1}$	1	1	e_1	-1	17984650808482467120744482	703167654594243921055775570812804531180	93115851572934093495709703243946643628787667233
					55018260184595355	8406610023558095993983	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$A_{1,2}$	1	2	e_2	1	17984650744814409511654202	703167654543441734981813230460502254870	93115851572934093495709703243946643628787667233
					69276825045372255	2094086634446884232283	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$B_{1,2}$	1	2	e_2	-1	17984650744814409511654202	703167654543441734981813230460502254870	93115851572934093495709703243946643628787667233
					69276825045372254	2094006842180586620276	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$A_{1,3}$	1	3	e_4	1	17984650617478294293473642	703167654441837362833888549755897702249	93115851572934093495709703243946643628787667233
					97793954766933253	0043384589034671935269	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$B_{1,3}$	1	3	e_4	-1	17984650617478294293473642	703167654441837362833888549755897702249	93115851572934093495709703243946643628787667233
					97793954766933252	0043304796768374323262	86916059704220044810998282712867300913296782539
							93070330393800538505254986715372378711309344915
$A_{2,0}$	2	0	-	-1	17984650426474121466202803	703167654289430802993382972489263810813	93115851572934092328731299276483073095571987932
					40569649349251262	4786504883090001113824	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
							31420186913716966330866475618791

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Table 27 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$B_{2,0}$	2	0	–	1	17984650426474121466202803	703167654289430802993382972489263810813	93115851572934092328731299276483073095571987932
					40569649349251259	4786265506291108277806	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$A_{2,1}$	2	1	e_1	1	17984650808482467120744482	703167654594243919437157010782994012131	93115851572934092328731299276483073095571987932
					55018260184595356	8932109480185516706388	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$B_{2,1}$	2	1	e_1	–1	17984650808482467120744482	703167654594243919437157010782994012131	93115851572934092328731299276483073095571987932
					55018260184595355	8932029687919219094382	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$A_{2,2}$	2	2	e_2	1	17984650744814409511654202	703167654543441733363194671067372311912	93115851572934092328731299276483073095571987932
					69276825045372255	1647492040243146555782	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$B_{2,2}$	2	2	e_2	–1	17984650744814409511654202	703167654543441733363194671067372311912	93115851572934092328731299276483073095571987932
					69276825045372254	1647412247976848943776	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$A_{2,3}$	2	3	e_5	1	17984650553810236684383363	703167654391035175141307651920507211253	93115851572934092328731299276483073095571987932
					12052519627709252	0296330998090991741764	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$B_{2,3}$	2	3	e_5	–1	17984650553810236684383363	703167654391035175141307651920507211253	93115851572934092328731299276483073095571987932
					12052519627709251	0296251205824694129758	69146064181327529702240701653258367944995299072
							31456820856109317655552783876264065507663817677
$A_{3,0}$	3	0	–	–1	17984650426474121466202803	703167654289430799935992365607912052650	93115851572934091161752895309019502562356308631
					40569649349251262	2509485161553545702072	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
$B_{3,0}$	3	0	–	1	17984650426474121466202803	703167654289430799935992365607912052650	93115851572934091161752895309019502562356308631
					40569649349251259	2509245784754652866057	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
$A_{3,1}$	3	1	e_2	1	17984650744814409511654202	703167654543441730305804061002617673294	93115851572934091161752895309019502562356308631
					69276825045371355	4158730571863144218537	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
$B_{3,1}$	3	1	e_2	–1	17984650744814409511654202	703167654543441730305804061002617673294	93115851572934091161752895309019502562356308631
					69276825045371354	4158650779596846606532	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375

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Table 27 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{3,2}$	3	2	e_3	1	17984650681146351902563922	703167654492639544231841721923676549165	93115851572934091161752895309019502562356308631
					83535389906148254	5902098873355913291032	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
							59349530195481177324869116467352
$B_{3,2}$	3	2	e_3	-1	17984650681146351902563922	703167654492639544231841721923676549165	93115851572934091161752895309019502562356308631
					83535389906148253	5902019081089615679027	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
							59349530195481177324869116467352
$A_{3,3}$	3	3	e_6	1	17984650490142179075293083	703167654340232986009954704686853176779	93115851572934091161752895309019502562356308631
					26311084488485251	163489505509176140017	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
							59349530195481177324869116467352
$B_{3,3}$	3	3	e_6	-1	17984650490142179075293083	703167654340232986009954704686853176779	93115851572934091161752895309019502562356308631
					26311084488485250	1634815263242878528012	51376068658435014593483120593649434976693815604
							69843311318318315356974194910327256795667343375
							59349530195481177324869116467352
$A_{4,0}$	4	0	-	-1	17984650426474121466202803	703167654289430795439829711875019055372	93115851572934089994774491341555932029140629330
					40569649349251262	7430026054119511778082	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$B_{4,0}$	4	0	-	1	17984650426474121466202803	703167654289430795439829711875019055372	93115851572934089994774491341555932029140629330
					40569649349251259	7429786677320618942070	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$A_{4,1}$	4	1	e_1	1	17984650808482467120744482	703167654594243911883603742528582343600	93115851572934089994774491341555932029140629330
					55018260184595356	3239801753993356682458	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$B_{4,1}$	4	1	e_1	-1	17984650808482467120744482	703167654594243911883603742528582343600	93115851572934089994774491341555932029140629330
					55018260184595355	3239721961727059070454	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$A_{4,2}$	4	2	e_3	1	17984650681146351902563922	703167654492639539735679065644061247524	93115851572934089994774491341555932029140629330
					83535389906148254	4710696800181322470050	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$B_{4,2}$	4	2	e_3	-1	17984650681146351902563922	703167654492639539735679065644061247524	93115851572934089994774491341555932029140629330
					83535389906148253	4710617007915024858046	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469
$A_{4,3}$	4	3	e_6	1	17984650490142179075293083	703167654340232981513792050317279603410	93115851572934089994774491341555932029140629330
					26311084488485251	7527450206640002982038	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
							14930298500590302150691676328469

Continued on next page

Table 27 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$B_{4,3}$	4	3	e_6	-1	17984650490142179075293083	703167654340232981513792050317279603410	93115851572934089994774491341555932029140629330
					26311084488485250	7527370414373705370034	33606073135542499484725539534040502008392332137
							08229801780636165542338792468698097455180910375
$A_{5,0}$	5	0	—	-1	17984650426474121466202803	703167654289430789504895010017223666799	14930298500590302150691676328469
					40569649349251262	1492156077917620893852	93115851572934088827796087374092361495924950029
							15836077612649984375967958474431569040090848669
$B_{5,0}$	5	0	—	1	17984650426474121466202803	703167654289430789504895010017223666799	46616292242619899065634471060463331858477284346
					40569649349251259	1491916701118728057843	52192512790339020367895665603487
							93115851572934088827796087374092361495924950029
$A_{5,1}$	5	1	e_4	1	17984650617478294293473642	703167654441837347726782023433963582640	15836077612649984375967958474431569040090848669
					97793954766923353	2316597563266220602125	46616292242619899065634471060463331858477284346
							52192512790339020367895665603487
$B_{5,1}$	5	1	e_4	-1	17984650617478294293473642	703167654441837347726782023433963582640	93115851572934088827796087374092361495924950029
					97793954766923352	2316517770999922990122	15836077612649984375967958474431569040090848669
							46616292242619899065634471060463331858477284346
$A_{5,2}$	5	2	e_5	1	17984650553810236684383363	703167654391035161652819685628383610693	52192512790339020367895665603487
					12052519627700252	2115937347629268120822	93115851572934088827796087374092361495924950029
							15836077612649984375967958474431569040090848669
$B_{5,2}$	5	2	e_5	-1	17984650553810236684383363	703167654391035161652819685628383610693	46616292242619899065634471060463331858477284346
					12052519627700251	211585755362970508819	52192512790339020367895665603487
							93115851572934088827796087374092361495924950029
$A_{5,3}$	5	3	e_6	1	17984650490142179075293083	703167654340232975578857347822803638746	15836077612649984375967958474431569040090848669
					26311084488485251	2561594489002972863819	46616292242619899065634471060463331858477284346
							52192512790339020367895665603487
$B_{5,3}$	5	3	e_6	-1	17984650490142179075293083	703167654340232975578857347822803638746	93115851572934088827796087374092361495924950029
					26311084488485250	2561514696736675251816	15836077612649984375967958474431569040090848669
							46616292242619899065634471060463331858477284346
$A_{6,0}$	6	0	—	-1	17984650426474121466202803	703167654289430782131188266401331647838	52192512790339020367895665603487
					40569649349251262	4975732647299265289392	93115851572934087660817683406628790962709270727
							98066082089757469267210377414822636071789365201
$B_{6,0}$	6	0	—	1	17984650426474121466202803	703167654289430782131188266401331647838	85002782705094980631204591092522938946853885293
					40569649349251259	4975493270500372453386	93454728492742386605619043634360
							93115851572934087660817683406628790962709270727

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Table 27 (continued)

job	block	pair	element	ϵ	p_j	r_j	s_j
$A_{6,1}$	6	1	e_3	1	17984650681146351902563922	703167654492639526427037615076929231262	93115851572934087660817683406628790962709270727
					83535389906147354	9960704422212111385576	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
$B_{6,1}$	6	1	e_3	-1	17984650681146351902563922	703167654492639526427037615076929231262	93115851572934087660817683406628790962709270727
					83535389906147353	9960624629945813773574	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
$A_{6,2}$	6	2	e_4	1	17984650617478294293473642	703167654441837340353075277908029835406	93115851572934087660817683406628790962709270727
					97793954766924253	8788029948010298127374	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
$B_{6,2}$	6	2	e_4	-1	17984650617478294293473642	703167654441837340353075277908029835406	93115851572934087660817683406628790962709270727
					97793954766924252	8787950155744000515372	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
$A_{6,3}$	6	3	e_5	1	17984650553810236684383363	703167654391035154279112940739130439550	93115851572934087660817683406628790962709270727
					12052519627709252	8261672830819142085372	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
$B_{6,3}$	6	3	e_5	-1	17984650553810236684383363	703167654391035154279112940739130439550	93115851572934087660817683406628790962709270727
					12052519627709251	8261593038552844473370	98066082089757469267210377414822636071789365201
							85002782705094980631204591092522938946853885293
							93454728492742386605619043634360
a	—	—	—	—	58348920198373178526660783	703167654594243921055775570812804531180	2
					96505888499776144625755437	8406689815824393605991	
					87905298044664841507417338		
					08067547688300195739126149		
					17677539698989119727232302		
					44986696367878010576917919		

Purpose. The table records the physical positive-integer input triples for the 49-job no-instance. **Verification rule.** Multiply p_j by the corresponding ratios in Table 27; rowwise equality, not visual digit matching, is the certificate.

Table 28: The 49 actual AWET job triples for the no-instance.

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$A_{1,0}$	17984650426474121466202803405696493	12646224453599219326700846974282670279597026129	16746560397026701387320214737714657601440291316
	49251262	61652139863025619242619940896586728246666059570	91622347556170164600179076515908772870088969207
		2049696984	99377573858641721319379161878828010394234847246
$B_{1,0}$	17984650426474121466202803405696493	12646224453599219326700846974282670279597004604	16746560397026701387320214737714657601440263382
	49251259	07608788610078260320034171580781338690647304593	16075159533365293309081758116599909240071953133
		2506888549	17586447257297391370897775859800611490758649325
$A_{1,1}$	17984650808482467120744482550182601	12646224727697089120677578014552695268877323897	16746560752737025523224271676598136296199056090
	84595356	06677604243098360398422866205515276580837374858	55690453902937175813565839389910969239894796735
		1847782440	84100078457208783975562154009684909796375007001
$B_{1,1}$	17984650808482467120744482550182601	12646224727697089120677578014552695268877316721	16746560752737025523224271676598136296199046778
	84595355	88662544199216353209476715472931085732778022271	97174724562002218716533399923474681363222458044
		3369748965	23503036256760673992735025336675776828549607694
$A_{1,2}$	17984650744814409511654202692768250	12646224682014110805509073626383899931194130407	16746560693451971500573595520117556513747778958
	45372255	43578993571280069608331329667096180384194455681	17399540020316765363672677544635247707892948736
		7623508165	79407593700652204492078681566895714466983546541
$B_{1,2}$	17984650744814409511654202692768250	12646224682014110805509073626383899931194123232	16746560693451971500573595520117556513747769646
	45372254	25564035131770210344065883539064610798655739370	58883810679381808266640238078198959831220610045
		4164222104	18810551500204094509251552893886581499158147234
$A_{1,3}$	17984650617478294293473642977939547	12646224590648154194578923907794858663251052837	16746560574881863455272243207156396948912268106
	66933253	02993452081541193864210131986795683361101664072	54068966986767043097450512195356516684727832301
		9659600057	68726467413930921880438182347074692151075636260
			48075063518058252362573441199077365810518007043
			1851425011713908255542189293

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Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{1,3}$	17984650617478294293473642977939547 66933252	12646224590648154194578923907794858663251045661 84978696850775630449306095067869356185703356844 0624908024	16746560574881863455272243207156396948912258794 95553237645832086000418072728920228808055493610 08129425213482811897611053674065559183250236953 44771125512673199812706287475290252717068848242 0347255674305453822095640612
$A_{2,0}$	17984650426474121466202803405696493 49251262	12646224453599219297590558067046475716505954146 19102212161113076027669268533458739645299407636 0437645888	16746560397026701177443228231719324542982894787 36154838606506236897932795438971435602519275693 19297573656337166047547726219713652450757455916 18956215866939917760932231003305618540810785727 1858750705334837713987664242
$B_{2,0}$	17984650426474121466202803405696493 49251259	12646224453599219297590558067046475716505932620 65058860908171112500264922622484846765382589662 3067257754	16746560397026701177443228231719324542982866852 60607650583736374958954500946778568442881294949 37372133830446098826498128468675302591785286479 14330532587410251177415943081340388626280466301 1298009554435845114560807869
$A_{2,1}$	17984650808482467120744482550182601 84595356	12646224727697089091567288420287836777135608535 77876021126482465908299131306189041081381904733 9640334128	16746560752737025313347280712647908095859694706 99698817145193542301473347343991750876697984491 29701187474604190081026123977833056451512885340 48594013695465367148051083905043023998404065364 2337340090814978976044934596
$B_{2,1}$	17984650808482467120744482550182601 84595355	12646224727697089091567288420287836777135601360 59860961082602257184421095208795106625606087537 0223795610	16746560752737025313347280712647908095859685395 41183087804270254988480582513260795156818657576 69059374199307167674009591394153606498522162194 80385452602288811620212321264387947360227292222 2150426373848648109569315805
$A_{2,2}$	17984650744814409511654202692768250 45372255	12646224682014110776398784146623818963575266928 18012769055069458791362566792415221995104465968 1312628410	16746560693451971290696605299159810837055411716 94828591219912453684932153868402530113959217584 95012460976569514359901527556950046120235084353 65084927282867248380052837073930221474493685991 1554654978438921302068043705
$B_{2,2}$	17984650744814409511654202692768250 45372254	12646224682014110776398784146623818963575259752 99997810615561397992163961938421726996252136759 6636391104	16746560693451971290696605299159810837055402405 36312861878989166371939389037671574394079890670 34370647701272491952884994973270596167244361207 96876366189690692852214074433275144836316912849 1367741261472590435592424914
$A_{2,3}$	17984650553810236684383363120525196 27709252	12646224544965175869706989441128863851248154798 16224697312660602198248810789068529808576355767 6857600528	16746560515596809222744579058695519060701225733 29312761260779259689727007046655903065502478886 34369915853706651400683015474835718967957499190 28492555057372717460262732707574634416433342589 6763016529195207076535754332

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Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{2,3}$	17984650553810236684383363120525196 27709251	12646224544965175869706989441128863851248147622 98210043686268985173088499665276353079956234989 6490991258	16746560515596809222744579058695519060701216421 70797031919855972376734242215924947345623151971 73728102578409628993666482891156269014966776044 60283993964196161932423970066919557778256569447 6576102812228876210060135541
$A_{3,0}$	17984650426474121466202803405696493 49251262	12646224453599219242604456785099799770310546667 12049380552042200908158865524147643426296932975 1322014864	16746560397026700967566241725723991484525498257 80687329656842309195686514362034098334949582178 39217573454014665430945401525558817466939120620 29356652302437677030335408562087900316785538033 6102942742513510495667798224
$B_{3,0}$	17984650426474121466202803405696493 49251259	12646224453599219242604456785099799770310525141 58006029299109949092087957892092226313588202357 1365615763	16746560397026700967566241725723991484525470323 05140141634107456608827243776957227645690636765 57157820403576860937327592042509516652470979667 30017102752976967804488098744384030296484270228 7512356298981535888318396168
$A_{3,1}$	17984650744814409511654202692768250 45371355	12646224682014110721412681834134147570295808288 76945374039094077907254731399668014135010523278 0439807635	16746560693451971080819615078202065160354664049 08101235599052365948410454669108605742341862586 92691413121127530797292995045404643967822282801 20549000629887314218367259248503232937558143746 2926482948349729176573501960
$B_{3,1}$	17984650744814409511654202692768250 45371354	12646224682014110721412681834134147570295801113 58930415599589254345173639444408253434242572556 8662084328	16746560693451971080819615078202065160354654737 49585506258140748419457364474082982179255547449 32004828770981595966087058551054877029666235816 87435817446733744476418155975935276264124387811 3396287467172404307457034608
$A_{3,2}$	17984650681146351902563922835353899 06148254	12646224636331132412713130785828245396347299279 03285998663360599502497306935728057599385628069 2040658128	16746560634166917058168940407706450425197375201 36651697631110599165221229563205885748866176773 28006444881614758838590803556781621119634363187 26967419826112981993258052019577209956218068516 6469148435495785341662803408
$B_{3,2}$	17984650681146351902563922835353899 06148253	12646224636331132412713130785828245396347292103 85271141828227923865094372862716554549944017049 4724789831	16746560634166917058168940407706450425197365889 78135968290198981636268139368180262185779861635 67319860531468824007384867062431854181478316202 93854236642959412251308948747009253282784312581 6938952954318460472546336056
$A_{3,3}$	17984650490142179075293083263110844 88485251	12646224499282197525428195756407636716421097776 86205250964024478951005316064866596798094188798 7415389267	16746560456311754990216916396219606219784171644 71397931470475731220121782906926173212224486255 59432946082465879559884143494444262958166605631 59276731282279359597281047510926183644860233031 7328676314080629270675025352

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Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{3,3}$	17984650490142179075293083263110844 88485250	12646224499282197525428195756407636716421090601 68190698942008247087636855638599866602093341348 3573823000	16746560456311754990216916396219606219784162333 12882202129564113691168692711900549649138171117 98746361732319944728678207000094496020010558647 26163548099125789855331944238358226971426477096 7798480832903304401558558000
$A_{4,0}$	17984650426474121466202803405696493 49251262	12646224453599219161742543197145508006937069848 44569673162900640116541286924575844936687242053 7002439484	16746560397026700757689255219728658426068101728 25219820707178381493440233285096761067379888663 59137573251711741553060792801238722346465436174 47461634940002956789087622993102697484489663881 0376836818678167994324777878
$B_{4,0}$	17984650426474121466202803405696493 49251259	12646224453599219161742543197145508006937048322 90526321909982416327944372195156246431918414560 7195566130	16746560397026700757689255219728658426068073793 49672632684478538258699986607135886848499978581 76943506976727199786874771586178470576501323705 53408215855036686625313562050179042057178409401 9481335047771715919295792471
$A_{4,1}$	17984650808482467120744482550182601 84595356	12646224727697088955719270527460750116389911770 95504351002964072273090705220599386405163978151 3313465048	16746560752737024893593298784747451695180971939 87715543629706275277288363252153314150304360002 20903405509378688340783480462752748182699631408 62002023865590180598867419727826505152748834940 9611989535848482030507989964
$B_{4,1}$	17984650808482467120744482550182601 84595355	12646224727697088955719270527460750116389904595 77489290959091776795497093184216398408269937225 6426141170	16746560752737024893593298784747451695180962628 29199814288806327532374947692833022744011056641 60172050084383841085388140057732664259378260585 63984217503934757210942732746851953343645083447 9313488945546331338831661495
$A_{4,2}$	17984650681146351902563922835353899 06148254	12646224636331132331851216007024271176285942779 93619309945057338805996428400352823728775983909 6774792700	16746560634166916848291950929741187272152002102 47501436798548414955539846391458610037906126013 73321103052034422092278934307579398977282184486 81703272988834574506949921801935383857082165443 3940153884257086816314843126
$B_{4,2}$	17984650681146351902563922835353899 06148253	12646224636331132331851216007024271176285935604 75604453109929339177756585406161987437702933746 3955893638	16746560634166916848291950929741187272151992790 88985707457648467210626430832138318631612822653 12589747627039574836883593902559315053960813663 83685466627179151119025234820960832047978413950 3641653293954936124638514657
$A_{4,3}$	17984650490142179075293083263110844 88485251	12646224499282197444566281870740924460484297829 51494847816803320668653847515771344411659603251 3580921538	16746560456311754780339929147231790637679780972 82509734436411873060546099900645812013079483676 96816650576599526493876213478461388577642809902 47473813337690082992929079368420041557281082875 3961916218322785318937910719

Continued on next page

Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{4,3}$	17984650490142179075293083263110844 88485250	12646224499282197444566281870740924460484290654	16746560456311754780339929147231790637679771661
		33480295794791764814444658084868735557235871796	23994005095511925315632684341325520606786180316
		3400998500	36085295151604679238480873073441304654321439079
			49456006976034659605004392387445489748177331382
$A_{5,0}$	17984650426474121466202803405696493 49251262	12646224453599219055004817280282645237743434971	3663415628020634627261582250
		48637747284659178241999014382769209060794132716	16746560397026700547812268713733325367610705198
		3379041224	69752311757514453791193952208159423799810195148
			79057573049348727961486485142760368238188605908
$B_{5,0}$	17984650426474121466202803405696493 49251259	48743454134004401335931020538503852972424114388	8754890892206262410826350594
			16746560397026700547812268713733325367610677263
		7292574337	94205123734849619908572729437314546051309320397
			96729193549817448922732252195689165512728521923
$A_{5,1}$	17984650617478294293473642977939547 66923353	59976175537032432301799206638548098653893718731	1216519875145158723829540133
		9383925125	16746560574881862615764288267265274431226567587
			29119937714251945198116971397224650701482408806
			25430200313822778056639512346254966771471701131
$B_{5,1}$	17984650617478294293473642977939547 66923352	9985085074027336391018531911	96930609781157607945103430966359660746116387587
			81284121149147204910153906583131864821011192010
		4227128944	16746560574881862615764288267265274431226558275
			11105182483502208275763955544189671323476265371
$A_{5,2}$	17984650553810236684383363120525196 27700252	7472294735006968495352928424	66914470972934500095765771422640007521971409548
			36154483281313848265518686650669259837629692926
		3535847144	18361694950156548565443301949813280048167726791
			1461508743015542769521978724
$B_{5,2}$	17984650553810236684383363120525196 27700251	1461508743015542769521978724	16746560515596808593113615082754624743363267024
		2284013569	20822038260559654940154223979725248316522177503
			72253158058688643511320166406772474197108699235
			46345313072749566736468183798993976434009176561
$A_{5,3}$	17984650490142179075293083263110844 88485251	8948718403995174873856375237	1461508743015542769521978724
			16746560515596808593113615082754624743363257712
		4813033569	62306308919671376979280483056110289067021885920
			11477031558844883831735422091082073288622004573

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Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{5,3}$	17984650490142179075293083263110844 88485250	12646224499282197337828555564561819552362234636	16746560456311754570462941898243975055575380989
		80565272763084063519667283369360313431814092309	35105808061459736940096675970750491564434189514
		6151714000	73424228570829324034381114623742245481841075418
			20136290621677664373969386262074163306231971827
$A_{6,0}$	17984650426474121466202803405696493 49251262	12646224453599218922391279149015987405940085629	4026888137963336143748066750
		64380316462463892338619639658598411376998615740	16746560397026700337935282207737992309153308669
		0951212704	14284802807850526088947671131222086532240501633
			98977572847074081597692560063405226914736675130
$B_{6,0}$	17984650426474121466202803405696493 49251259	12646224453599218922391279149015987405940064104	10408112074003801776661668159494944945637414697
		10336965209586673553384734438689387776722152082	7696301974016820173296562320
		7979312974	16746560397026700337935282207737992309153280734
			38737614785220701558445472267493205254118662214
$A_{6,1}$	17984650681146351902563922835353899 06147354	12646224636331132092499948477949527526974189281	16514880122996065286370115384323073233780619629
		16203350670331324614118940092741528017622370785	26926959224584865638928892471326539329078596661
		6966165904	3510823746857003316165659240
			16746560634166916428537971973810660966052875478
$B_{6,1}$	17984650681146351902563922835353899 06147353	12646224636331132092499948477949527526974182105	05044508344476687385517420929299675179434198547
		98188493835216993320343287157605004487186569464	25142602169437043002872299102700389216681254948
		1021449622	09836236274229056332826792157986289356703443246
			3407778122974287325857483440
$A_{6,2}$	17984650617478294293473642977939547 66924253	12646224590648153790269351616841868401079688785	16746560634166916428537971973810660966052866166
		64464116097174977451760615431133473742492582443	46528779003600079208683354641390048086726918740
		9603801622	64321704594744370899098150876339671323029236447
			82009185324422744286915866928596820817850503900
$B_{6,2}$	17984650617478294293473642977939547 66924252	12646224590648153790269351616841868401079681610	8679285380587681706813849080
		46449360866432794082659300294788662446591675962	16746560574881862405887299532292493801836569057
		7285601744	33857034248552886102384100932456457493747756011
			70468908705489982331342029047458496588356167250
$A_{6,3}$	17984650553810236684383363120525196 27709252	12646224544965175494507707774983725339005033981	16520180945592449023658572655453446076347012635
		80157095836053327318370527486369821505345187775	1650217566185301295948133080
		8578261744	16746560574881862405887299532292493801836559745
			75341304907676277925550034644546830401040476205

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Table 28 (continued)

job	p_j	$w_j^E = p_j r_j$	$w_j^L = p_j s_j$
$B_{6,3}$	17984650553810236684383363120525196	12646224544965175494507707774983725339005026806	16746560515596808383236627090774326637695677164
	27709251	62142442209683291873943550148816722314715704732	81561491912278708998353646715683063644320467196
		8572145870	04244672677494290126638244287670824647728913305
			94489768098277101547073785978226298965946341008
a			5955377598515223900433464360
	58348920198373178526660783965058884	41029073363996773689167059673802225267845491754	11669784039674635705332156793011776999552289251
	99776144625755437879052980446648415	1960127998313592455401871503232330208645156563	51087575810596089329683014834676161350953766003
	07417338080675476883001957391261491	97088250697587781926300915200104309273202097967	91478252298353550793979782394544646048997339273
	76775396989891197272323024498669636	73124955288367843185783767946665701407970207230	5756021153835838
	7878010576917919	10119519188595586515233652729	

B.2.5 Master identity and decision threshold for the no-instance

The master decomposition is unchanged in form. For a coded state $x \in \{0,1\}^6$, with $\sigma = 2x - \mathbf{1}$, write

$$\hat{\mathcal{P}}(x) = \sum_{k=1}^6 \frac{\hat{\Delta}_k}{2} (\hat{C}_k(x) - k)^2$$

and

$$\hat{R}(\sigma) = \sum_{v=1}^6 \hat{e}_v \sigma_v + \sum_{1 \leq u < v \leq 6} (\hat{J}_{uv} - \hat{J}_{uv}^{\text{main}}) \sigma_u \sigma_v.$$

Then the exact identity is

$$\hat{\Phi}_{\text{can}}(\sigma) = \hat{A} + \hat{\mathcal{P}}(x) + \hat{R}(\sigma). \quad (\text{B.5})$$

There is no asymptotic term hidden in (B.5); all within-band and rounding effects are included in $\hat{R}$. The exact objective constant is

$$\hat{C}_0 = \frac{\begin{array}{c} 534897527366821322739105551596309127297 \\ 799059216238170750587653268897525195329 \\ 410461989192755406232054368774062046392 \\ 616855931712806514066090688975889142996 \\ 984381068616059330074138423269936085437 \\ 173807968096703513233692016041062418063 \\ 754462248811152089393704635 \end{array}}{2}.$$

Moreover,

$$\sum_{v=1}^6 \hat{D}_v = \frac{80221752587453752620342082754}{2356930},$$

and therefore

$$\hat{\Gamma} = \frac{9}{8}(6^2)\hat{W} + \frac{1}{8} \sum_v \hat{D}_v = \frac{\begin{array}{c} 18180323615823999998373485997 \\ 2834788526506620165 \end{array}}{4}.$$

The shifted constant is

$$\hat{A} = \frac{\begin{array}{c} 106979505473364264547821110319261825459 \\ 559811843247634150117530653779505039065 \\ 882092397838551081246410873754812409278 \\ 523371186342561302813218137795177828599 \\ 396876213723211866014827684653987217087 \\ 434761593619328510124156440620550243730 \\ 2657948362994797686001789057 \end{array}}{4}.$$

The decisive gap comparison is

$$\frac{\hat{\Delta}_{\min}}{2} = \frac{10160434876787887161601621019 \quad 18180323615823999998373485997}{63550374940926910024249} > \frac{2834788526506620167}{2} = 2\hat{\Gamma} + 1,$$

with positive margin

$$\frac{\hat{\Delta}_{\min}}{2} - (2\hat{\Gamma} + 1) = \frac{50793084222131523808008918355}{1788770076200201702041}.$$

Finally, the no-instance decision threshold is

$$\hat{B} = \left\lfloor \hat{A} + \hat{\Gamma} \right\rfloor = \frac{267448763683410661369552775798154563648 \quad 899529608119085375293826634448762597664 \quad 705230994596377703116027184387031023196 \quad 308427965856403257033045344487944571498 \quad 492190534308029665037069211634968042718 \quad 586903984048321275355841910590935609321 \quad 598202083957396553127102305}{1788770076200201702041}.$$

Thus a single nonzero integer prefix error costs more than the most favorable possible residual swing on both sides of the threshold comparison. This is the numerical fact that will stop the attempted forward construction and make the reverse supposition contradictory.

B.2.6 The “if” route: where the forward construction fails

The forward implication proved in [lemma I.17](#) starts with an exact-cover vector. The present source instance has no such vector, so there is no legitimate witness with which to begin that proof. To see the obstruction at the closest possible point, nevertheless try the former yes-witness

$$x^\dagger = (1, 0, 0, 0, 1, 0), \quad \sigma^\dagger = (+1, -1, -1, -1, +1, -1).$$

This produces the same formal early–tardy orientation pattern as in [Section B.1.10](#): blocks 1 and 5 use their +1 states and all other blocks use their −1 states. Every block is arithmetically synchronized and has its target tardy load. The construction fails only when the logical prefix equations are evaluated.

For this attempted witness,

$$\hat{C}(x^\dagger) = (1, 2, 2, 4, 5, 6), \quad \hat{C}(x^\dagger) - (1, \dots, 6) = (0, 0, -1, 0, 0, 0).$$

Therefore all terms in $\widehat{\mathcal{P}}$ vanish except the third one, and

$$\widehat{\mathcal{P}}(x^\dagger) = \frac{\widehat{\Delta}_3}{2} = \frac{10160434876787887161601621019}{2} \cdot \frac{63550374940926910024249}{2}.$$

The exact residual for this state is

$$\widehat{R}(\sigma^\dagger) = \frac{-3294787992173267513354522350}{4} \cdot \frac{016919741241278067}{4}.$$

Substitution in (B.5) gives the integer schedule cost

$$\widehat{\Phi}_\dagger = \widehat{A} + \widehat{\mathcal{P}}(x^\dagger) + \widehat{R}(\sigma^\dagger) = \frac{267448763683410661369552775798154563648}{376838278431939949645139872} \cdot \frac{899529608119085375293826634448762597664}{308427965856403257033045344487944571498} \cdot \frac{705230994596377703116027184387031023196}{492190534308029665037069211634968042718} \cdot \frac{586903984048321783331311243947690372528}{308427965856403257033045344487944571498}.$$

It misses the required threshold by

$$\widehat{\Phi}_\dagger - \widehat{B} = \frac{50797546933335675476320677863}{6194474543396518037567} > 0.$$

This calculation is the direct numerical failure of the attempted “if” construction: the former cover still passes the block and load gadgets, but the uncovered element e_3 creates one unit of third-prefix error, and the main gap converts that single unit into an insurmountable cost.

The former witness is not quite the least expensive coded state. Exact enumeration shows that the minimum coded cost occurs at

$$\widehat{x}^{\text{best}} = (0, 1, 0, 0, 0, 1), \quad \widehat{C}(\widehat{x}^{\text{best}}) = (1, 2, 3, 4, 6, 6),$$

so its sole error is at the fifth prefix. For this state,

$$\widehat{\mathcal{P}}(\widehat{x}^{\text{best}}) = \frac{10160435380358105851691128636}{2} \cdot \frac{89946286355926164549607}{2},$$

$$\widehat{R}(2\widehat{x}^{\text{best}} - \mathbf{1}) = \frac{-5625598711919279964842548215}{4} \cdot \frac{293427898608008223}{4},$$

and

$$\hat{\Phi}_{\text{can}}(2\hat{x}^{\text{best}} - \mathbf{1}) - \hat{B} = \frac{50797491180918775276456928749}{2926111123856803617707} > 0.$$

Hence even the most favorable coded assignment is above the threshold. [Table 29](#) supplies the exhaustive $2^6 = 64$ -state check. The last column is the exact value of $\hat{\Phi}_{\text{can}} - \hat{B}$; every entry is positive.

Purpose. The table exhausts all 64 no-instance coded states and certifies that none reaches the decision threshold. **Verification rule.** Compute the hatted prefix totals, penalty, residual, and $\hat{A} + \hat{P} + \hat{R} - \hat{B}$; every final entry must be positive.

Table 29: Exact audit of all 64 coded states for the no-instance.

x	$\widehat{C}(x)$	$\widehat{C}(x) - (1, \dots, 6)$	RPE? $\widehat{P}(x)$	$\widehat{R}(\sigma)$	$\widehat{\Phi}_{\text{can}} - \widehat{B}$	
000000	0,0,0,0,0,0	-1,-2,-3,-4,-5,-6	no	92459962716614084396639698718108 541805605806098669265/2	47695293424111348295371716423956 535537330164009/4	462299478313213586661569272682683836832 39655755220594
000001	0,0,1,2,3,3	-1,-2,-2,-2,-2,-3	no	26417132704720167841790484498028 594418372890703566529/2	18927046305926658913194663649671 590772513624417/4	132085256333126208425599746139652164183 87006853534328
000010	0,0,0,1,2,3	-1,-2,-3,-3,-3,-3	no	41657785361610361398637486958024 441764142315834388849/2	27077689966468620937625082912483 704892294337173/4	208288539994186327564739819515679557943 00249364123677
000011	0,0,1,3,5,6	-1,-2,-2,-1,0,0	no	10160435966657714682655147378690 040914454940867145461/2	92297226979195694573689028056761 15237483890897/4	508017483995049226121994209785572866755 9148177890414
000100	1,1,2,2,2,3	0,-1,-1,-2,-3,-3	no	12192522571531490496380005774337 010690946276991319485	13423743216239965821231612833885 425029852265211/4	121924804766582549963714651485252259536 05402827730747
000101	1,1,3,4,5,6	0,-1,0,0,0,0	no	50802186073962339078941124128825 6631698507230927505	-3201267808843462710537441503053 251376084059453/4	507975609613631619923737673212887659688 531583257601
000110	1,1,2,3,4,6	0,-1,-1,-1,-1,0	no	20320871264286427464741114872878 41366275449744009737	14531597713551122267180440923504 97392238893467/4	203204203890954602525217223308387124520 2666177078063
000111	1,1,3,5,7,9	0,-1,0,1,2,3	no	76203267241074105603169542746118 60575800450197547431	-4251571404092678403130137577110 823903691337881/4	762028021040551997714735755836247308939 7342648057920
001000	0,1,2,2,2,3	-1,-1,-1,-2,-3,-3	no	25401088767425114274419095062858 134935138697410676787/2	14071190638337216628067617653802 726142846305193/4	127005024507011771615137086146184877095 53752790259651
001001	0,1,3,4,5,6	-1,-1,0,0,0,0	no	2032087345841380632379059967606 26816643157889892827/2	-3676062584512840776305108646670 405820944609687/4	10159973030960043430876298838815853202 2992082138951
001010	0,1,2,3,4,6	-1,-1,-1,-1,-1,0	no	50802178772194187746073064887597 96285797042916057291/2	79132462931489434638552328564870 3132731770317/4	254006368563182715602724390704572634637 7173014316184
001011	0,1,3,5,7,9	-1,-1,0,1,2,3	no	16256697072576954402292992063407 834704847043823132679/2	-6035648743899525156066330347347 073721053051247/4	812830157656725166626521108140633730695 7960021648487
001100	1,2,4,4,4,6	0,0,1,0,-1,0	no	10160435128572996506646374828267 48330648426537286928	-5309068861776614951098067720473 748007851072041/4	101599673478104464651090377459482500351 4292947863877
001101	1,2,5,6,7,9	0,0,2,2,2,3	no	10668457097220523716908055526769 213905769344714054336	-1091308599152541150506279558360 0611612761771993/4	106684089181399862755551833273553247969 19309896956297
001110	1,2,4,5,6,9	0,0,1,1,1,3	no	60962613001534656498539547887704 59702038517174807640	-9941914858041063469547097318689 868010543151825/4	609621336386571157958809146828113682087 4382912364643
001111	1,2,5,7,9,12	0,0,2,3,4,6	no	33021415192987324635530810153085 698545932205565504722	-4625652138154222121590952514339 376505447758461/4	330213685857652505369752838216325767523 90947576910066
010000	1,2,2,2,3,3	0,0,-1,-2,-2,-3	no	18288783731404681285335624747647 776686715908698035925/2	11560533412276668010621067110899 167951266827885/4	914434930502665415183481909537567285945 2810539069893
010001	1,2,3,4,6,6	0,0,0,0,1,0	no	10160435380358105851691128636899 46286355926164549607/2	-5625598711919279964842548215293 427898608008223/4	507974911809187752764569287492926111123 856803617707
010010	1,2,2,3,5,6	0,0,-1,-1,0,0	no	20320869934422281262002876283092 23182797958861614857/2	-1719332597407802070195571912815 137741390730999/4	101599761602892515114963033154664017891 7412456469638

Continued on next page

Table 29 (continued)

x	$\widehat{C}(x)$	$\widehat{C}(x) - (1, \dots, 6)$	RPE? $\widehat{P}(x)$	$\widehat{R}(\sigma)$	$\widehat{\Phi}_{\text{can}} - \widehat{B}$	
010011	1,2,3,5,8,9	0,0,0,1,3,3	no	19304827416914627264900650385096 939319983516755987887/2	-7985184871968112143738314571530 378481259473791/4	965236626135205608042229332425483356870 0006436470455
010100	2,3,4,4,5,6	1,1,1,0,0,0	no	15240654167600843896990340493620 88595792042399958538	-6927687417108371279413779974891 006987194163565/4	152401823402919055260621826220210166434 3163974762606
010101	2,3,5,6,8,9	1,1,2,2,3,3	no	14224609615230740211449790684374 393029980739070374767	-1197058344820305840446939686373 7334478188784745/4	142245611717758386006851936333101838869 49987896523540
010110	2,3,4,5,7,9	1,1,1,1,2,3	no	86363702801278715592265770826856 92539893985366578464	-1156053341403496759699735422866 7409672429267357/4	863632193918547849048468189963214216434 4435632606584
010111	2,3,5,7,10,12	1,1,2,3,5,6	no	38609654787069162300410771038070 770242855452250924367	-5683149595494016820132098450036 382053417795221/4	386096079154727238669065700713311645250 62807269820521
011000	1,3,4,4,5,6	0,1,1,0,0,0	no	20320872091580354977389845845400 63638337941371879259/2	-5675955734925997583451600810440 681012604957967/4	101599673478104445737010049565483600030 1585908045097
011001	1,3,5,6,8,9	0,1,2,2,3,3	no	27433175606099347141240497854564 672506715334712711717/2	-1184109396378731358111088966282 1464061454169363/4	137165393919671430637918577158449273392 94520366158477
011010	1,3,4,5,7,9	0,1,1,1,2,3	no	16256696935893609836794070651187 271526541827305119111/2	-1161808429599006258820370069083 6179070341224891/4	812830011261669136088139234095346984552 9014440598292
011011	1,3,5,7,10,12	0,1,2,3,5,6	no	76203265949776191319162458561957 426932464761073810917/2	-6862942675215740683942116875739 607009184342971/4	381015858083433872956460623617345013226 33496614164675
011100	2,4,6,6,7,9	1,2,3,2,2,3	no	15748675071557055711295647503923 173146538762626843887	-9272885856553485978512029680436 219594285427853/4	157486273025265520129241569422007598287 86732428831883
011101	2,4,7,8,10,12	1,2,4,4,5,6	no	49786133565182802704880416715926 697215096147234419504	-3294787992313541125763320095470 734284254424321/4	497860872906767650664951393413816801387 15444544158383
011110	2,4,6,7,9,12	1,2,3,3,4,6	no	38101633167323856629918402130239 657786701623478294273	-6567994404859677220031064790893 814644599239685/4	381015860745162158549991011887584668545 50830701829311
011111	2,4,7,9,12,15	1,2,4,5,7,9	no	89411831969420238542936608662615 955124031778299799564	10330383309015905534638517461549 025775437857163/4	894117891012070262369129963885303273025 91090532608814
100000	1,2,2,3,3,3	0,0,-1,-1,-2,-3	no	15240653214414363055215248168610 758263144812843264101/2	10330383309213982479634994521720 348166130939217/4	762028373899396927110324805933901635296 2316327711814
100001	1,2,3,5,6,6	0,0,0,1,1,0	no	20320870436992499952092383900356 19094212958116140215/2	-6567994404643902772347586756808 658162946971883/4	101599642904198427662893017440612713624 4806694672096
100010	1,2,2,4,5,6	0,0,-1,0,0,0	no	10160434876787887161601621019635 50374940926910024249/2	-3294787992173267513354522350016 919741241278067/4	507975469333356754763206778636194474543 396518037567
100011	1,2,3,6,8,9	0,0,0,2,3,3	no	22352957933904945495021026964133 957743554612610759711/2	-9272885856395514863416230961068 570960313095851/4	111764311979219690886318016942942453959 37434600450852
100100	2,3,4,5,5,6	1,1,1,1,0,0	no	20320871695918040947190968125349 24999720558375753842	-6862942675050471822135199869054 723229534963585/4	203204000304709577210114534501996452734 2619365357905
100101	2,3,5,7,8,9	1,1,2,3,3,3	no	16764718379389338736550104500238 575049623318949351287	-1161808429580709622370978271023 7461199732659757/4	167646700240592252247760526390779042815 60887389531307
100110	2,3,4,6,7,9	1,1,1,2,2,3	no	10160435538623030674286765372204 201751679533293964376	-1184109396367984805189165197085 4088129484725777/4	101603871275405001943247564655762158294 60369296127891
100111	2,3,5,8,10,12	1,1,2,4,5,6	no	42165807056891200235551210380280 625070355064081491495	-5675955734800834551545362144559 470989676328633/4	421657601870932269753425765602250957217 90180535754296
101000	1,3,4,5,5,6	0,1,1,1,0,0	no	30481307148214749077791101108857 36446194973323469867/2	-5683149595541749087696004172059 881370077953007/4	152401848581429900845228719772683199943 0012515591641

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Table 29 (continued)

x	$\widehat{C}(x)$	$\widehat{C}(x) - (1, \dots, 6)$	RPE? $\widehat{P}(x)$	$\widehat{R}(\sigma)$	$\widehat{\Phi}_{\text{can}} - \widehat{B}$	
101001	1,3,5,7,8,9	0,1,2,3,3,3	no	32513393134416544191441125486293 036546000494470664757/2	-1156053341406500236187425897677 7074898130239395/4	162566482262658790194699763408667808700 34391076117489
101010	1,3,4,6,7,9	0,1,1,2,2,3	no	19304827452883928066914447230224 289950112923159890935/2	-1197058344830859400462098190047 8341642528878331/4	965236528298706239630872652615167664677 3919321070844
101011	1,3,5,8,10,12	0,1,2,4,5,6	no	83315570489420267189443337246377 136587463984734945173/2	-6927687417196209376878364037718 180060575071403/4	416577380619792397356693284698825656554 89845597049695
101100	2,4,6,7,7,9	1,2,3,3,2,3	no	18288783835715654236395961319787 355166181342505820407	-7985184872048652494491780347040 316408963334245/4	182887363886103966642328417631272751974 05108638331805
101101	2,4,7,9,10,12	1,2,4,5,5,6	no	54358329340668280050060981584482 224850452791016577240	-1719332597470644918262036714411 241578135405705/4	543582834600260911223997560852580530389 45264856070773
101110	2,4,6,8,9,12	1,2,3,4,4,6	no	41657785437145894565058841472449 512614201235308861401	-5625598712057623648183693305520 873673991804477/4	416577385799371769906529334928111930252 85685184255241
101111	2,4,7,10,12,15	1,2,4,6,7,9	no	95000071250569155298157299057517 155567245454033547908	11560533412156021829966922994585 556266842217379/4	950000286898934687771627606155329110049 37389117447212
110000	2,4,4,5,6,6	1,2,1,1,1,0	no	81283490416571591193437739364893 36075988086301622321/2	-4625652138075915195571937172404 334310574036979/4	406412791360650548069309214154538172821 3333880646875
110001	2,4,5,7,9,9	1,2,2,3,4,3	no	43689872689467091914020466494036 313893929164436114853/2	-9941914857945059040841081002840 991724710244595/4	218448884084457919107454771030329130280 19519413841237
110010	2,4,4,6,8,9	1,2,1,2,3,3	no	28449219931862854619155562510587 674725329740796241817/2	-1091308599150490791163145955638 3077265567986311/4	142245617868508898733508074137139550581 98422379469290
110011	2,4,5,8,11,12	1,2,2,4,6,6	no	98556224196614057252699129708880 199080816359358593697/2	-5309068861738413854979730719342 379569698100611/4	492780653202307736317461051757924264961 16155628116655
110100	3,5,6,7,8,9	2,3,3,3,3,3	no	24893067421046497575754424956729 674370432909213888675	-6035648743854026313236875058898 470137223032433/4	248930204613252720522478507137959164371 18243281475526
110101	3,5,7,9,11,12	2,3,4,5,6,6	no	64010743540106555144926783812494 382913772136218294329	79132462937809069190197954801114 0807520974879/4	640106982871286729294494608542742767078 60206471883008
110110	3,5,6,8,10,12	2,3,3,4,5,6	no	50294156098548359074755530836771 724391164654346028883	-3676062584525145266063332672939 310084794526673/4	502941097287236733834692183872235629476 40001520742174
110111	3,5,7,10,13,15	2,3,4,6,8,9	no	10668457252607905156336132701290 9206203276651564364211	14071190638342609640996394601447 655969955573955/4	106684530593067671589013741328292863356 493512426602659
111000	2,5,6,7,8,9	1,3,3,3,3,3	no	46738003969006595306531599370907 008081127388143663899/2	-4251571404260180689671505017370 603416170856239/4	233689554708014070282206313338622564892 15708402462849
111001	2,5,7,9,11,12	1,3,4,5,6,6	no	12497335620712671044487631708243 6425167805842152475207/2	14531597712053074428636776260045 51970718560857/4	62486633016044258463765023324226258763 43782129222777
111010	2,5,6,8,10,12	1,3,3,4,5,6	no	97540181324010318304533811130991 108122590878407944315/2	-3201267809068768329666488258030 538736243513611/4	487700444108791673250748272151584963449 63623516438714
111011	2,5,7,10,13,15	1,3,4,6,8,9	no	21032101417907170328174540348326 6071746814872844614971/2	13423743216032357704789567052821 971760651996801/4	105160464994662616088962132005309805870 203244958651645
111100	3,6,8,9,10,12	2,4,5,5,5,6	no	66550852800641509884743271210053 451447938264024873146	92297226977192590876379753119298 97113960669871/4	665508096572631447545580471858322862217 15410888385573
111101	3,6,9,11,13,15	2,4,6,7,8,9	no	12700544321485665862574964264280 9379625646178966438188	27077689966286008070581156392651 320859730301895/4	127005404533470110637251664354383484579 779262272358621
111110	3,6,8,10,12,15	2,4,5,6,7,9	no	10719259471054238513283958868308 8382164730927041843814	18927046305668545210876056841207 864801310467591/4	107192553991494921989975895468387599257 999995742805671

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Table 29 (continued)					
x	$\widehat{C}(x)$	$\widehat{C}(x) - (1, \dots, 6)$	RPE? $\widehat{\mathcal{P}}(x)$	$\widehat{R}(\sigma)$	$\widehat{\Phi}_{\text{can}} - \widehat{B}$
111111	3, 6, 9, 12, 15, 18	2, 4, 6, 8, 10, 12	no	18491992543322816879327939743621 7083611211612197338530	47695293423870932095740110589406 184919891906242485201012425437529737754 175394842231722

The table gives a stronger check than the reduction itself needs. Source-level reasoning already ruled out every exact cover. The numerical audit additionally verifies, without appealing to an asymptotic estimate, that no one of the 64 canonical schedules reaches the constructed bound.

B.2.7 The “only if” route: the hypothetical target witness is impossible

Now run the reverse implication in full. Suppose, for contradiction, that the constructed no-instance admitted a schedule of cost at most $\hat{B}$. Choose an optimal schedule and replace it, using [lemma I.3](#), by an optimal canonical schedule.

Gate 1: the anchor must be early in the selected canonical schedule. The all-rejected coded state provides a feasible benchmark of cost $\hat{C}^*$, and

$$\hat{C}^* - \hat{B} = \frac{46229947831321358666156927268}{268383683239655755220594} > 0.$$

If the anchor were tardy, its tardy self-cost alone would be

$$\hat{p}_a^2 \hat{s}_a = \frac{680919297663224297469522772039887703131}{818185701605521353164494680402051534719} \\ \frac{634615271115978950656748767054982151116}{098783494343969724943804661695902292754} \\ \frac{150322500412307014293134310459733809136}{889835337744566351718894261991040961104} \\ \frac{909767454545739800961024637863943069853}{588098070640688084360565913330526581122}.$$

This value exceeds $\hat{C}^*$, and hence $\hat{B}$, by a vast margin. Therefore the selected optimal canonical schedule keeps the anchor early.

Gate 2: every block must have its target load and one of its two coded states. If the vector of block-load errors is a nonzero integer vector, positive definiteness of the minimum kernel makes the load-lock loss at least $\hat{H}$. All other assignment-dependent oscillation is bounded in absolute value by $\hat{U}$. The exact slack is

$$\hat{H} - \hat{U} = \frac{35131387587510277603106815554480}{53742461249340243125211289344295} > 0.$$

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Thus a low-cost schedule cannot carry any block-load error. The arithmetic synchronization argument then leaves, in each of the six blocks, only the displayed state and its complement.

Consequently the whole schedule decodes to a binary vector $x \in \{0, 1\}^6$.

Gate 3: every prefix equation would have to be exact. Because the source audit in Table 21 proves that no binary vector is prefix-exact, the decoded vector has at least one nonzero integer error $\hat{C}_k(x) - k$. Therefore

$$\hat{\mathcal{P}}(x) \geq \frac{\hat{\Delta}_{\min}}{2}.$$

Using $\hat{R}(\sigma) \geq -\hat{\Gamma}$ in (B.5) yields

$$\begin{aligned} \hat{\Phi}_{\text{can}}(\sigma) - \hat{B} &\geq \hat{A} + \frac{\hat{\Delta}_{\min}}{2} - \hat{\Gamma} - \hat{B} \\ &= \frac{10158616844426304761601783671}{2} \\ &= \frac{03577540152400403404085}{2} > 0. \end{aligned}$$

This contradicts the supposed cost bound. Equivalently, gap dominance would force $\hat{\mathcal{P}}(x) = 0$; then $\hat{C}_k(x) = k$ for every k , and successive differences would give

$$\hat{A}^{\text{inc}}x = \mathbf{1}.$$

But $\hat{C}_6(x) = 6$ forces exactly two selected columns, whereas every pair of columns intersects. The decoded exact cover demanded by the reverse implication therefore cannot exist. This closes the contradiction from the hypothetical “only if” starting assumption.

B.2.8 Side-by-side conclusion

The two examples differ only by exchanging e_3 and e_4 between columns S_1 and S_3 , yet the reduction records that local combinatorial change exactly.

Table 30: What changes between the closely related yes- and no-instances.

feature	yes-instance in Section B.1	no-instance in Section B.2
source structure	$S_1 \cap S_5 = \emptyset$ and $S_1 \cup S_5 = X$	every pair of subset-columns intersects
closest vector	$x = (1, 0, 0, 0, 1, 0)$	$x^\dagger = (1, 0, 0, 0, 1, 0)$
prefix totals	$(1, 2, 3, 4, 5, 6)$	$(1, 2, 2, 4, 5, 6)$
main penalty	$\mathcal{P}(x) = 0$	$\hat{\mathcal{P}}(x^\dagger) = \hat{\Delta}_3/2$
threshold test	$\Phi_{\text{cover}} - B < 0$	$\hat{\Phi}_\dagger - \hat{B} > 0$, and all 64 coded states are above $\hat{B}$
logical outcome	exact cover decodes to a low-cost schedule	a low-cost schedule would decode to a nonexistent exact cover

In the yes-instance, the forward route begins with a genuine exact cover, annihilates every prefix square, and reaches the decision bound. In the no-instance, the nearest analogous assignment incurs one third-prefix square; more generally, the reverse route shows that any schedule reaching the bound would have to annihilate all prefix squares and hence create an exact cover that the source incidence matrix forbids. The two examples therefore exhibit both sides of the reduction, including the precise numerical mechanism by which the no-instance is rejected.

C Glossary and Notation

Purpose. The table fixes the principal terminology and symbol meanings used across the five Parts and appendices. **Verification rule.** Treat construction-specific symbols as local unless the scope column or an explicit notation reset states otherwise.

Term or symbol	Meaning in this tutorial
AWET	Single-machine asymmetric weighted earliness–tardiness scheduling with common due date $d = \sum_j p_j$.
p_j, w_j^E, w_j^L	Processing time, earliness weight, and tardiness weight of job j .
r_j, s_j	Smith-type ratios w_j^E/p_j and w_j^L/p_j for the early and tardy arms.
E, T	Early/on-time and tardy job sets in a canonical compact V-shaped schedule.
Canonical side objective	The exact quadratic set function in (2.1); it is the common starting point of all four technical Parts.
RXC3	Restricted Exact Cover by 3-Sets, the strong-hardness source in Part I.
RPE	Regular prefix exactness, the intermediate source whose squared prefix errors are compiled into Part I’s scheduling objective.
Anchored SDP	The Part II relaxation obtained after forcing one enumerated job to have early marginal one.
e_i, t_i, e_{ij}, t_{ij}	Part II pseudo-marginal early/tardy masses and pseudo-joint masses.
Minimum kernel	A matrix with entries $\min\{q_i, q_j\}$; its Gram representation is the positive-semidefinite engine of the tardy payment in Part II.
Strict antithetical	The class with $r_1 < \dots < r_n$ and $s_1 > \dots > s_n$ after one common indexing.
Q, S, U, M, D in Part III	Total tagged load, selected early tagged load, residual budget $3Q^3$, dominant scale $8Q^3$, and due-date scale $M + Q$.
Ratio-order refinements	Fixed total orders $\prec_E, \prec_L$ refining nondecreasing r_j and s_j ; ties may be broken deterministically because equal-ratio swaps do not alter pair coefficients.
Ratio permutation	After indexing jobs by $\prec_E$, the permutation of their ranks in $\prec_L$.
Inversion graph $G_\times$	The graph joining exactly those job pairs whose relative orders in $\prec_E$ and $\prec_L$ disagree.
Separable permutation	A permutation recursively constructed by direct and skew sums; equivalently, its inversion graph is a cograph.
Separating-tree block $J(M)$	The exact leaf set below separating-tree node M ; it is an interval in both ratio orders.
Ordered separating tree	A binary direct/skew decomposition tree for a separable ratio permutation, with left child earlier in $\prec_E$.
z_M	Part IV’s additive block signature: early processing load, tardy processing load, earliness weight of early jobs, and tardiness weight of tardy jobs.
(e_M, t_M, u_M, v_M)	

Term or symbol	Meaning in this tutorial
$\mathcal{D}_M(z)$	Minimum internal canonical cost of a side assignment on separating-tree block $J(M)$ having signature z ; the exact Part IV dynamic-programming table.
Geometric box	A four-tuple of logarithmic coordinate buckets, with zero kept in a singleton bucket, used to trim the exact separating-tree table.

D Citing This Document

The title is *Asymmetric Weighted Earliness Tardiness: Scheduling with a Nonrestrictive Common Due Date*, version dated September 11, 2026. Citations to a particular result should include its Part-based theorem number and, where helpful, the descriptive Part title. The front-matter provenance note states the tutorial’s policy for historical priority claims.

References

- [1] F. Alizadeh. Interior point methods in semidefinite programming with applications to combinatorial optimization. *SIAM Journal on Optimization*, 5(1):13–51, 1995. doi: [10.1137/0805002](https://doi.org/10.1137/0805002).
- [2] M. H. Albert and M. D. Atkinson. Simple permutations and pattern restricted permutations. *Discrete Mathematics*, 300(1–3):1–15, 2005. doi: [10.1016/j.disc.2005.06.016](https://doi.org/10.1016/j.disc.2005.06.016).
- [3] K. R. Baker and G. D. Scudder. Sequencing with earliness and tardiness penalties: A review. *Operations Research*, 38(1):22–36, 1990. doi: [10.1287/opre.38.1.22](https://doi.org/10.1287/opre.38.1.22).
- [4] P. Bose, J. F. Buss, and A. Lubiw. Pattern matching for permutations. *Information Processing Letters*, 65(5):277–283, 1998. doi: [10.1016/S0020-0190\(97\)00209-3](https://doi.org/10.1016/S0020-0190(97)00209-3).
- [5] D. G. Corneil, Y. Perl, and L. K. Stewart. A linear recognition algorithm for cographs. *SIAM Journal on Computing*, 14(4):926–934, 1985. doi: [10.1137/0214065](https://doi.org/10.1137/0214065).
- [6] M. R. Garey and D. S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, San Francisco, 1979.
- [7] T. F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985. doi: [10.1016/0304-3975\(85\)90224-5](https://doi.org/10.1016/0304-3975(85)90224-5).
- [8] M. X. Goemans and D. P. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM*, 42(6):1115–1145, 1995. doi: [10.1145/227683.227684](https://doi.org/10.1145/227683.227684).
- [9] V. S. Gordon, J.-M. Proth, and C. Chu. A survey of the state-of-the-art of common due date assignment and scheduling research. *European Journal of Operational Research*, 139(1):1–25, 2002. doi: [10.1016/S0377-2217\(01\)00181-3](https://doi.org/10.1016/S0377-2217(01)00181-3).
- [10] M. Habib and C. Paul. A survey of the algorithmic aspects of modular decomposition. *Computer Science Review*, 4(1):41–59, 2010. doi: [10.1016/j.cosrev.2010.01.001](https://doi.org/10.1016/j.cosrev.2010.01.001).
- [11] N. G. Hall, W. Kubiak, and S. P. Sethi. Earliness–tardiness scheduling problems, II: Deviation of completion times about a restrictive common due date. *Operations Research*, 39(5):847–856, 1991. doi: [10.1287/opre.39.5.847](https://doi.org/10.1287/opre.39.5.847).

-
- [12] N. G. Hall and M. E. Posner. Earliness–tardiness scheduling problems, I: Weighted deviation of completion times about a common due date. *Operations Research*, 39(5):836–846, 1991. doi: [10.1287/opre.39.5.836](https://doi.org/10.1287/opre.39.5.836).
- [13] J. A. Hoogeveen and S. L. van de Velde. Earliness–tardiness scheduling around almost equal due dates. *INFORMS Journal on Computing*, 9(1):92–99, 1997. doi: [10.1287/ijoc.9.1.92](https://doi.org/10.1287/ijoc.9.1.92).
- [14] J. J. Kanet. Minimizing the average deviation of job completion times about a common due date. *Naval Research Logistics Quarterly*, 28(4):643–651, 1981. doi: [10.1002/nav.3800280411](https://doi.org/10.1002/nav.3800280411).
- [15] H. Kellerer, K. Rustogi, and V. A. Strusevich. A fast FPTAS for single machine scheduling problem of minimizing total weighted earliness and tardiness about a large common due date. *Omega*, 90:101992, 2020. doi: [10.1016/j.omega.2018.11.001](https://doi.org/10.1016/j.omega.2018.11.001).
- [16] M. Y. Kovalyov and W. Kubiak. A fully polynomial approximation scheme for the weighted earliness–tardiness problem. *Operations Research*, 47(5):757–761, 1999. doi: [10.1287/opre.47.5.757](https://doi.org/10.1287/opre.47.5.757).
- [17] S. S. Panwalkar, M. L. Smith, and A. Seidmann. Common due date assignment to minimize total penalty for the one machine scheduling problem. *Operations Research*, 30(2):391–399, 1982. doi: [10.1287/opre.30.2.391](https://doi.org/10.1287/opre.30.2.391).
- [18] G. A. Rolim and M. S. Nagano. Structural properties and algorithms for earliness and tardiness scheduling against common due dates and windows: A review. *Computers & Industrial Engineering*, 149:106803, 2020. doi: [10.1016/j.cie.2020.106803](https://doi.org/10.1016/j.cie.2020.106803).
- [19] M. Grötschel, L. Lovász, and A. Schrijver. *Geometric Algorithms and Combinatorial Optimization*. Springer, Berlin, 1988.
- [20] L. Vandenberghe and S. Boyd. Semidefinite programming. *SIAM Review*, 38(1):49–95, 1996.